

Canonical GHZ configurations - cluster P1 - helicity flipping

species: electron
polarization cluster: P1
source cluster configurations: 111
canonical configurations collected: 46

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity flipping
classification: complete-chain R_flip/conserved > 1
per-page labels: isolated flip/conserved and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 72 configurations
- 3 components: 1 configurations
- 4 components: 9 configurations
- 5 components: 7 configurations
- 6 components: 18 configurations
- 7 components: 4 configurations

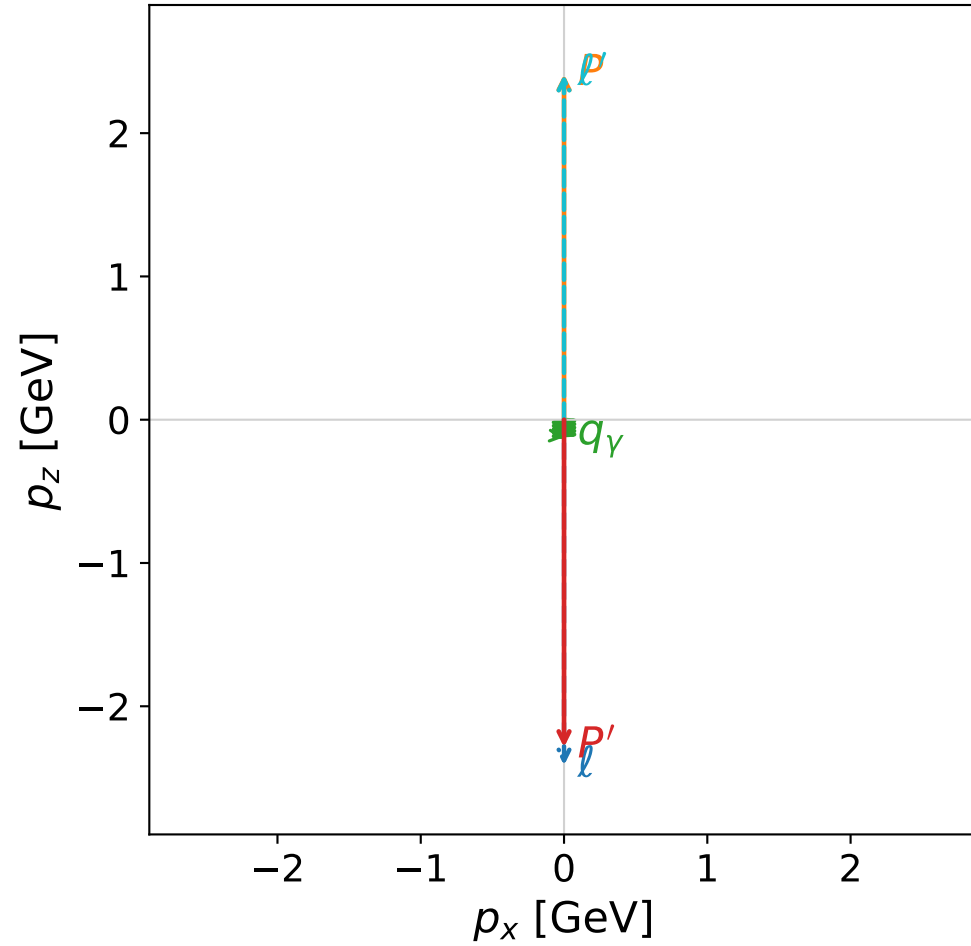
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_01\combined\
helicity_flipping\canonical_dghz_polarization_cluster_01_helicity_flipping_configurati
ons_electron.csv

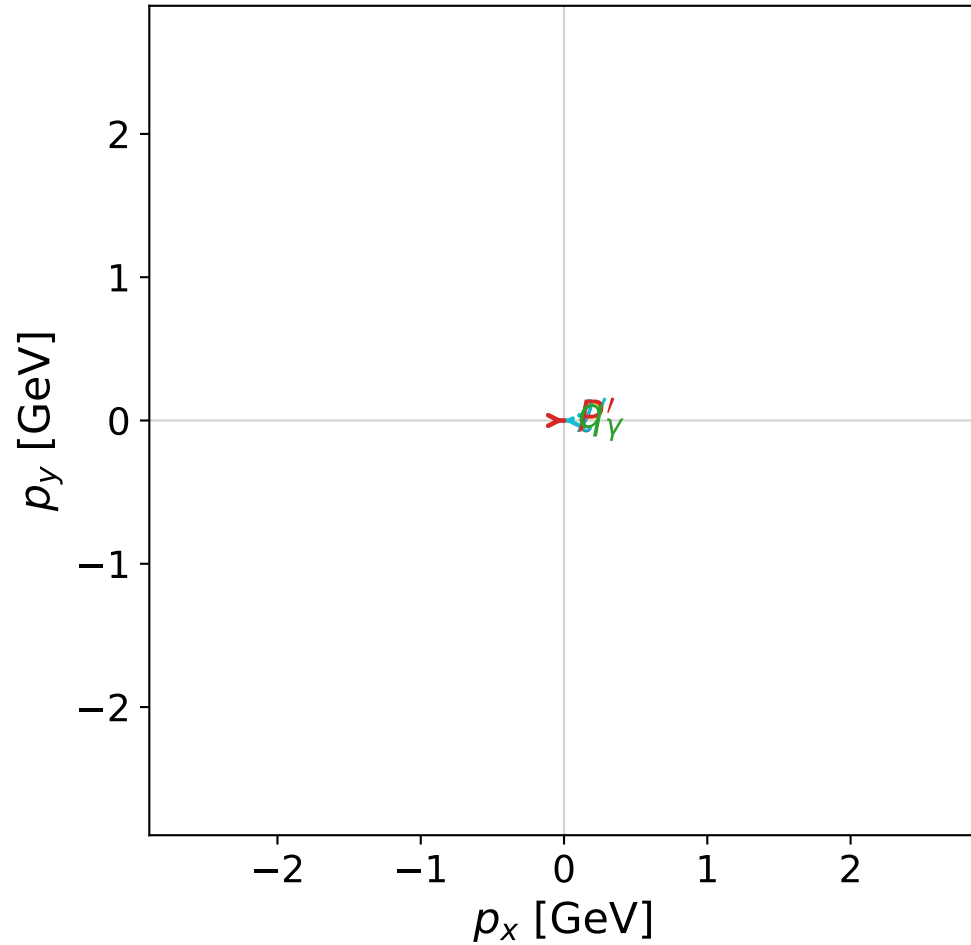
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

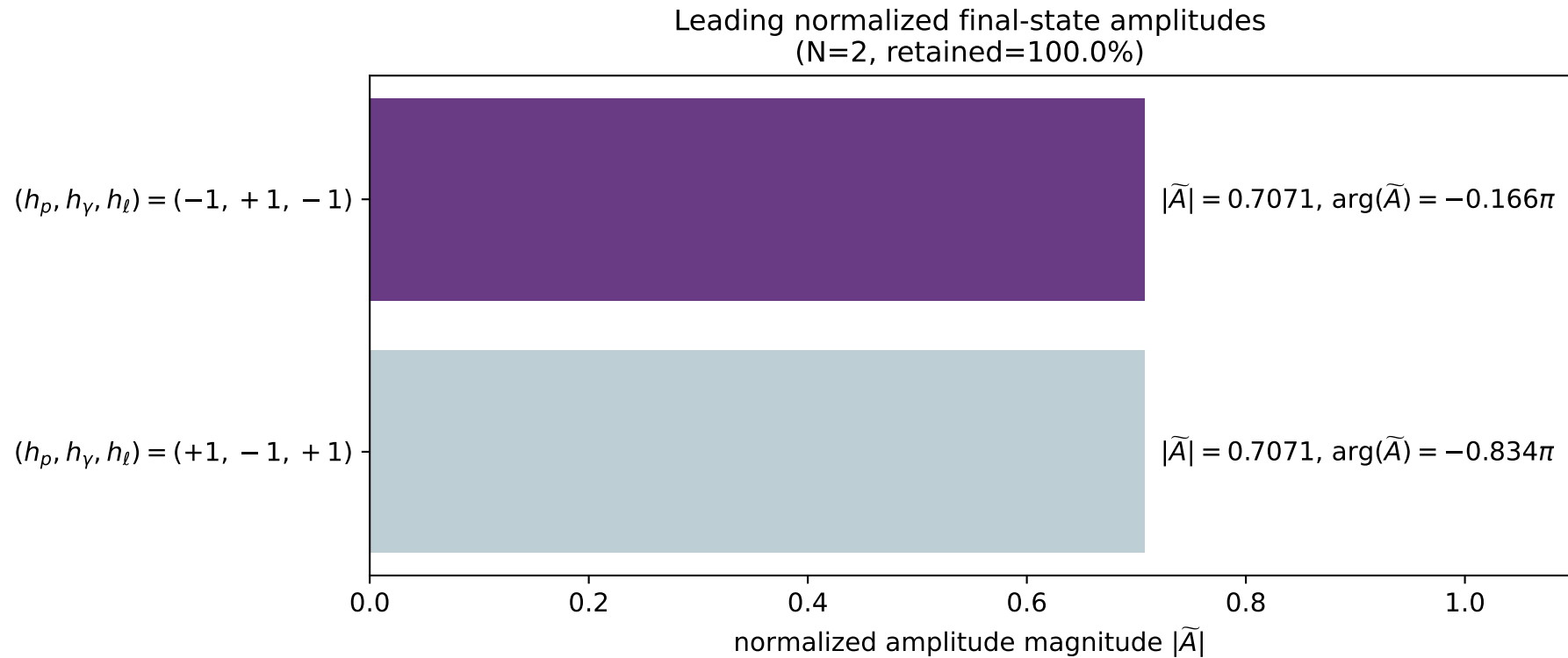


dghz_mixing_angles_polarization_01_local_minimum_2
 $d_{\text{GHZ}}=2.29125\text{e-}08$
 region: polarization_cluster_01_local_minimum_2
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0002
 lepton mixing angle: $\alpha_e=0.87696633$ rad
 incoming lepton: $0.639486|+\rangle +0.768802|-\rangle$
 proton mixing angle: $\alpha_p=0.87697731$ rad
 incoming proton: $0.639478|+\rangle +0.768809|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28627$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0335096$, $\phi_\gamma=0.522884$
 $E_\gamma=0.121256$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.19891$
 $Q^2=23.228$, $x_B=0.998073$, $t=-22.0603$, $W^2=0.924694$, $y=0.964872$

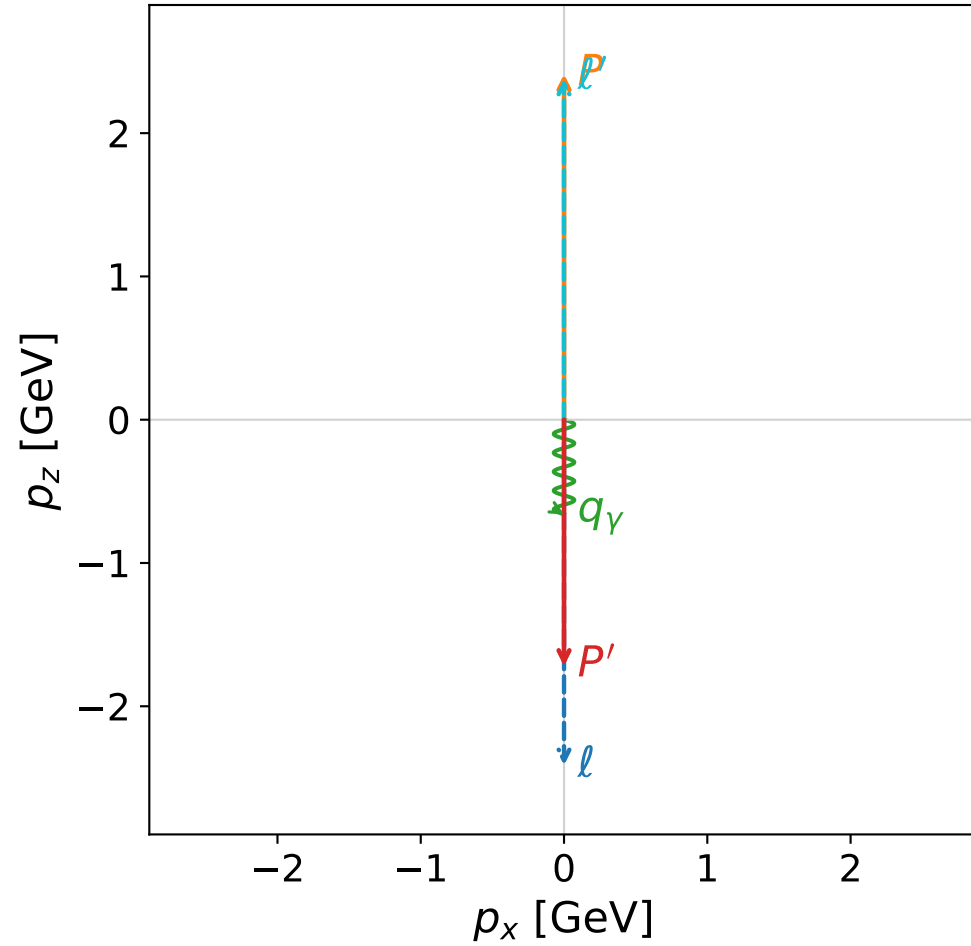
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.4075$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4075$
 P' $E= 2.4712$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2863$
 q_γ $E= 0.1213$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1213$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.736\text{e}+21$; $I_h=+3.505\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.619\text{e-}25$; $I_{TL}=+1.952\text{e-}17$

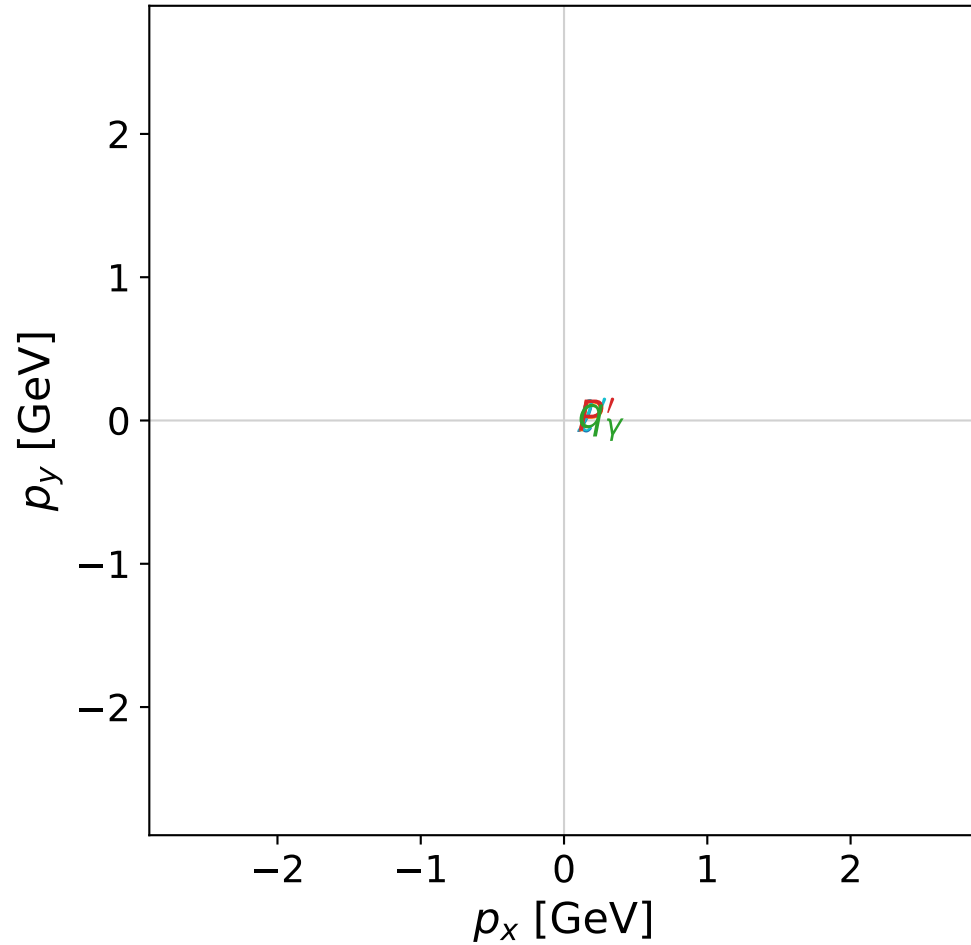


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

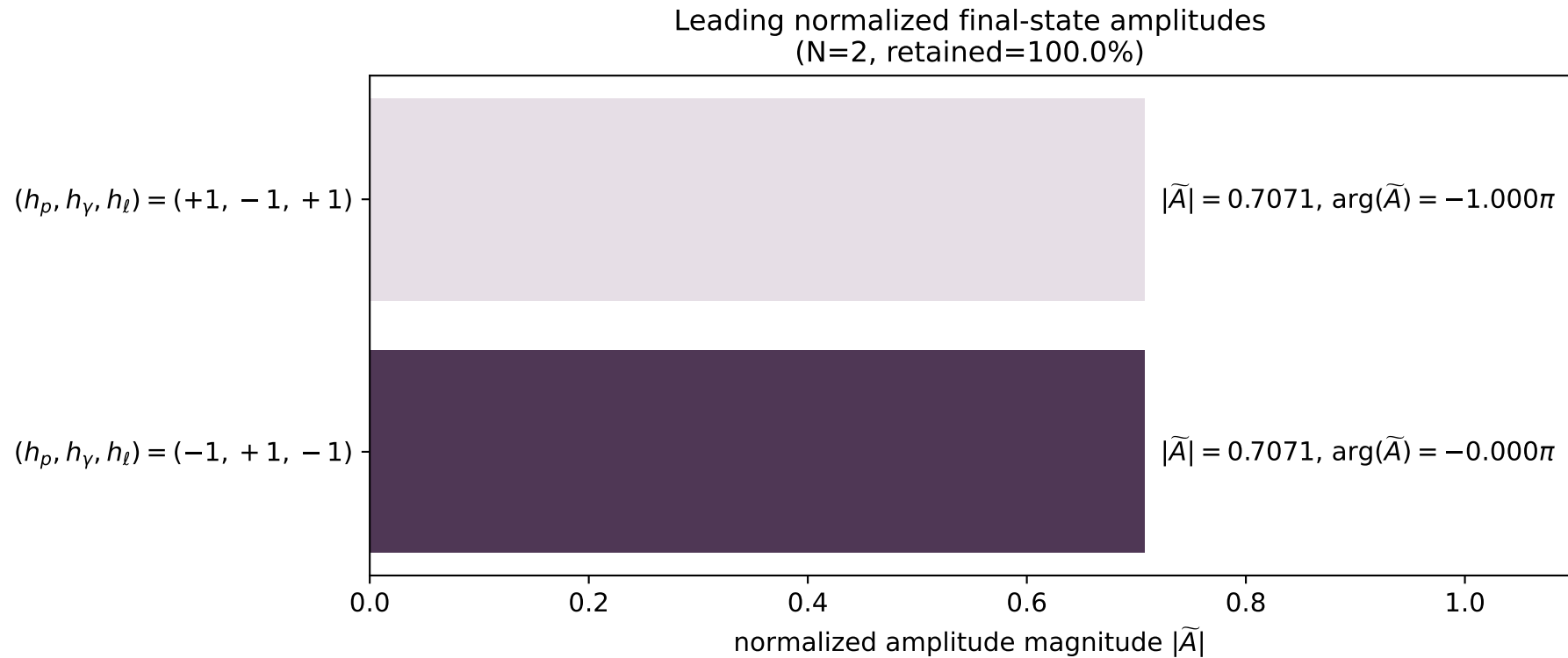


dghz_mixing_angles_polarization_01_local_minimum_11
 $d_{\{\rm GHZ\}}=2.44872\text{e-}08$
 region: polarization_cluster_01_local_minimum_11
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0011
 lepton mixing angle: $\alpha_e=0.59862965$ rad
 incoming lepton: $0.826109|+\rangle +0.563511|-\rangle$
 proton mixing angle: $\alpha_p=0.59861028$ rad
 incoming proton: $0.82612|+\rangle +0.563495|-\rangle$

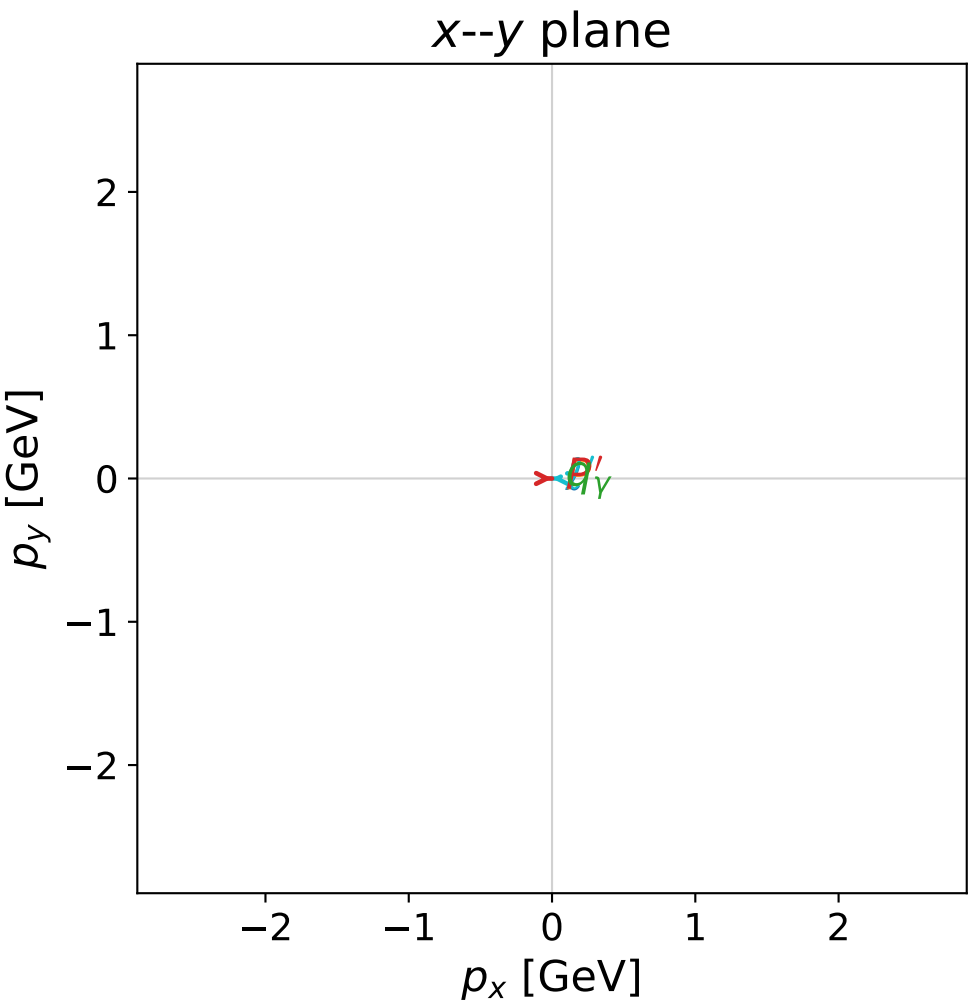
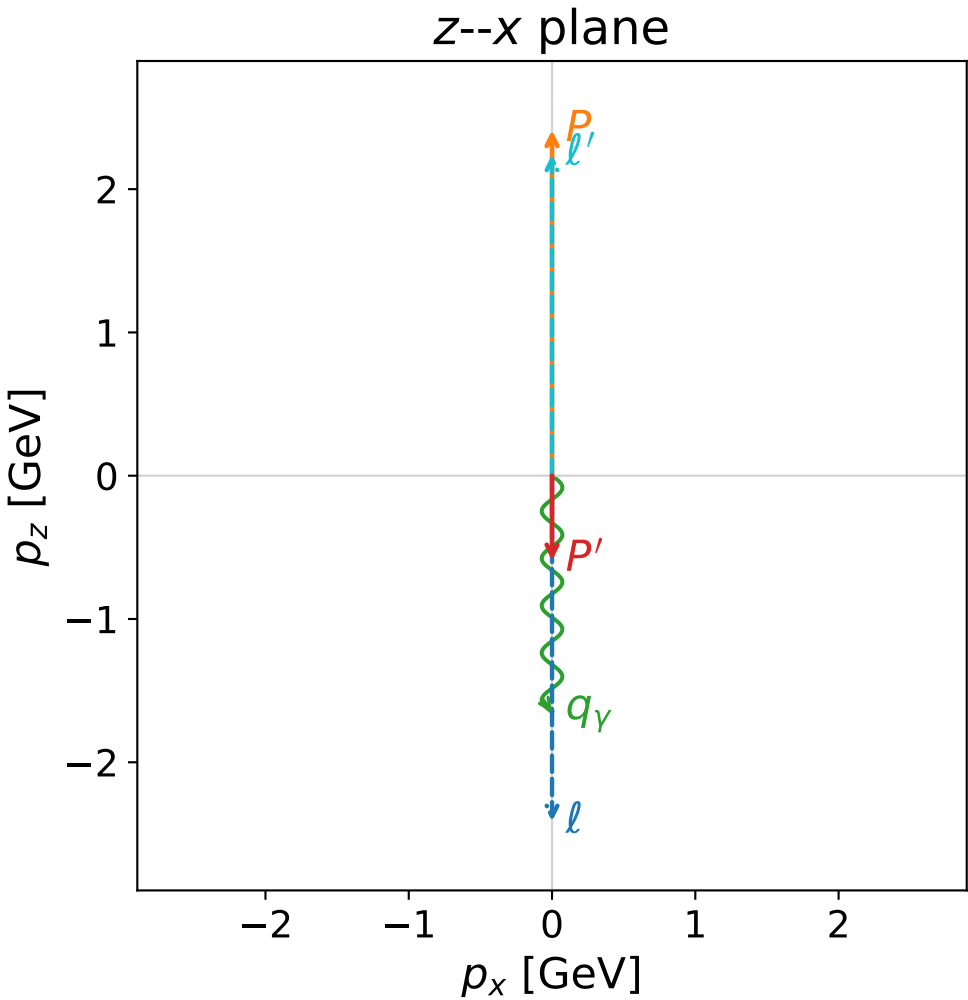
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.72071$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.57447$, $\phi_\gamma=1.17554\text{e-}06$
 $E_\gamma=0.659762$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.674751$
 $Q^2=22.9669$, $x_B=0.986451$, $t=-16.6848$, $W^2=1.19529$, $y=0.965266$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3805$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3805$
 P' $E= 1.9598$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.7207$
 q_γ $E= 0.6598$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6598$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.117\text{e+}23$; $I_h=-1.204\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=7.142\text{e-}28$; $I_{TL}=-6.921\text{e-}18$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

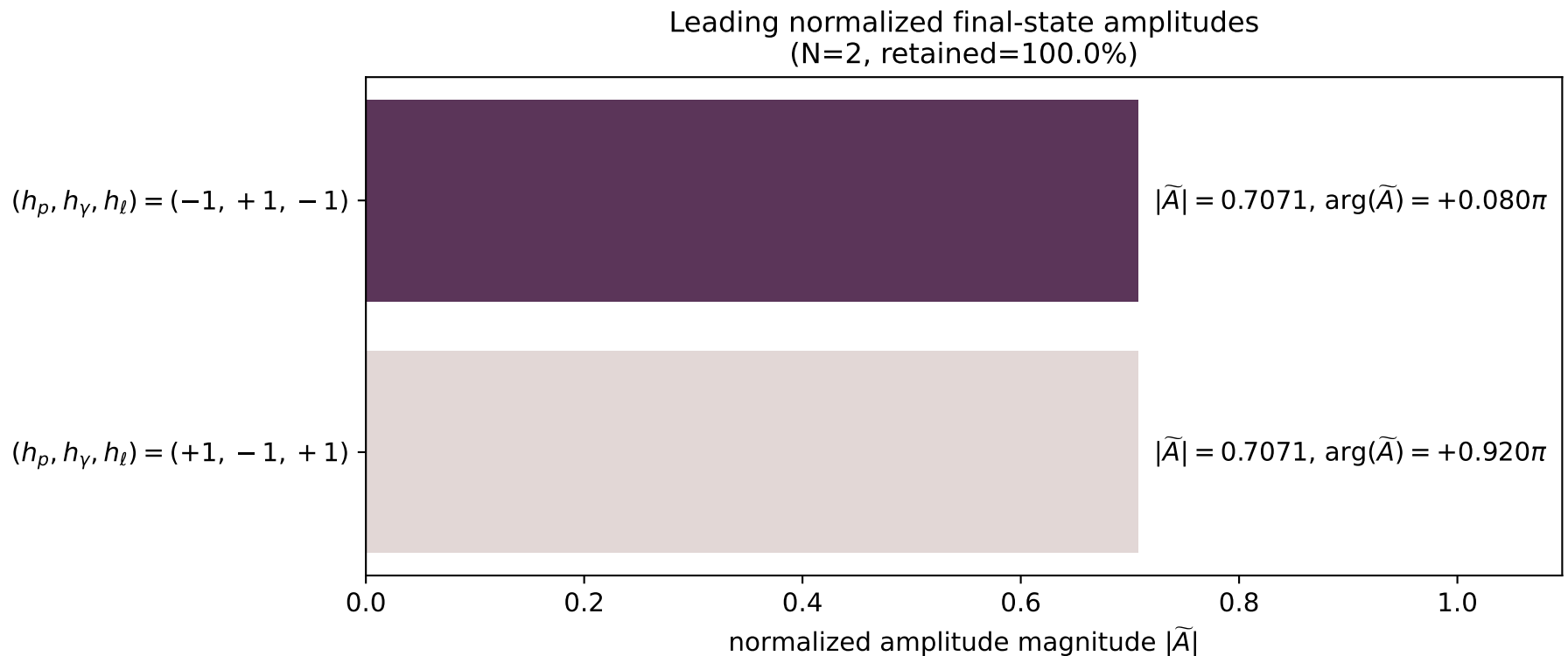


dghz_mixing_angles_polarization_01_local_minimum_16
d_{\rm GHZ}=2.47608e-08
region: polarization_cluster_01_local_minimum_16
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0016
lepton mixing angle: $\alpha_e=0.89433114$ rad
incoming lepton: $0.626041|+\rangle +0.779791|-\rangle$
proton mixing angle: $\alpha_p=0.89433384$ rad
incoming proton: $0.626038|+\rangle +0.779792|-\rangle$

$s=24.9994$, $\sqrt{s}=4.99994$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.59265$
 $\theta_{p'}=3.1415$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.28079$, $\phi_\gamma=6.02949$
 $E_\gamma=1.64887$, $\theta_{\ell'}=2.53716\text{e-}05$, $\phi_{\ell'}=3.1392$
 $Q^2=21.6261$, $x_B=0.926939$, $t=-6.84211$, $W^2=2.58442$, $y=0.967293$

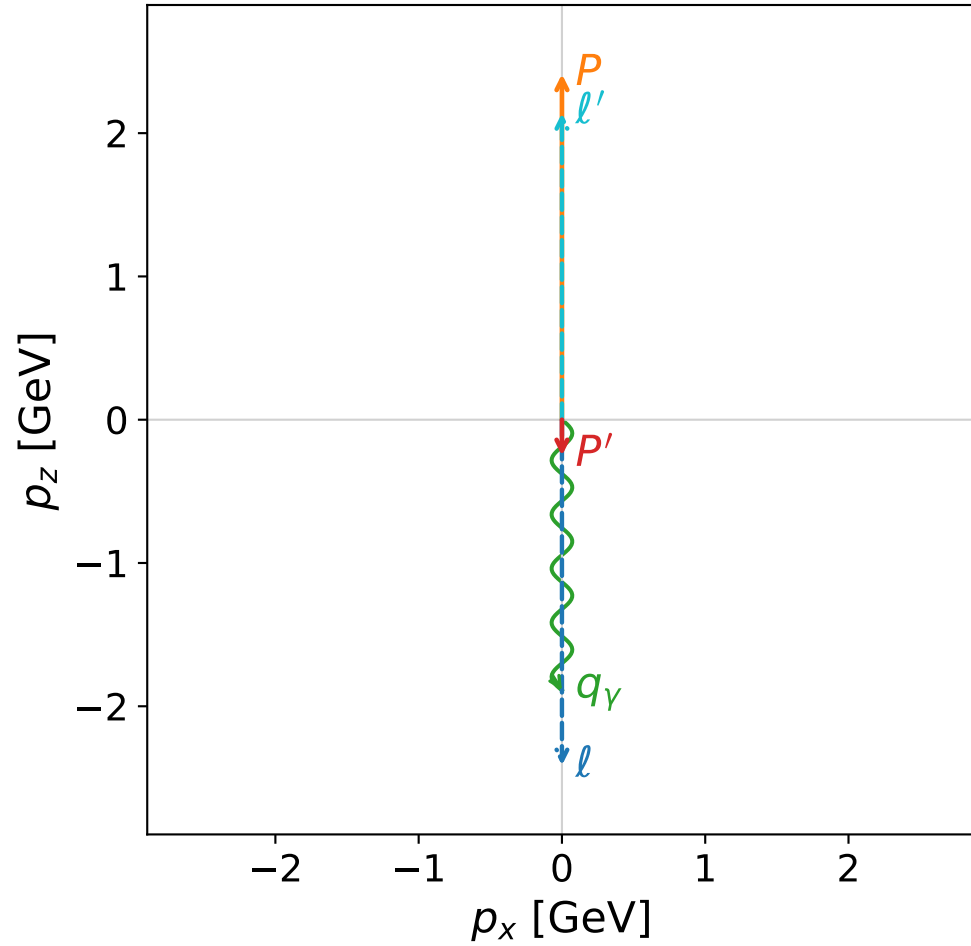
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2415$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.2415$
 P' $E= 1.1095$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.5927$
 q_γ $E= 1.6489$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.6489$

GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=9.459e+16; I_h=-8.703e-10
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=3.300e-13; I_TL=+1.759e-07

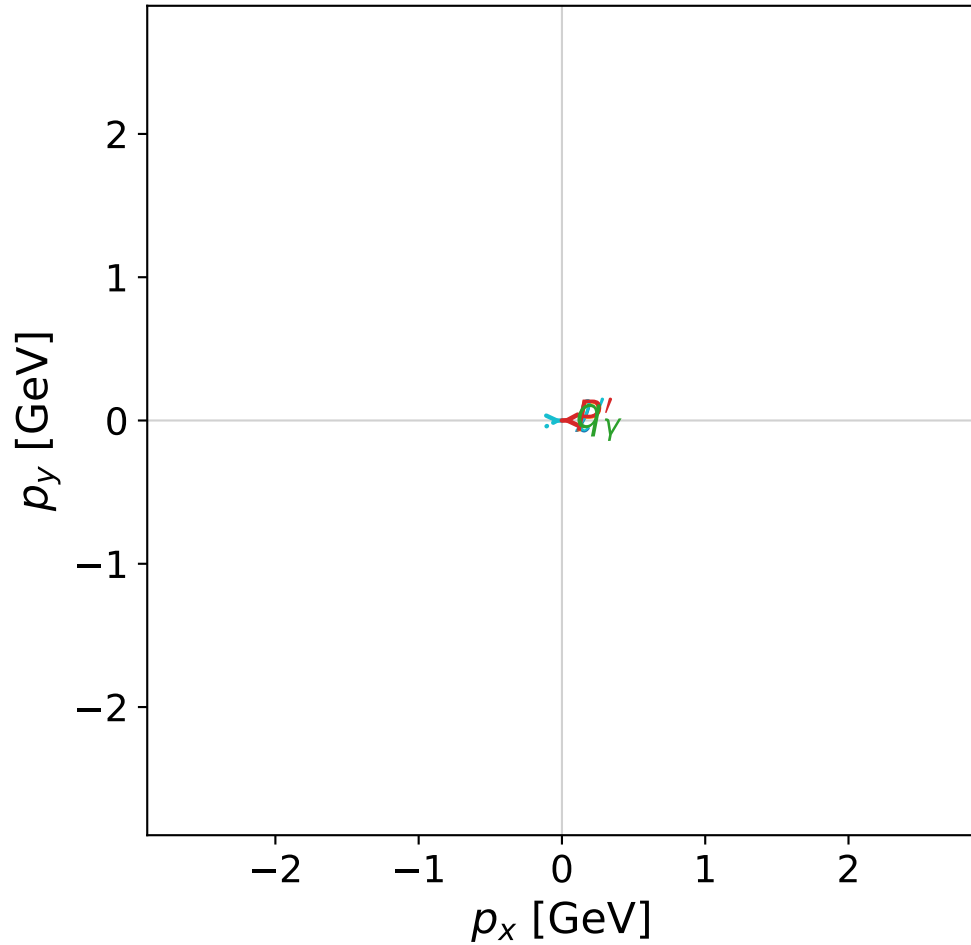


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



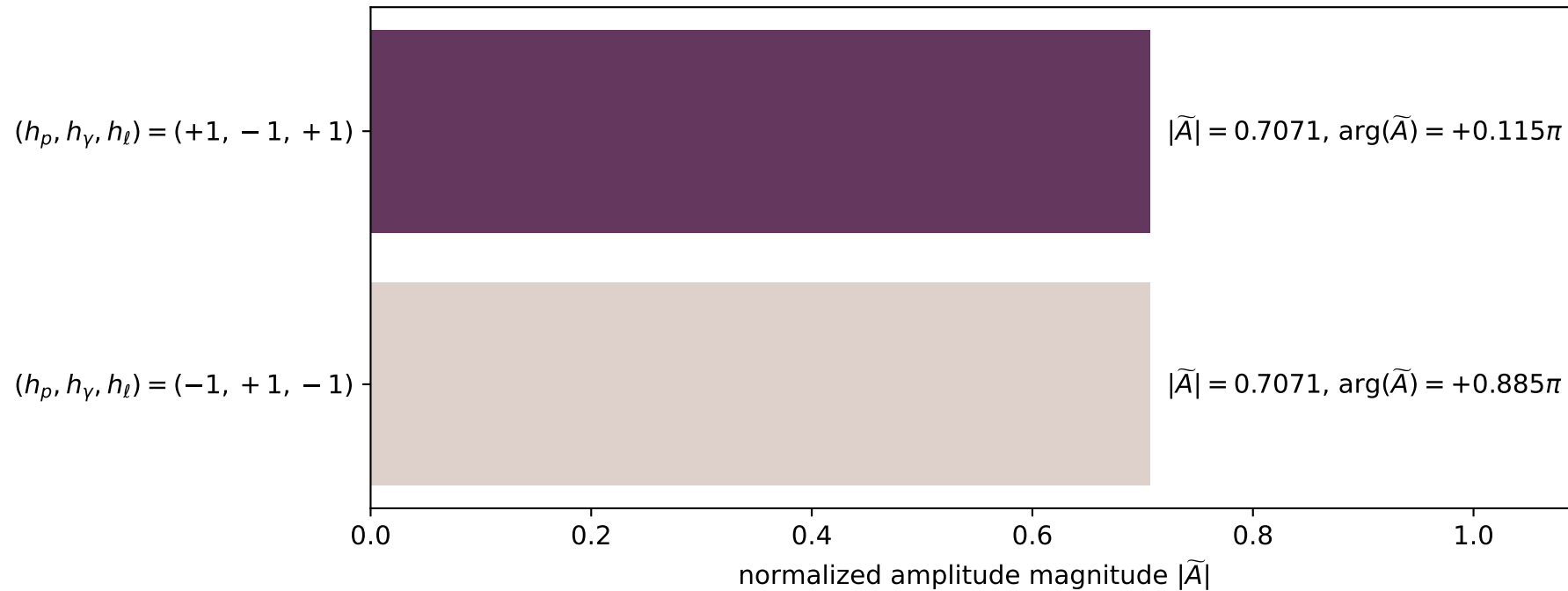
dghz_mixing_angles_polarization_01_local_minimum_19
 $d_{\text{GHZ}}=2.53539\text{e-}08$
 region: polarization_cluster_01_local_minimum_19
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0019
 lepton mixing angle: $\alpha_e=0.78519375$ rad
 incoming lepton: $0.707251|+\rangle + 0.706962|-\rangle$
 proton mixing angle: $\alpha_p=0.7851987$ rad
 incoming proton: $0.707248|+\rangle + 0.706966|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.251318$
 $\theta_{P'}=3.14131$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.17031$, $\phi_\gamma=0.388985$
 $E_\gamma=1.8888$, $\theta_{\ell'}=3.33062\text{e-}05$, $\phi_{\ell'}=0.0287211$
 $Q^2=20.648$, $x_B=0.88364$, $t=-4.47898$, $W^2=3.59883$, $y=0.968774$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1401$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.1401$
 P' $E= 0.9711$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.2513$
 q_γ $E= 1.8888$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8888$

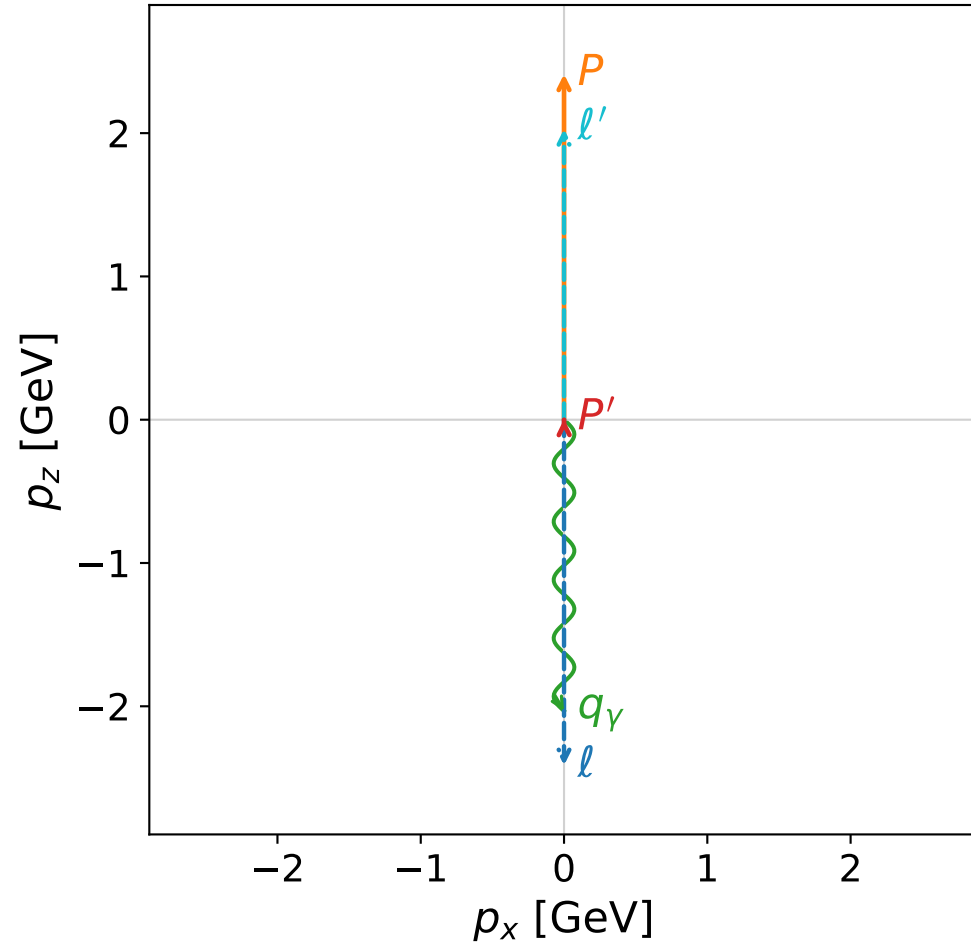
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.089\text{e+}16$; $I_h=-1.866\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.027\text{e-}12$; $I_{TL}=+1.441\text{e-}09$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

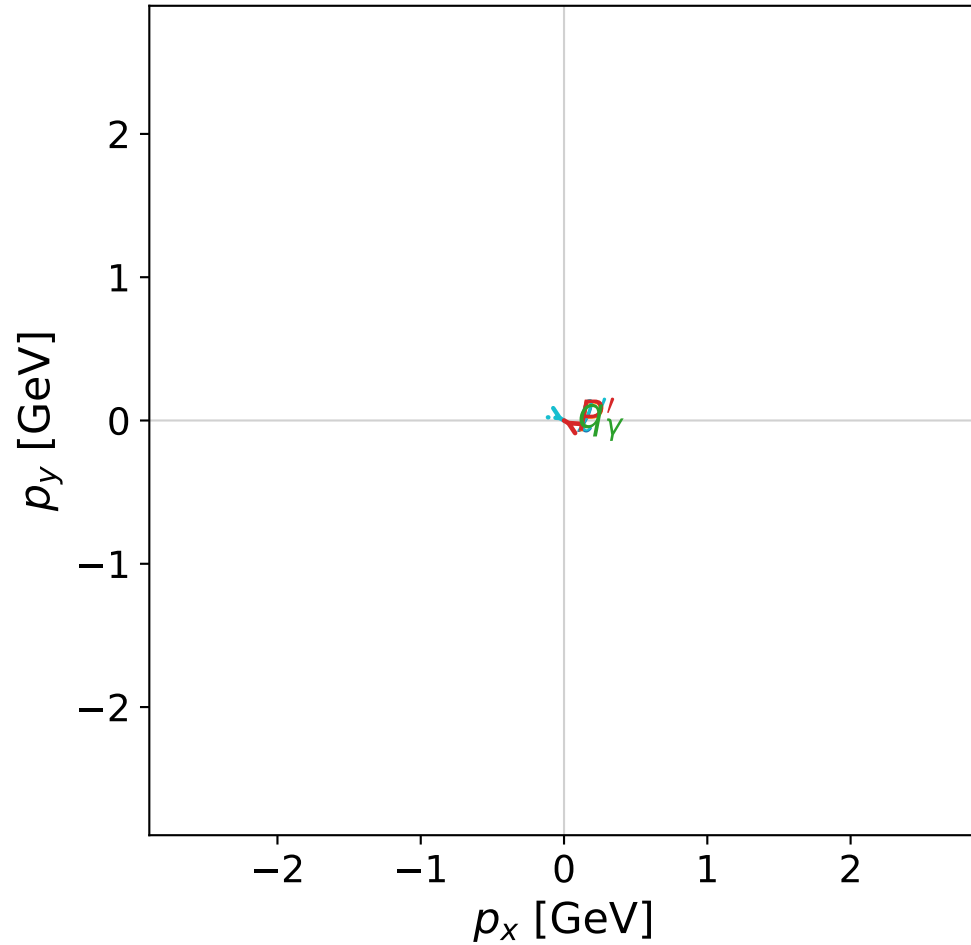


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

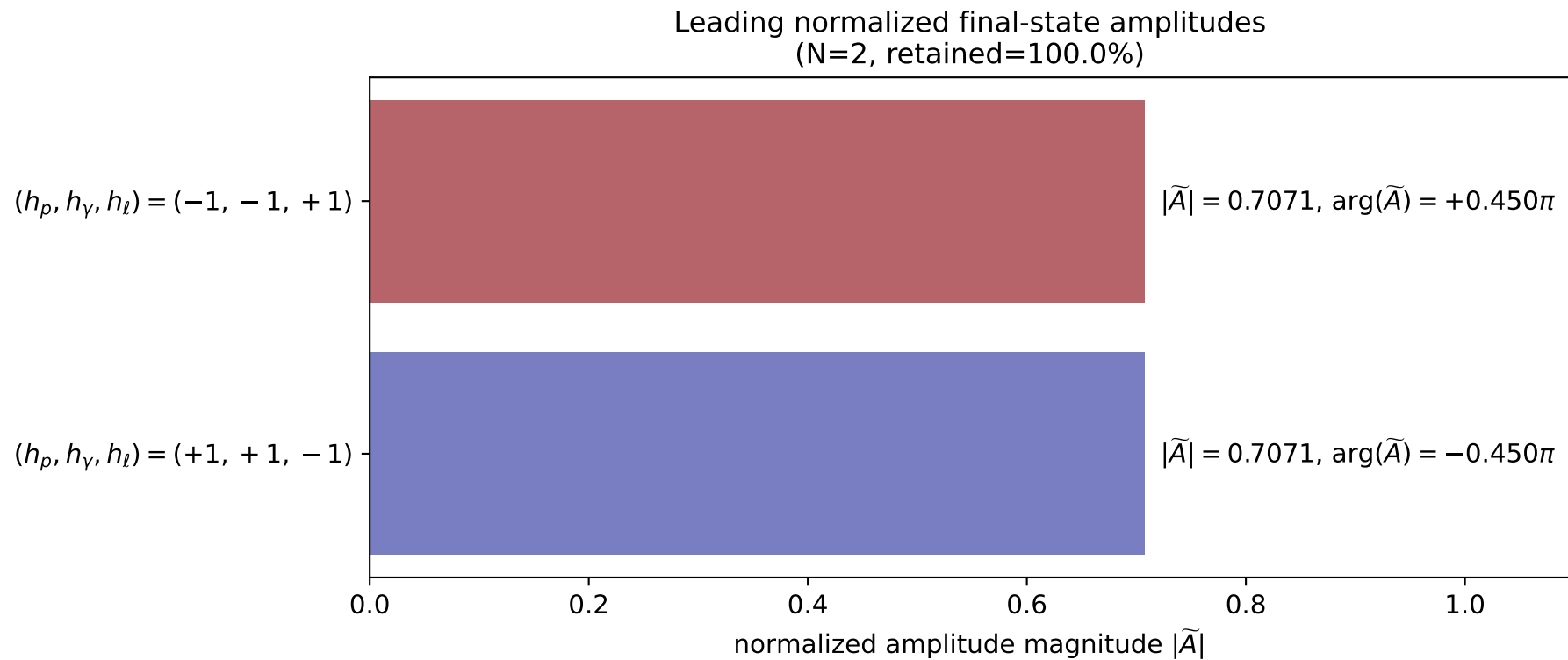


dghz_mixing_angles_polarization_01_local_minimum_23
 $d_{\text{GHZ}}=2.57818\text{e-}08$
 region: polarization_cluster_01_local_minimum_23
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0023
 lepton mixing angle: $\alpha_e=0.77571838$ rad
 incoming lepton: $0.713918|+\rangle + 0.700229|-\rangle$
 proton mixing angle: $\alpha_p=0.77571056$ rad
 incoming proton: $0.713924|+\rangle + 0.700224|-\rangle$

$s=24.9977$, $\sqrt{s}=4.99977$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.00408939$
 $\theta_{P'}=0.0106052$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.61744$, $\phi_\gamma=4.55652$
 $E_\gamma=2.03292$, $\theta_{\ell'}=2.13758\text{e-}05$, $\phi_{\ell'}=5.75903$
 $Q^2=19.5733$, $x_B=0.836333$, $t=-3.07548$, $W^2=4.71027$, $y=0.970393$

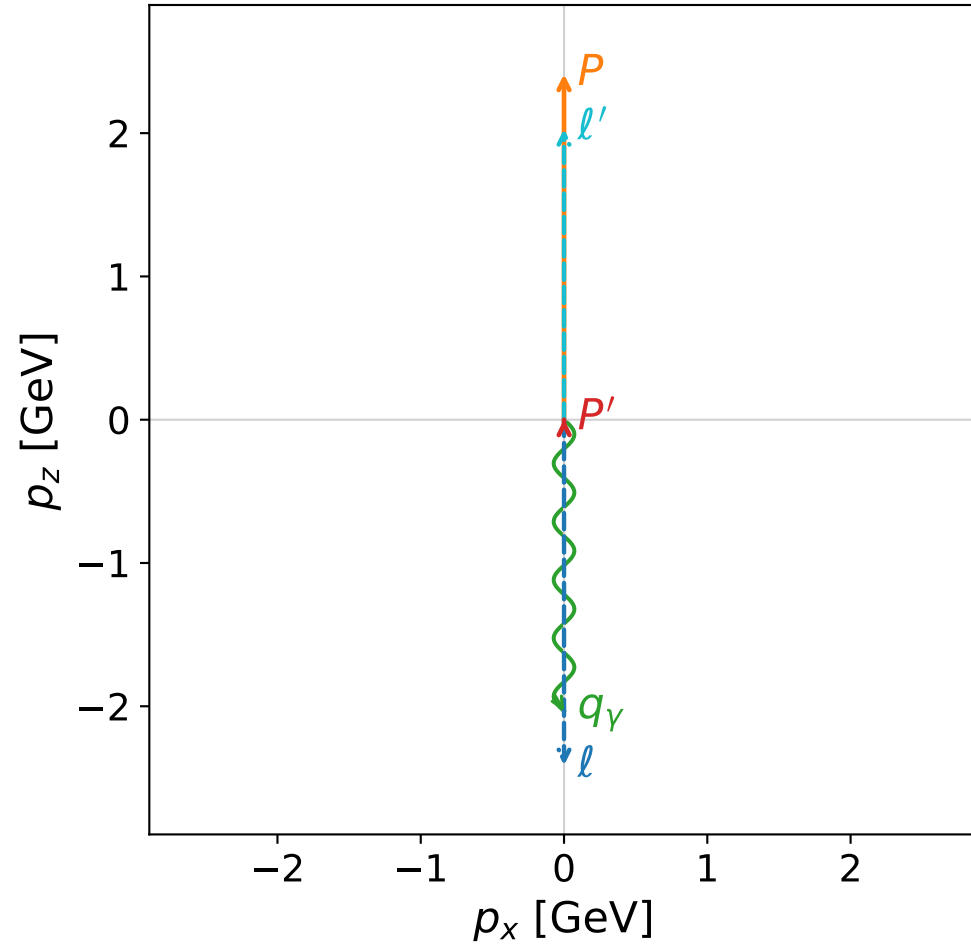
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0288$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0288$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0041$
 q_γ $E= 2.0329$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0329$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.844\text{e+}17$; $I_h=-4.509\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.724\text{e-}12$; $I_{TL}=+3.140\text{e-}08$

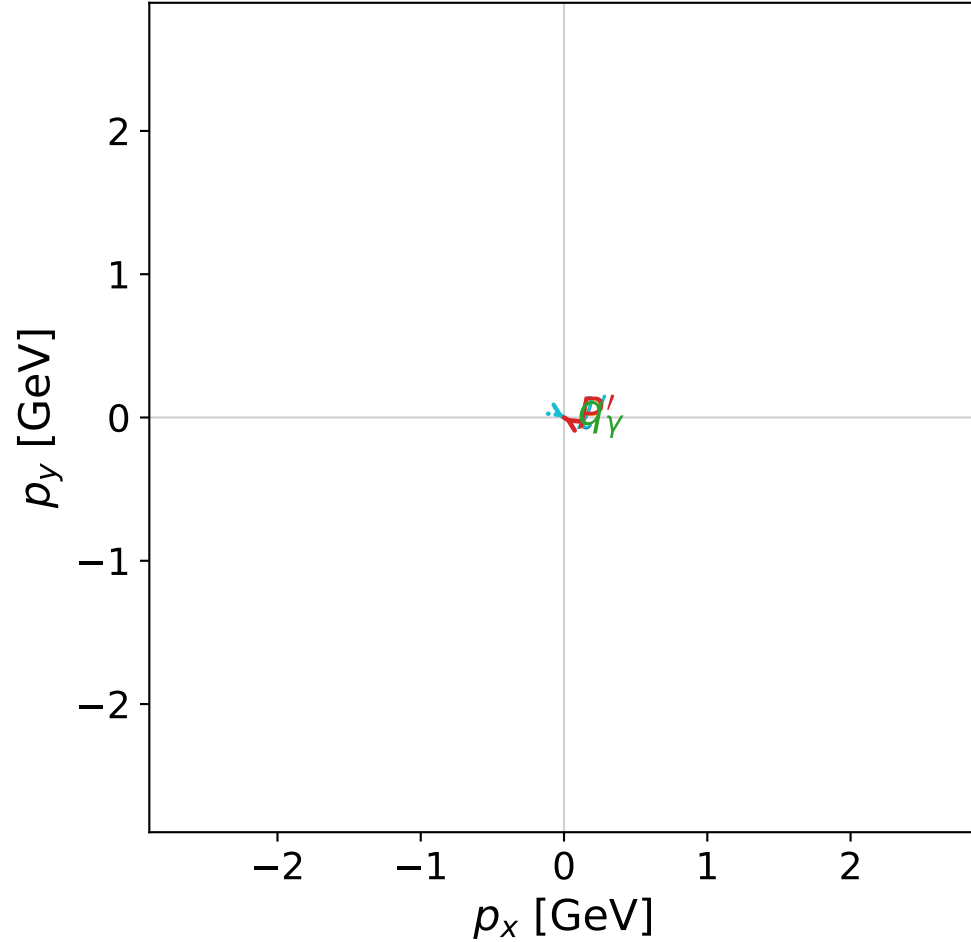


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

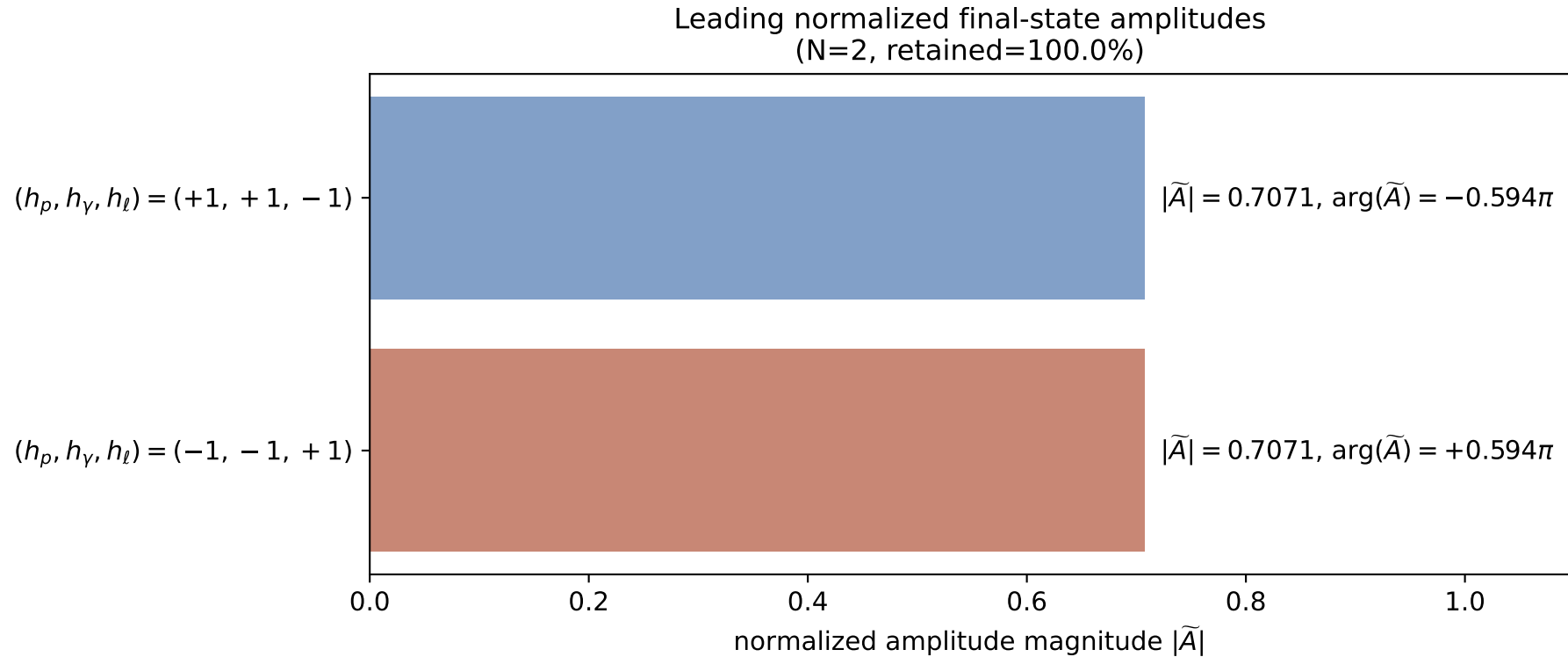


dghz_mixing_angles_polarization_01_local_minimum_24
 $d_{\text{GHZ}}=2.58712\text{e-}08$
 region: polarization_cluster_01_local_minimum_24
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0024
 lepton mixing angle: $\alpha_e=0.742527$ rad
 incoming lepton: $0.736762|+\rangle + 0.676152|-\rangle$
 proton mixing angle: $\alpha_p=0.74253335$ rad
 incoming proton: $0.736758|+\rangle + 0.676157|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0041485$
 $\theta_{P'}=0.00973758$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.58165$, $\phi_\gamma=5.00822$
 $E_\gamma=2.03307$, $\theta_\ell=1.99099\text{e-}05$, $\phi_\ell=5.72325$
 $Q^2=19.5752$, $x_B=0.836327$, $t=-3.07541$, $W^2=4.71079$, $y=0.970396$

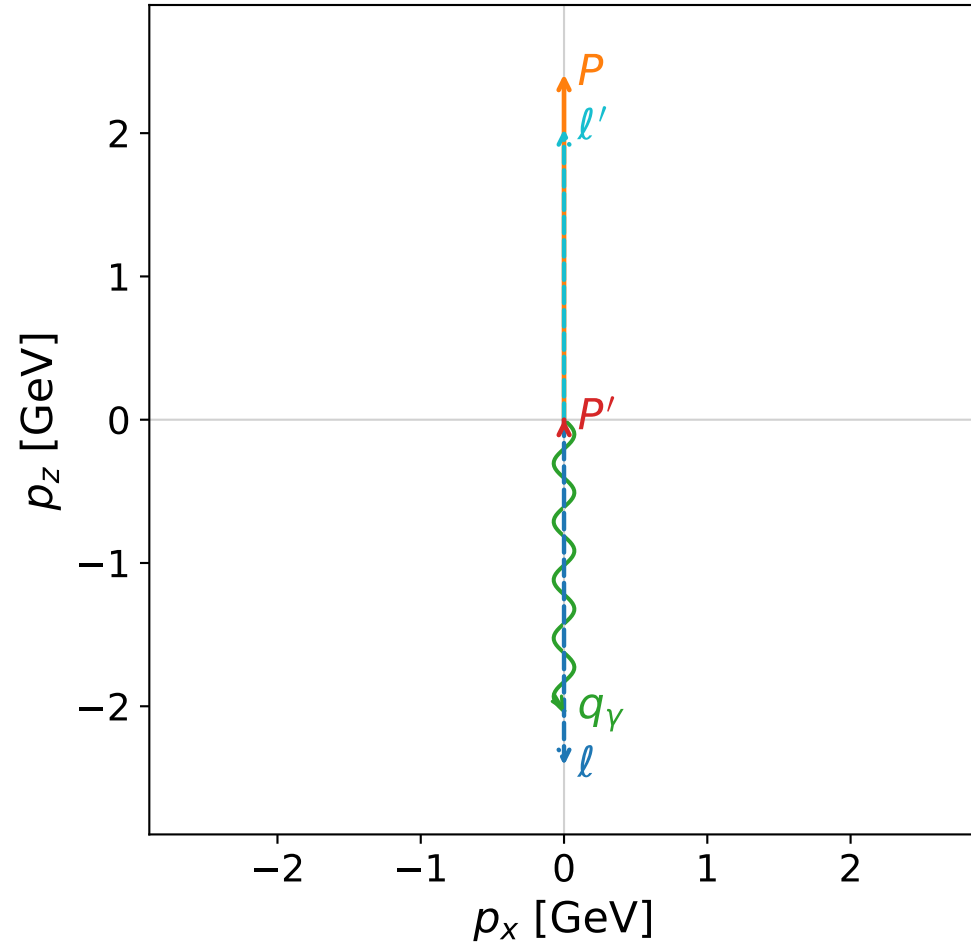
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0289$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0289$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0041$
 q_γ $E= 2.0331$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0331$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.105\text{e+}17$; $I_h=-1.822\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.506\text{e-}12$; $I_{TL}=+1.262\text{e-}07$

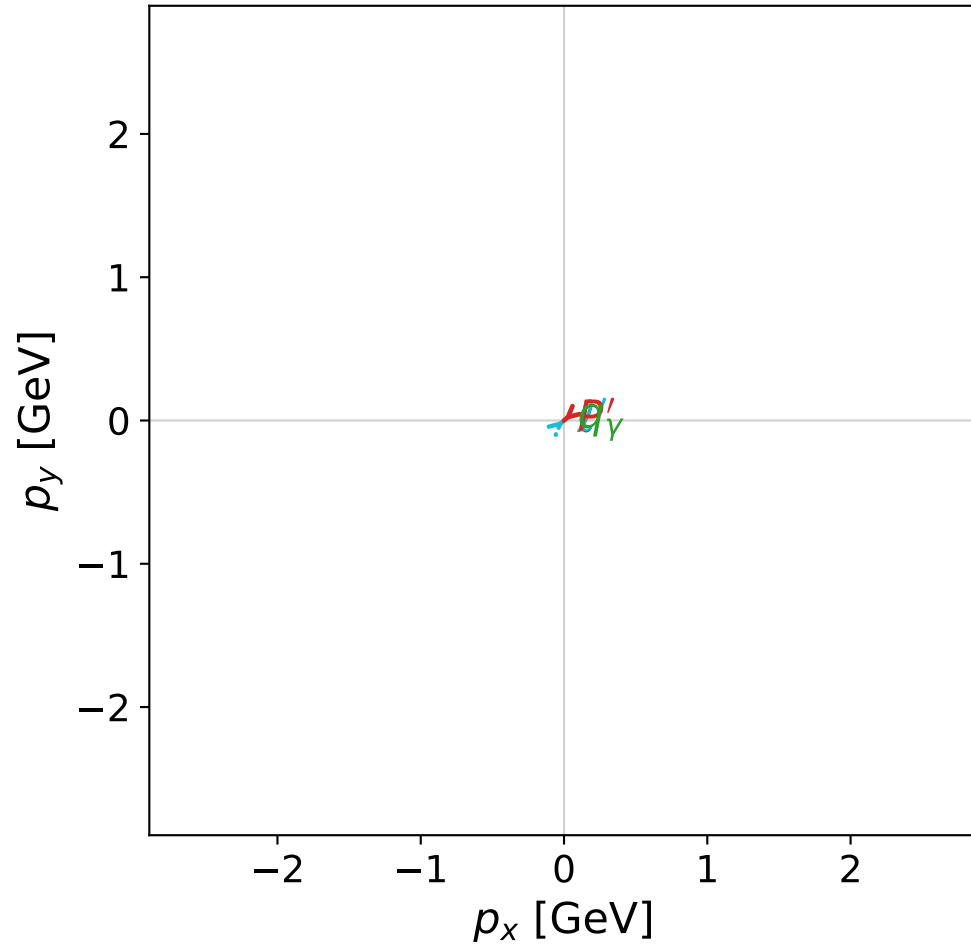


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

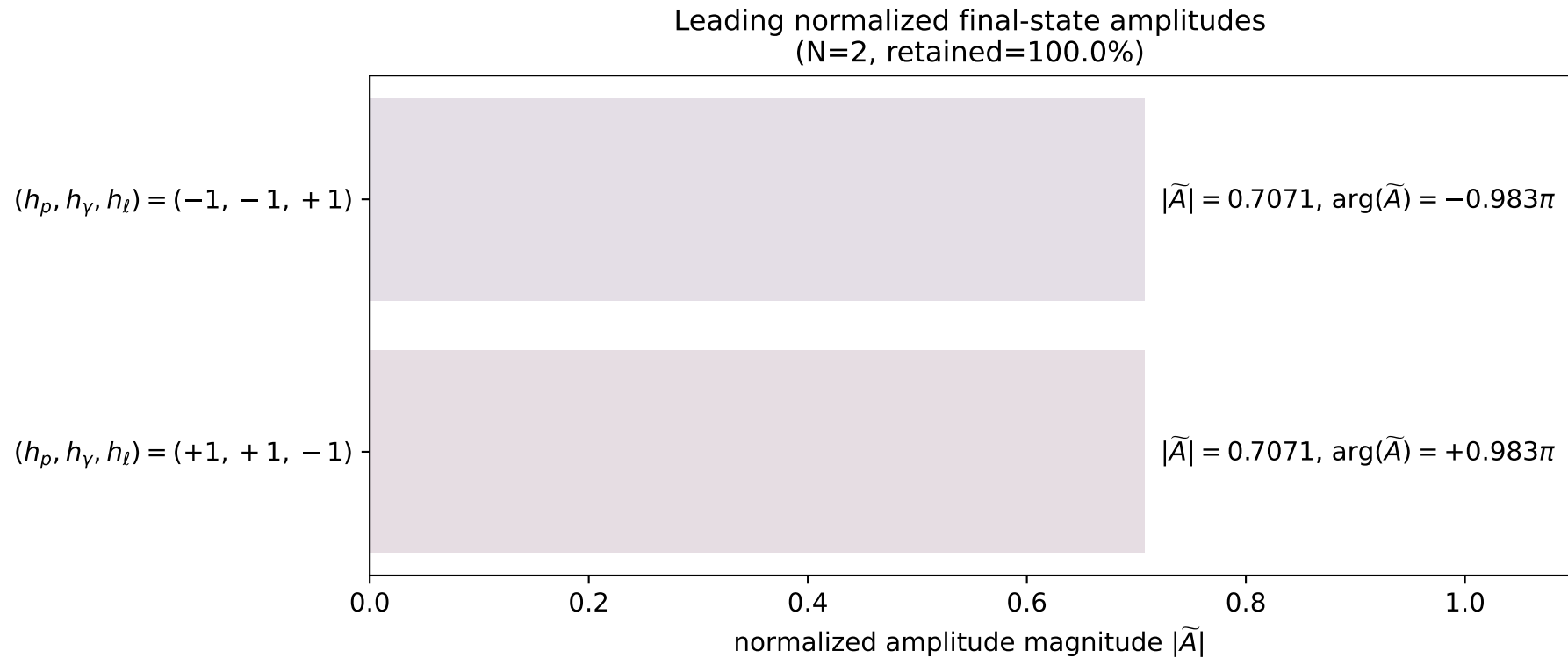


dghz_mixing_angles_polarization_01_local_minimum_25
 $d_{\text{GHZ}}=2.59286\text{e-}08$
 region: polarization_cluster_01_local_minimum_25
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0025
 lepton mixing angle: $\alpha_e=0.768686$ rad
 incoming lepton: $0.718825|+\rangle +0.695191|-\rangle$
 proton mixing angle: $\alpha_p=0.76868388$ rad
 incoming proton: $0.718826|+\rangle +0.69519|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00436041$
 $\theta_{P'}=0.00736311$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.8561$, $\phi_\gamma=0.0545742$
 $E_\gamma=2.03318$, $\theta_{\ell'}=1.5825\text{e-}05$, $\phi_{\ell'}=0.714507$
 $Q^2=19.5741$, $x_B=0.836282$, $t=-3.07439$, $W^2=4.71185$, $y=0.970398$

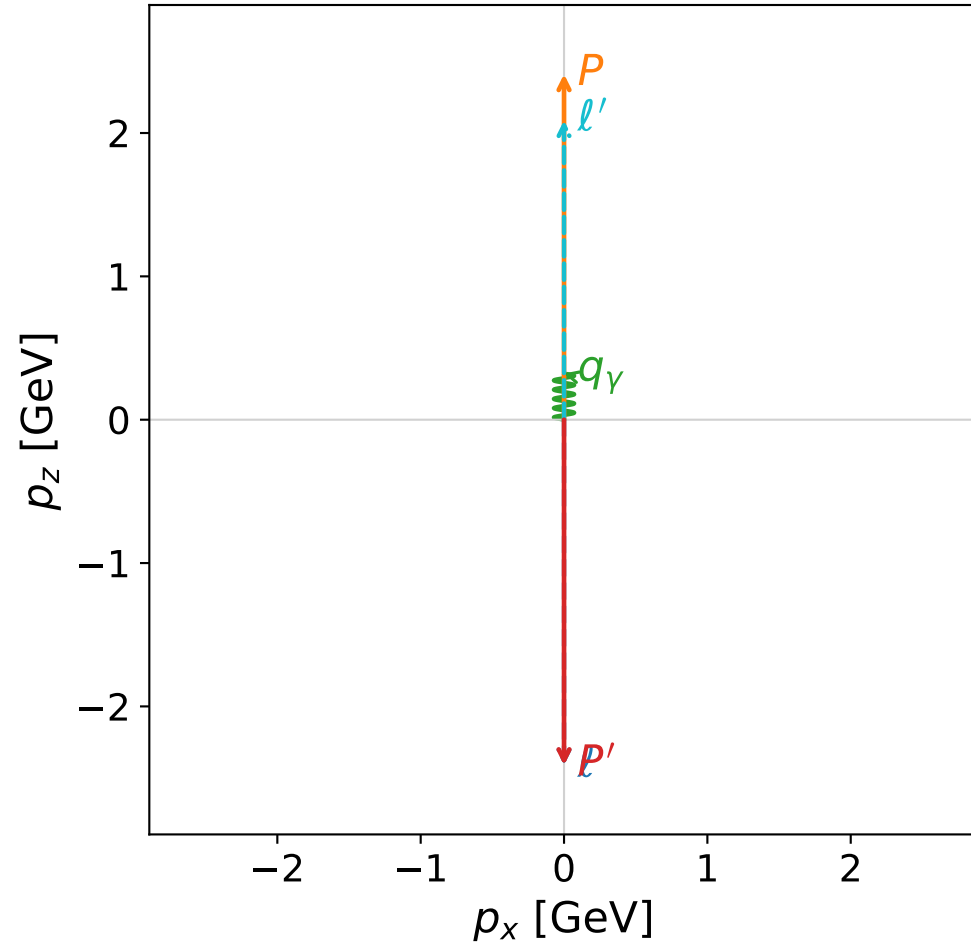
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0288$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0288$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0044$
 q_γ $E= 2.0332$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0332$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.360\text{e+}17$; $I_h=-5.025\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.460\text{e-}13$; $I_{TL}=+3.488\text{e-}08$

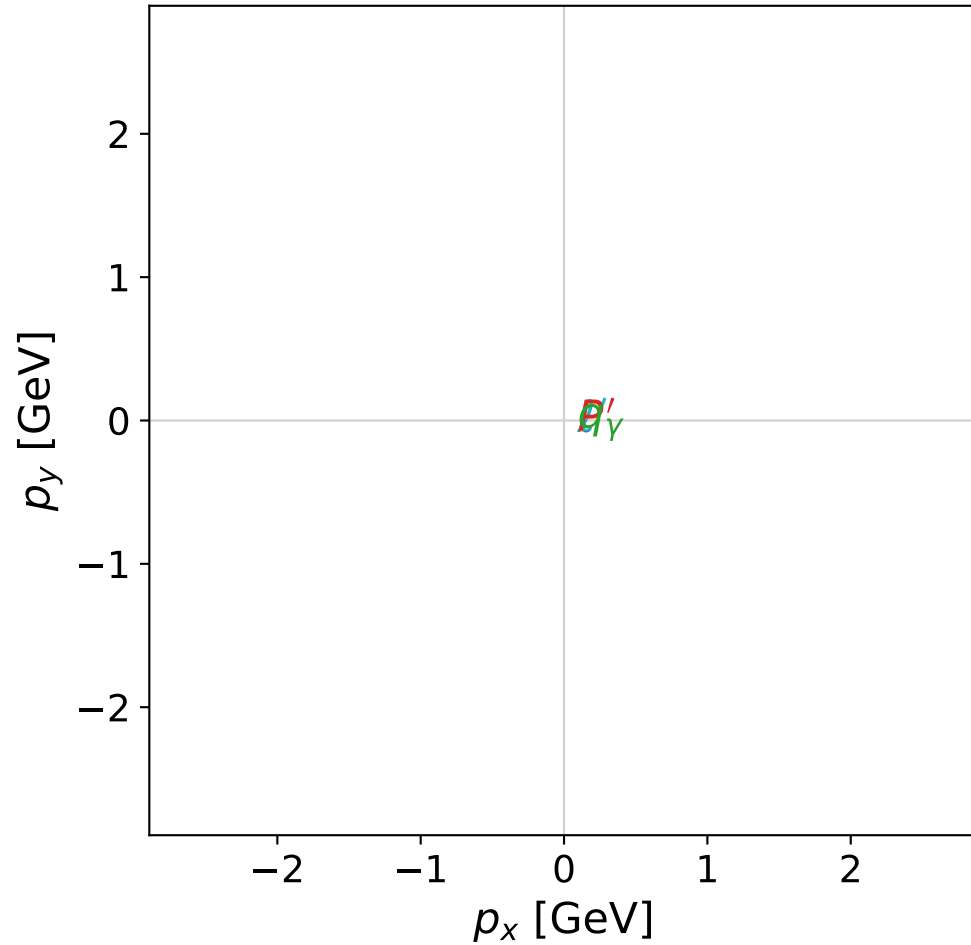


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

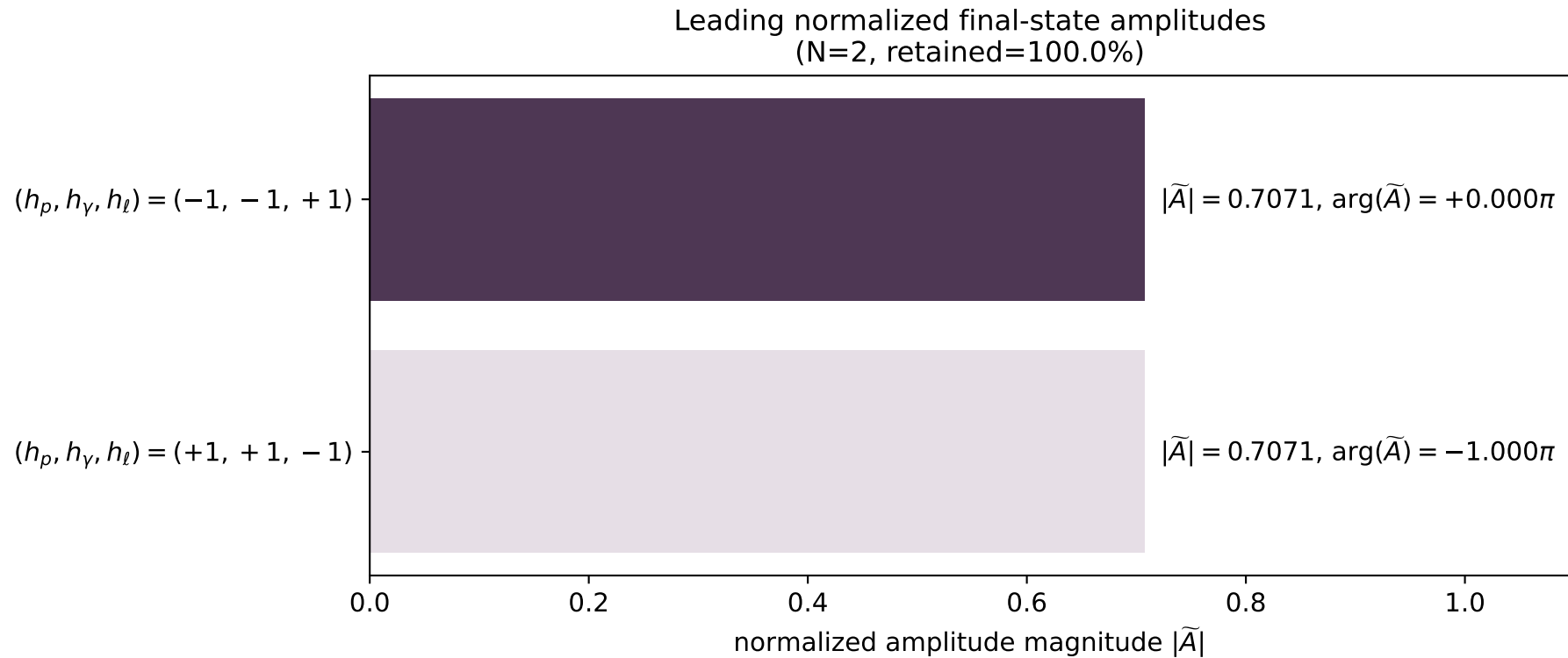


dghz_mixing_angles_polarization_01_local_minimum_27
 $d_{\text{GHZ}}=2.59874\text{e-}08$
 region: polarization_cluster_01_local_minimum_27
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0027
 lepton mixing angle: $\alpha_e=0.75349531$ rad
 incoming lepton: $0.729302|+\rangle + 0.684192|-\rangle$
 proton mixing angle: $\alpha_p=0.81730854$ rad
 incoming proton: $0.684187|+\rangle + 0.729307|-\rangle$

$s=24.9779$, $\sqrt{s}=4.99779$, $|\vec{P}|=2.41087$, $|\vec{P}'|=2.41087$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.0939447$, $\phi_\gamma=0.966166$
 $E_\gamma=0.322487$, $\theta_{l'}=0$, $\phi_{l'}=3.23554$
 $Q^2=20.1393$, $x_B=0.862026$, $t=-23.2492$, $W^2=4.10329$, $y=0.969487$

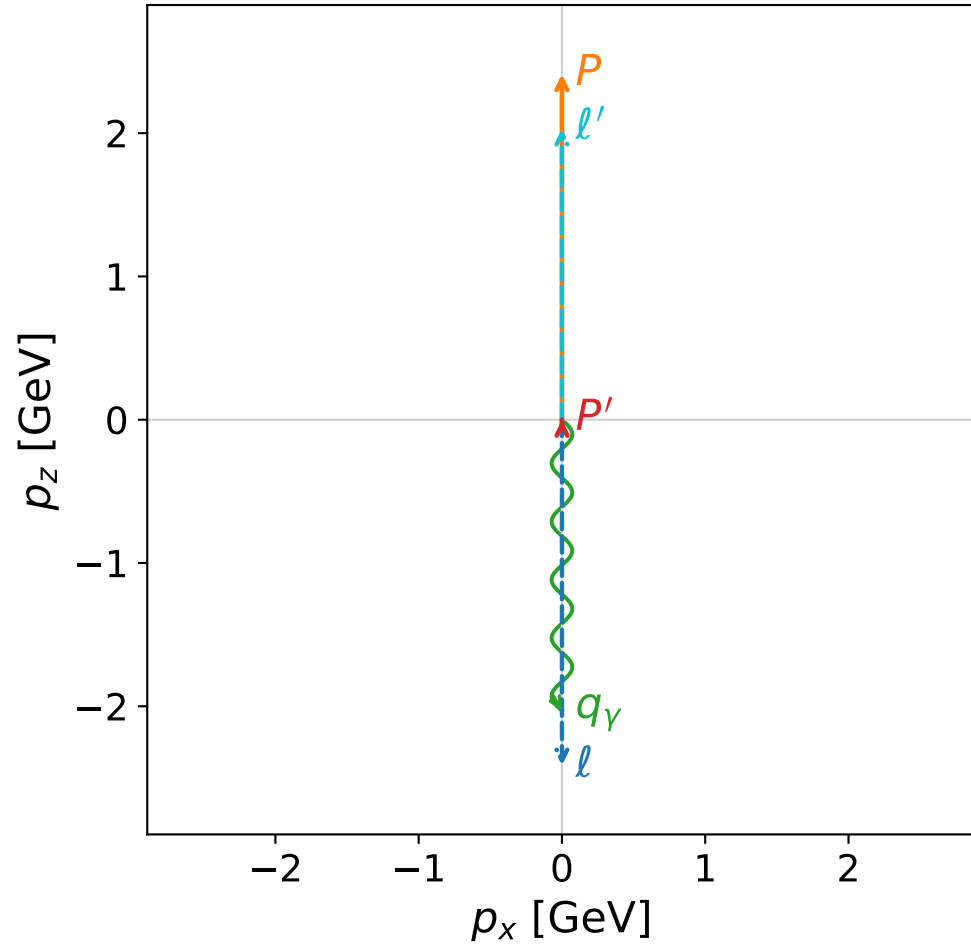
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4109$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4109$
 P $E= 2.5869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4109$
 l' $E= 2.0884$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0884$
 P' $E= 2.5869$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4109$
 q_γ $E= 0.3225$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3225$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.064\text{e+}22$; $I_h=+7.642\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.932\text{e-}27$; $I_{TL}=+7.642\text{e-}14$

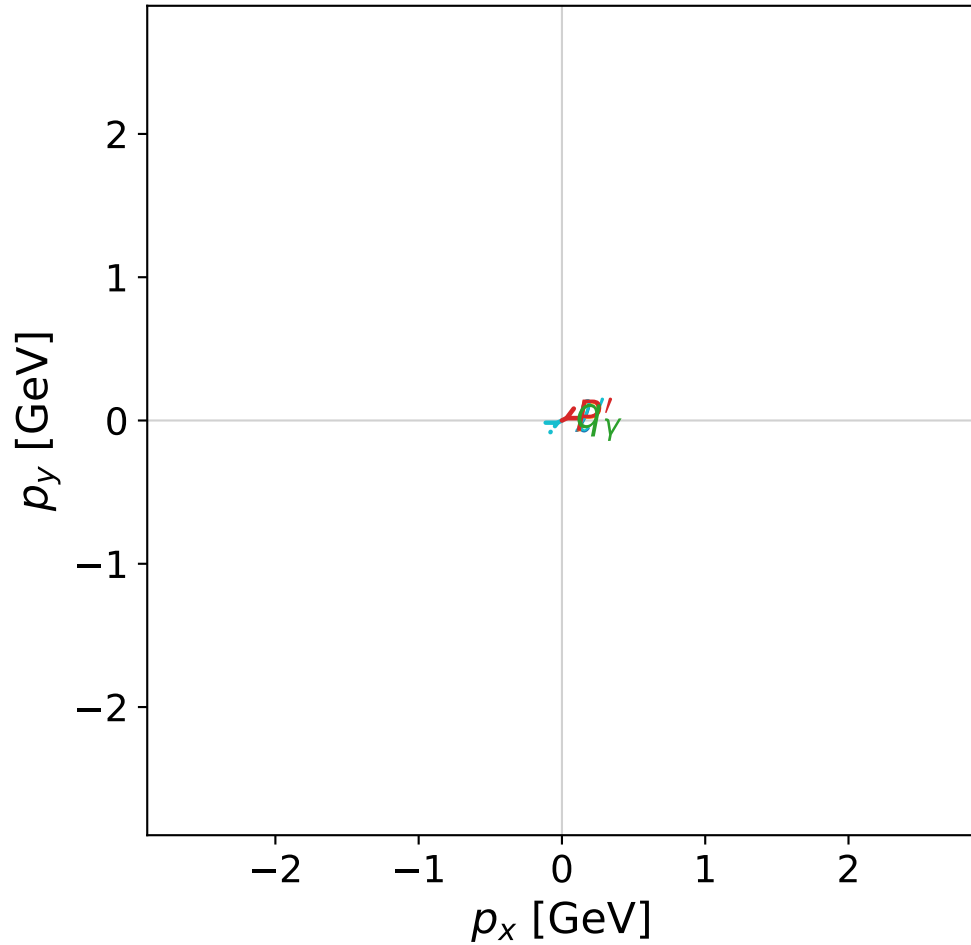


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



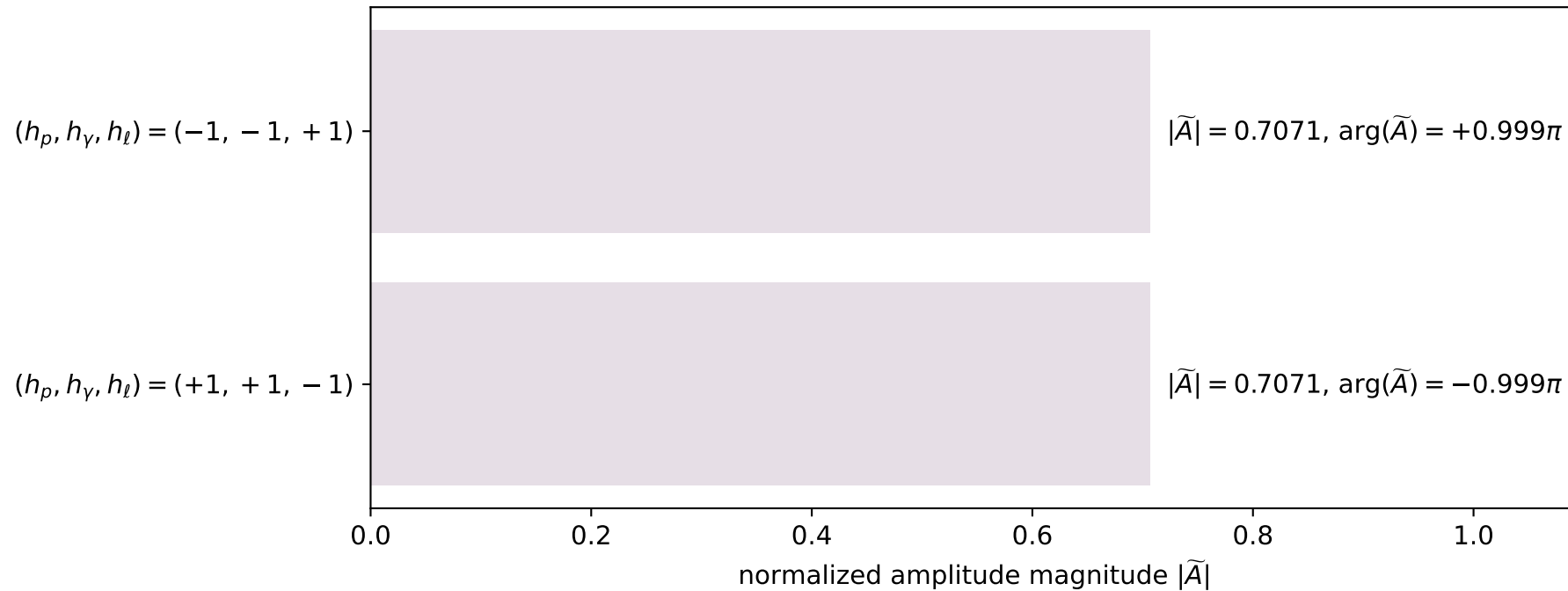
dghz_mixing_angles_polarization_01_local_minimum_28
 $d_{\text{GHZ}}=2.60293\text{e-}08$
 region: polarization_cluster_01_local_minimum_28
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0028
 lepton mixing angle: $\alpha_e=0.7050247$ rad
 incoming lepton: $0.761596|+\rangle +0.648053|-\rangle$
 proton mixing angle: $\alpha_p=0.70501577$ rad
 incoming proton: $0.761601|+\rangle +0.648046|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000139487$
 $\theta_{P'}=0.0108069$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.60549$, $\phi_\gamma=6.28094$
 $E_\gamma=2.03107$, $\theta_{\ell'}=7.42072\text{e-}07$, $\phi_{\ell'}=0.463893$
 $Q^2=19.5945$, $x_B=0.837181$, $t=-3.0947$, $W^2=4.6907$, $y=0.970367$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0309$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0001$
 q_γ $E= 2.0311$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0311$

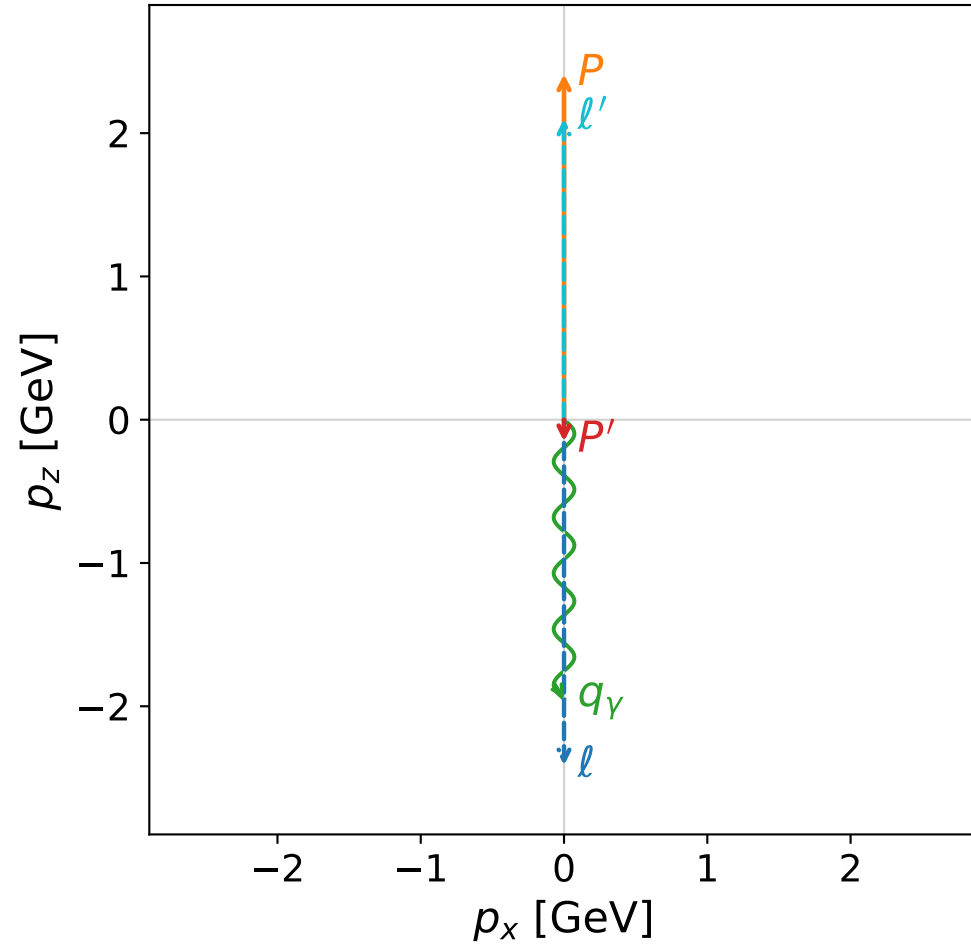
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.462\text{e+}20$; $I_h=-1.358\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.111\text{e-}15$; $I_{TL}=+9.301\text{e-}09$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

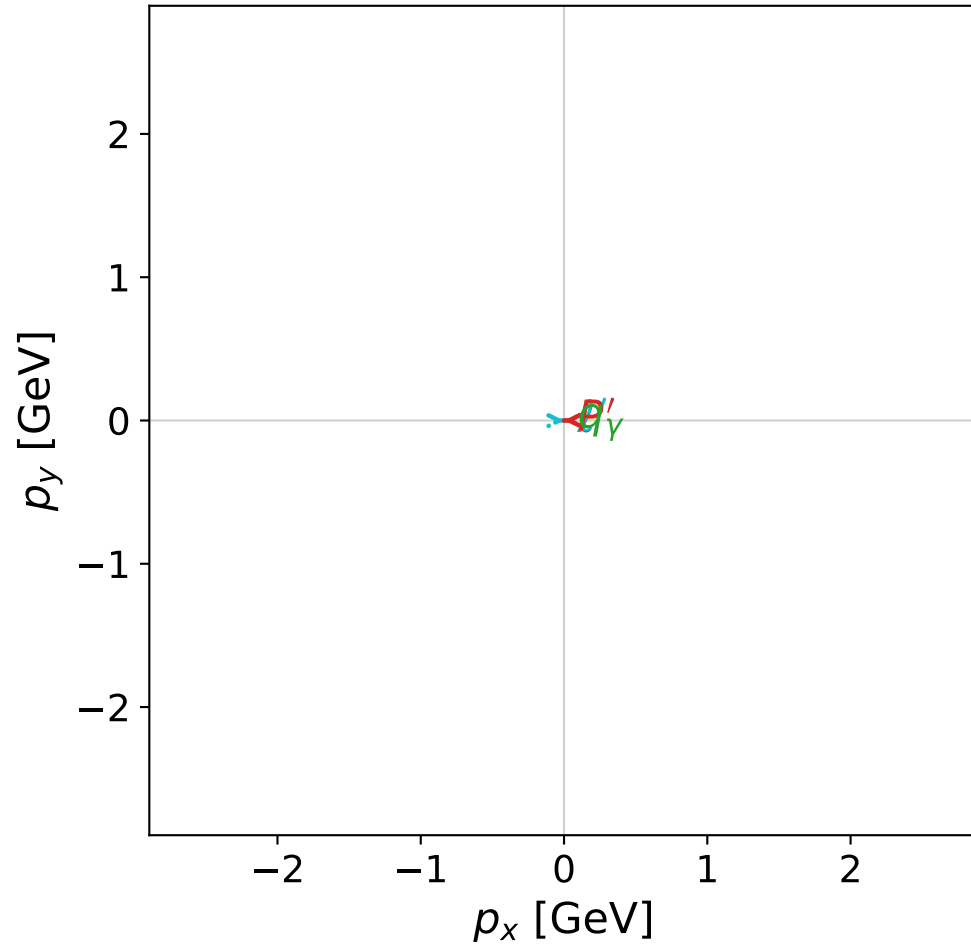


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

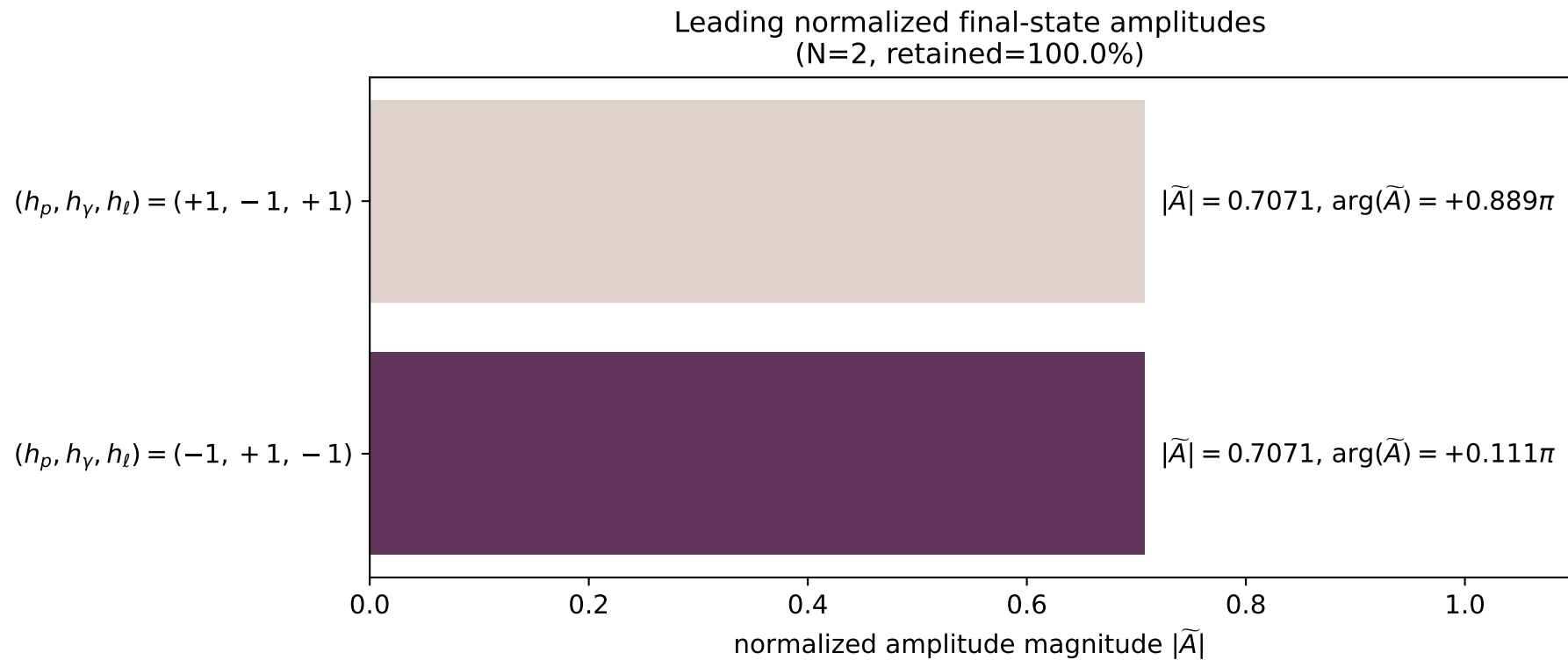


dghz_mixing_angles_polarization_01_local_minimum_32
 $d_{\text{GHZ}}=2.62922\text{e-}08$
 region: polarization_cluster_01_local_minimum_32
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0032
 lepton mixing angle: $\alpha_e=0.6915138$ rad
 incoming lepton: $0.770282|+\rangle + 0.637704|-\rangle$
 proton mixing angle: $\alpha_p=0.69151265$ rad
 incoming proton: $0.770282|+\rangle + 0.637703|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.152285$
 $\theta_{p'}=3.14116$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.14913$, $\phi_\gamma=2.80196$
 $E_\gamma=1.94872$, $\theta_{\ell'}=3.10422\text{e-}05$, $\phi_{\ell'}=0.00753968$
 $Q^2=20.2706$, $x_B=0.866979$, $t=-3.89357$, $W^2=3.98998$, $y=0.969344$

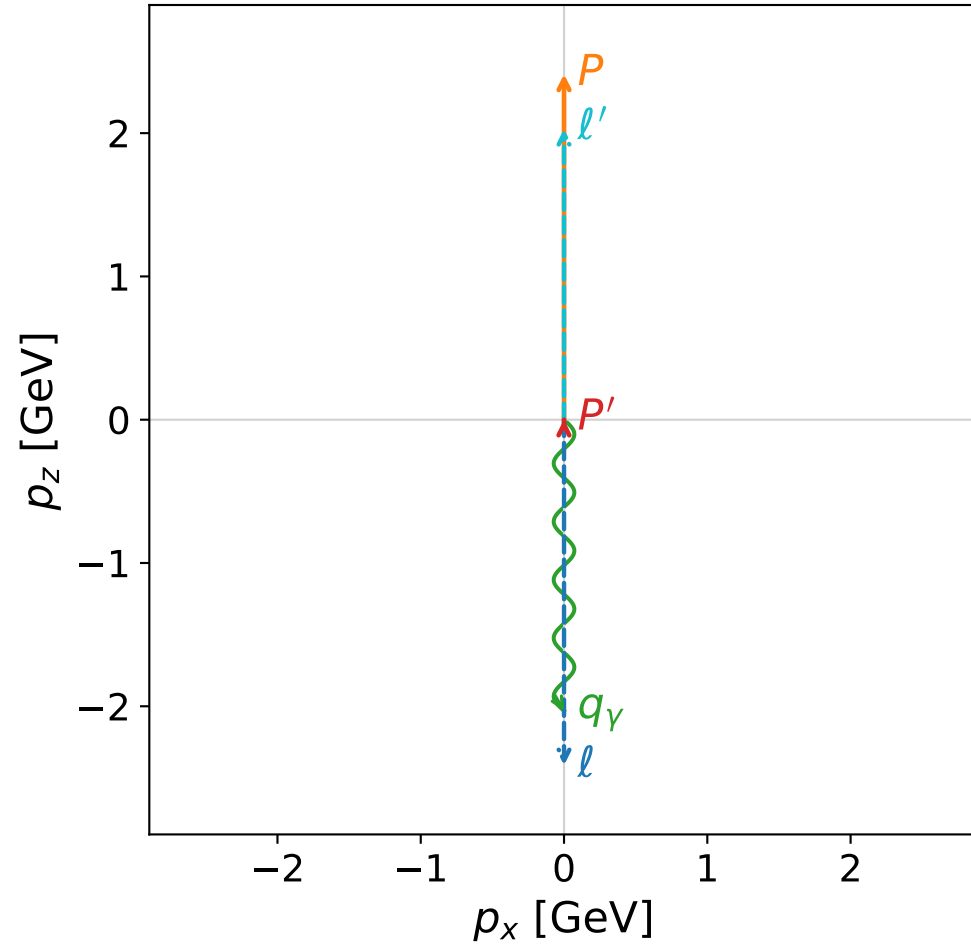
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1010$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.1010$
 P' $E= 0.9503$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.1523$
 q_γ $E= 1.9487$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9487$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.039\text{e+}16$; $I_h=-7.733\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.468\text{e-}12$; $I_{TL}=+4.153\text{e-}07$

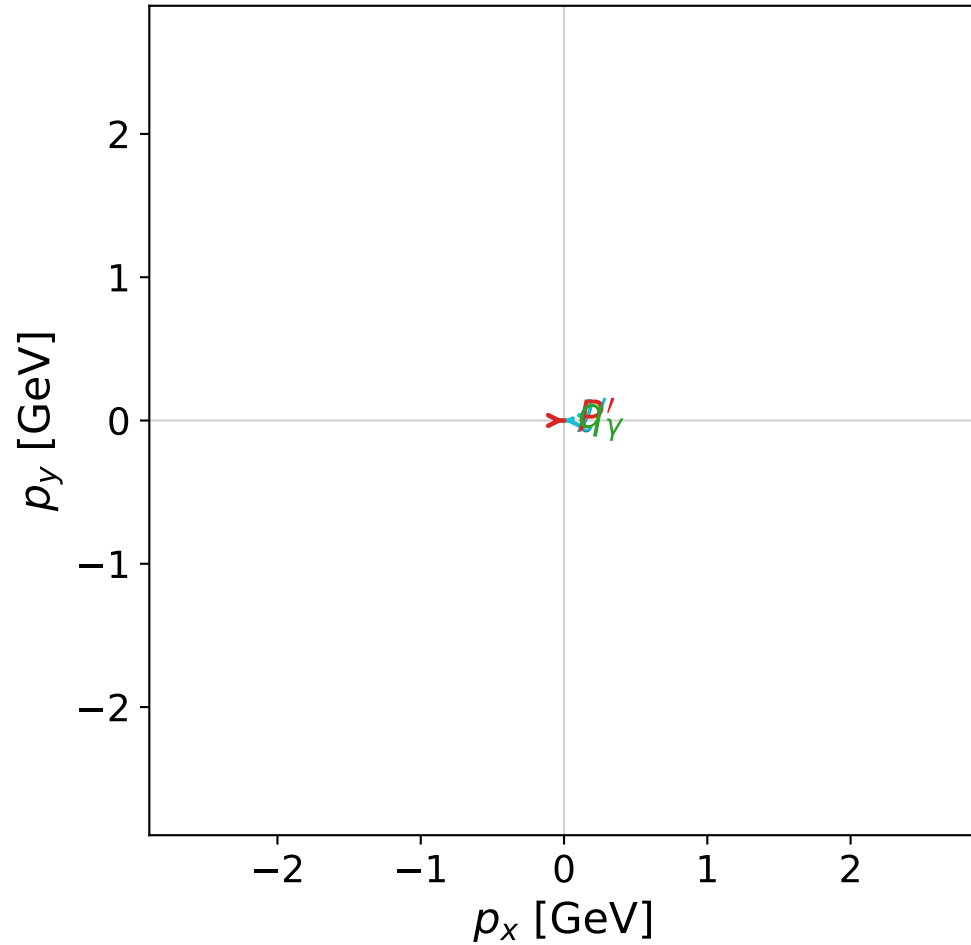


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

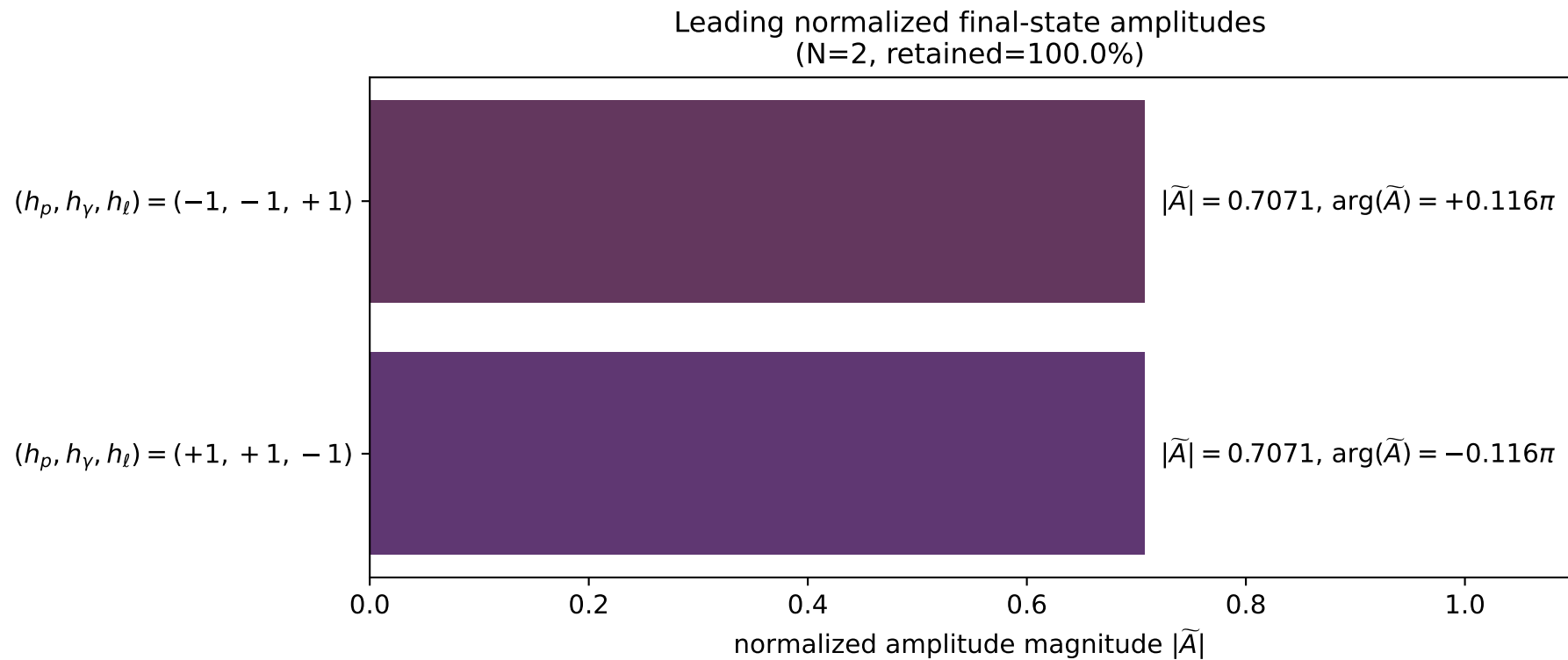


dghz_mixing_angles_polarization_01_local_minimum_35
 $d_{\text{GHZ}}=2.63636\text{e-}08$
 region: polarization_cluster_01_local_minimum_35
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0035
 lepton mixing angle: $\alpha_e=0.86247052$ rad
 incoming lepton: $0.650563|+\rangle + 0.759452|-\rangle$
 proton mixing angle: $\alpha_p=0.86246408$ rad
 incoming proton: $0.650568|+\rangle + 0.759448|-\rangle$

$s=24.999$, $\sqrt{s}=4.9999$, $|\vec{P}|=2.41196$, $|\vec{P}'|=0.00239249$
 $\theta_{P'}=0.00487673$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.28293$, $\phi_\gamma=3.50677$
 $E_\gamma=2.03215$, $\theta_{\ell'}=5.74822\text{e-}06$, $\phi_{\ell'}=3.14134$
 $Q^2=19.5828$, $x_B=0.836699$, $t=-3.08376$, $W^2=4.70189$, $y=0.970382$

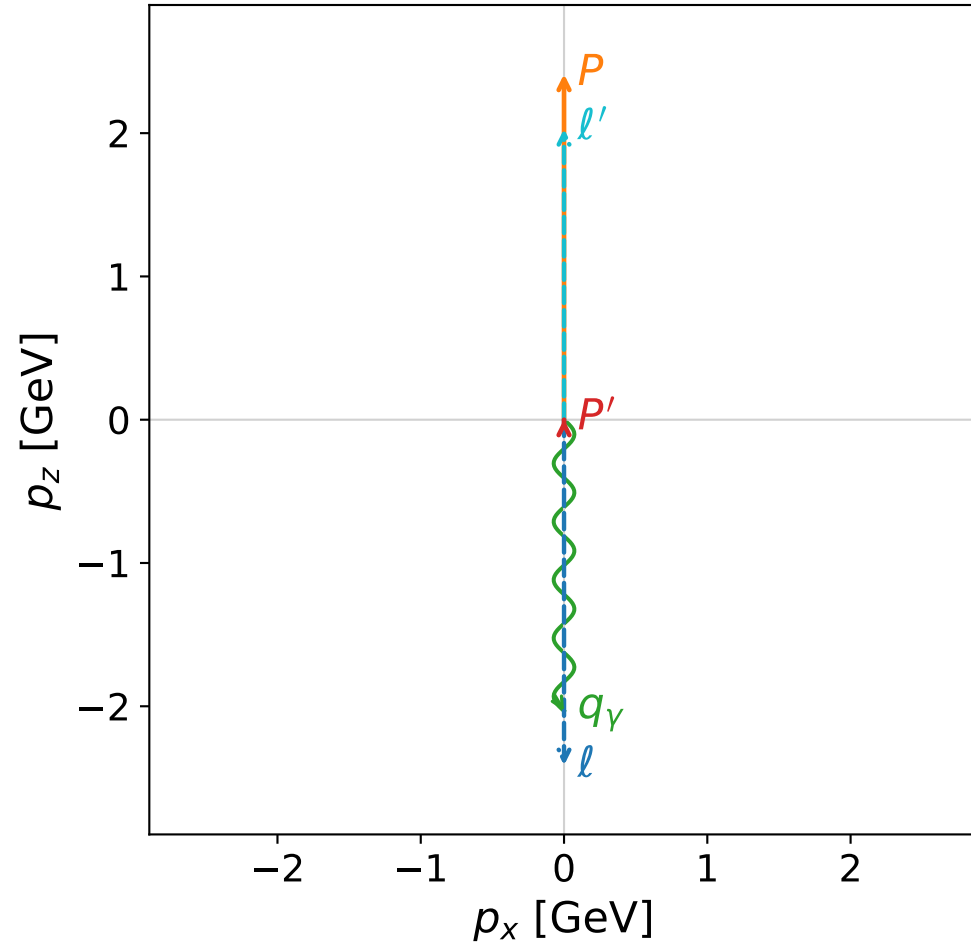
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0298$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0298$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0024$
 q_γ $E= 2.0321$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0321$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.468\text{e+}18$; $I_h=-1.123\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.271\text{e-}13$; $I_{TL}=+7.742\text{e-}08$

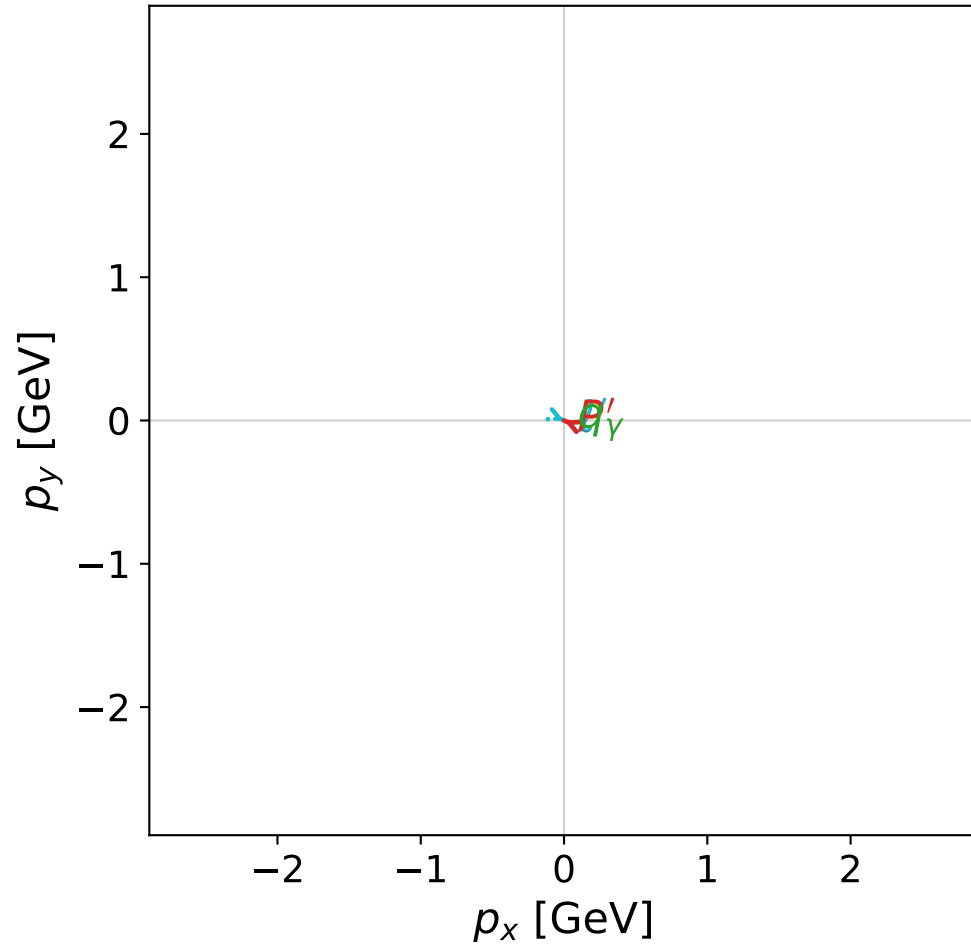


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

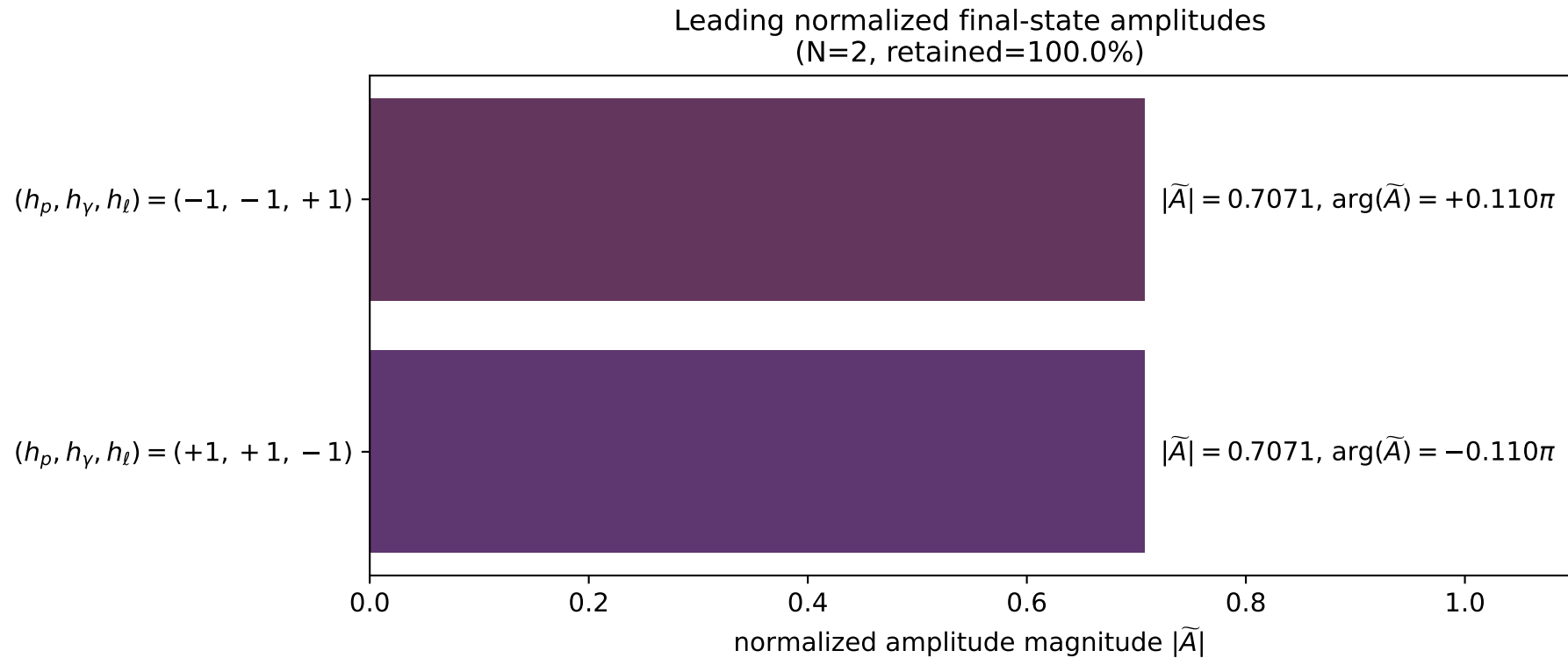


dghz_mixing_angles_polarization_01_local_minimum_44
 $d_{\text{GHZ}}=2.65428\text{e-}08$
 region: polarization_cluster_01_local_minimum_44
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0044
 lepton mixing angle: $\alpha_e=0.70477583$ rad
 incoming lepton: $0.761757|+\rangle +0.647863|-\rangle$
 proton mixing angle: $\alpha_p=0.70475776$ rad
 incoming proton: $0.761769|+\rangle +0.647849|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00358496$
 $\theta_{P'}=2.38484\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=2.72855$, $\phi_Y=3.48858$
 $E_Y=2.03279$, $\theta_{\ell'}=4.21468\text{e-}08$, $\phi_{\ell'}=5.87014$
 $Q^2=19.5779$, $x_B=0.836447$, $t=-3.07811$, $W^2=4.70796$, $y=0.970392$

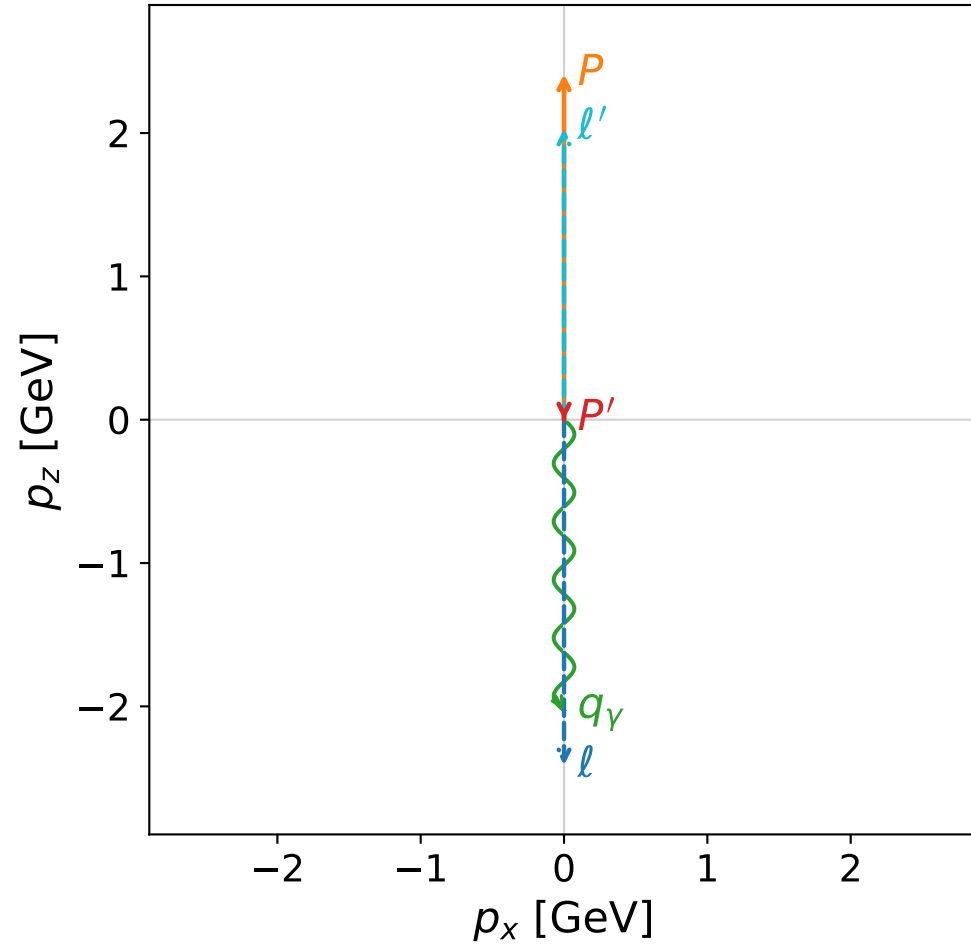
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0292$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0292$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0036$
 q_Y $E= 2.0328$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0328$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.236\text{e+}22$; $I_h=-7.561\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.862\text{e-}18$; $I_{TL}=+5.447\text{e-}10$

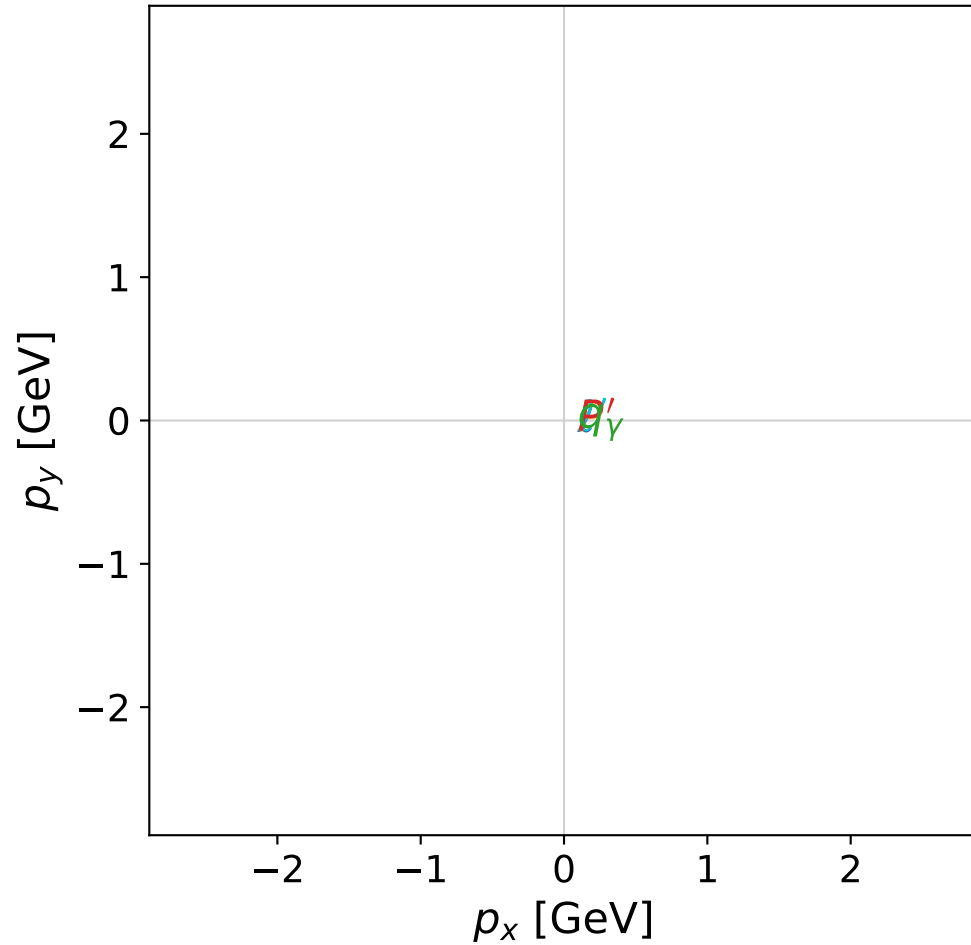


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

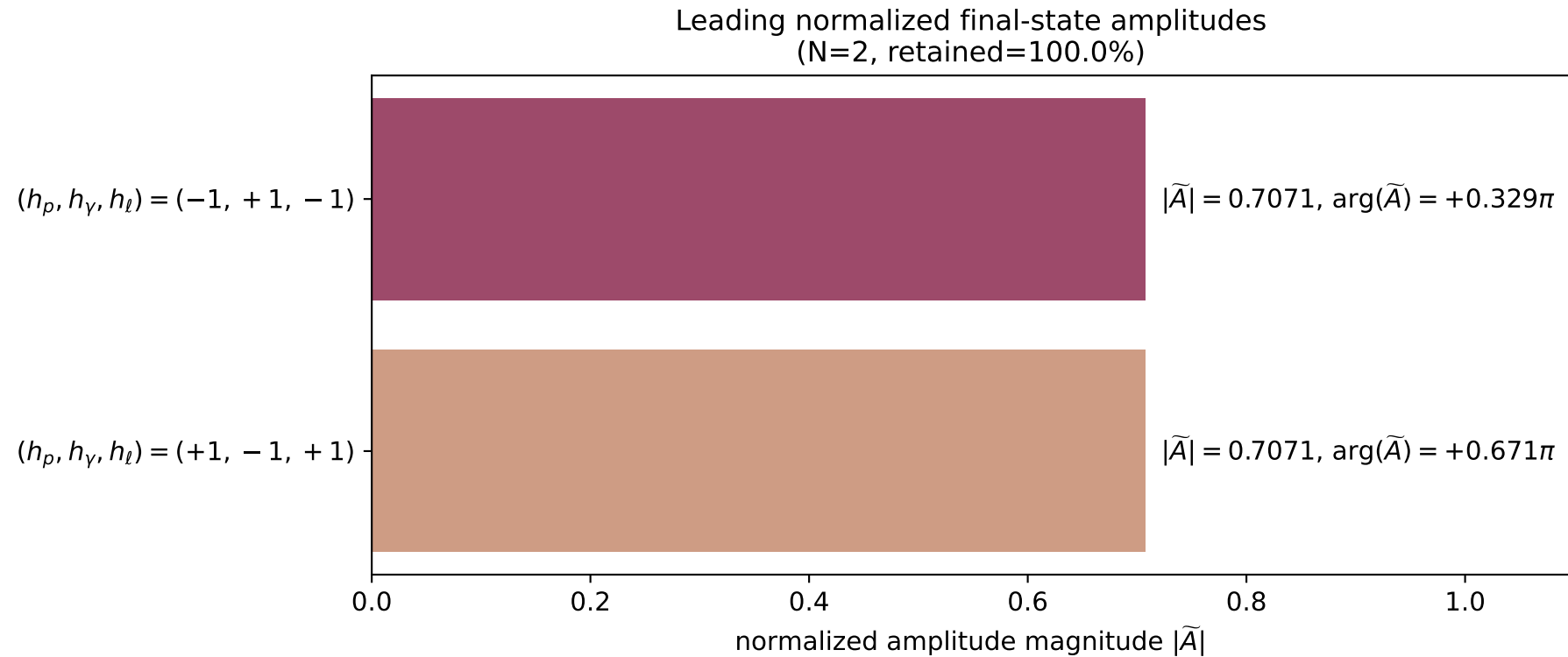


dghz_mixing_angles_polarization_01_local_minimum_46
 $d_{\text{GHZ}}=2.66651\text{e-}08$
 region: polarization_cluster_01_local_minimum_46
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0046
 lepton mixing angle: $\alpha_e=0.78323852$ rad
 incoming lepton: $0.708632|+\rangle +0.705578|-\rangle$
 proton mixing angle: $\alpha_p=0.78324505$ rad
 incoming proton: $0.708628|+\rangle +0.705583|-\rangle$

$s=24.9844$, $\sqrt{s}=4.99844$, $|\vec{P}|=2.41121$, $|\vec{P}'|=0.000949457$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14724$, $\phi_\gamma=5.24835$
 $E_\gamma=2.02974$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.10635$
 $Q^2=19.5857$, $x_B=0.837366$, $t=-3.09854$, $W^2=4.68379$, $y=0.970342$

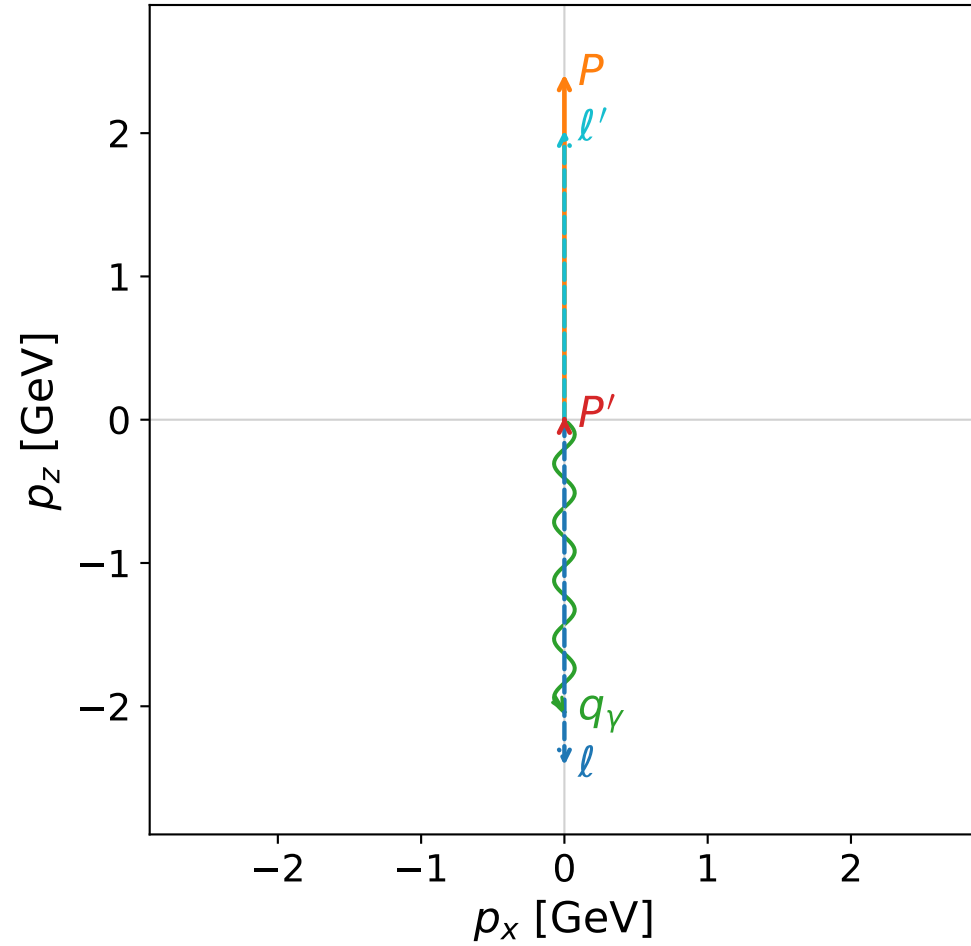
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4112$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4112$
 P $E= 2.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4112$
 ℓ' $E= 2.0307$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0009$
 q_γ $E= 2.0297$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0297$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.070\text{e}+24$; $I_h=-4.776\text{e-}16$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=8.739\text{e-}31$; $I_{TL}=+6.065\text{e-}20$

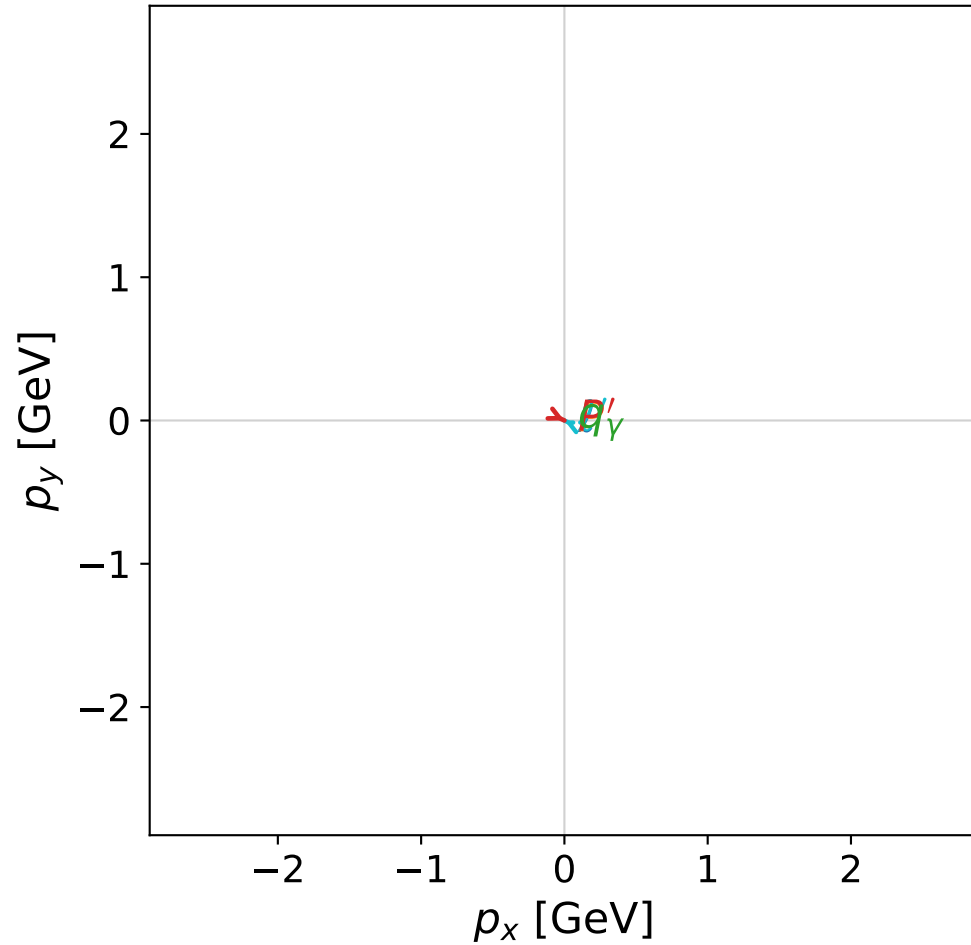


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

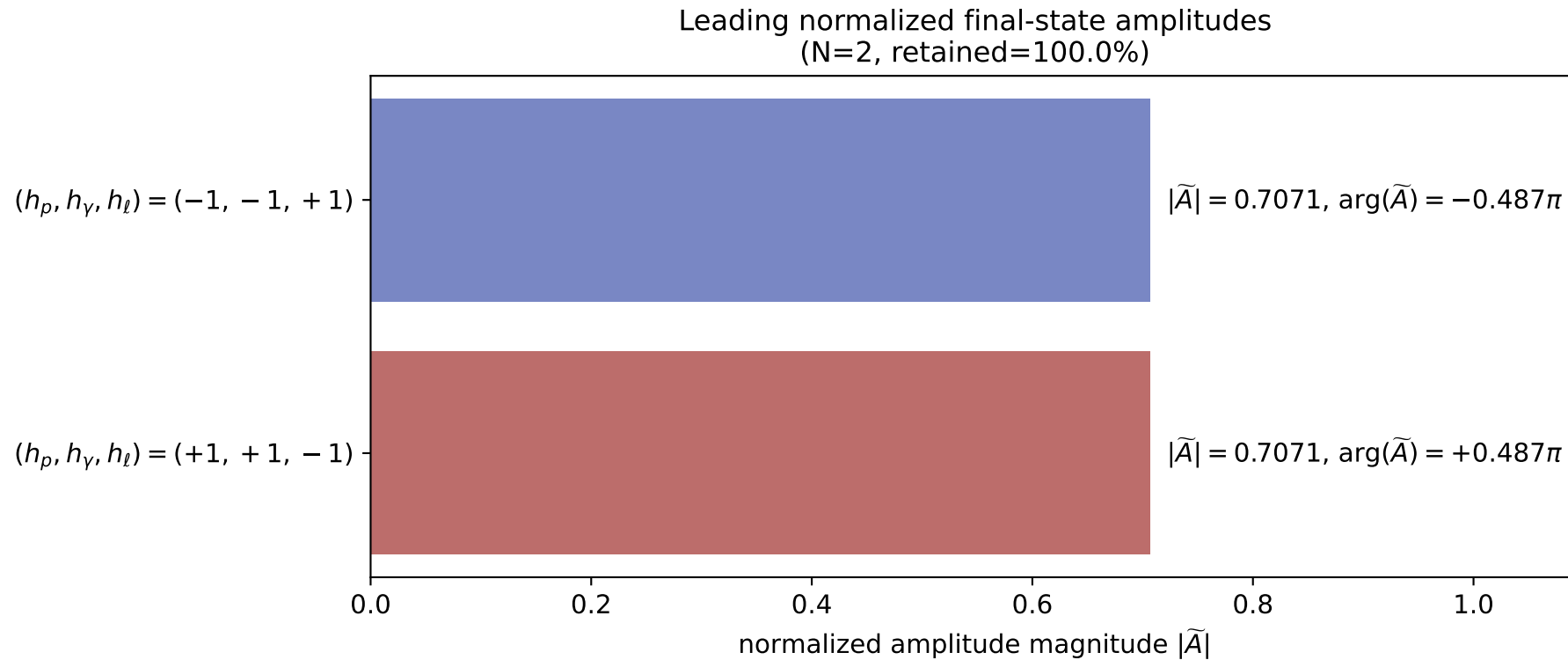


dghz_mixing_angles_polarization_01_local_minimum_47
 $d_{\text{GHZ}}=2.6669\text{e-}08$
 region: polarization_cluster_01_local_minimum_47
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0047
 lepton mixing angle: $\alpha_e=0.77596456$ rad
 incoming lepton: $0.713746|+\rangle +0.700405|-\rangle$
 proton mixing angle: $\alpha_p=0.77595102$ rad
 incoming proton: $0.713755|+\rangle +0.700395|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0205162$
 $\theta_{P'}=0.000153398$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.83508$, $\phi_\gamma=1.61218$
 $E_\gamma=2.04115$, $\theta_{\ell'}=1.55751\text{e-}06$, $\phi_{\ell'}=2.69349$
 $Q^2=19.4952$, $x_B=0.832806$, $t=-2.99756$, $W^2=4.7937$, $y=0.970517$

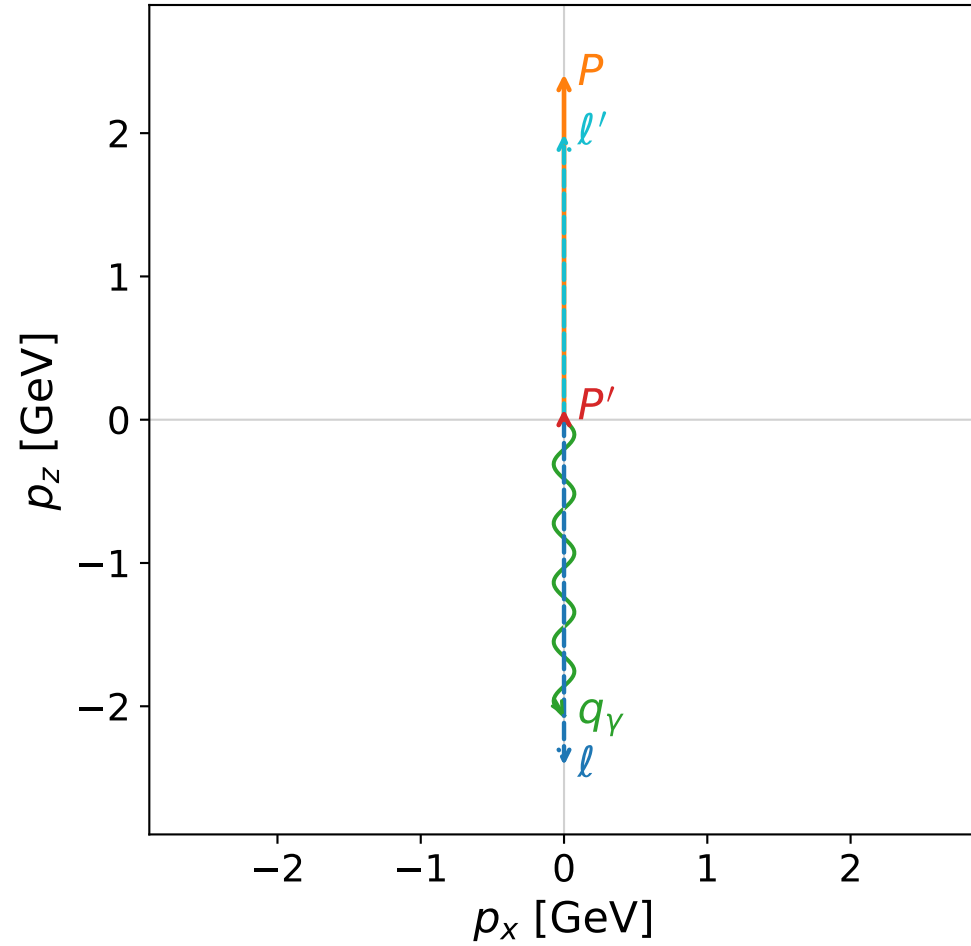
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0206$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0205$
 q_γ $E= 2.0411$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0411$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.474\text{e+}19$; $I_h=+3.320\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.532\text{e-}15$; $I_{TL}=-2.346\text{e-}09$

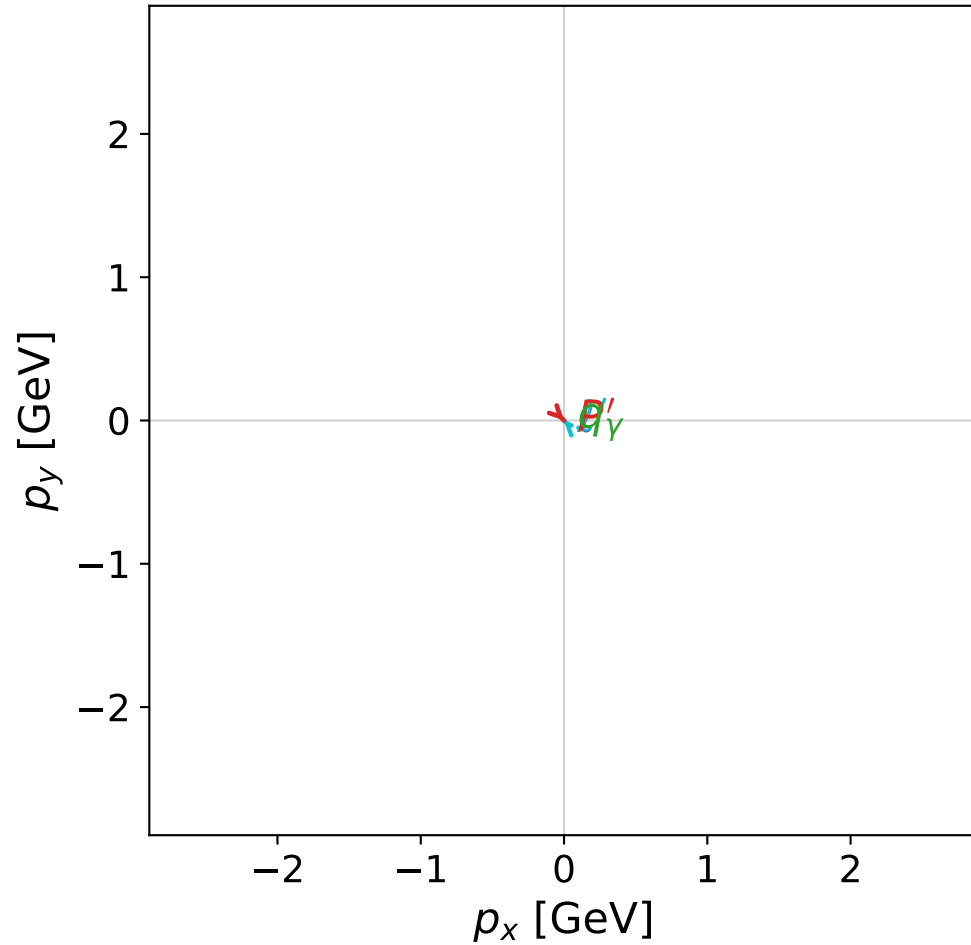


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

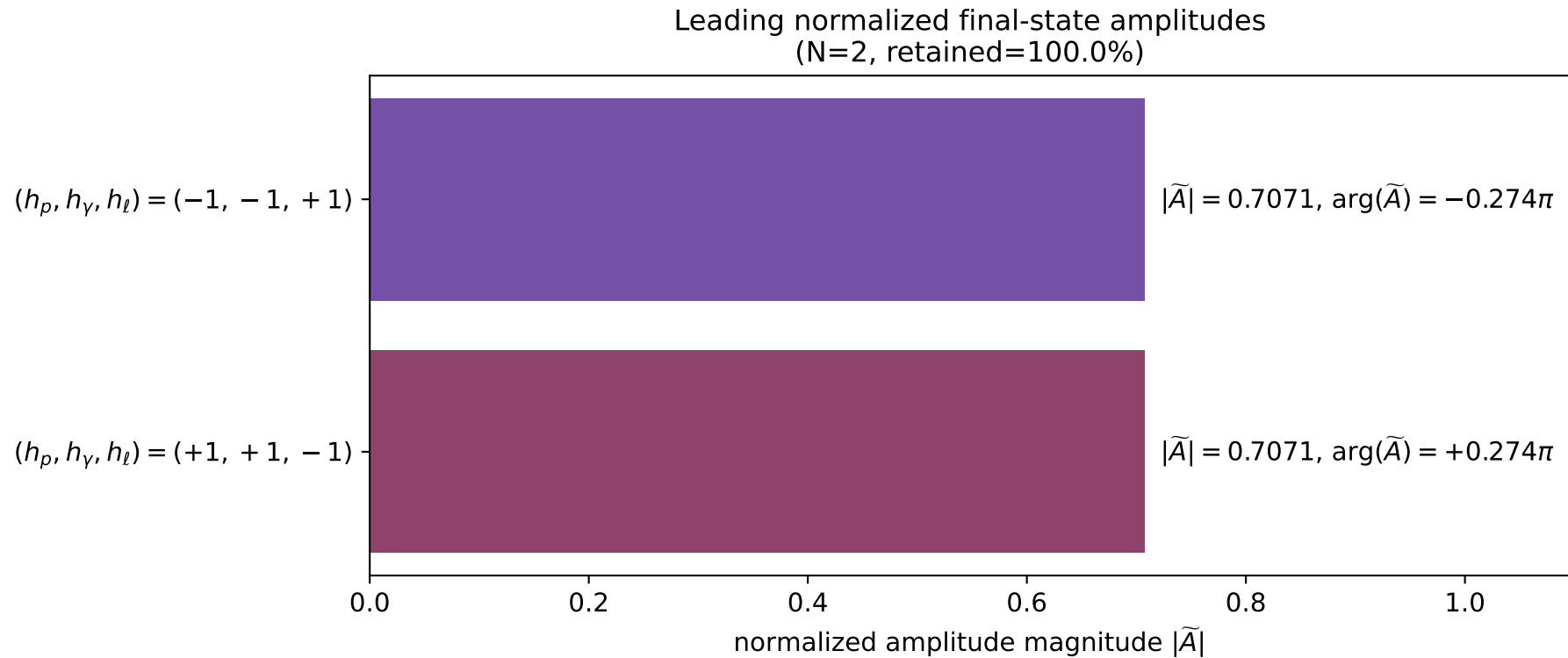


dghz_mixing_angles_polarization_01_local_minimum_57
 $d_{\text{GHZ}}=2.72176\text{e-}08$
 region: polarization_cluster_01_local_minimum_57
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0057
 lepton mixing angle: $\alpha_e=0.84620821$ rad
 incoming lepton: $0.662827|+\rangle +0.748772|-\rangle$
 proton mixing angle: $\alpha_p=0.84620257$ rad
 incoming proton: $0.662831|+\rangle +0.748769|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0736114$
 $\theta_{P'}=0.000405073$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.49139$, $\phi_\gamma=2.28109$
 $E_\gamma=2.06636$, $\theta_{\ell'}=1.49632\text{e-}05$, $\phi_{\ell'}=2.34979$
 $Q^2=19.2262$, $x_B=0.820972$, $t=-2.75519$, $W^2=5.07248$, $y=0.970924$

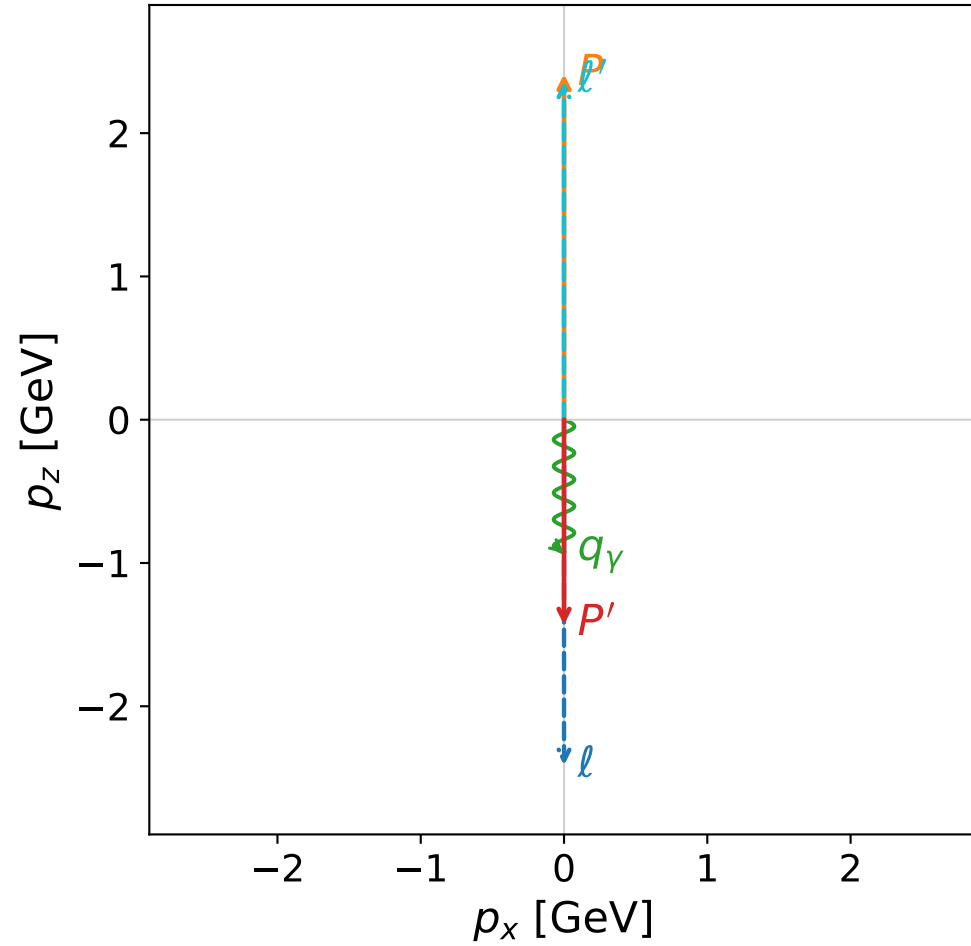
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9928$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9928$
 P' $E= 0.9409$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0736$
 q_γ $E= 2.0664$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0664$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.725\text{e+}17$; $I_h=-1.589\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.014\text{e-}12$; $I_{TL}=+1.215\text{e-}07$

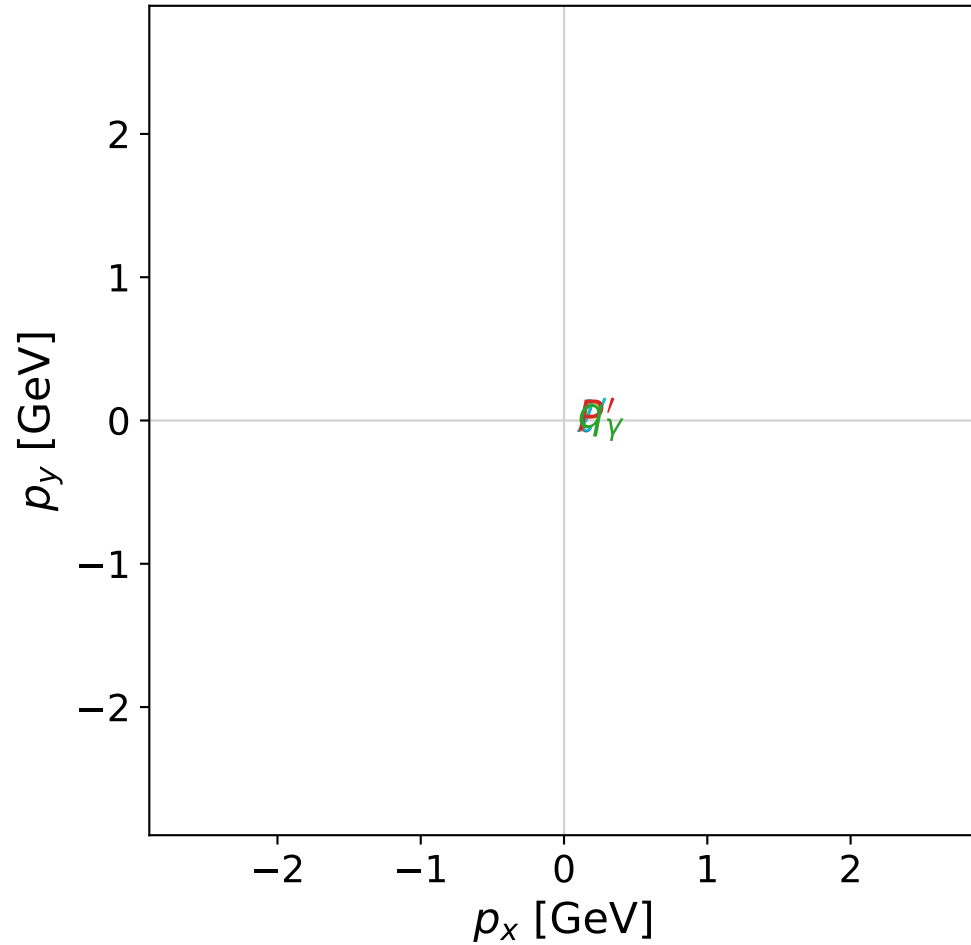


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

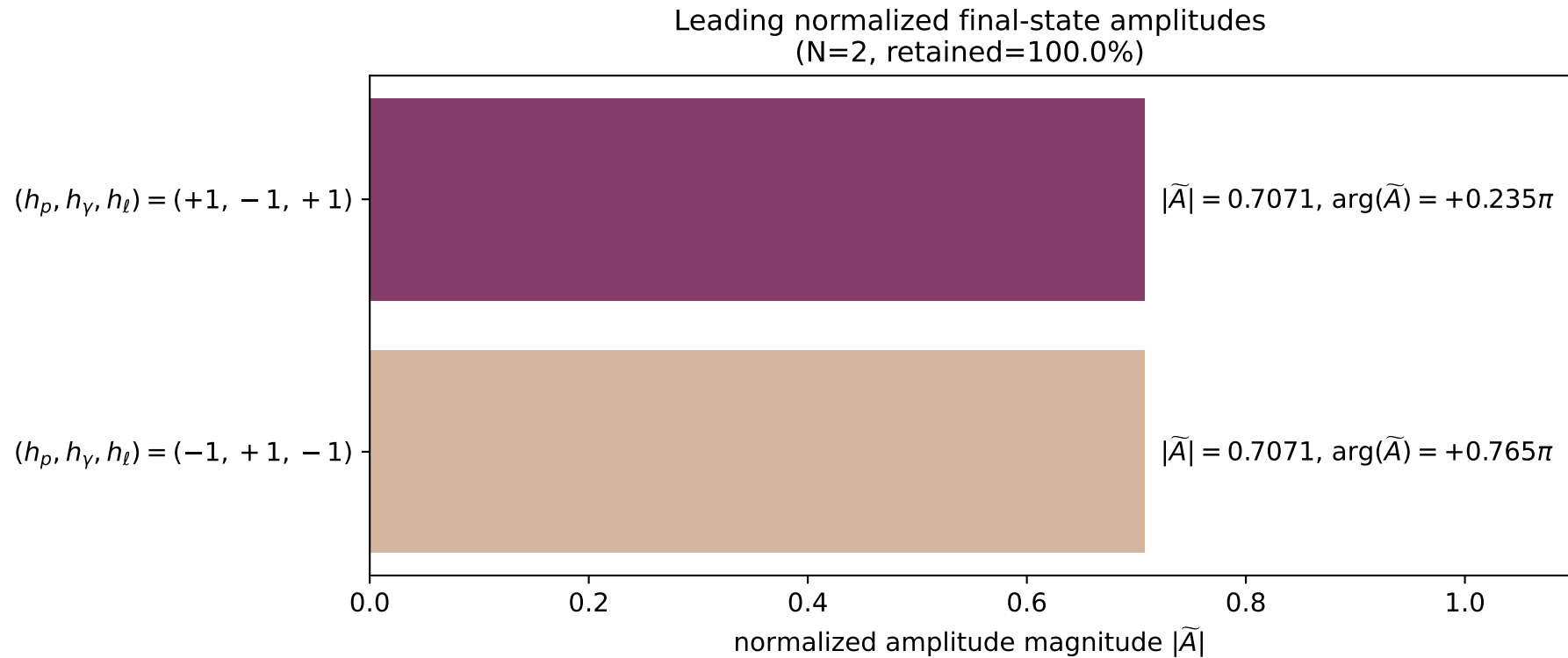


dghz_mixing_angles_polarization_01_local_minimum_58
 $d_{\text{GHZ}}=2.7258\text{e-}08$
 region: polarization_cluster_01_local_minimum_58
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0058
 lepton mixing angle: $\alpha_e=1.0707816$ rad
 incoming lepton: $0.479438|+\rangle + 0.877576|-\rangle$
 proton mixing angle: $\alpha_p=1.0707773$ rad
 incoming proton: $0.479442|+\rangle + 0.877573|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.43163$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.31472$, $\phi_\gamma=3.87889$
 $E_\gamma=0.92841$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.393495$
 $Q^2=22.7698$, $x_B=0.977683$, $t=-14.0055$, $W^2=1.39961$, $y=0.965565$

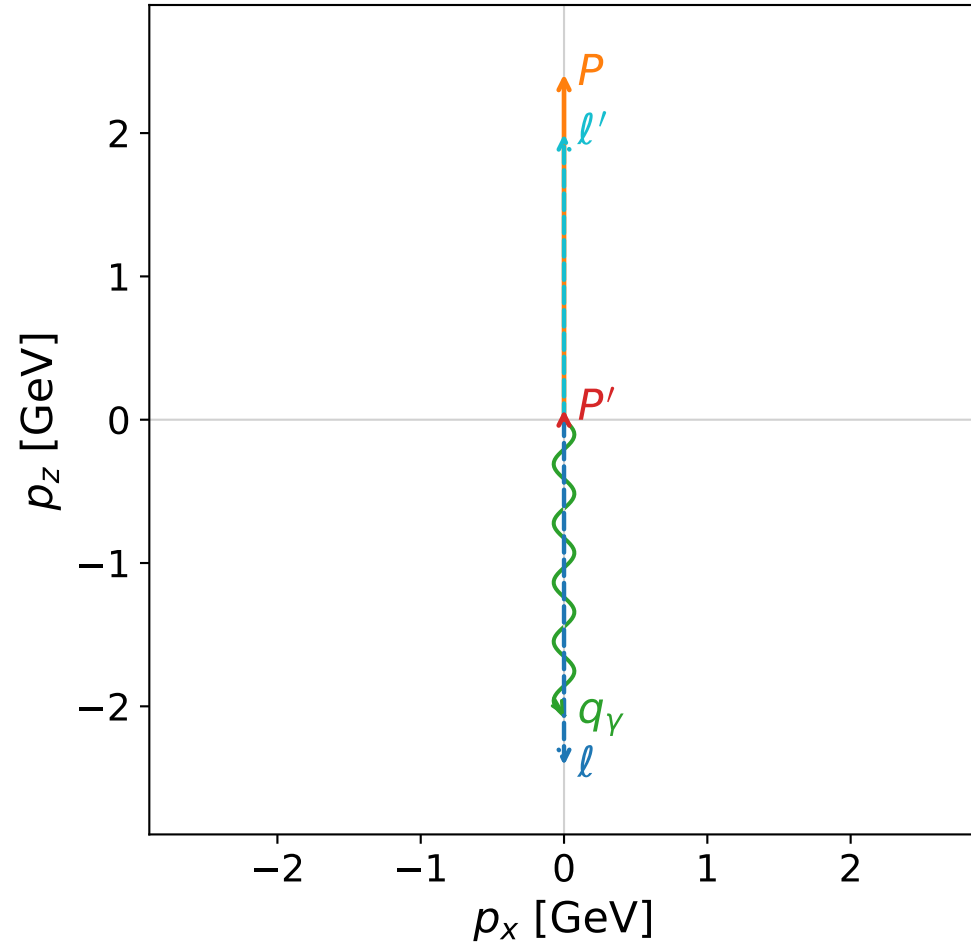
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3600$
 P' $E= 1.7116$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4316$
 q_γ $E= 0.9284$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9284$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.758\text{e}+23$; $I_h=-8.781\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.222\text{e-}28$; $I_{TL}=-2.762\text{e-}18$

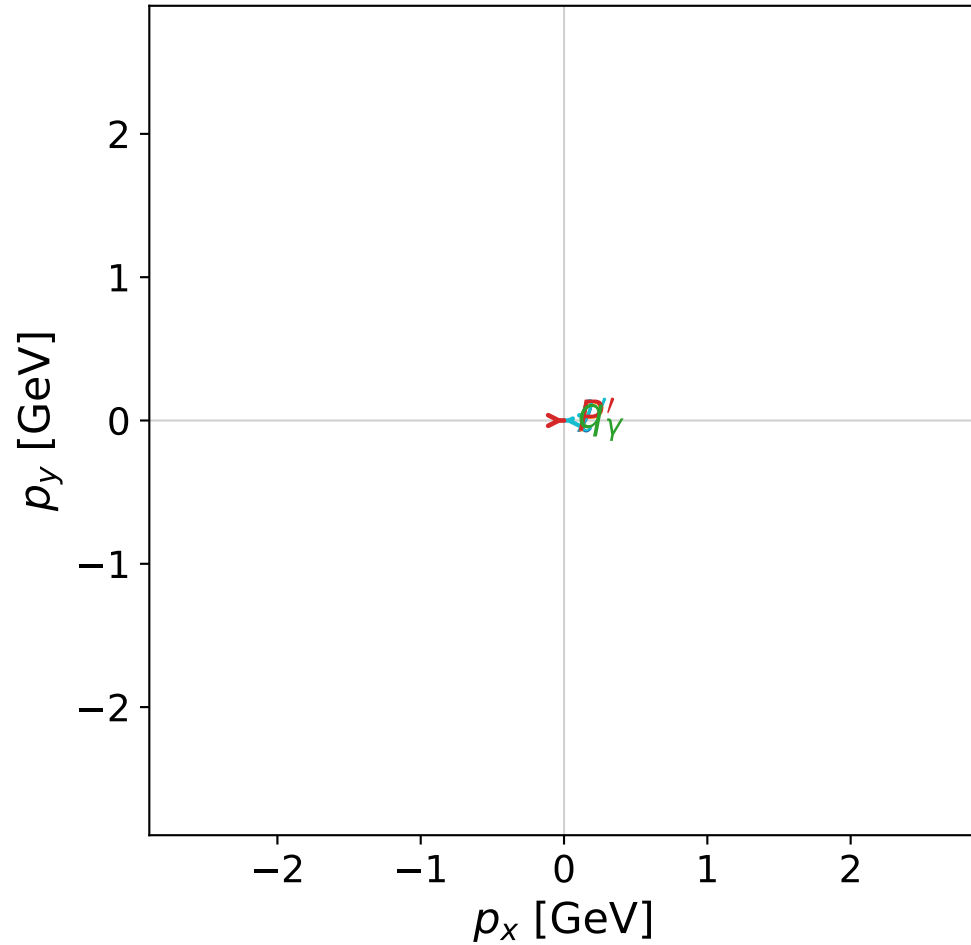


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

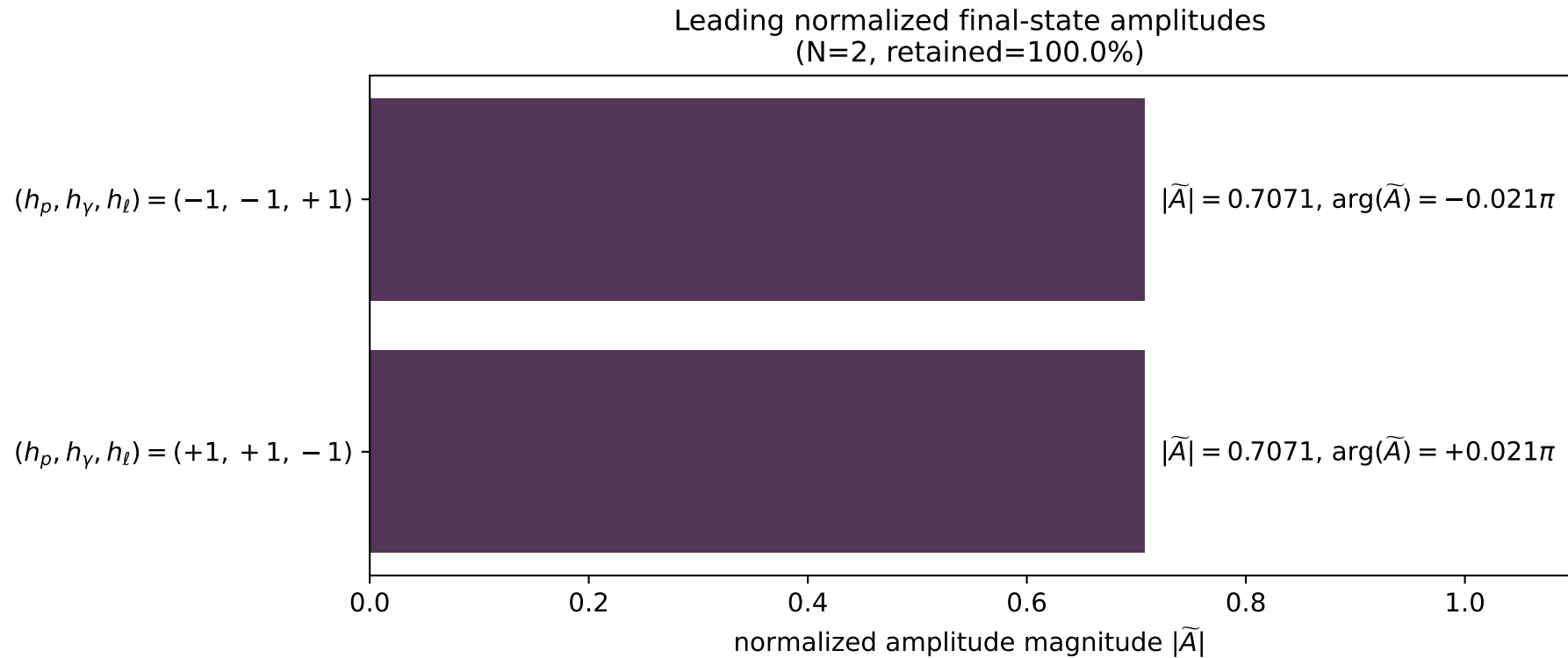


dghz_mixing_angles_polarization_01_local_minimum_61
 $d_{\text{GHZ}}=2.75163\text{e-}08$
 region: polarization_cluster_01_local_minimum_61
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0061
 lepton mixing angle: $\alpha_e=0.88093079$ rad
 incoming lepton: $0.636433|+\rangle +0.771332|-\rangle$
 proton mixing angle: $\alpha_p=0.88091145$ rad
 incoming proton: $0.636448|+\rangle +0.771319|-\rangle$

$s=24.9904$, $\sqrt{s}=4.99903$, $|\vec{P}|=2.41152$, $|\vec{P}'|=0.0704794$
 $\theta_{P'}=0.000153398$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.000314416$, $\phi_\gamma=3.07524$
 $E_\gamma=2.06444$, $\theta_\ell=5.4221\text{e-}06$, $\phi_\ell=3.14191$
 $Q^2=19.2338$, $x_B=0.821656$, $t=-2.76826$, $W^2=5.05464$, $y=0.970889$

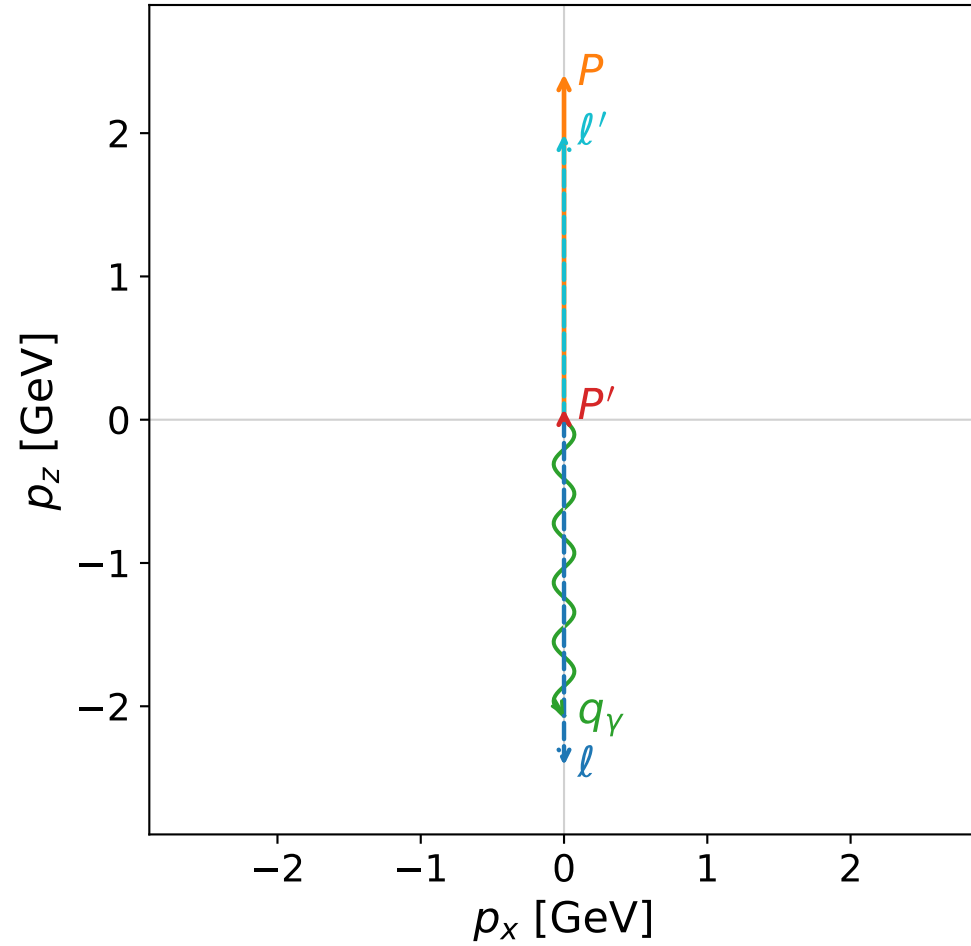
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4115$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4115$
 P $E= 2.5875$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4115$
 ℓ' $E= 1.9940$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9940$
 P' $E= 0.9406$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0705$
 q_γ $E= 2.0644$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0644$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.753\text{e+}18$; $I_h=-1.298\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.352\text{e-}13$; $I_{TL}=+9.878\text{e-}08$

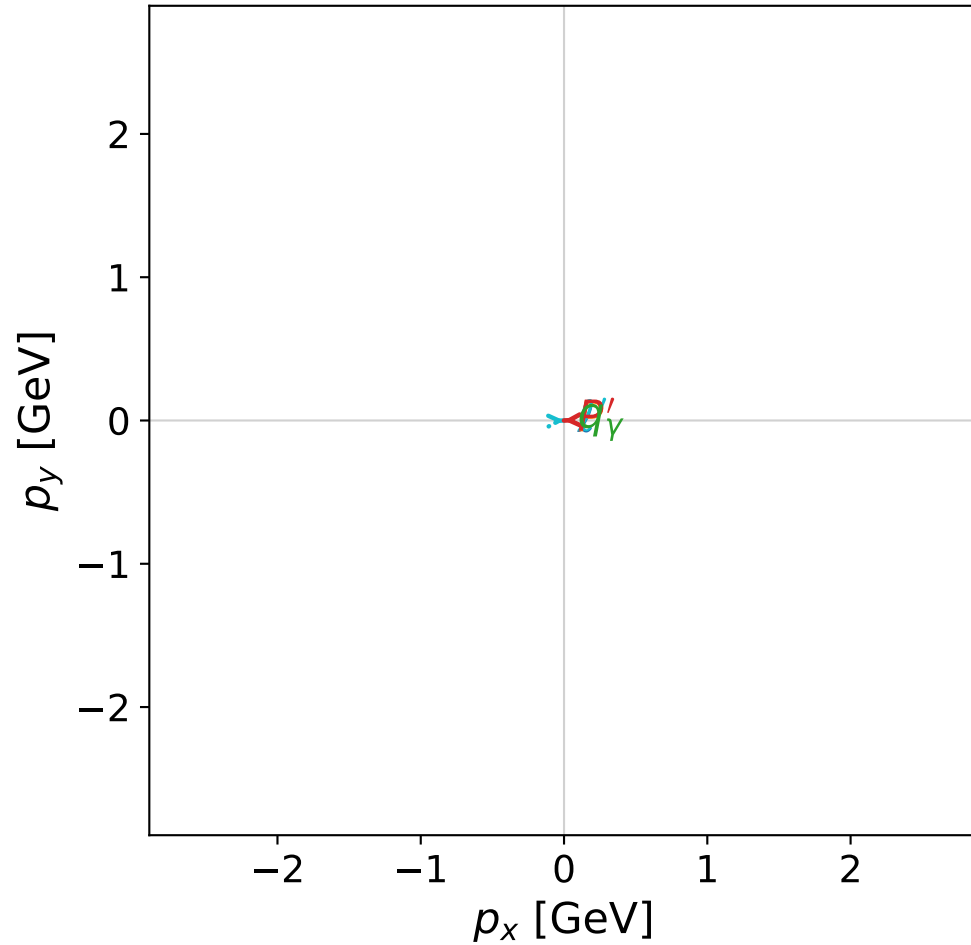


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

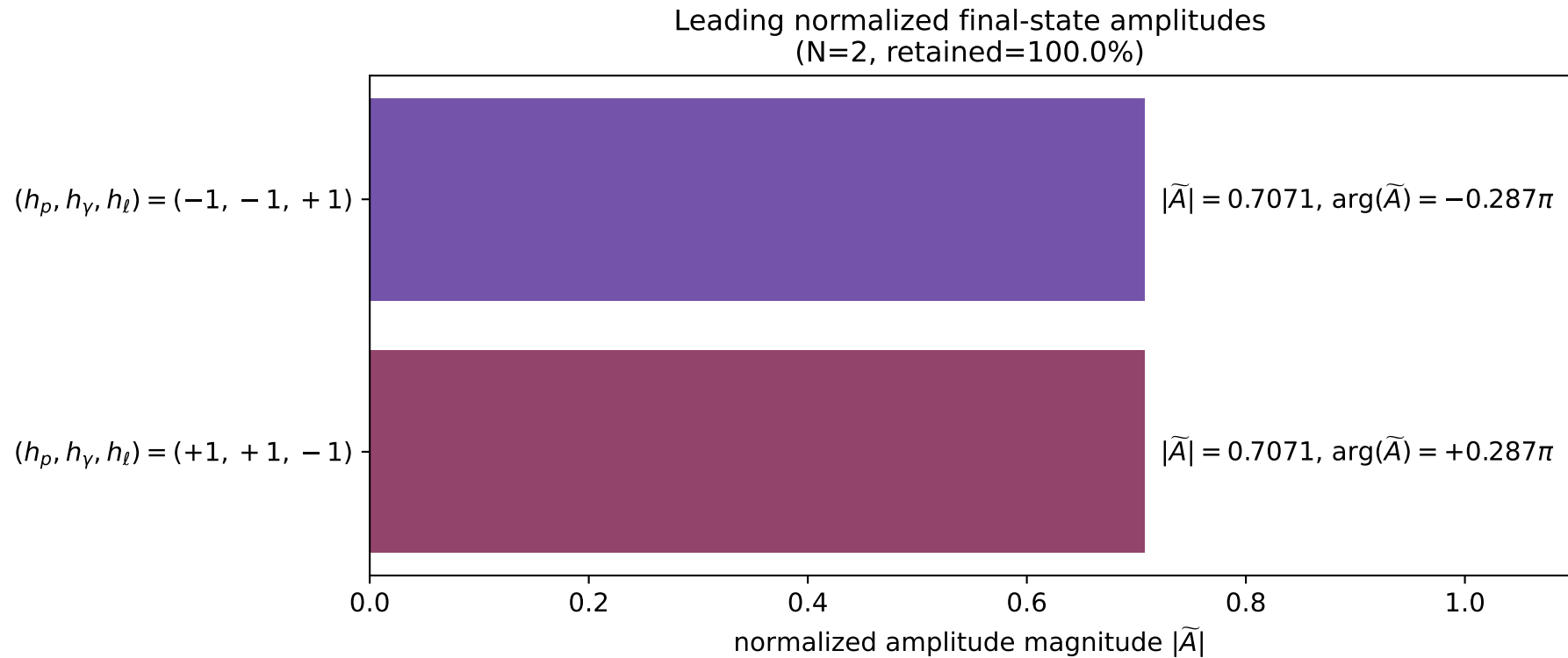


dghz_mixing_angles_polarization_01_local_minimum_65
 $d_{\text{GHZ}}=2.78329\text{e-}08$
 region: polarization_cluster_01_local_minimum_65
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0065
 lepton mixing angle: $\alpha_e=0.5734809$ rad
 incoming lepton: $0.840017|+\rangle + 0.542559|-\rangle$
 proton mixing angle: $\alpha_p=0.57346771$ rad
 incoming proton: $0.840025|+\rangle + 0.542548|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0755206$
 $\theta_{P'}=2.17755\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.18$, $\phi_\gamma=2.24128$
 $E_\gamma=2.06724$, $\theta_\ell=8.1617\text{e-}08$, $\phi_\ell=0.0384033$
 $Q^2=19.2163$, $x_B=0.820535$, $t=-2.74677$, $W^2=5.08278$, $y=0.970939$

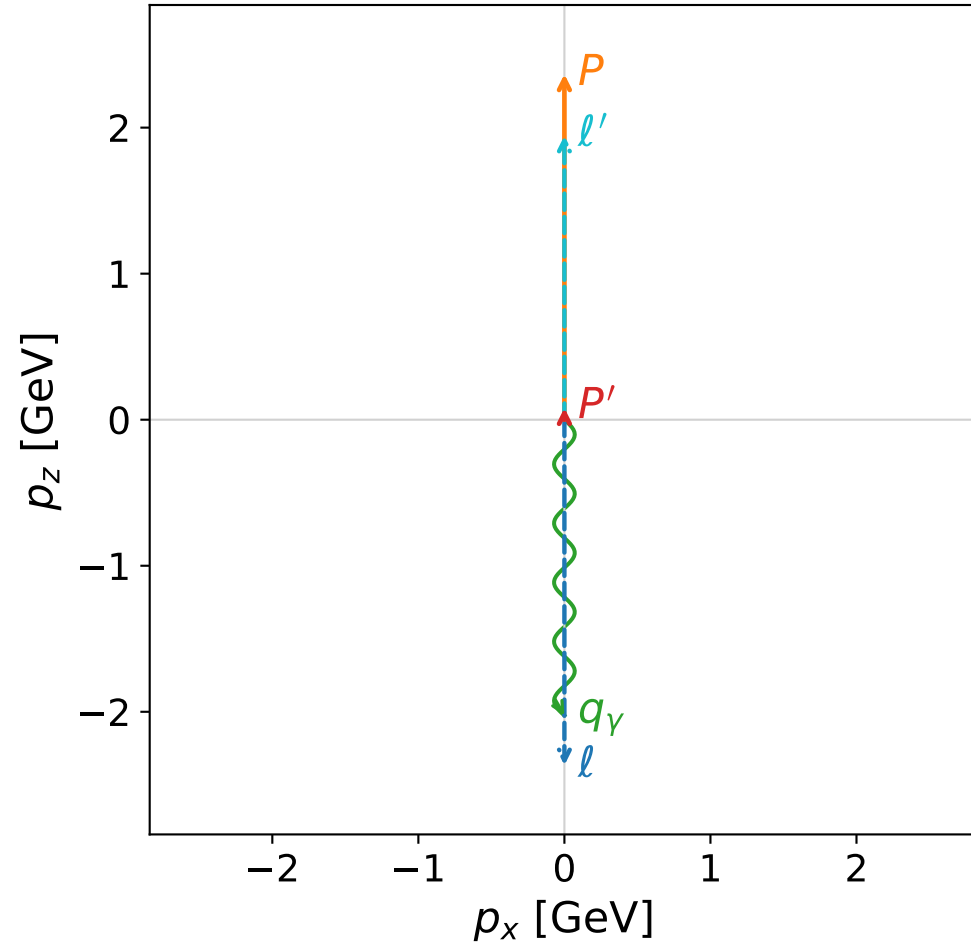
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9917$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9917$
 P' $E= 0.9410$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0755$
 q_γ $E= 2.0672$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0672$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.048\text{e+}22$; $I_h=-4.547\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.679\text{e-}17$; $I_{TL}=+3.525\text{e-}09$

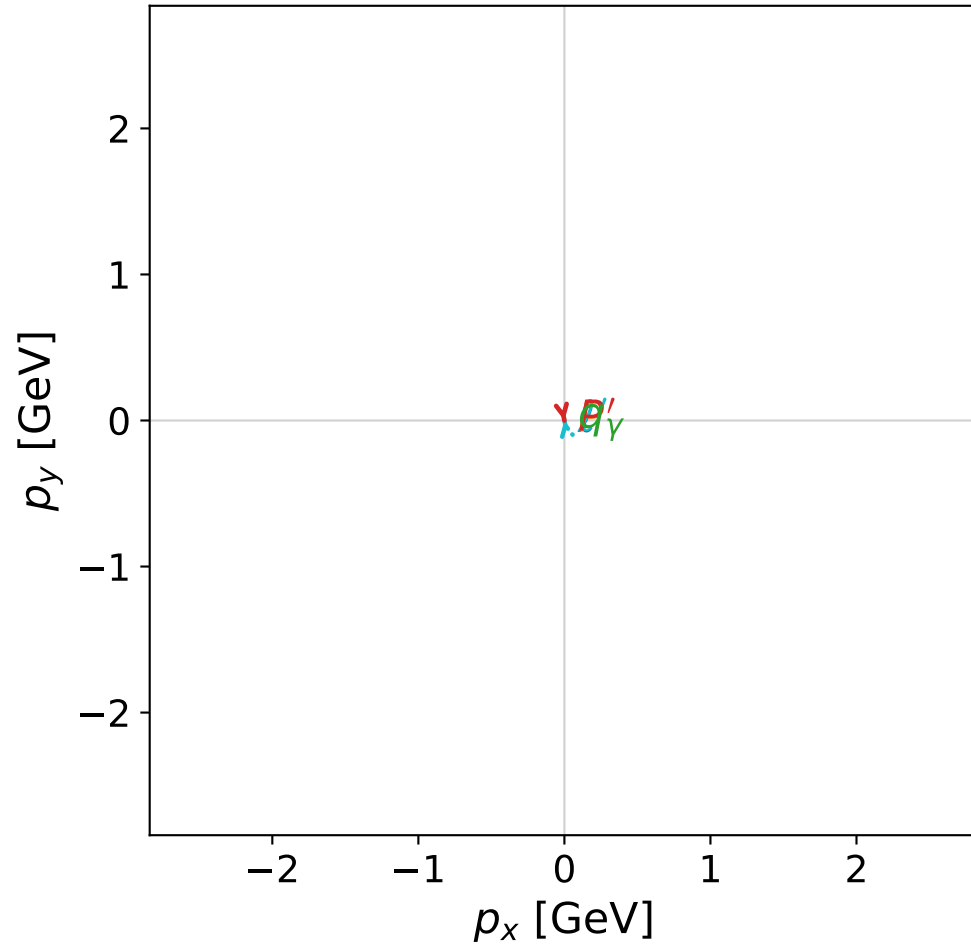


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

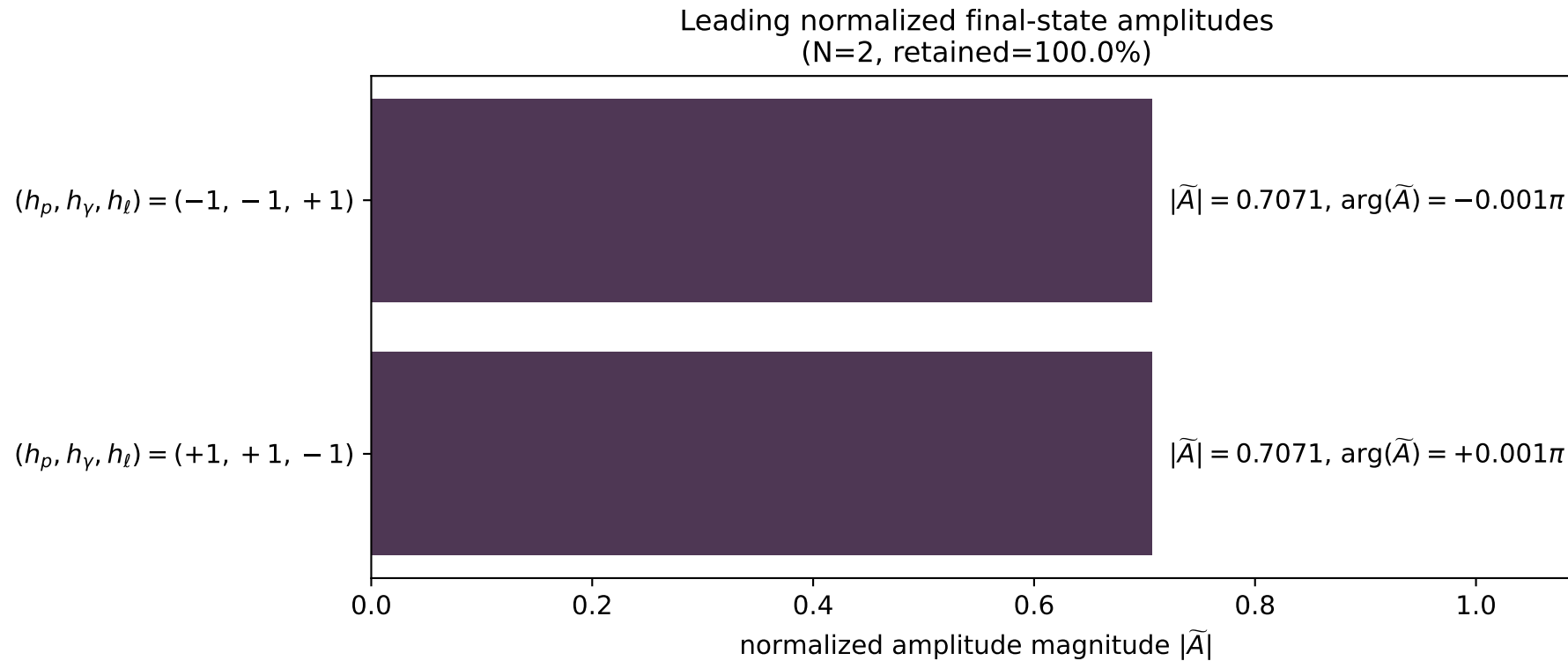


dghz_mixing_angles_polarization_01_local_minimum_69
 $d_{\text{GHZ}}=2.83483\text{e-}08$
 region: polarization_cluster_01_local_minimum_69
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0069
 lepton mixing angle: $\alpha_e=0.82158482$ rad
 incoming lepton: $0.681062|+\rangle + 0.732226|-\rangle$
 proton mixing angle: $\alpha_p=0.82156325$ rad
 incoming proton: $0.681077|+\rangle + 0.732211|-\rangle$

$s=24.1272$, $\sqrt{s}=4.91194$, $|\vec{P}|=2.36641$, $|\vec{P}'|=0.080731$
 $\theta_{P'}=0.00091855$, $\theta_\gamma=3.14159$, $\phi_P=4.90382$, $\phi_\gamma=3.13744$
 $E_\gamma=2.0256$, $\theta_\ell=3.81287\text{e-}05$, $\phi_{\ell'}=1.76223$
 $Q^2=18.4095$, $x_B=0.816362$, $t=-2.6513$, $W^2=5.02098$, $y=0.970029$

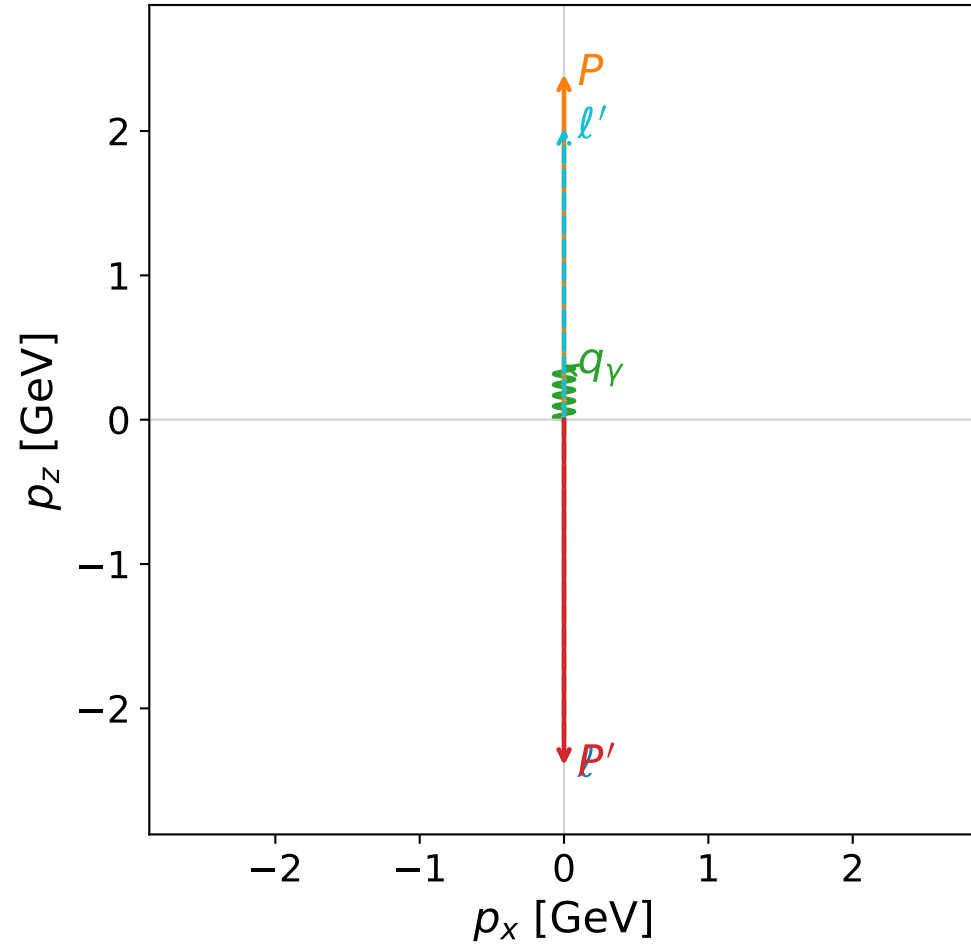
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3664$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3664$
 P $E= 2.5455$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3664$
 ℓ' $E= 1.9449$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 1.9449$
 P' $E= 0.9415$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 0.0807$
 q_γ $E= 2.0256$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0256$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.568\text{e+}16$; $I_h=-6.632\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=7.070\text{e-}12$; $I_{TL}=+5.264\text{e-}08$

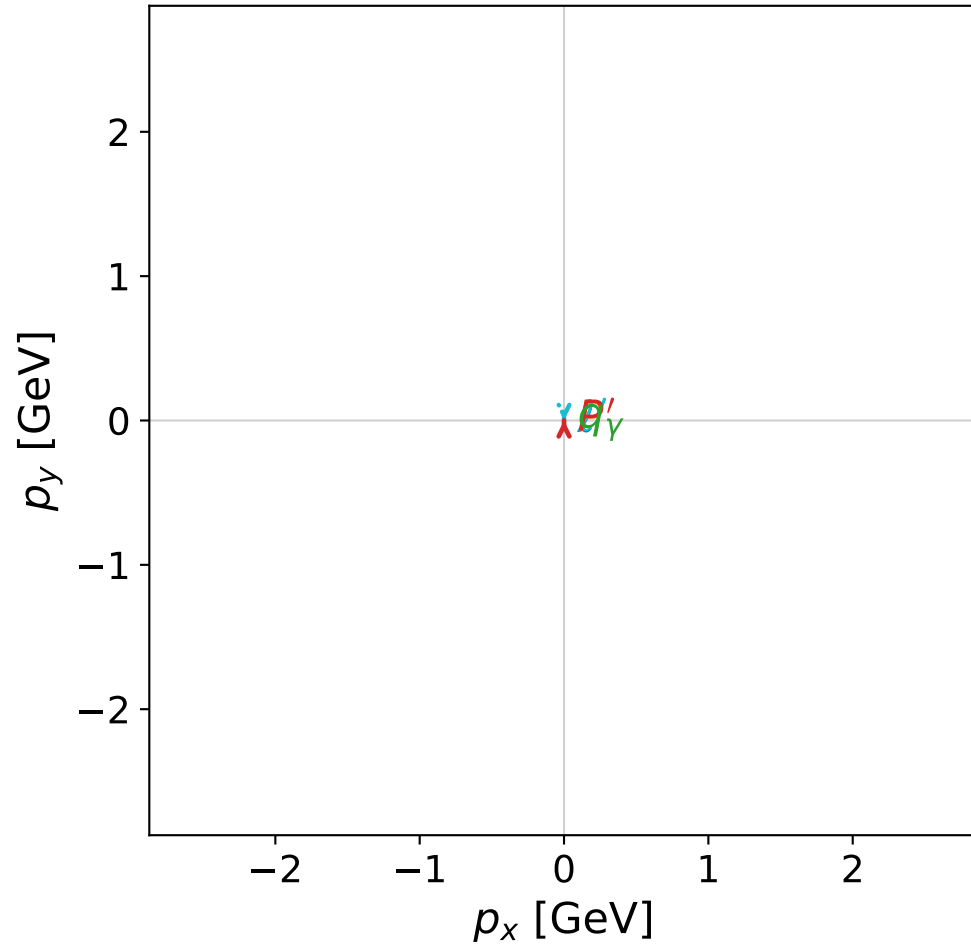


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

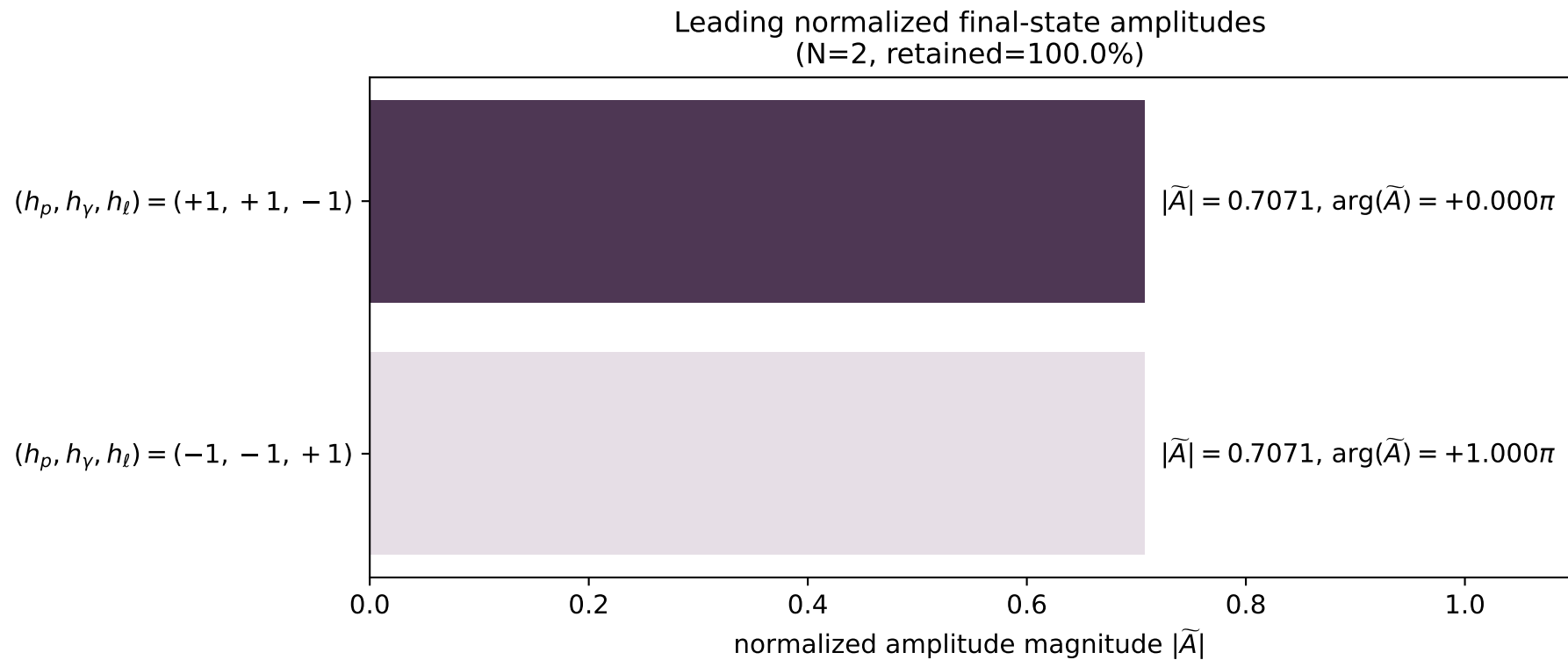


dghz_mixing_angles_polarization_01_local_minimum_71
 $d_{\text{GHZ}}=2.87516\text{e-}08$
 region: polarization_cluster_01_local_minimum_71
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0071
 lepton mixing angle: $\alpha_e=0.61281827$ rad
 incoming lepton: $0.81803|+\rangle + 0.575175|-\rangle$
 proton mixing angle: $\alpha_p=0.95795616$ rad
 incoming proton: $0.575193|+\rangle + 0.818018|-\rangle$

$s=24.6579$, $\sqrt{s}=4.96567$, $|\vec{P}|=2.39424$, $|\vec{P}'|=2.39424$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.69602$, $\phi_\gamma=3.17247$
 $E_\gamma=0.373172$, $\theta_{l'}=0$, $\phi_{l'}=4.83761$
 $Q^2=19.3557$, $x_B=0.839297$, $t=-22.9296$, $W^2=4.58594$, $y=0.969879$

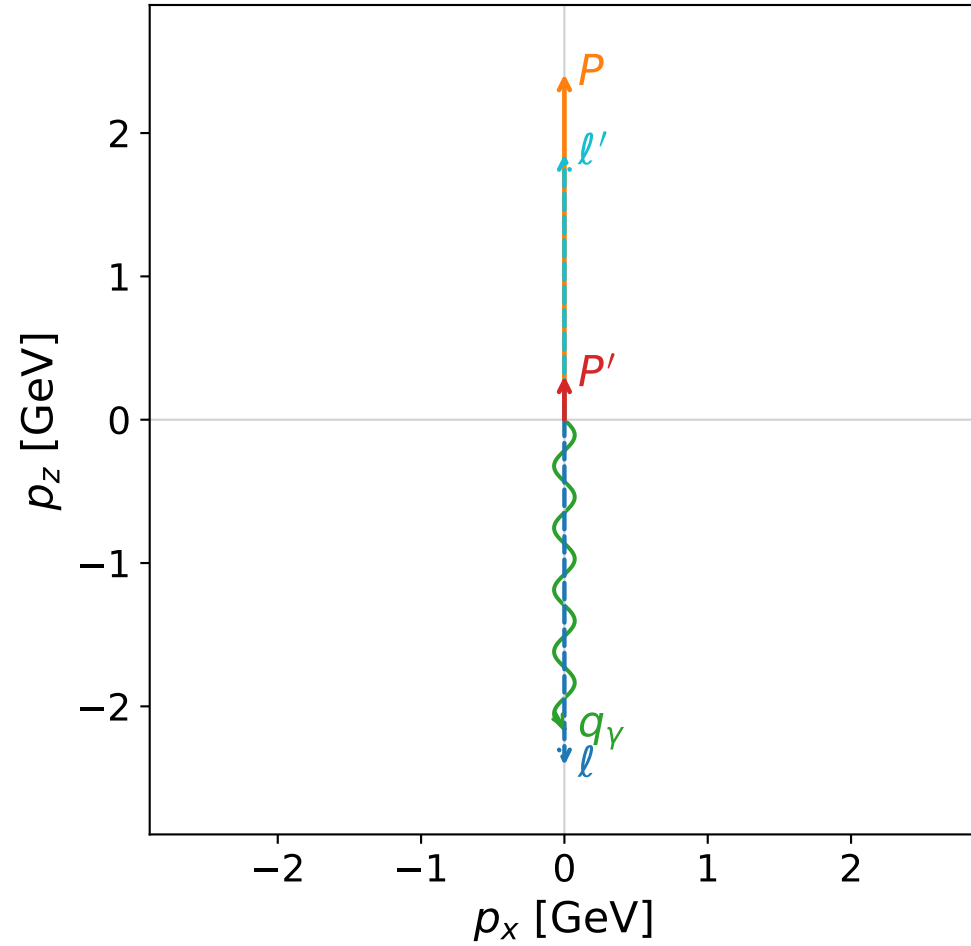
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3942$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3942$
 P $E= 2.5714$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3942$
 l' $E= 2.0211$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0211$
 P' $E= 2.5714$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.3942$
 q_γ $E= 0.3732$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.3732$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.310\text{e}+22$; $I_h=-1.427\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.149\text{e-}26$; $I_{TL}=-1.427\text{e-}13$

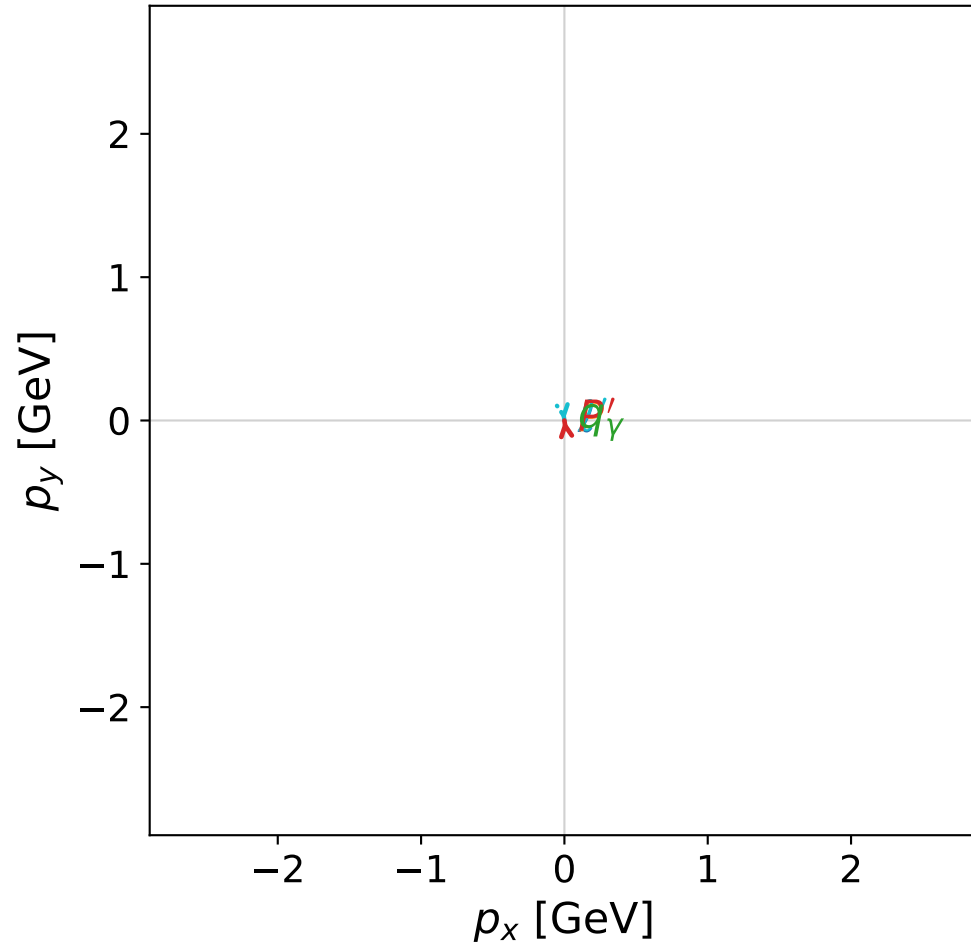


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

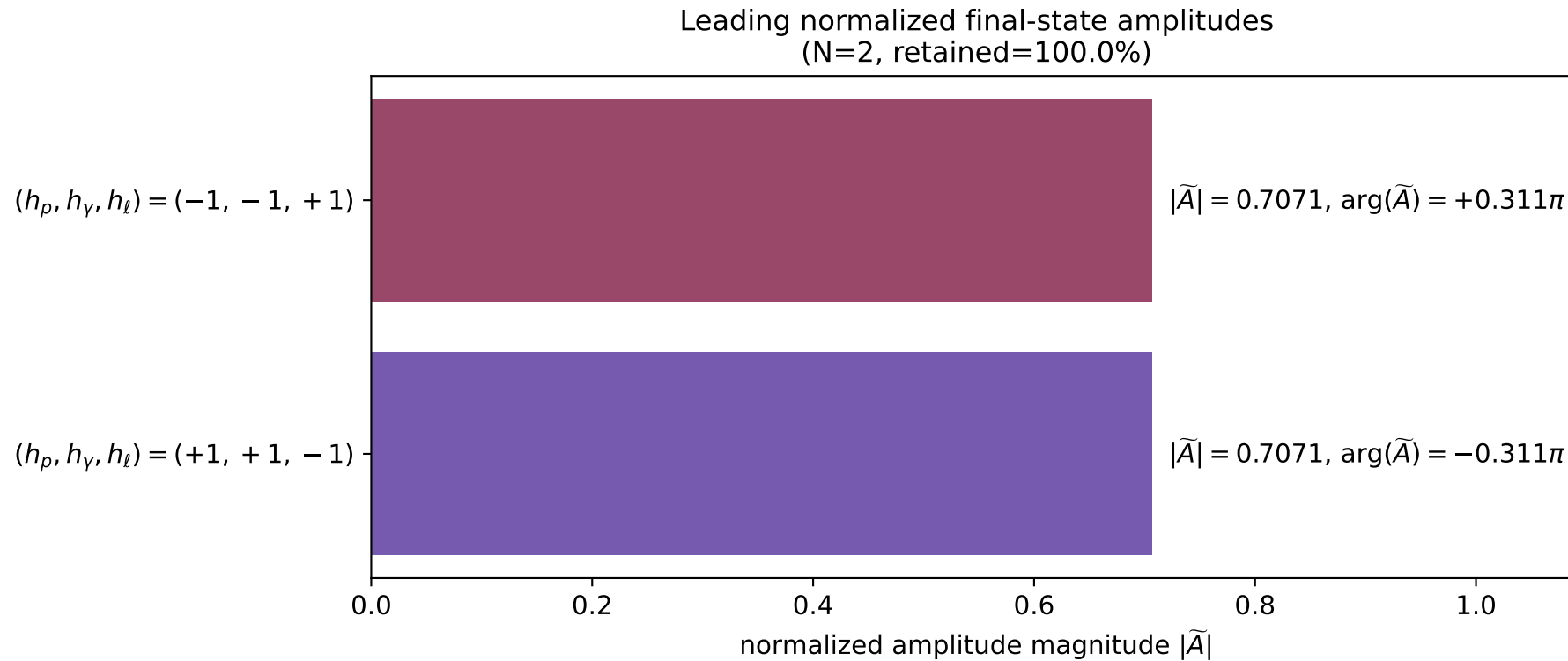


dghz_mixing_angles_polarization_01_local_minimum_72
 $d_{\text{GHZ}}=2.90848\text{e-}08$
 region: polarization_cluster_01_local_minimum_72
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0072
 lepton mixing angle: $\alpha_e=0.79108284$ rad
 incoming lepton: $0.703076|+\rangle +0.711115|-\rangle$
 proton mixing angle: $\alpha_p=0.7910723$ rad
 incoming proton: $0.703083|+\rangle +0.711108|-\rangle$

$s=24.9857$, $\sqrt{s}=4.99857$, $|\vec{P}|=2.41128$, $|\vec{P}'|=0.305039$
 $\theta_{P'}=0.000101732$, $\theta_Y=3.14159$, $\phi_{P'}=1.70875$, $\phi_Y=4.11801$
 $E_Y=2.15863$, $\theta_{l'}=1.67417\text{e-}05$, $\phi_{l'}=4.85035$
 $Q^2=17.8781$, $x_B=0.762282$, $t=-1.87322$, $W^2=6.45512$, $y=0.97293$

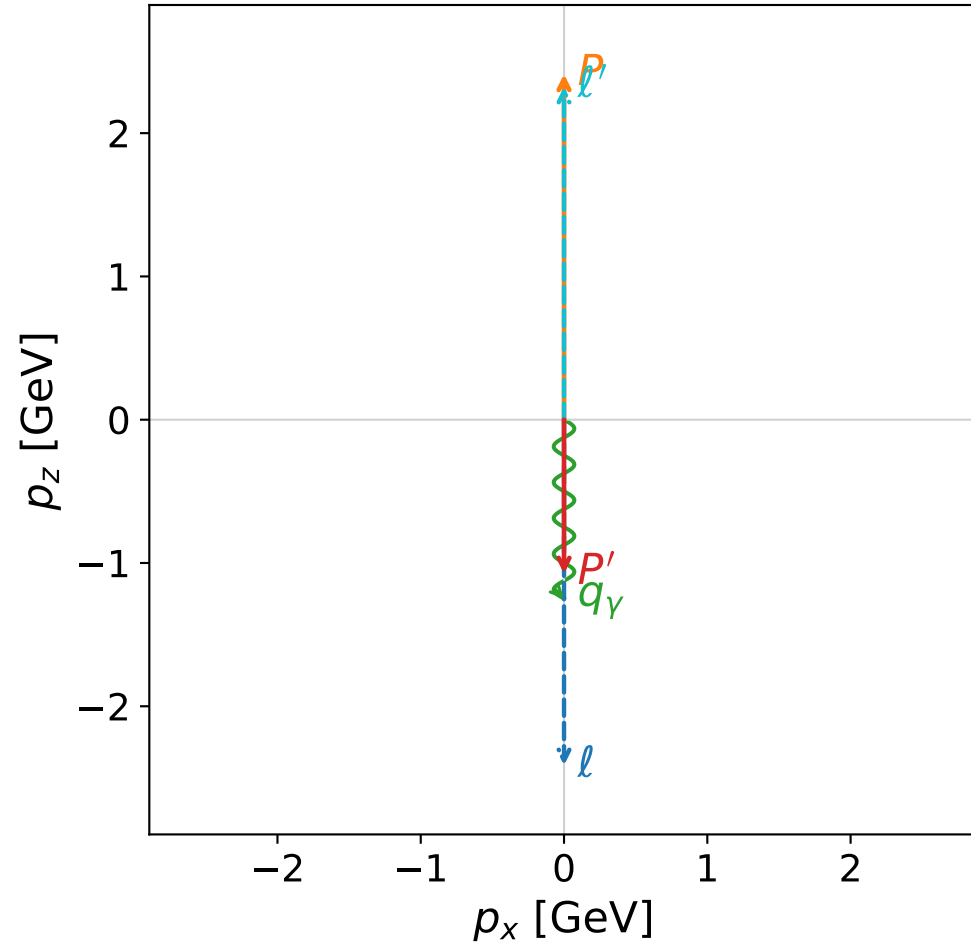
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 1.8536$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.8536$
 P' $E= 0.9864$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3050$
 q_Y $E= 2.1586$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1586$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.034\text{e+}17$; $I_h=+3.091\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.051\text{e-}12$; $I_{TL}=-2.977\text{e-}09$

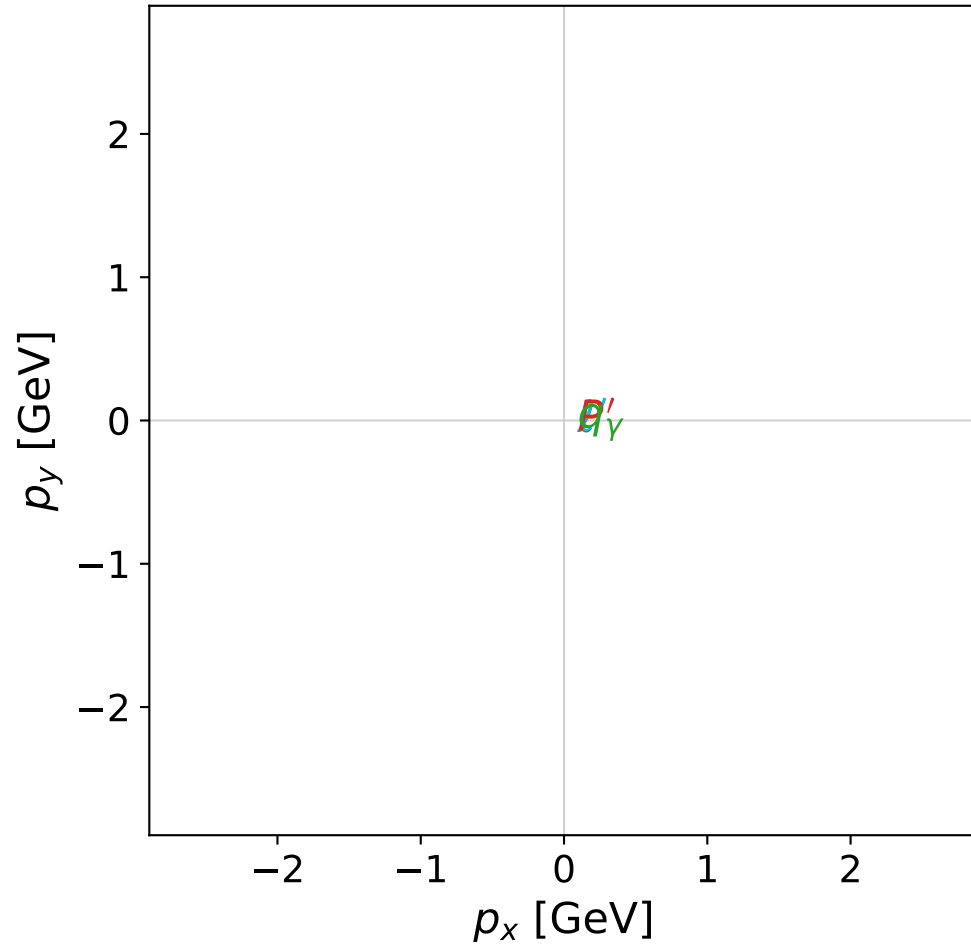


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

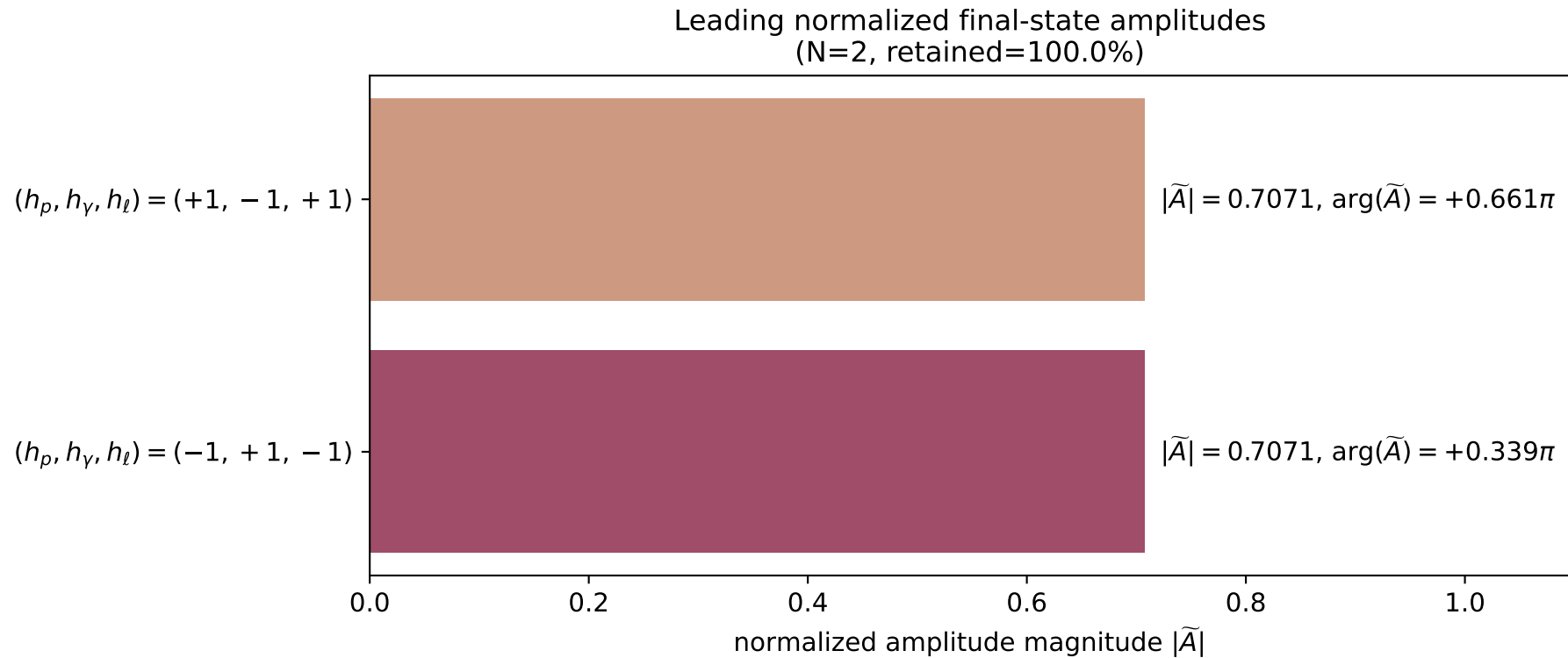


dghz_mixing_angles_polarization_01_local_minimum_75
 $d_{\text{GHZ}}=2.94743\text{e-}08$
 region: polarization_cluster_01_local_minimum_75
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0075
 lepton mixing angle: $\alpha_e=1.1150181$ rad
 incoming lepton: $0.440161|+\rangle + 0.897919|-\rangle$
 proton mixing angle: $\alpha_p=1.1150063$ rad
 incoming proton: $0.440172|+\rangle + 0.897914|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.07529$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.49134$, $\phi_\gamma=5.21955$
 $E_\gamma=1.2489$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.11298$
 $Q^2=22.4239$, $x_B=0.962308$, $t=-10.8132$, $W^2=1.75814$, $y=0.966088$

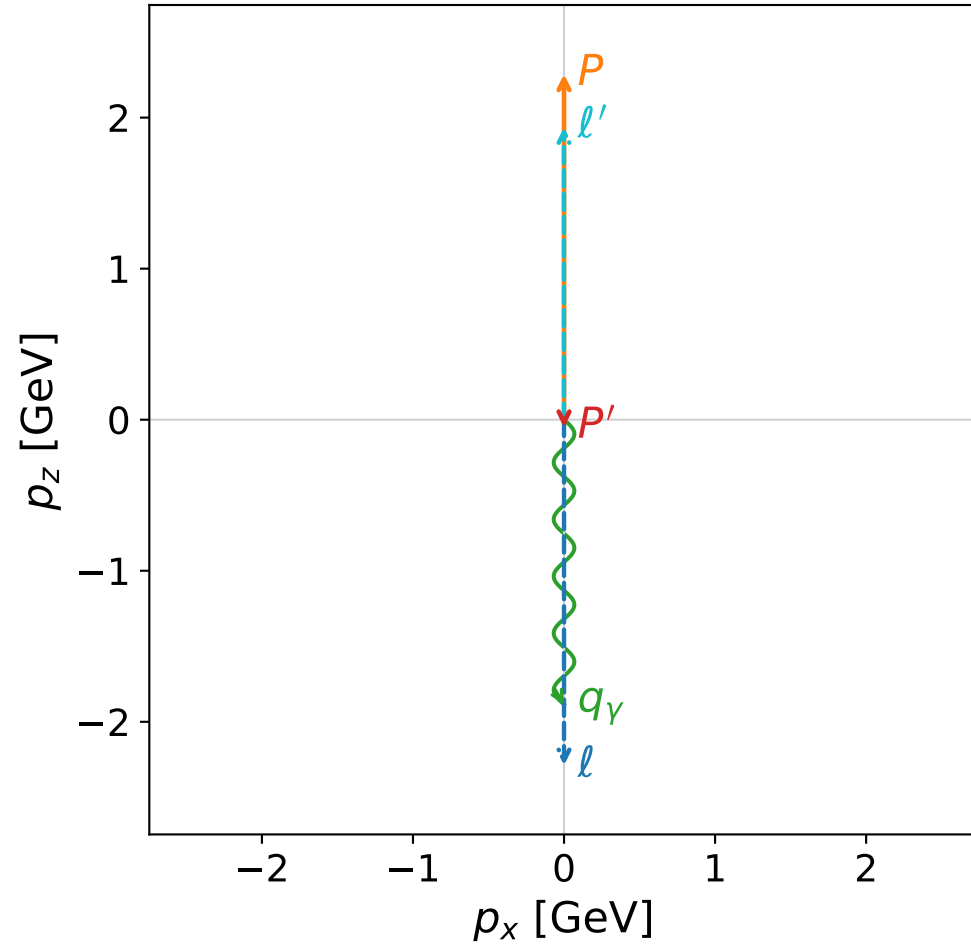
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3242$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.3242$
 P' $E= 1.4269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0753$
 q_γ $E= 1.2489$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2489$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.955\text{e}+23$; $I_h=+4.052\text{e-}13$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.728\text{e-}29$; $I_{TL}=-6.742\text{e-}19$

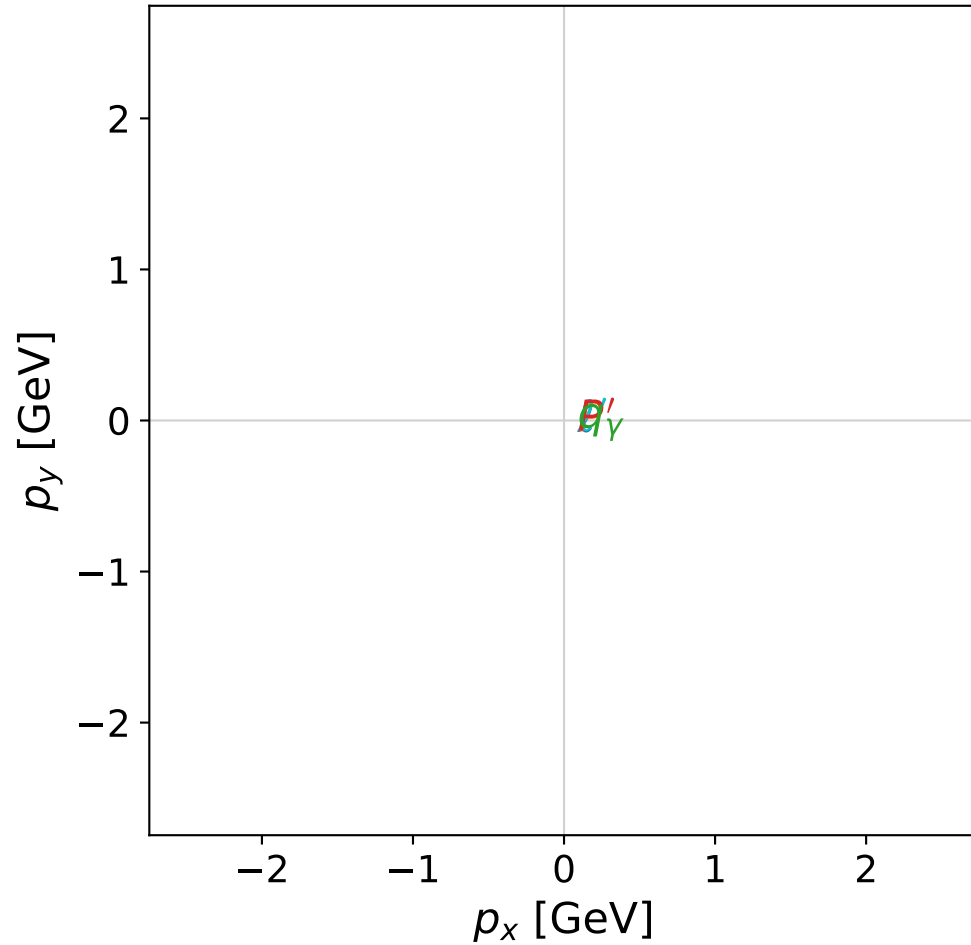


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

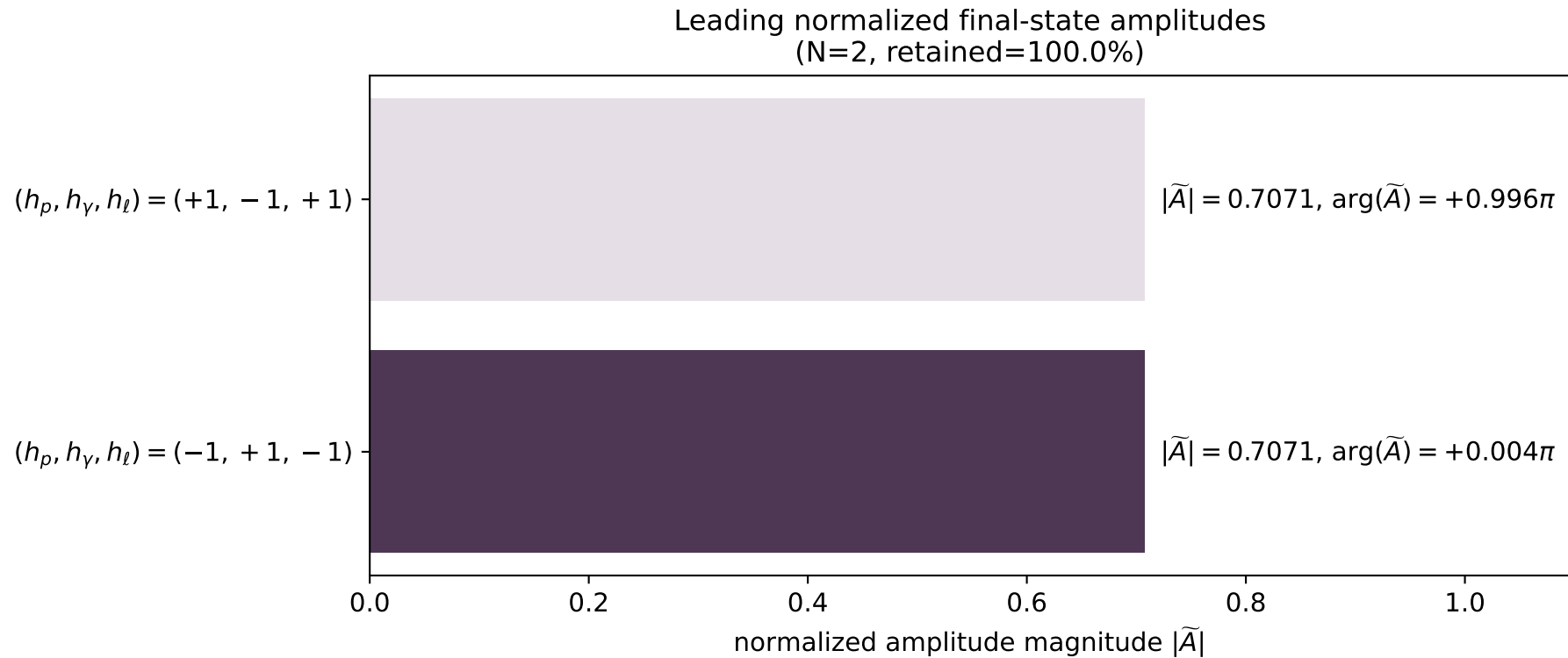


dghz_mixing_angles_polarization_01_local_minimum_76
 $d_{\text{GHZ}}=2.95168\text{e-}08$
 region: polarization_cluster_01_local_minimum_76
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0076
 lepton mixing angle: $\alpha_e=0.77617426$ rad
 incoming lepton: $0.713599|+\rangle + 0.700555|-\rangle$
 proton mixing angle: $\alpha_p=0.77615415$ rad
 incoming proton: $0.713613|+\rangle + 0.70054|-\rangle$

$s=22.6644$, $\sqrt{s}=4.76071$, $|\vec{P}|=2.28795$, $|\vec{P}'|=0.0517317$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.27601$, $\phi_\gamma=6.27171$
 $E_\gamma=1.88478$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.13024$
 $Q^2=17.7225$, $x_B=0.841177$, $t=-3.12298$, $W^2=4.22605$, $y=0.967142$

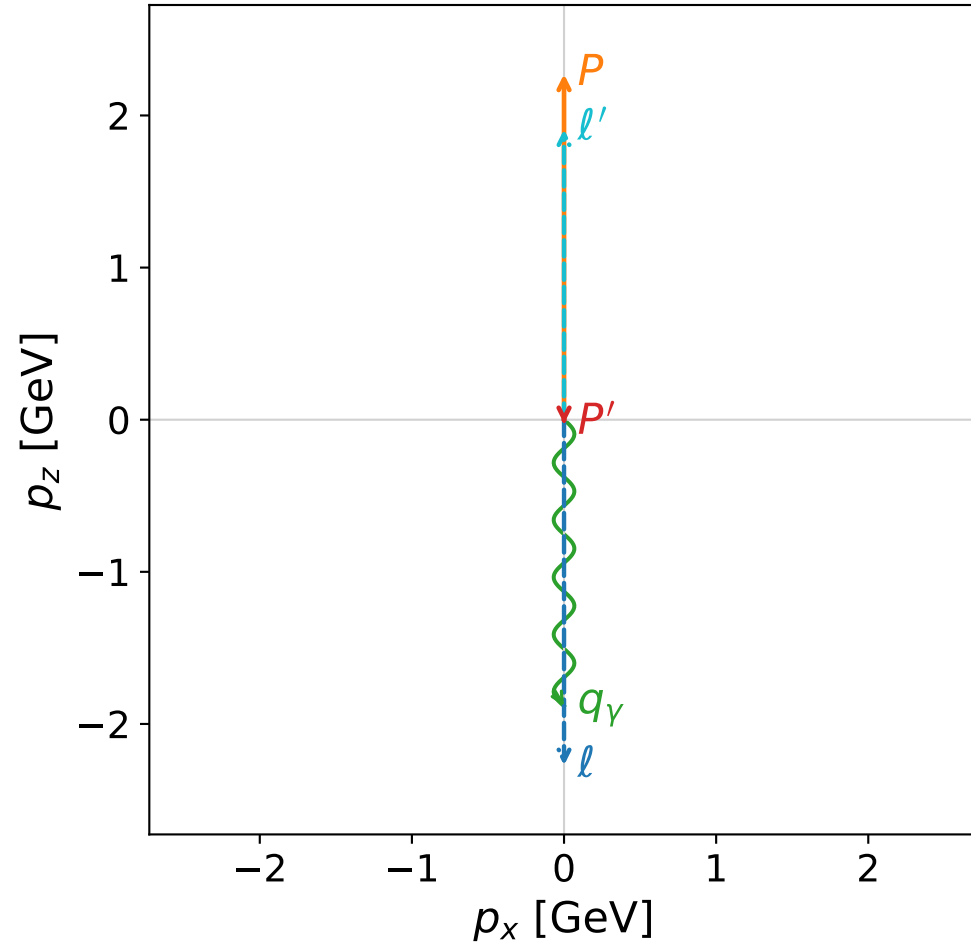
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2879$
 P $E= 2.4728$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2879$
 ℓ' $E= 1.9365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9365$
 P' $E= 0.9394$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0517$
 q_γ $E= 1.8848$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8848$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.188\text{e}+24$; $I_h=-4.337\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.225\text{e-}30$; $I_{TL}=+5.303\text{e-}19$

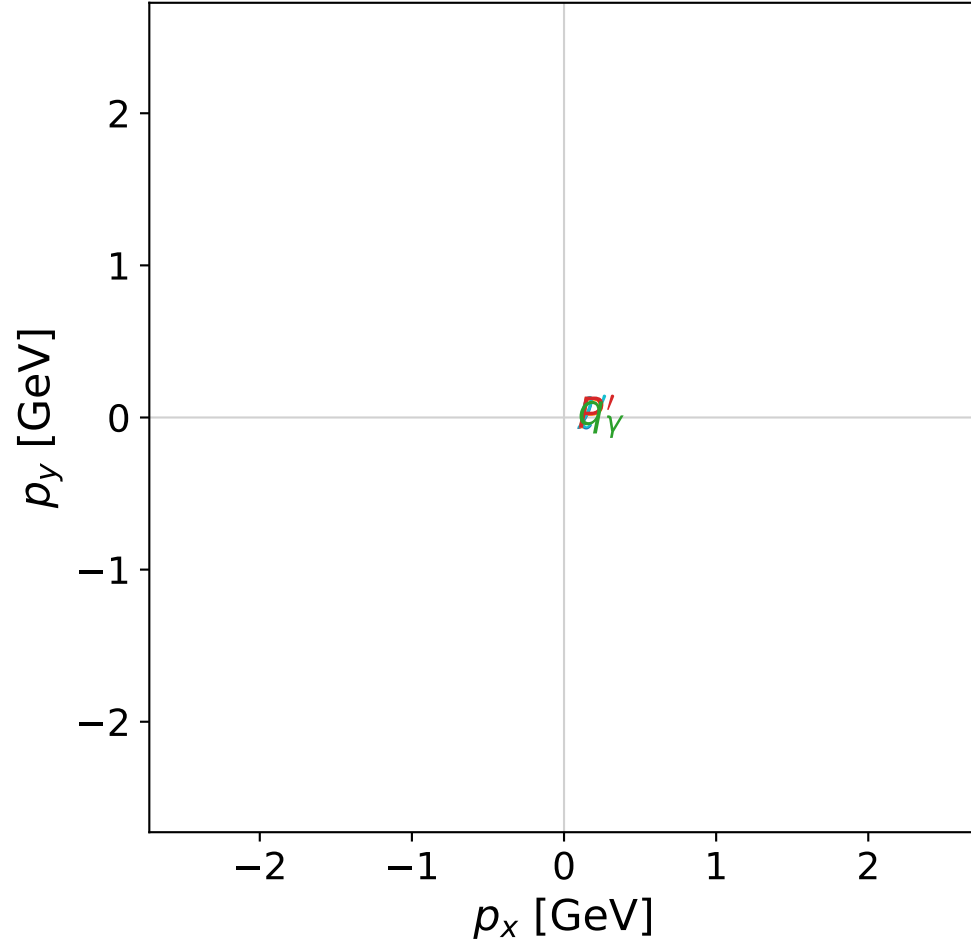


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

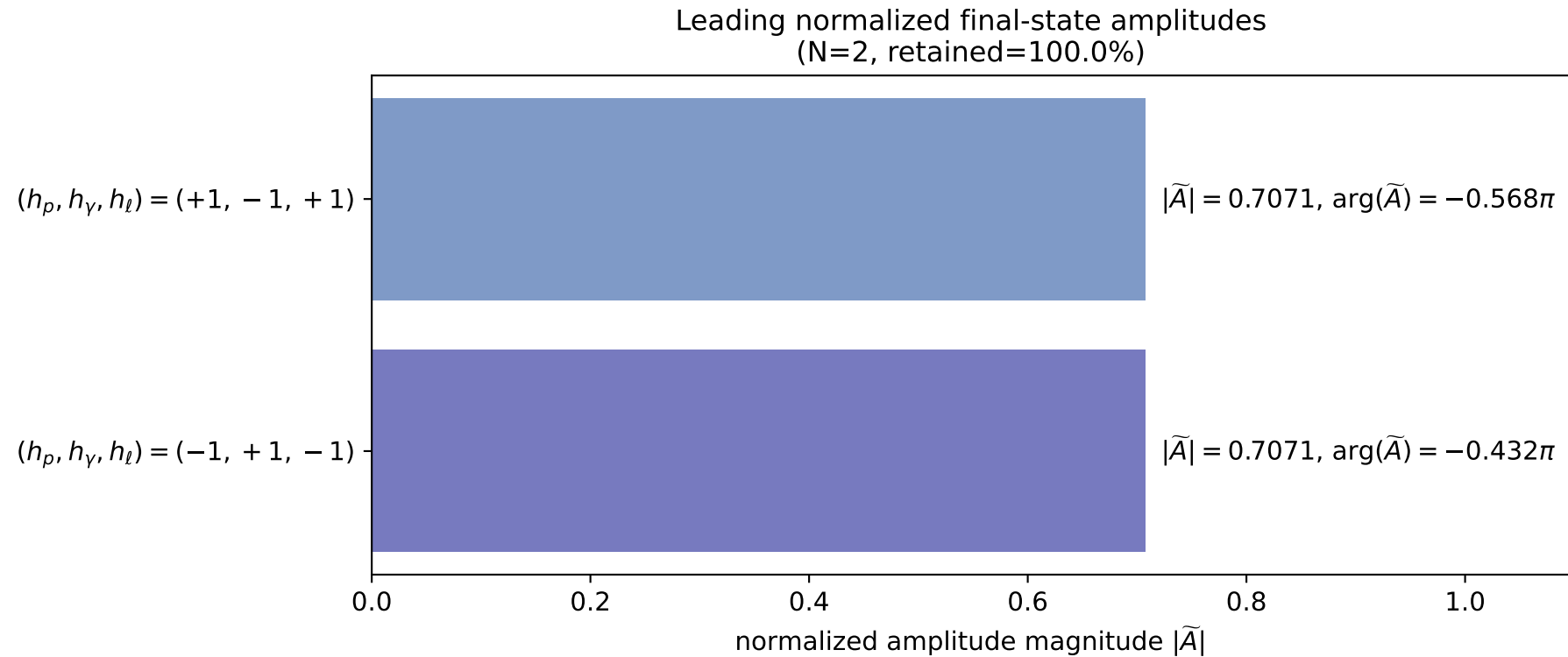


dghz_mixing_angles_polarization_01_local_minimum_79
 $d_{\text{GHZ}}=3.01399\text{e-}08$
 region: polarization_cluster_01_local_minimum_79
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0079
 lepton mixing angle: $\alpha_e=0.76405178$ rad
 incoming lepton: $0.722039|+\rangle + 0.691853|-\rangle$
 proton mixing angle: $\alpha_p=0.76404422$ rad
 incoming proton: $0.722044|+\rangle + 0.691847|-\rangle$

$s=22.3692$, $\sqrt{s}=4.72961$, $|\vec{P}|=2.27179$, $|\vec{P}'|=0.0259335$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0568276$, $\phi_\gamma=1.35777$
 $E_\gamma=1.88266$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.48614$
 $Q^2=17.3437$, $x_B=0.834663$, $t=-2.97077$, $W^2=4.31541$, $y=0.966955$

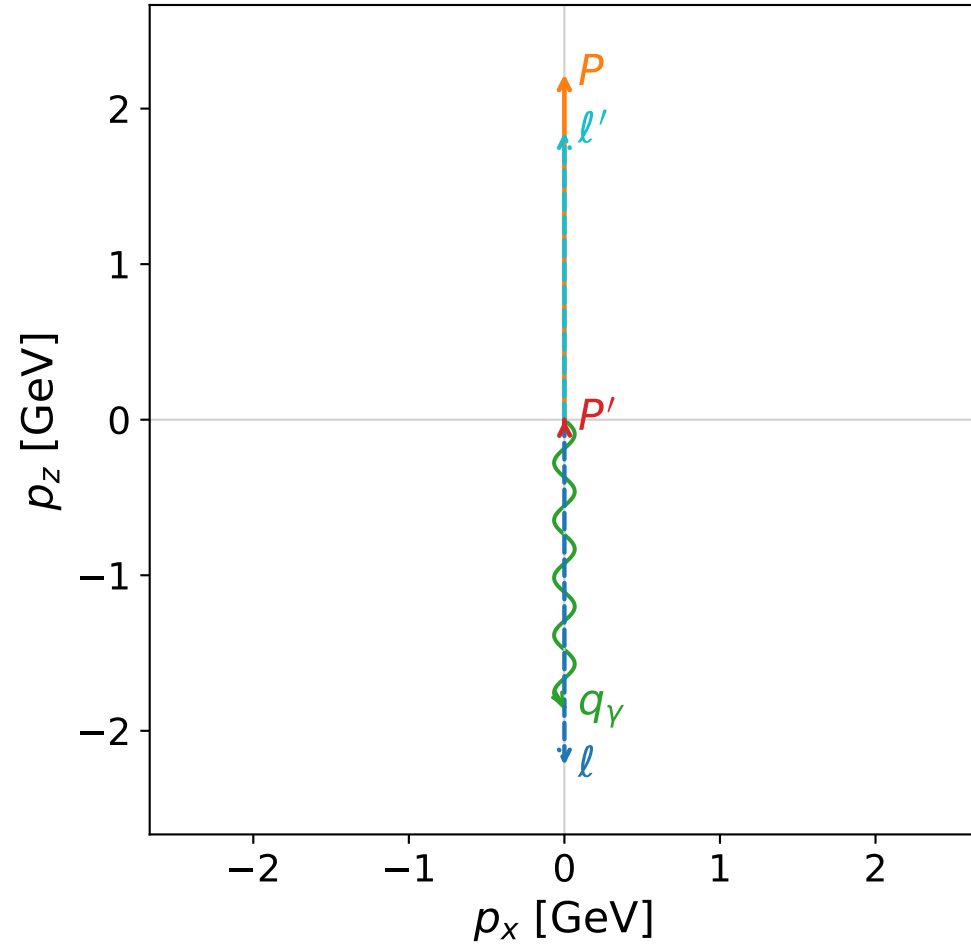
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2718$
 P $E= 2.4578$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2718$
 ℓ' $E= 1.9086$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9086$
 P' $E= 0.9384$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0259$
 q_γ $E= 1.8827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8827$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.221\text{e}+24$; $I_h=-2.033\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.853\text{e-}27$; $I_{TL}=+2.671\text{e-}19$

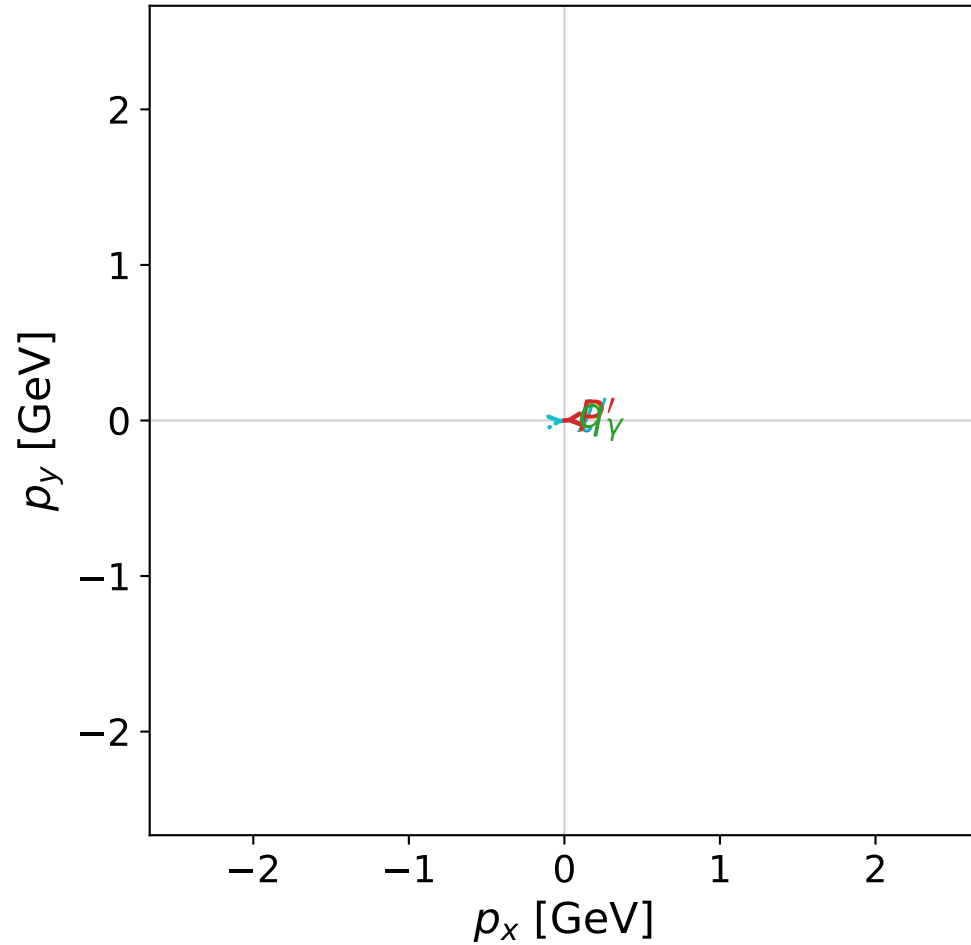


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

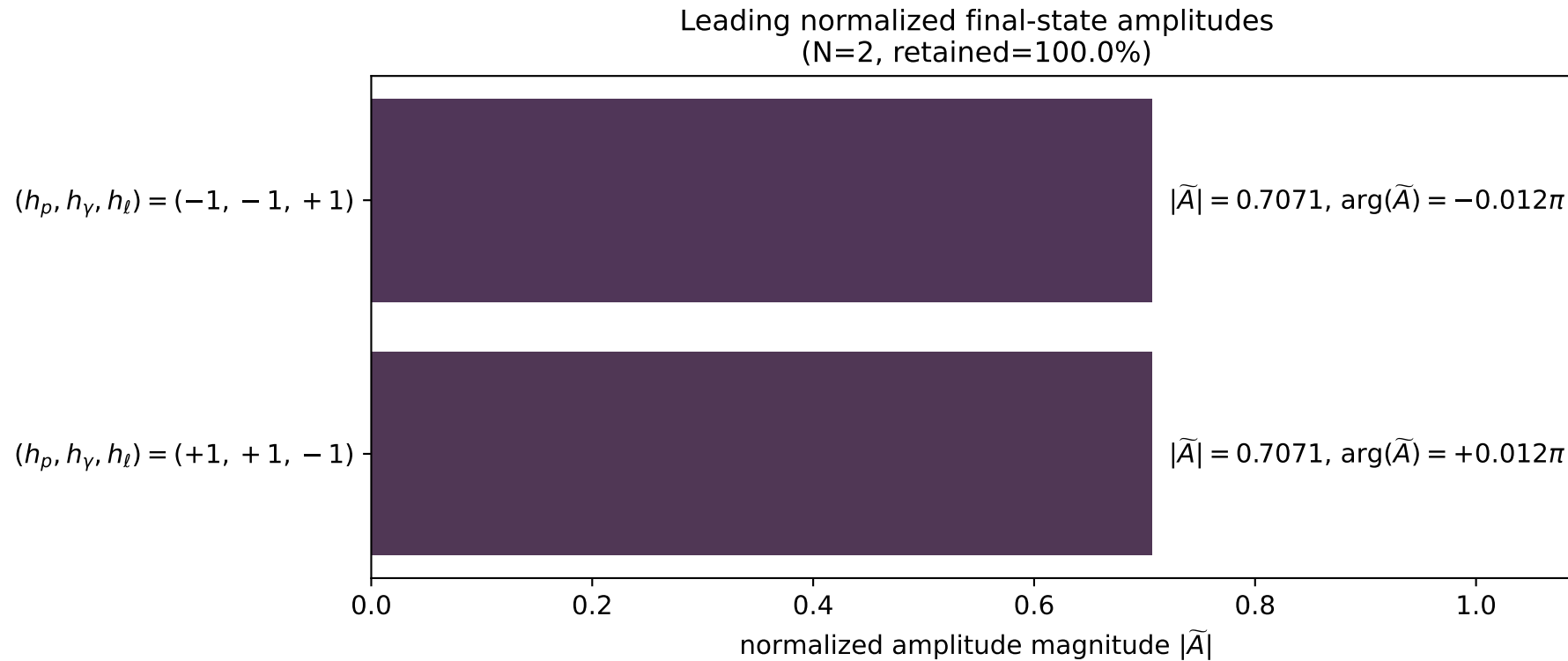


dghz_mixing_angles_polarization_01_local_minimum_83
 $d_{\text{GHZ}}=3.05691\text{e-}08$
 region: polarization_cluster_01_local_minimum_83
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0083
 lepton mixing angle: $\alpha_e=0.74952575$ rad
 incoming lepton: $0.732012|+\rangle +0.681292|-\rangle$
 proton mixing angle: $\alpha_p=0.74952252$ rad
 incoming proton: $0.732014|+\rangle +0.681289|-\rangle$

$s=21.4675$, $\sqrt{s}=4.63331$, $|\vec{P}|=2.22171$, $|\vec{P}'|=8.39622\text{e-}05$
 $\theta_{P'}=0.0161201$, $\theta_Y=3.14159$, $\phi_{P'}=3.23565$, $\phi_Y=3.1028$
 $E_Y=1.8477$, $\theta_{\ell'}=7.32586\text{e-}07$, $\phi_{\ell'}=0.0940553$
 $Q^2=16.4194$, $x_B=0.825677$, $t=-2.7641$, $W^2=4.34643$, $y=0.965916$

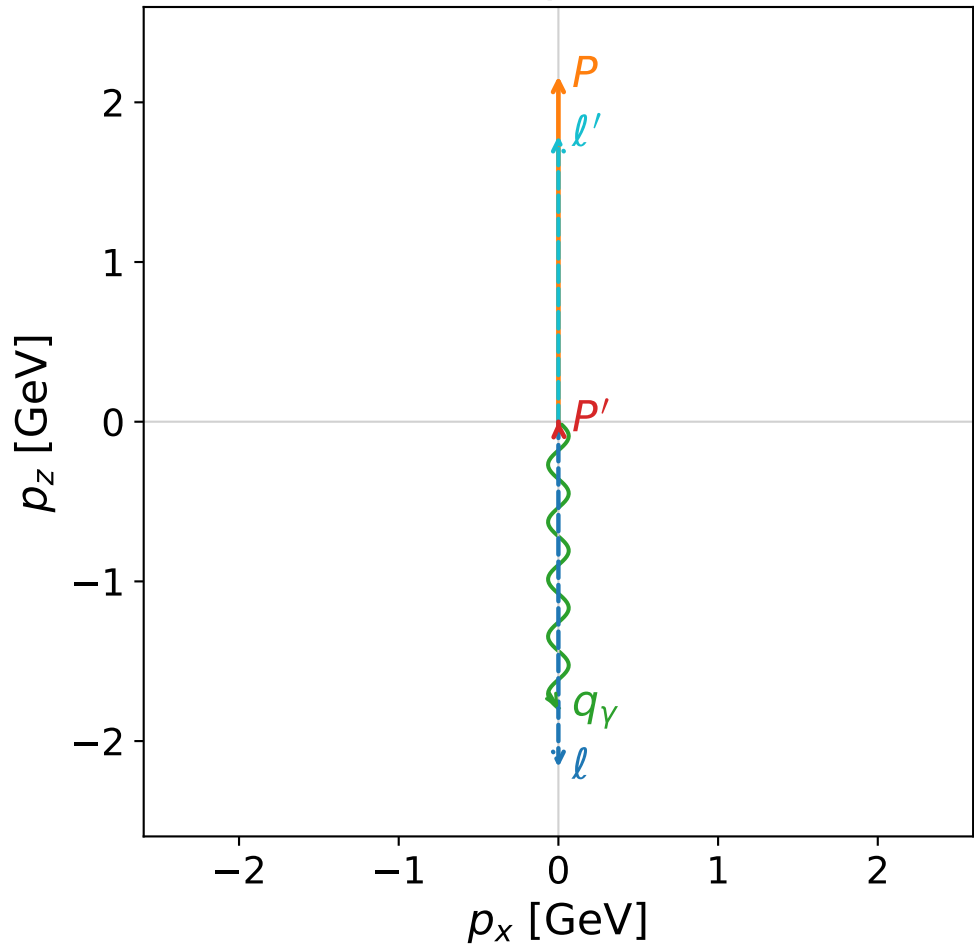
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2217$
 P $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2217$
 ℓ' $E= 1.8476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8476$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0001$
 q_Y $E= 1.8477$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.8477$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.303\text{e+}20$; $I_h=-7.233\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.659\text{e-}15$; $I_{TL}=+5.205\text{e-}09$

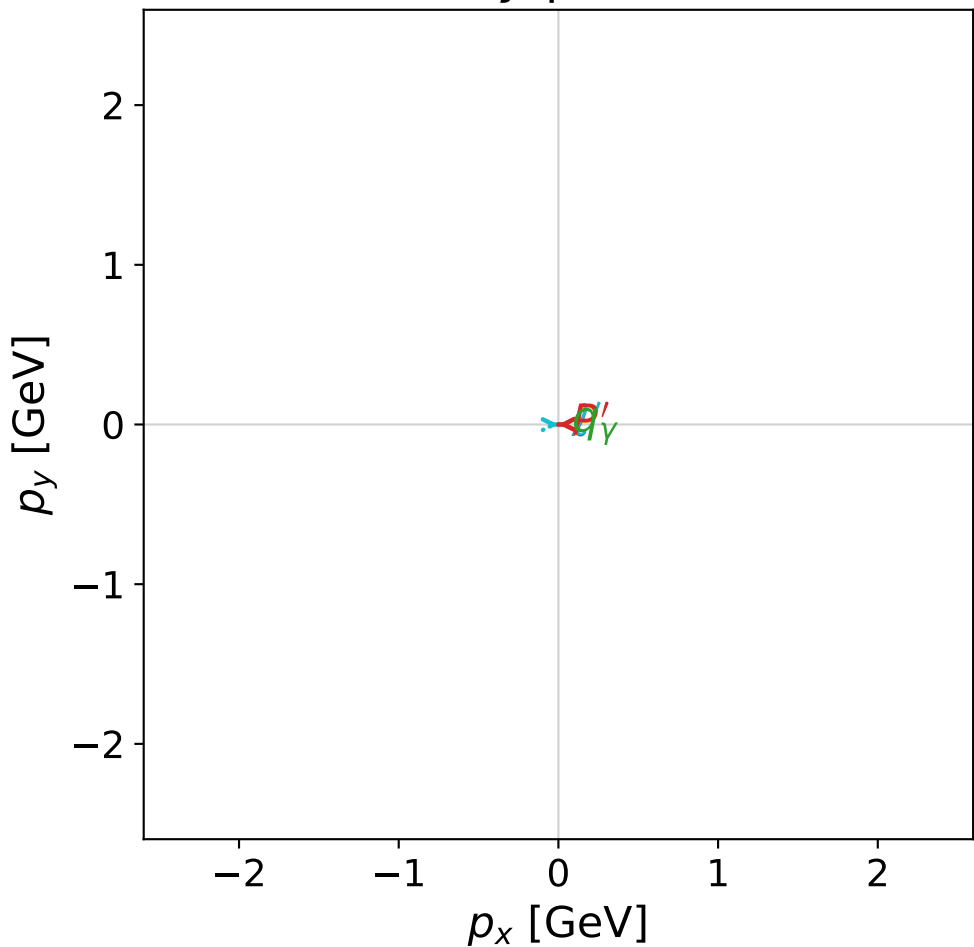


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

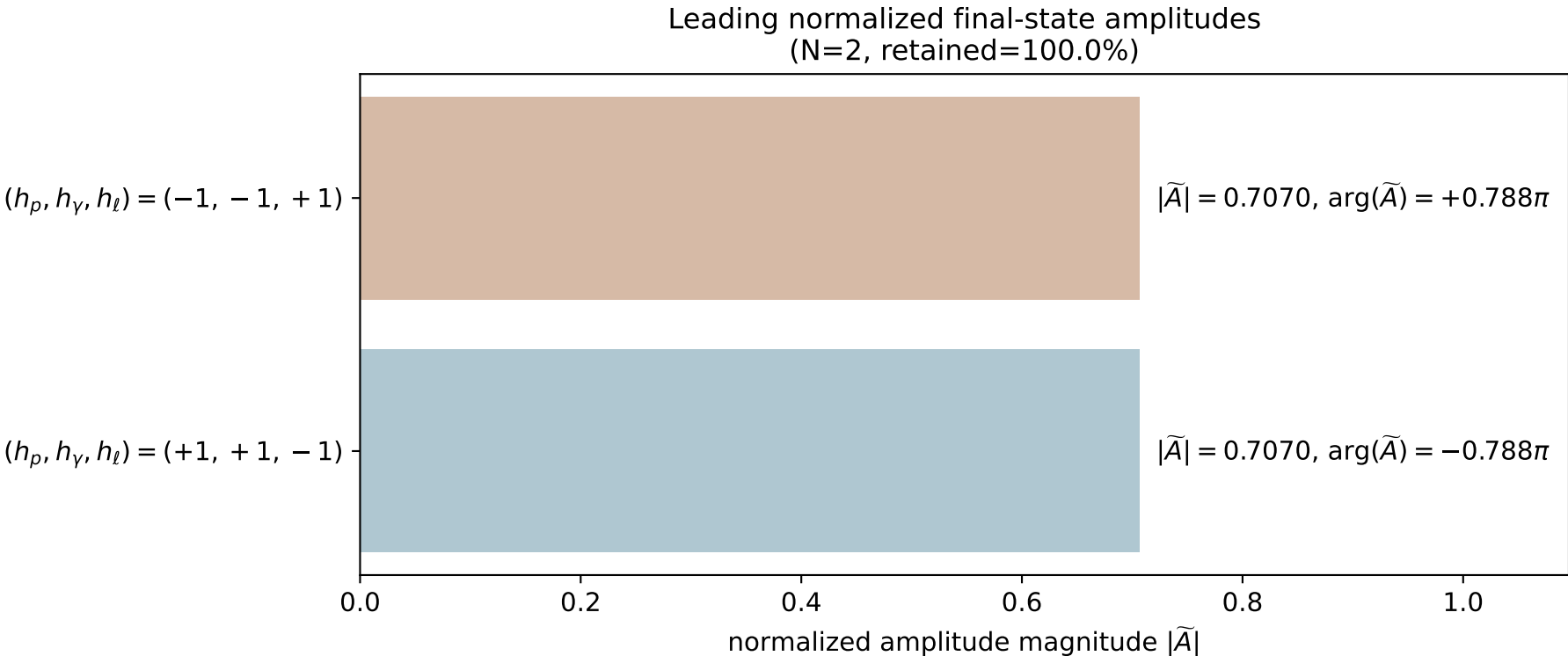


```
dghz_mixing_angles_polarization_01_local_minimum_95
d_{\rm GHZ}=3.40442e-08
region: polarization_cluster_01_local_minimum_95
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0095
lepton mixing angle: alpha_e=0.83191558 rad
incoming lepton: 0.673461|+> +0.739223|->
proton mixing angle: alpha_p=0.83190695 rad
incoming proton: 0.673467|+> +0.739217|->
```

```
s=20.4568, sqrt(s)=4.52291, |P|=2.16419, |P'|=0.00379415
theta_{p'}=0.0374492, theta_gamma=3.14159, phi_{p'}=3.15038, phi_gamma=5.61715
E_gamma=1.79435, theta_{l'}=7.93354e-05, phi_{l'}=0.00878615
Q^2=15.5004, x_B=0.820986, t=-2.6489, W^2=4.25968, y=0.964415
```

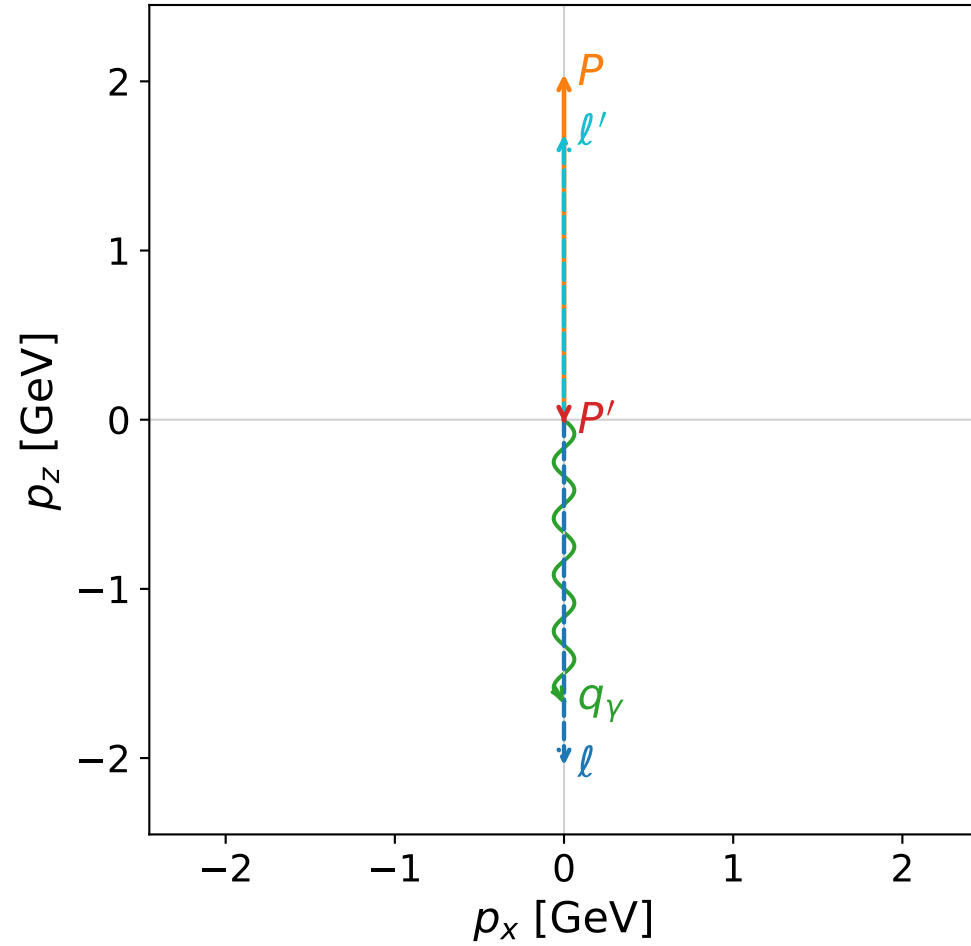
```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.1642 p_x=  0.0000 p_y=  0.0000 p_z= -2.1642
P      E=  2.3587 p_x=  0.0000 p_y=  0.0000 p_z=  2.1642
l'     E=  1.7906 p_x=  0.0001 p_y=  0.0000 p_z=  1.7906
P'     E=  0.9380 p_x= -0.0001 p_y= -0.0000 p_z=  0.0038
q_gamma E=  1.7943 p_x=  0.0000 p_y= -0.0000 p_z= -1.7943
```

```
GHZ mechanism: helicity flipping (full H G C)
helicity powers: flip 100.000%, conserve 0.000%
R_flip/conserve=1.031e+16; I_h=+1.053e-09
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=3.441e-11; I_TL=-7.670e-07
```

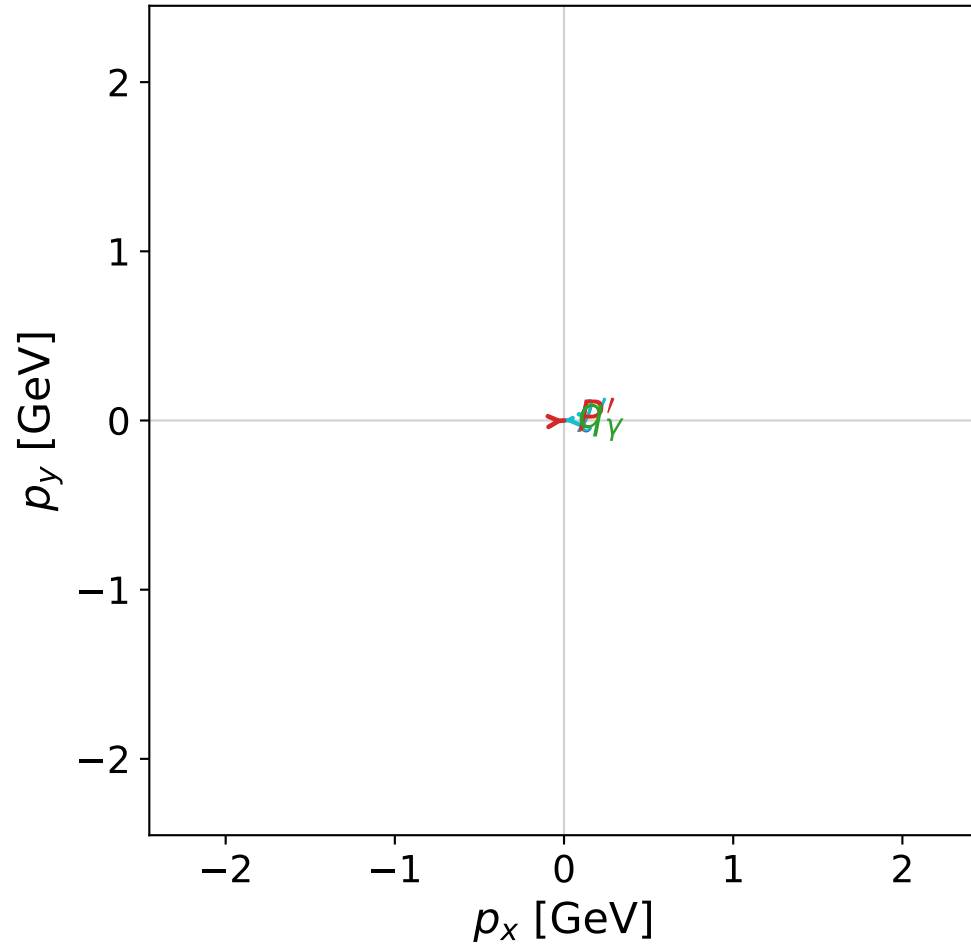


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

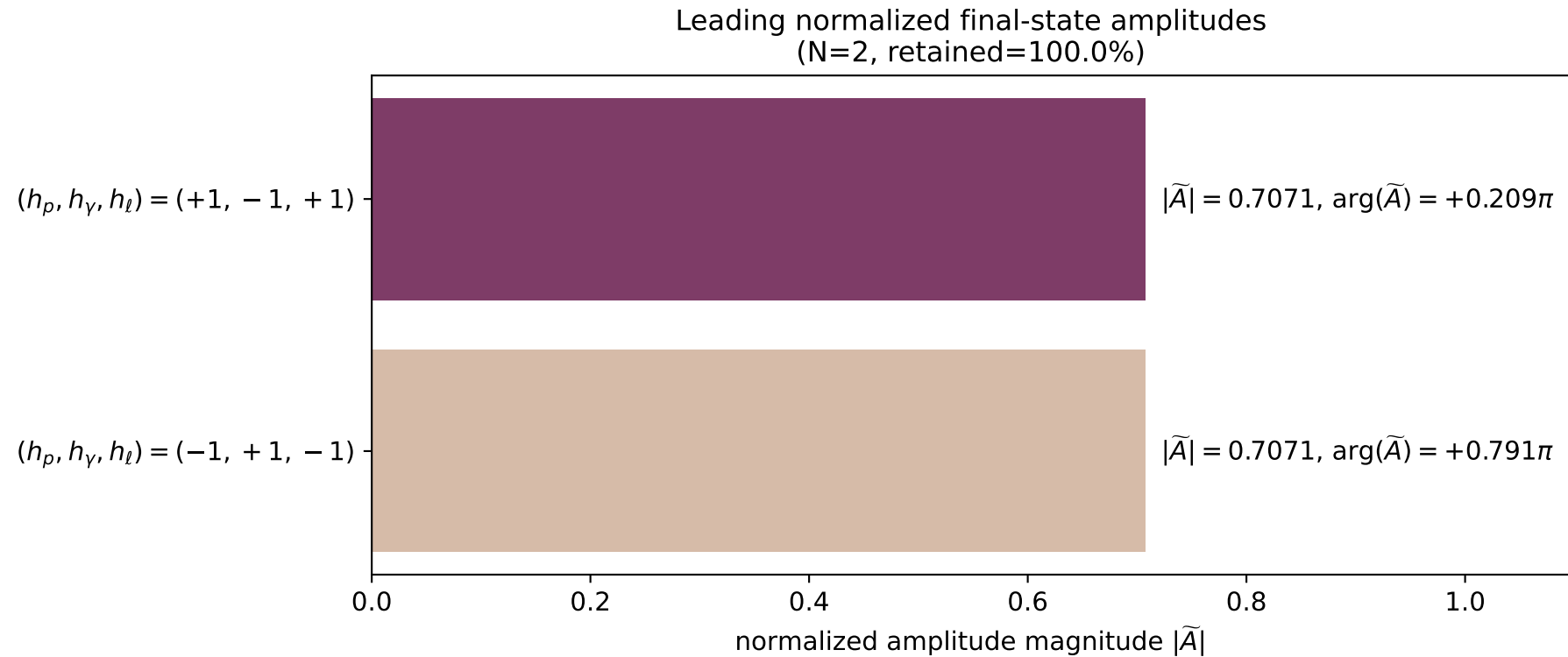


dghz_mixing_angles_polarization_01_local_minimum_104
 $d_{\text{GHZ}}=3.80072\text{e-}08$
 region: polarization_cluster_01_local_minimum_104
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0104
 lepton mixing angle: $\alpha_e=0.79818758$ rad
 incoming lepton: $0.698006|+\rangle +0.716092|-\rangle$
 proton mixing angle: $\alpha_p=0.79817765$ rad
 incoming proton: $0.698013|+\rangle +0.716085|-\rangle$

$s=18.4089$, $\sqrt{s}=4.29056$, $|\vec{P}|=2.04275$, $|\vec{P}'|=0.0189369$
 $\theta_{P'}=3.13802$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.06771$, $\phi_\gamma=3.86678$
 $E_\gamma=1.66672$, $\theta_{\ell'}=4.01396\text{e-}05$, $\phi_{\ell'}=3.2093$
 $Q^2=13.7735$, $x_B=0.818012$, $t=-2.53544$, $W^2=3.94412$, $y=0.960561$

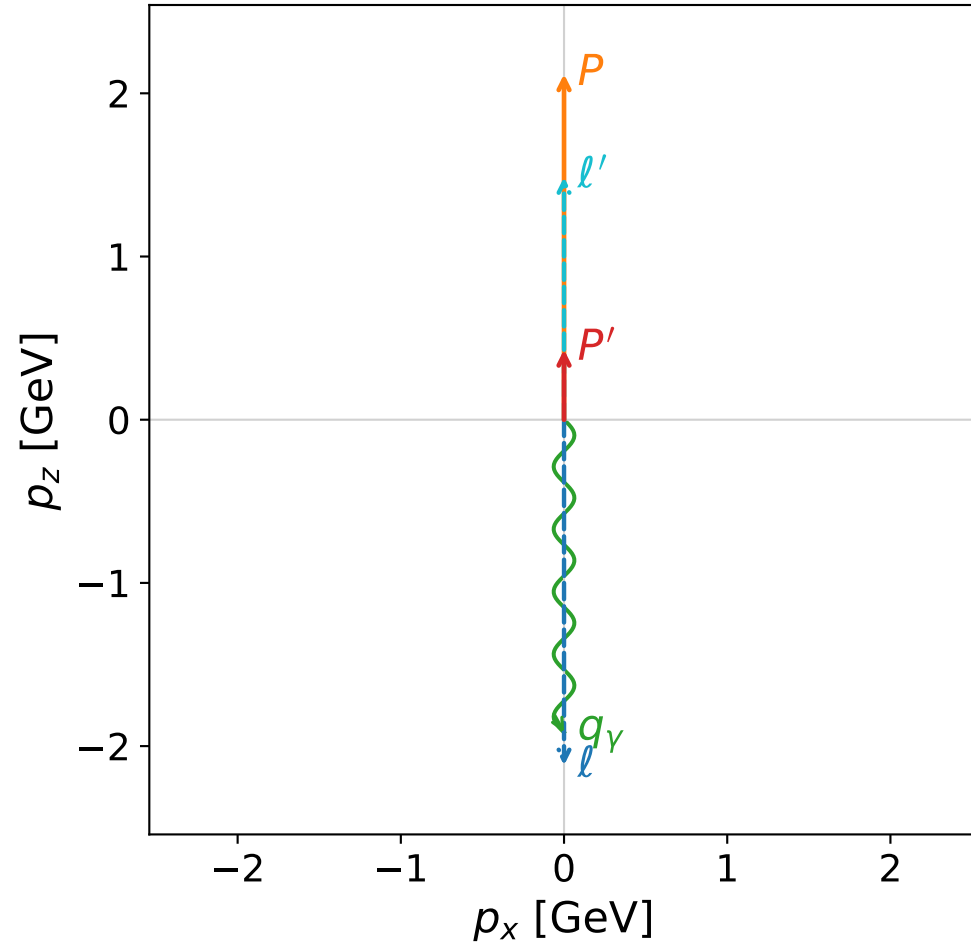
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0427$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0427$
 P $E= 2.2478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0427$
 ℓ' $E= 1.6857$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 1.6857$
 P' $E= 0.9382$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.0189$
 q_γ $E= 1.6667$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.6667$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.544\text{e+}16$; $I_h=-1.570\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.964\text{e-}12$; $I_{TL}=+1.148\text{e-}07$

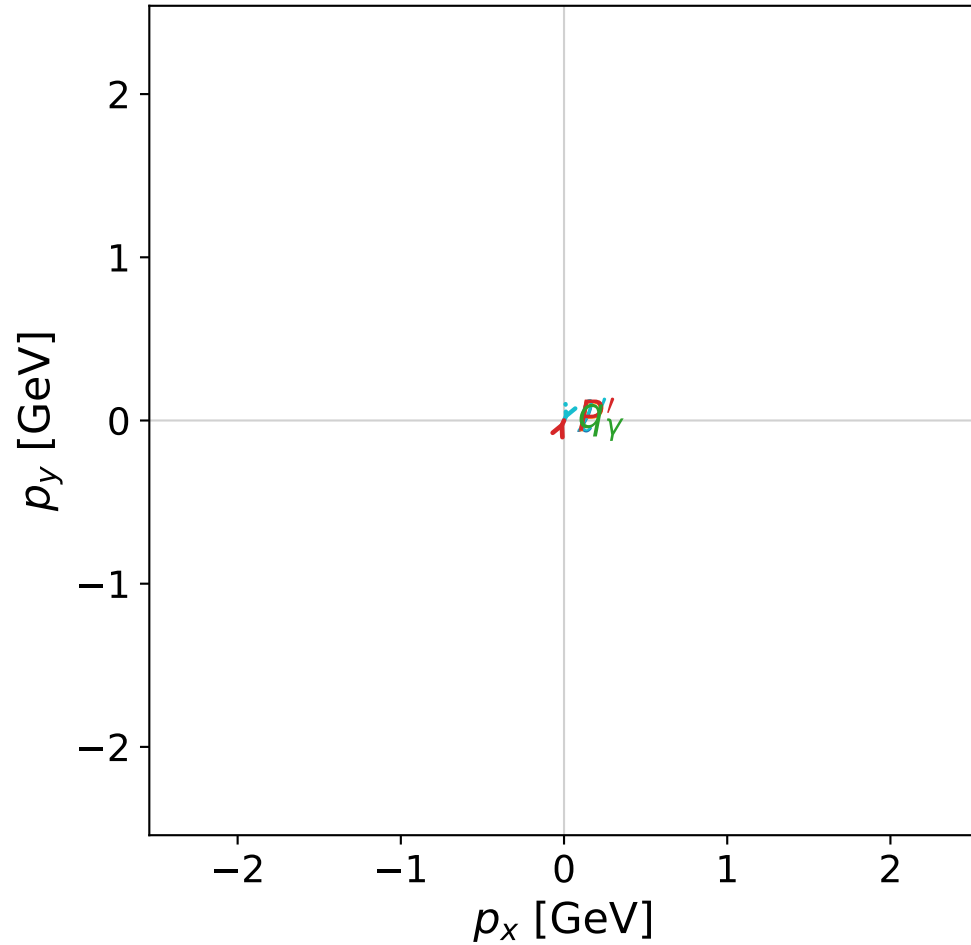


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

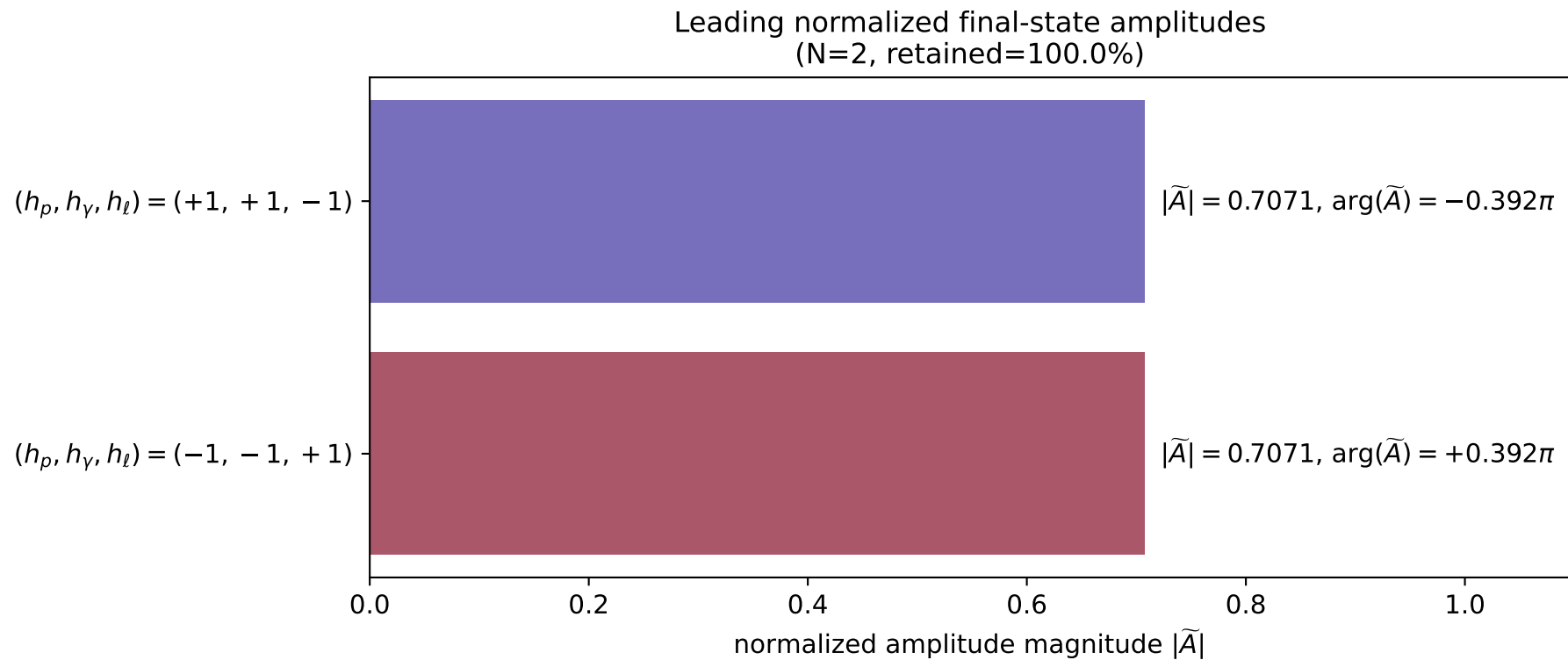


dghz_mixing_angles_polarization_01_local_minimum_112
 $d_{\text{GHZ}}=4.14029\text{e-}08$
 region: polarization_cluster_01_local_minimum_112
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0112
 lepton mixing angle: $\alpha_e=0.7523389$ rad
 incoming lepton: $0.730093|+\rangle + 0.683348|-\rangle$
 proton mixing angle: $\alpha_p=0.75235965$ rad
 incoming proton: $0.730078|+\rangle + 0.683363|-\rangle$

$s=19.6572$, $\sqrt{s}=4.43364$, $|\vec{P}|=2.1176$, $|\vec{P}'|=0.429483$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.14894$, $\phi_\gamma=4.37324$
 $E_\gamma=1.91574$, $\theta_{l'}=2.21637\text{e-}05$, $\phi_{l'}=4.29053$
 $Q^2=12.5892$, $x_B=0.692189$, $t=-1.20006$, $W^2=6.47813$, $y=0.968585$

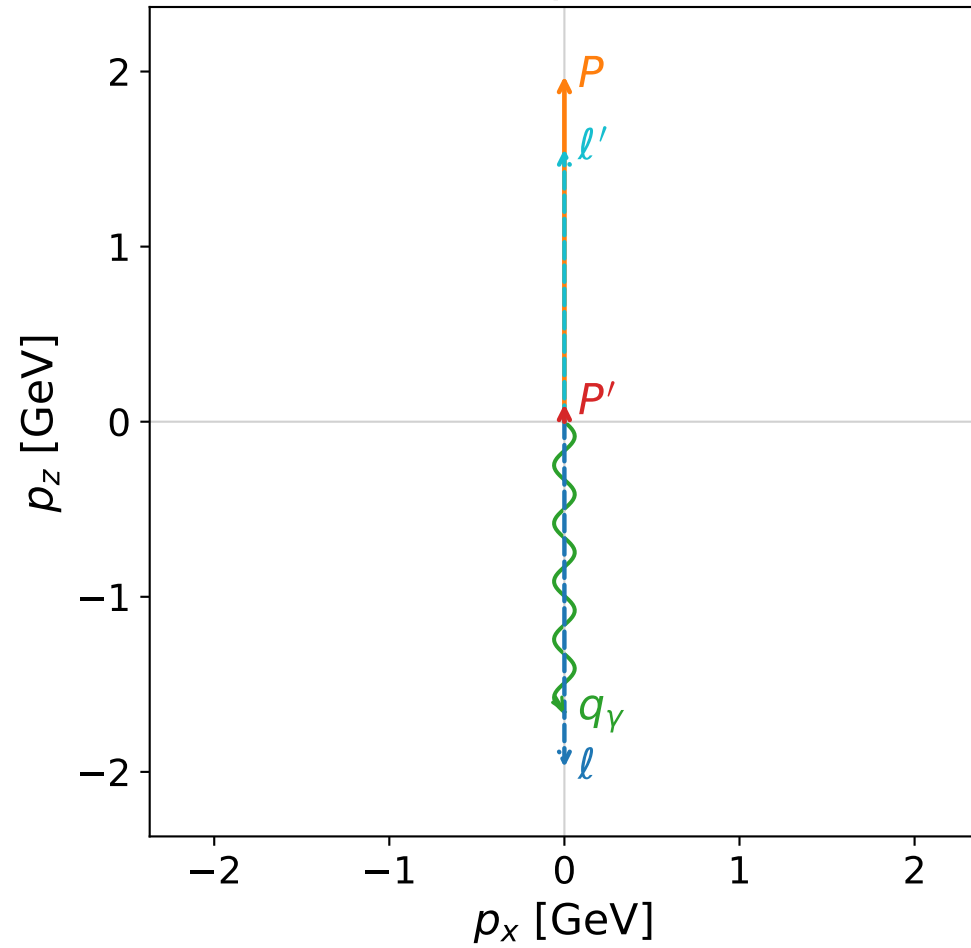
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1176$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1176$
 P $E= 2.3160$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1176$
 l' $E= 1.4863$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4863$
 P' $E= 1.0316$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4295$
 q_γ $E= 1.9157$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9157$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.244\text{e+}17$; $I_h=+8.009\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.882\text{e-}12$; $I_{TL}=-1.000\text{e-}07$

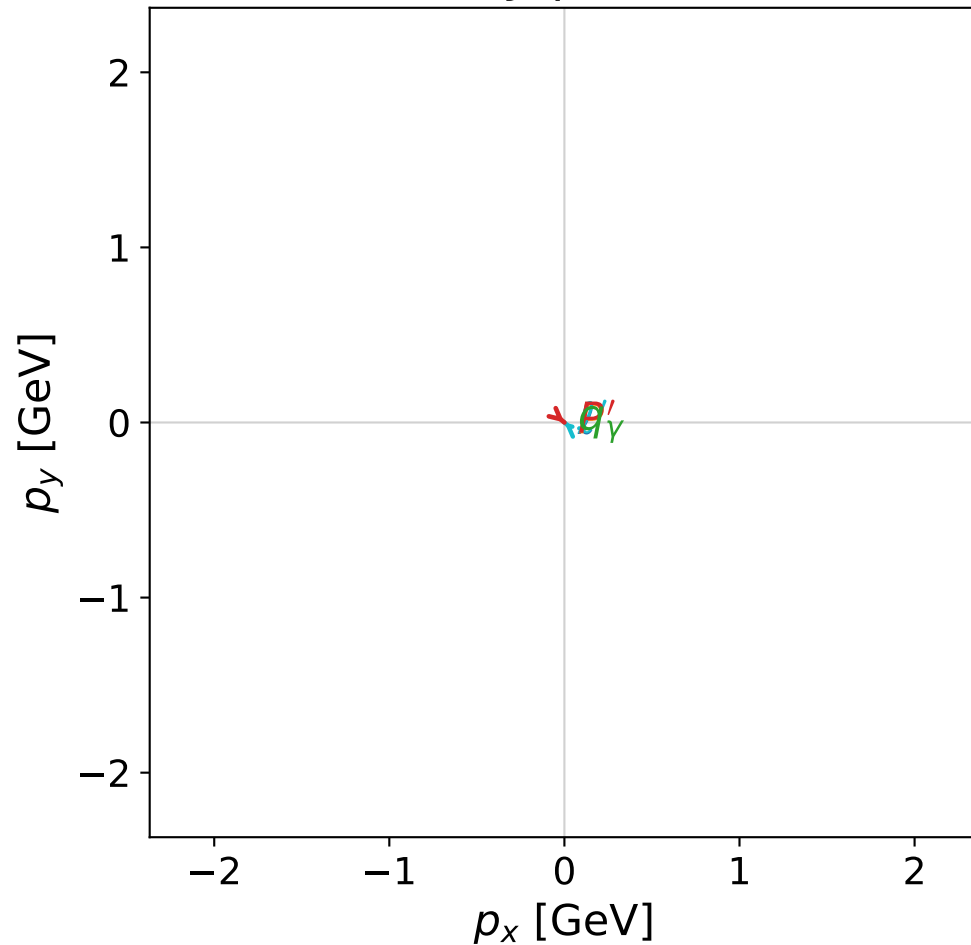


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

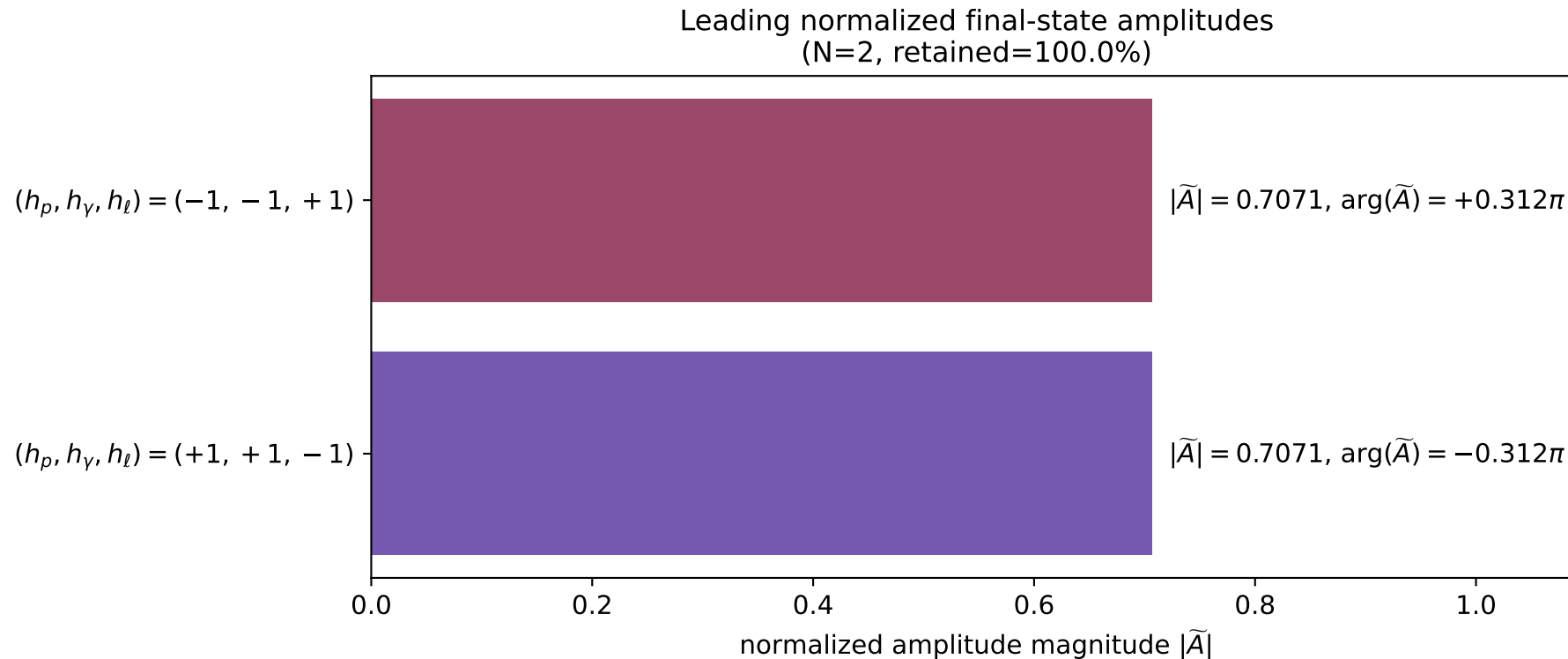


dghz_mixing_angles_polarization_01_local_minimum_113
 $d_{\text{GHZ}}=4.24901\text{e-}08$
 region: polarization_cluster_01_local_minimum_113
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0113
 lepton mixing angle: $\alpha_e=0.83948165$ rad
 incoming lepton: $0.667849|+\rangle + 0.744297|-\rangle$
 proton mixing angle: $\alpha_p=0.8394633$ rad
 incoming proton: $0.667862|+\rangle + 0.744285|-\rangle$

$s=17.2983$, $\sqrt{s}=4.15912$, $|\vec{P}|=1.97379$, $|\vec{P}'|=0.101055$
 $\theta_{P'}=0.000400551$, $\theta_Y=3.14159$, $\phi_{P'}=5.58175$, $\phi_Y=4.12102$
 $E_Y=1.65837$, $\theta_{\ell'}=2.59918\text{e-}05$, $\phi_{\ell'}=2.44016$
 $Q^2=12.2953$, $x_B=0.780179$, $t=-1.9648$, $W^2=4.34413$, $y=0.959869$

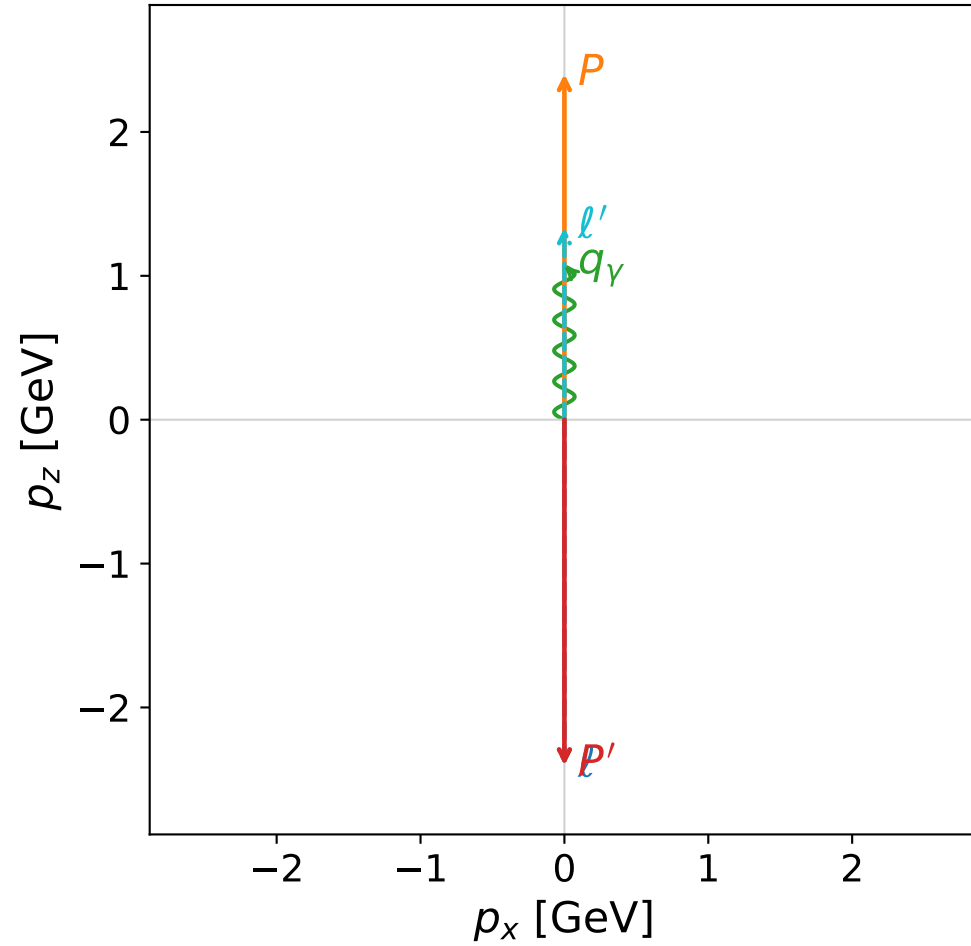
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9738$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9738$
 P $E= 2.1853$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9738$
 ℓ' $E= 1.5573$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5573$
 P' $E= 0.9434$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1011$
 q_Y $E= 1.6584$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.6584$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.841\text{e+}16$; $I_h=-3.321\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.150\text{e-}12$; $I_{TL}=+2.896\text{e-}07$

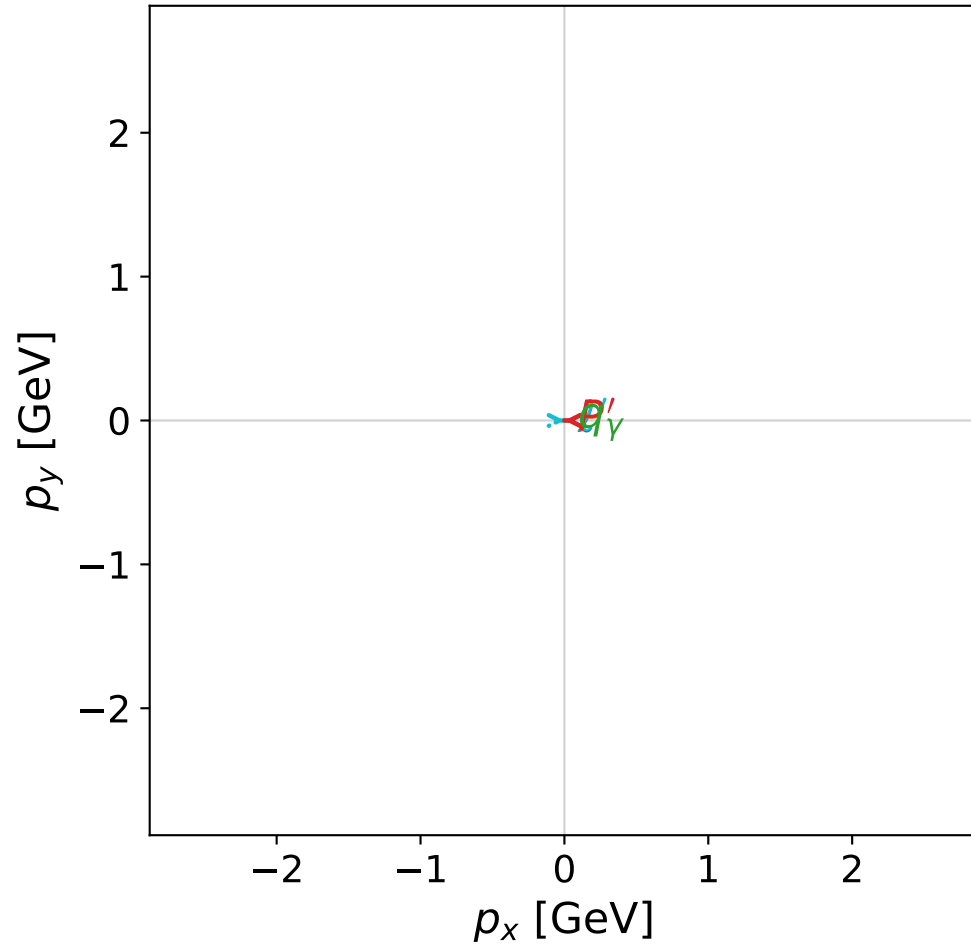


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

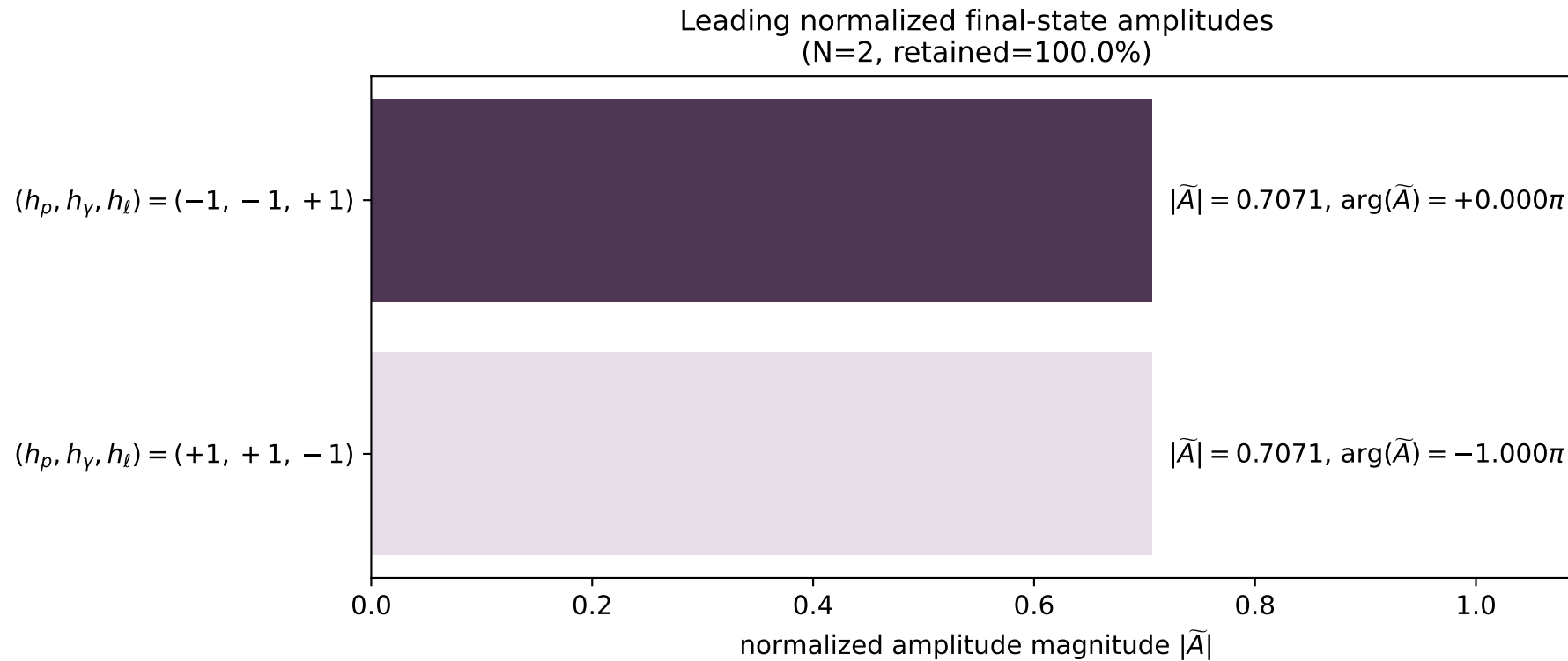


dghz_mixing_angles_polarization_01_local_minimum_114
 $d_{\text{GHZ}}=4.29606\text{e-}08$
 region: polarization_cluster_01_local_minimum_114
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0114
 lepton mixing angle: $\alpha_e=0.94445649$ rad
 incoming lepton: $0.586183|+\rangle + 0.810178|-\rangle$
 proton mixing angle: $\alpha_p=0.62636419$ rad
 incoming proton: $0.810164|+\rangle + 0.586203|-\rangle$

$s=24.8065$, $\sqrt{s}=4.98061$, $|\vec{P}|=2.40198$, $|\vec{P}'|=2.40198$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.70525$, $\phi_\gamma=5.75553$
 $E_\gamma=1.06741$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.84684$
 $Q^2=12.8224$, $x_B=0.546678$, $t=-23.078$, $W^2=11.5126$, $y=0.980293$

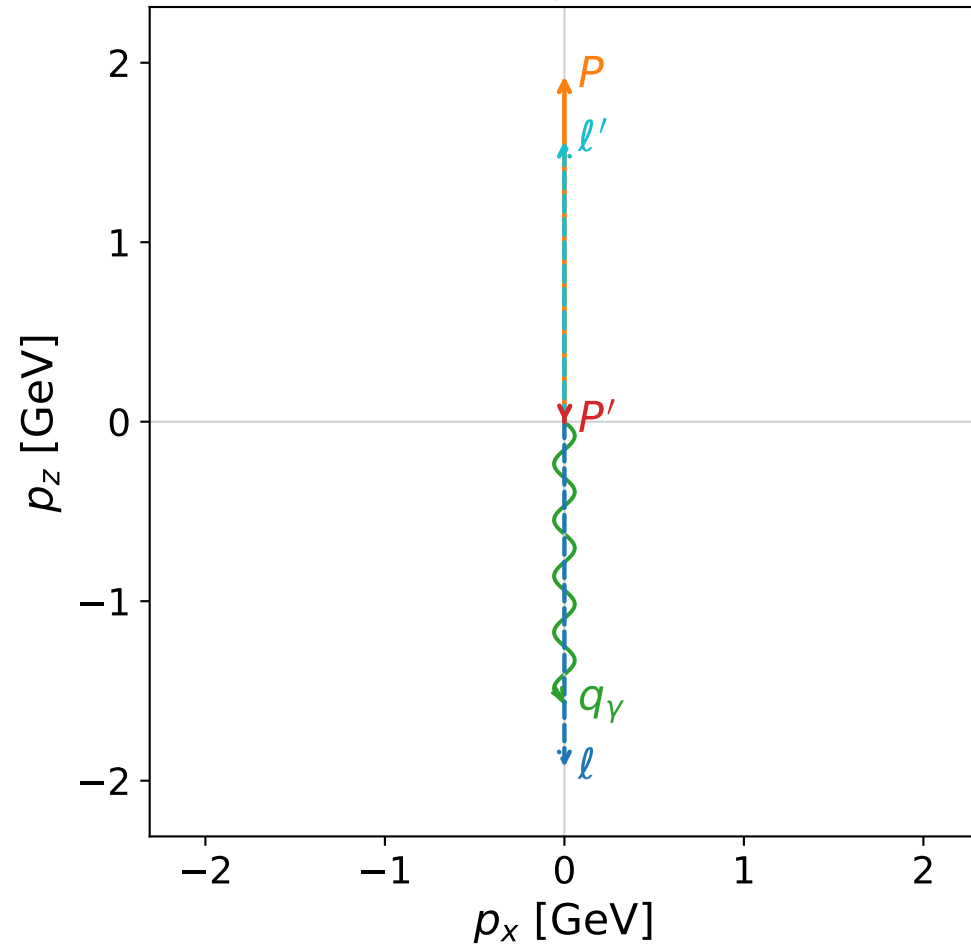
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4020$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4020$
 P $E= 2.5786$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4020$
 ℓ' $E= 1.3346$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.3346$
 P' $E= 2.5786$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4020$
 q_γ $E= 1.0674$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0674$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=4.568\text{e+}23$; $I_h=+3.525\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.888\text{e-}28$; $I_{TL}=+3.525\text{e-}14$

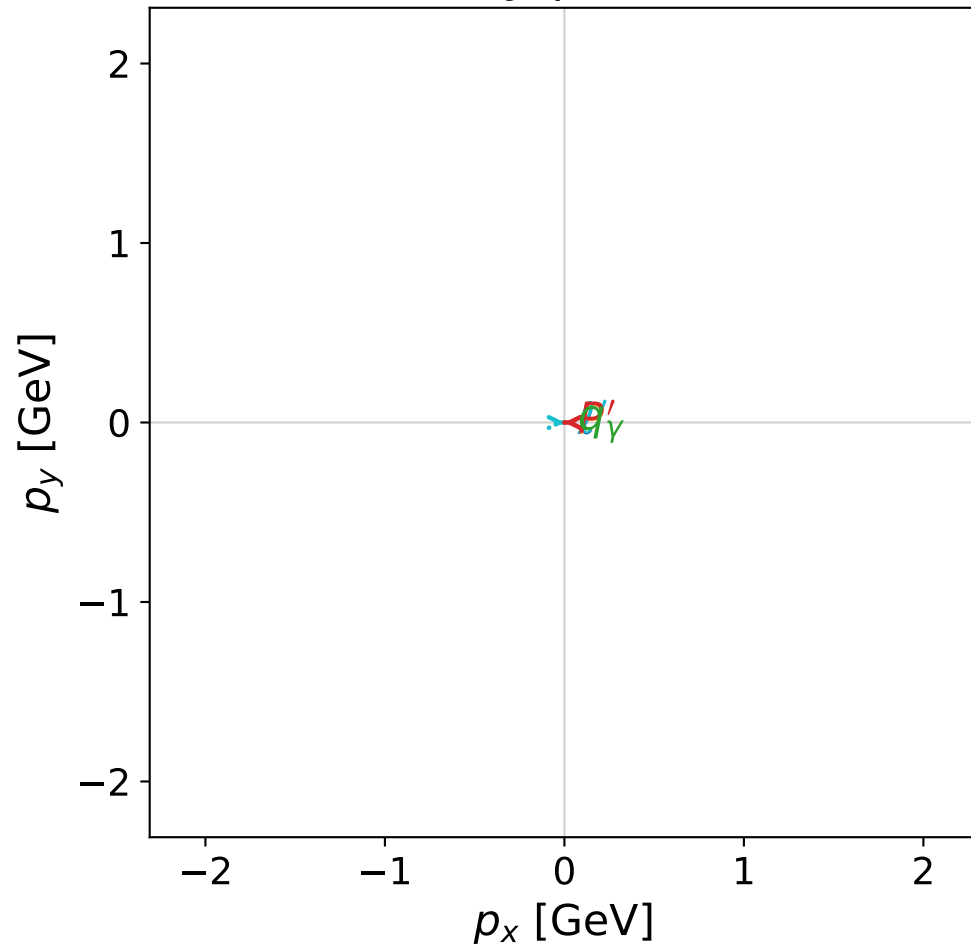


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

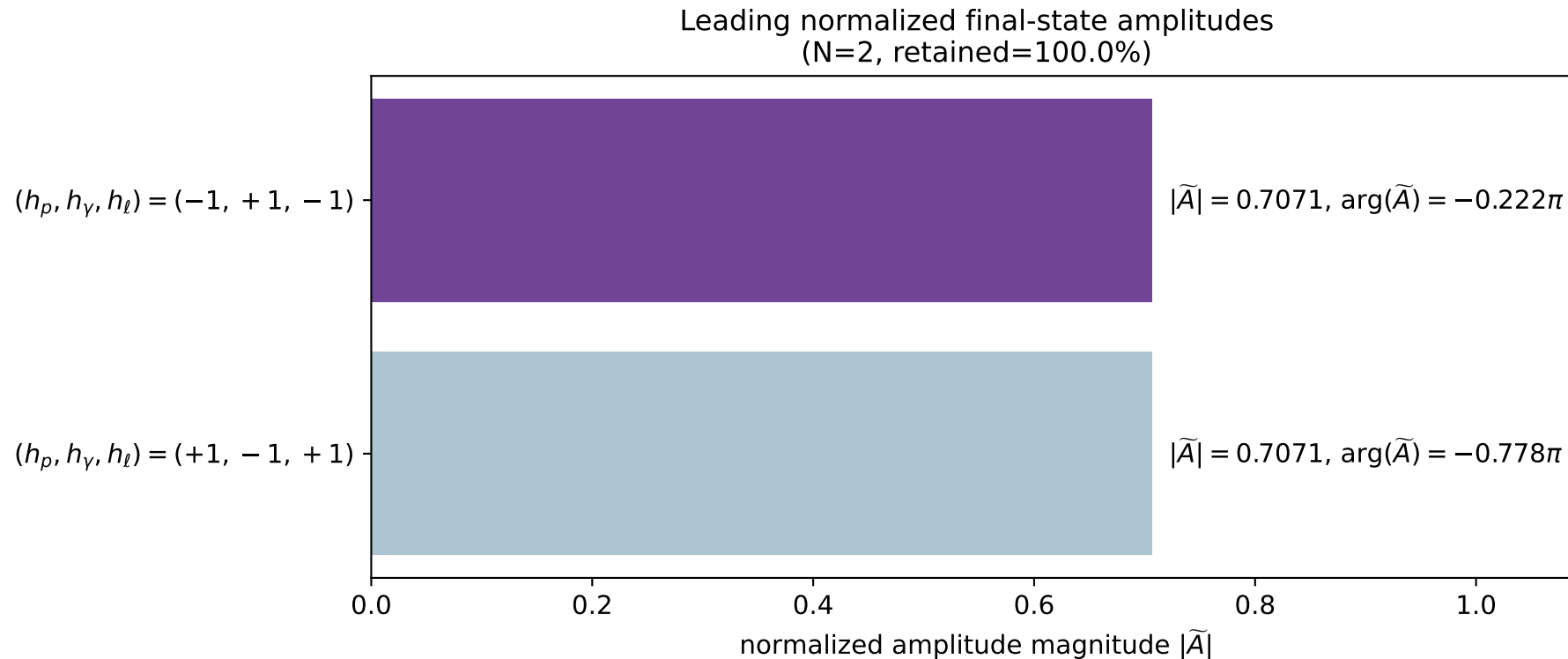


dghz_mixing_angles_polarization_01_local_minimum_115
 $d_{\text{GHZ}}=4.30756\text{e-}08$
 region: polarization_cluster_01_local_minimum_115
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0115
 lepton mixing angle: $\alpha_e=0.76452518$ rad
 incoming lepton: $0.721711|+\rangle +0.692194|-\rangle$
 proton mixing angle: $\alpha_p=0.76454044$ rad
 incoming proton: $0.721701|+\rangle +0.692205|-\rangle$

$s=16.5413$, $\sqrt{s}=4.0671$, $|\vec{P}|=1.92538$, $|\vec{P}'|=0.000570618$
 $\theta_{P'}=3.11934$, $\theta_Y=3.14159$, $\phi_{P'}=3.14223$, $\phi_Y=3.8405$
 $E_Y=1.56427$, $\theta_{\ell'}=8.11432\text{e-}06$, $\phi_{\ell'}=0.000639486$
 $Q^2=12.0516$, $x_B=0.804278$, $t=-2.26037$, $W^2=3.81262$, $y=0.95677$

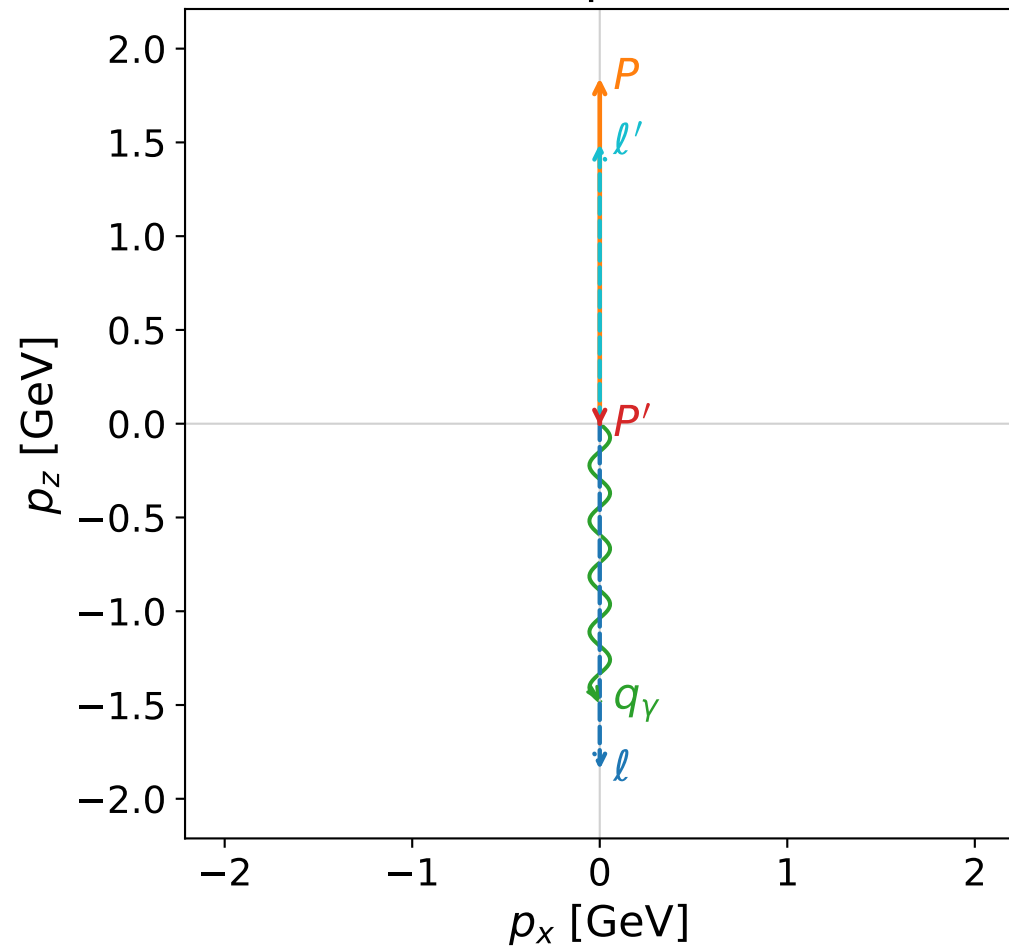
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9254$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9254$
 P $E= 2.1417$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9254$
 ℓ' $E= 1.5648$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5648$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0006$
 q_Y $E= 1.5643$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.5643$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.579\text{e+}17$; $I_h=-5.532\text{e-}11$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.109\text{e-}13$; $I_{TL}=+4.222\text{e-}08$

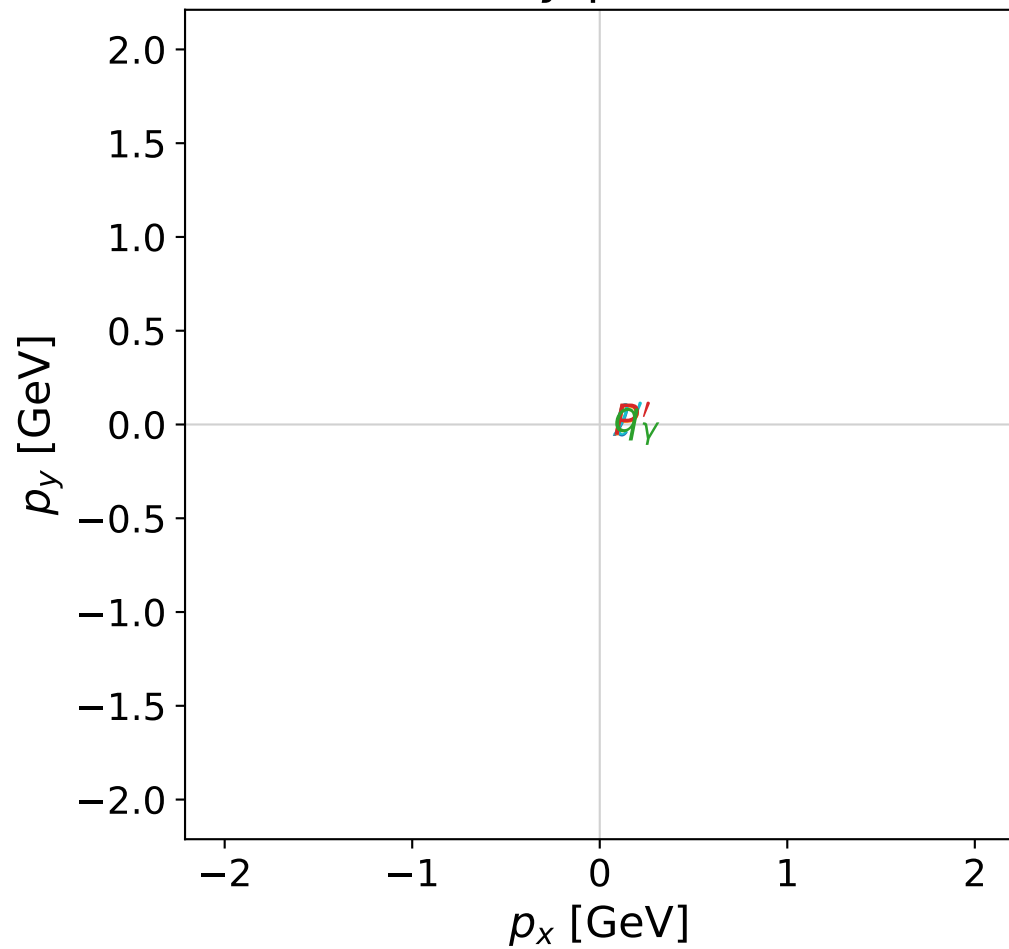


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

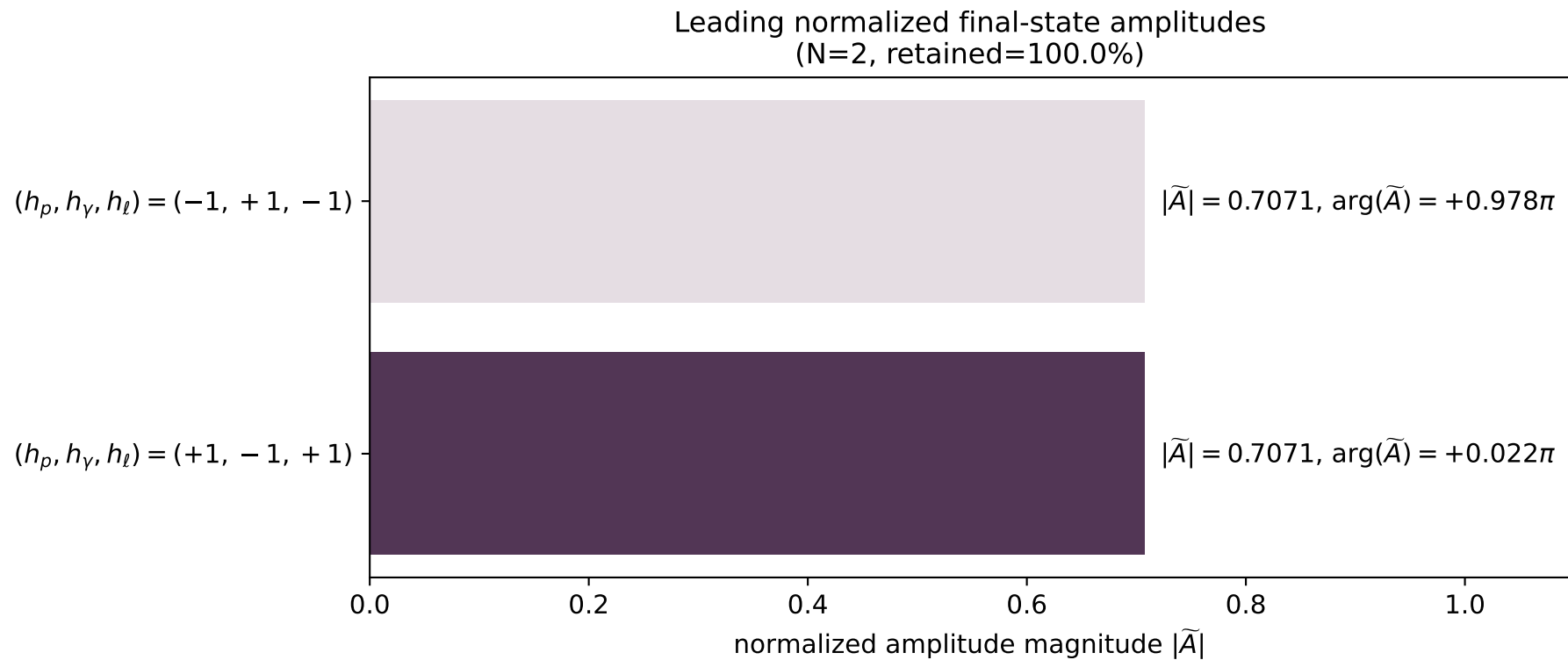


dghz_mixing_angles_polarization_01_local_minimum_121
 $d_{\text{GHZ}}=4.75198\text{e-}08$
 region: polarization_cluster_01_local_minimum_121
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0121
 lepton mixing angle: $\alpha_e=0.80409735$ rad
 incoming lepton: $0.693762|+\rangle + 0.720205|-\rangle$
 proton mixing angle: $\alpha_p=0.80410559$ rad
 incoming proton: $0.693756|+\rangle + 0.72021|-\rangle$

$s=15.2928$, $\sqrt{s}=3.9106$, $|\vec{P}|=1.8428$, $|\vec{P}'|=0.0114504$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=3.39183$, $\phi_Y=3.21079$
 $E_Y=1.48054$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.0705831$
 $Q^2=10.9978$, $x_B=0.800329$, $t=-2.16199$, $W^2=3.62364$, $y=0.953419$

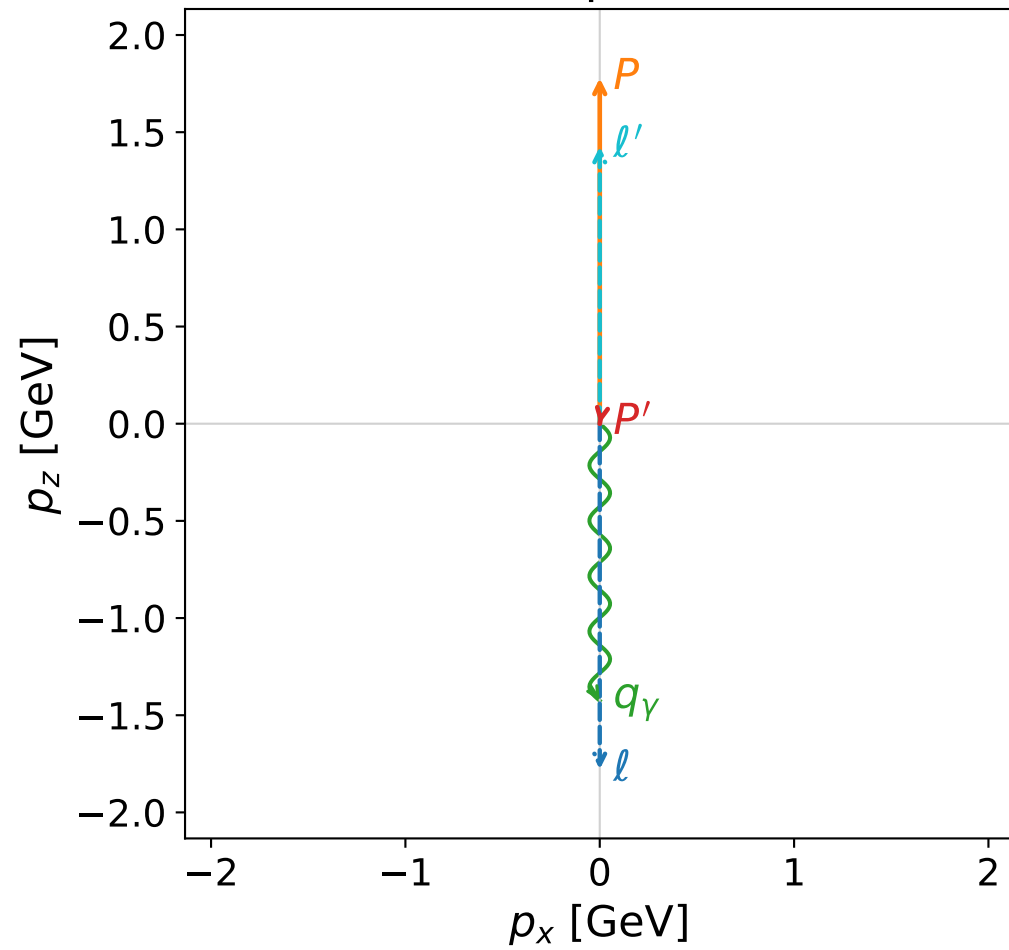
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8428$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8428$
 P $E= 2.0678$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8428$
 ℓ' $E= 1.4920$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4920$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0115$
 q_Y $E= 1.4805$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4805$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.174\text{e}+24$; $I_h=-8.071\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.496\text{e-}27$; $I_{TL}=+1.669\text{e-}18$

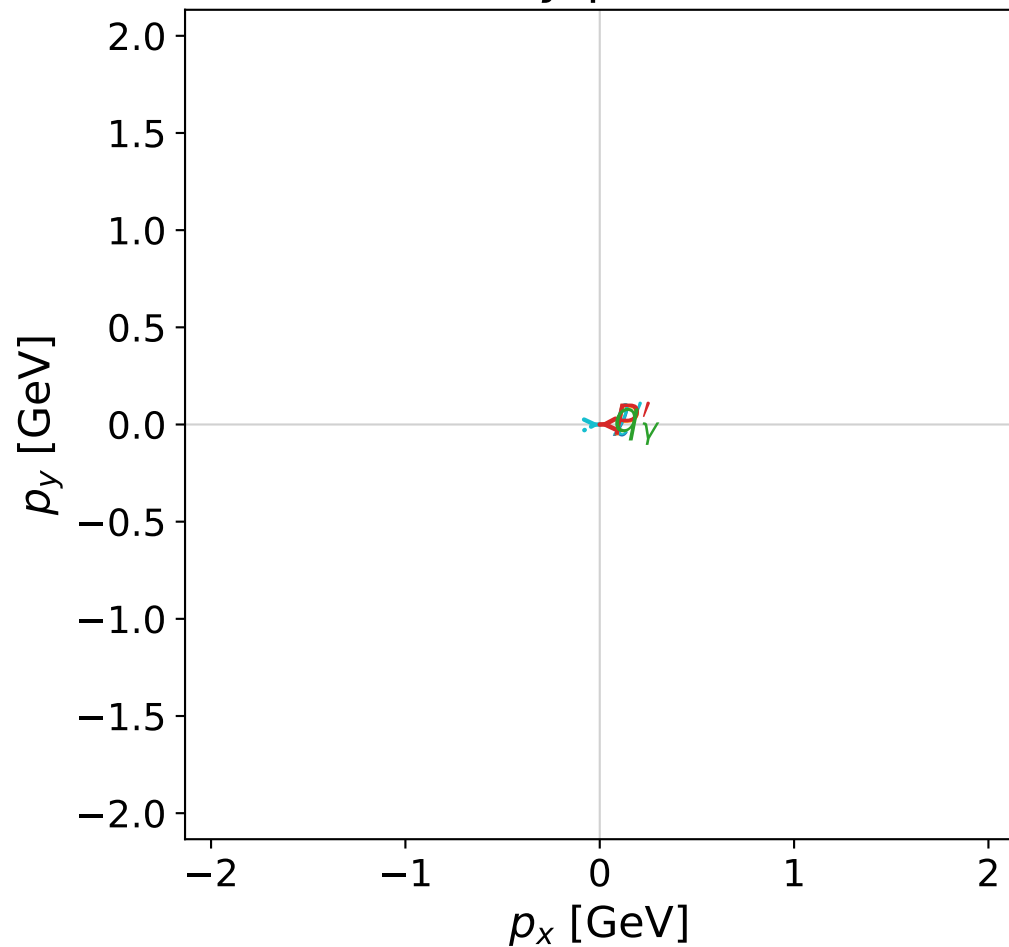


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

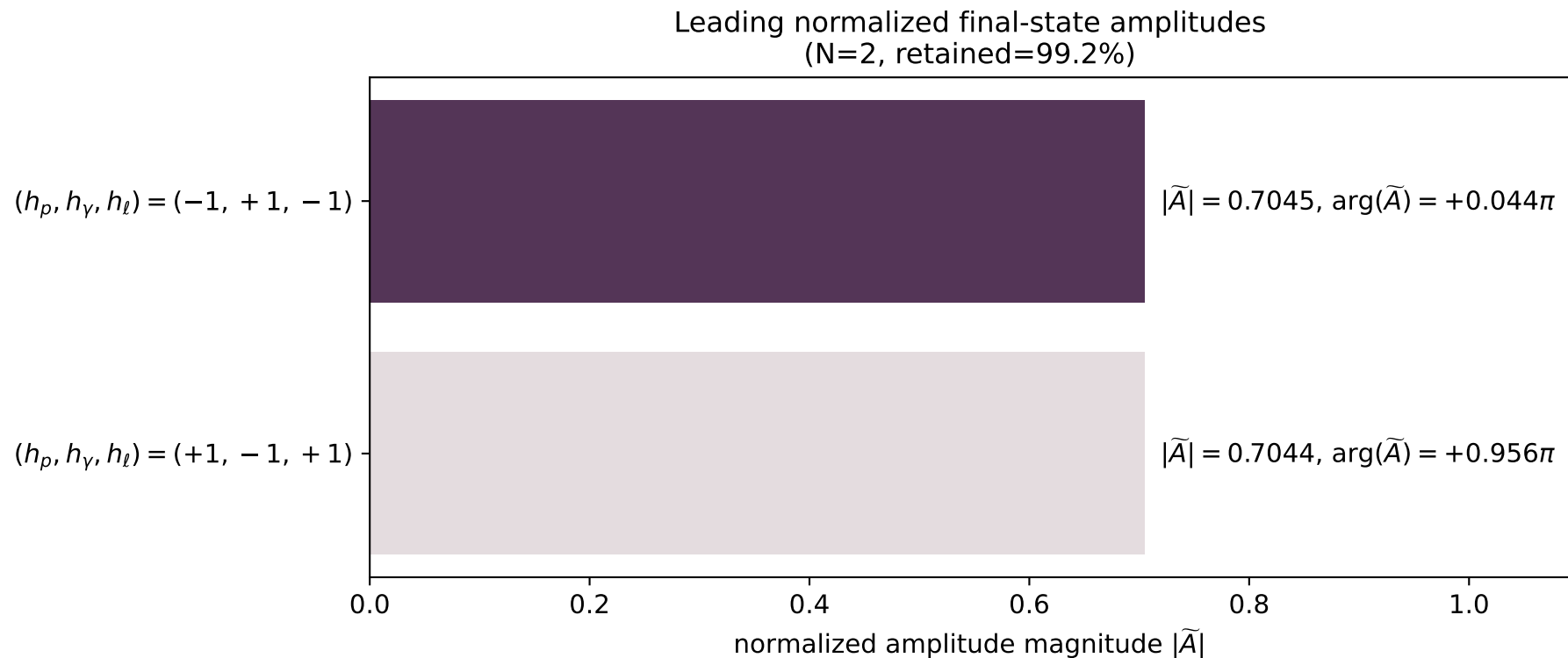


dghz_mixing_angles_polarization_01_local_minimum_126
 $d_{\text{GHZ}}=4.95818\text{e-}08$
 region: polarization_cluster_01_local_minimum_126
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0126
 lepton mixing angle: $\alpha_e=0.79324484$ rad
 incoming lepton: $0.701537|+\rangle + 0.712633|-\rangle$
 proton mixing angle: $\alpha_p=0.79327627$ rad
 incoming proton: $0.701514|+\rangle + 0.712655|-\rangle$

$s=14.3543$, $\sqrt{s}=3.78871$, $|\vec{P}|=1.77824$, $|\vec{P}'|=0.000403116$
 $\theta_{p'}=2.96794$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.16594$, $\phi_\gamma=3.0289$
 $E_\gamma=1.42516$, $\theta_{\ell'}=4.88583\text{e-}05$, $\phi_{\ell'}=0.0243458$
 $Q^2=10.1399$, $x_B=0.791416$, $t=-2.01337$, $W^2=3.55231$, $y=0.950862$

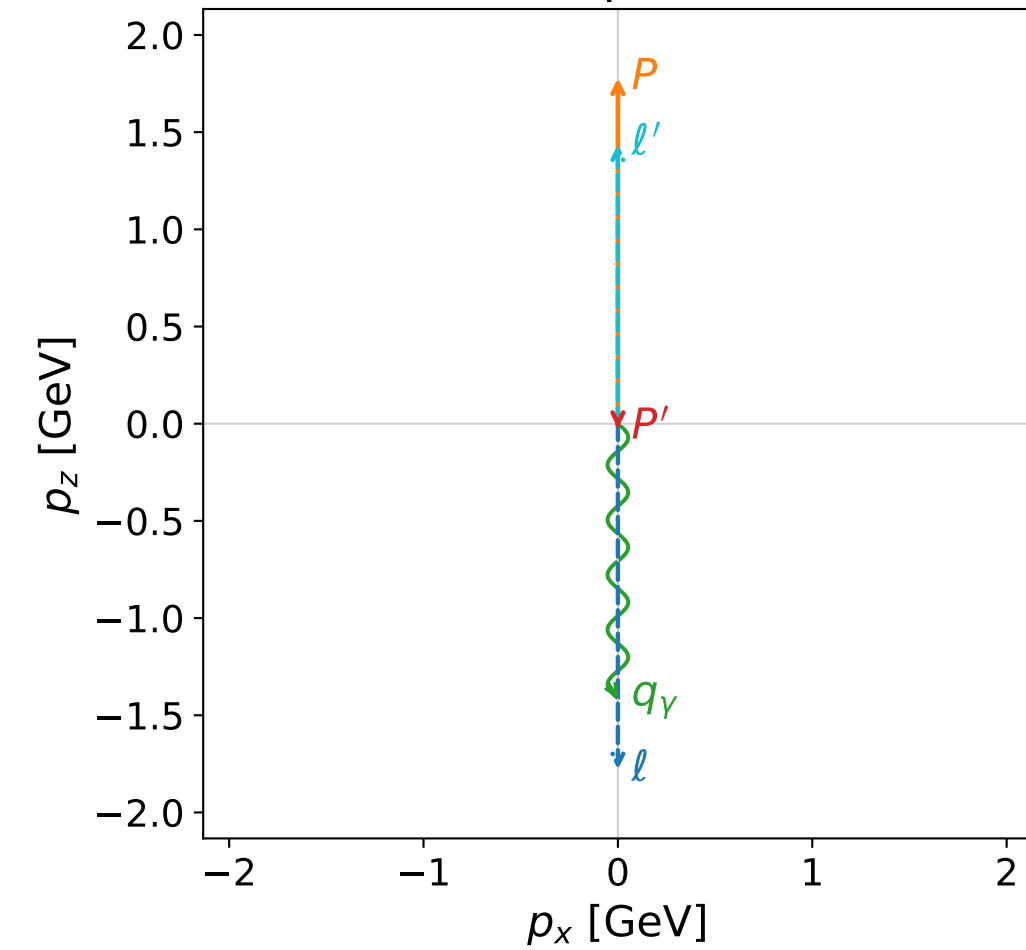
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7782$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7782$
 P $E= 2.0105$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7782$
 ℓ' $E= 1.4256$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.4256$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.0004$
 q_γ $E= 1.4252$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.4252$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.738\text{e+}16$; $I_h=+1.379\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{\text{L/T}}=2.346\text{e-}11$; $I_{\text{TL}}=-1.062\text{e-}07$

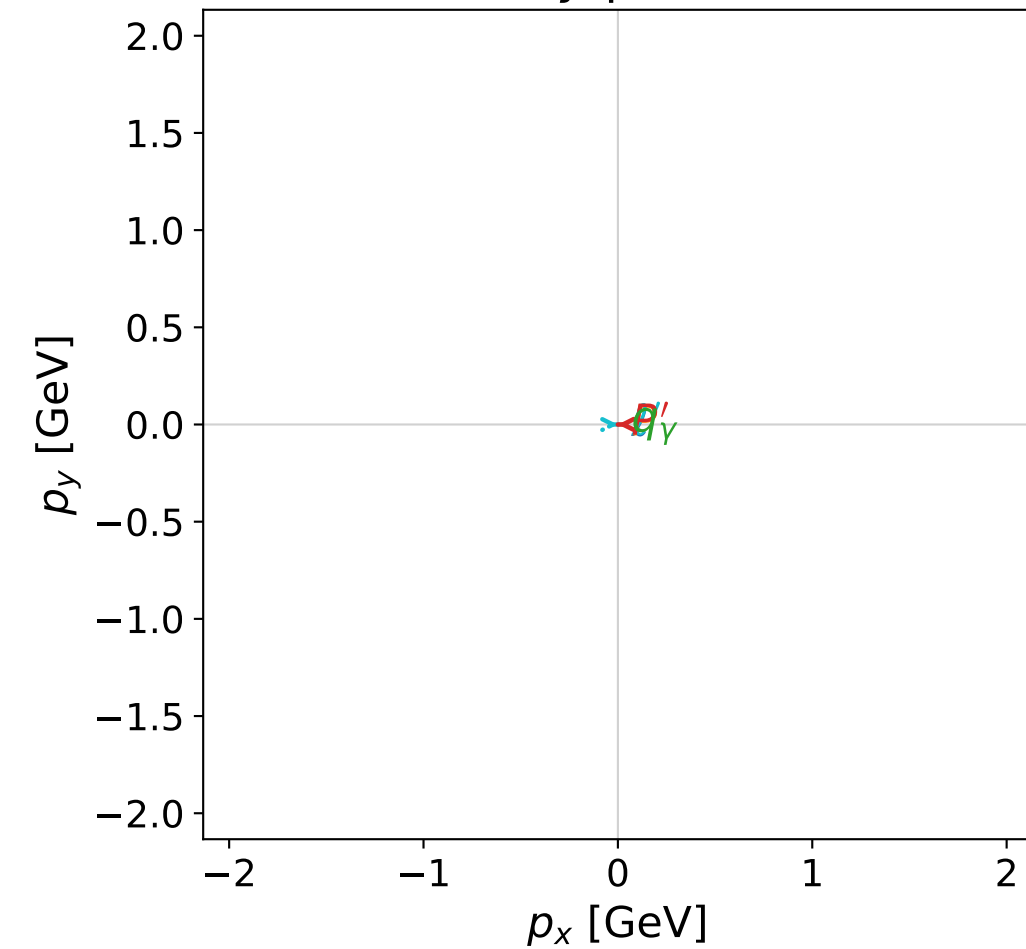


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

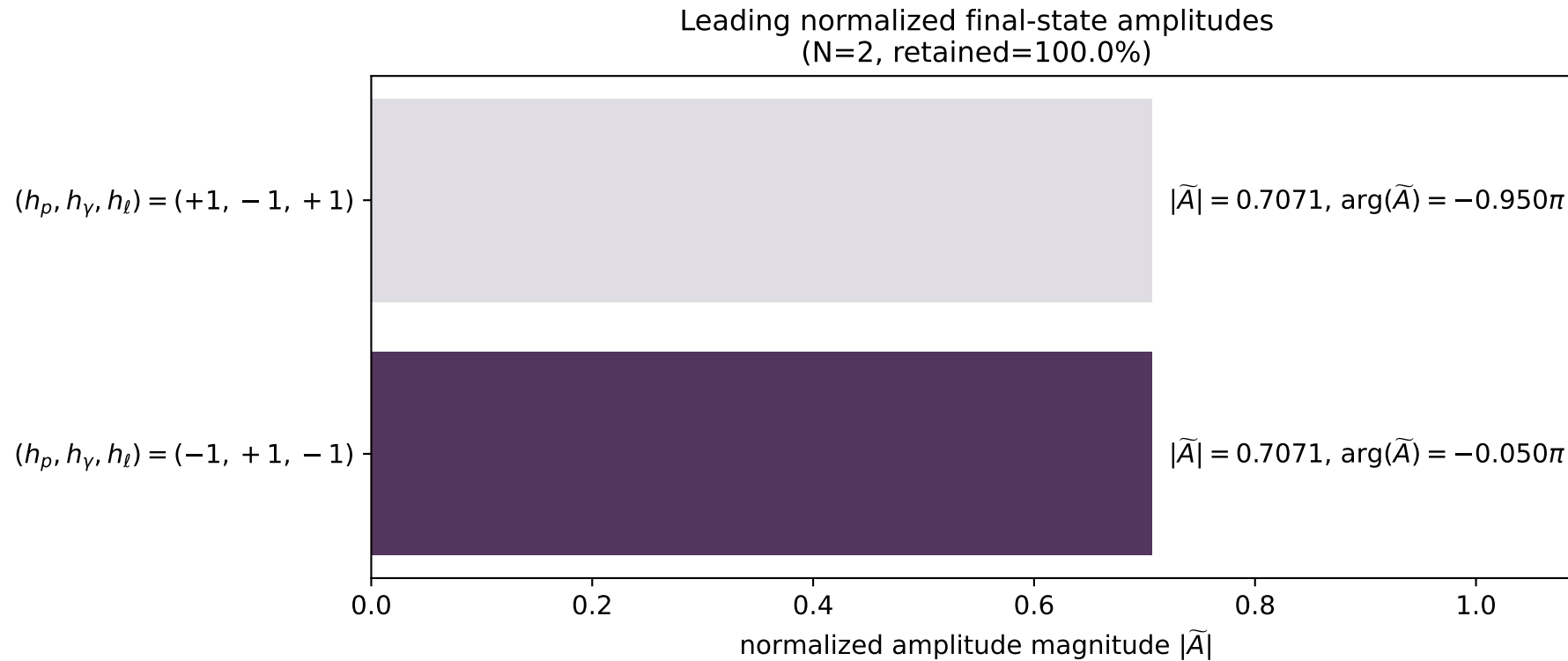


dghz_mixing_angles_polarization_01_local_minimum_129
 $d_{\text{GHZ}}=5.2277\text{e-}08$
 region: polarization_cluster_01_local_minimum_129
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0129
 lepton mixing angle: $\alpha_e=0.89040334$ rad
 incoming lepton: $0.629099|+\rangle + 0.777326|-\rangle$
 proton mixing angle: $\alpha_p=0.89039462$ rad
 incoming proton: $0.629105|+\rangle + 0.77732|-\rangle$

$s=14.3507$, $\sqrt{s}=3.78823$, $|\vec{P}|=1.77798$, $|\vec{P}'|=0.024118$
 $\theta_{P'}=3.13955$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14474$, $\phi_\gamma=3.30063$
 $E_\gamma=1.4129$, $\theta_{\ell'}=3.42918\text{e-}05$, $\phi_{\ell'}=0.00314395$
 $Q^2=10.22$, $x_B=0.79823$, $t=-2.09854$, $W^2=3.46317$, $y=0.950447$

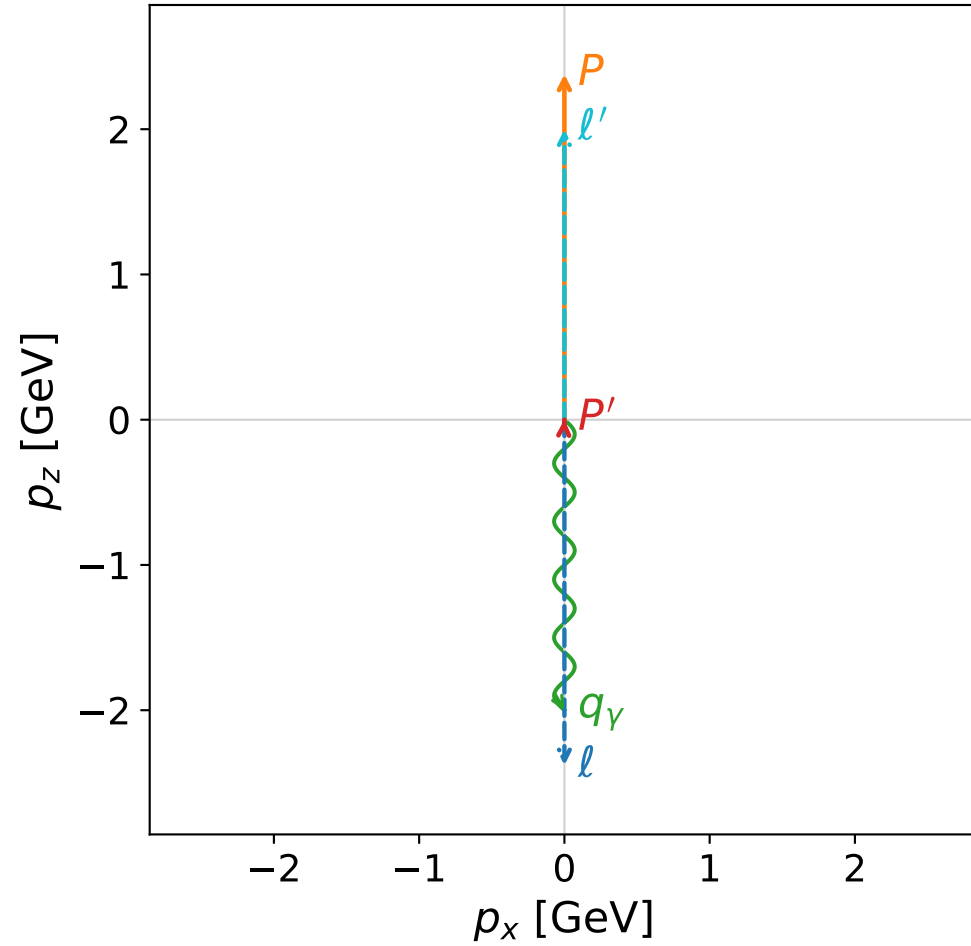
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7780$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7780$
 P $E= 2.0102$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7780$
 ℓ' $E= 1.4370$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4370$
 P' $E= 0.9383$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0241$
 q_γ $E= 1.4129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4129$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.304\text{e+}16$; $I_h=+1.322\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.145\text{e-}11$; $I_{TL}=-9.971\text{e-}07$

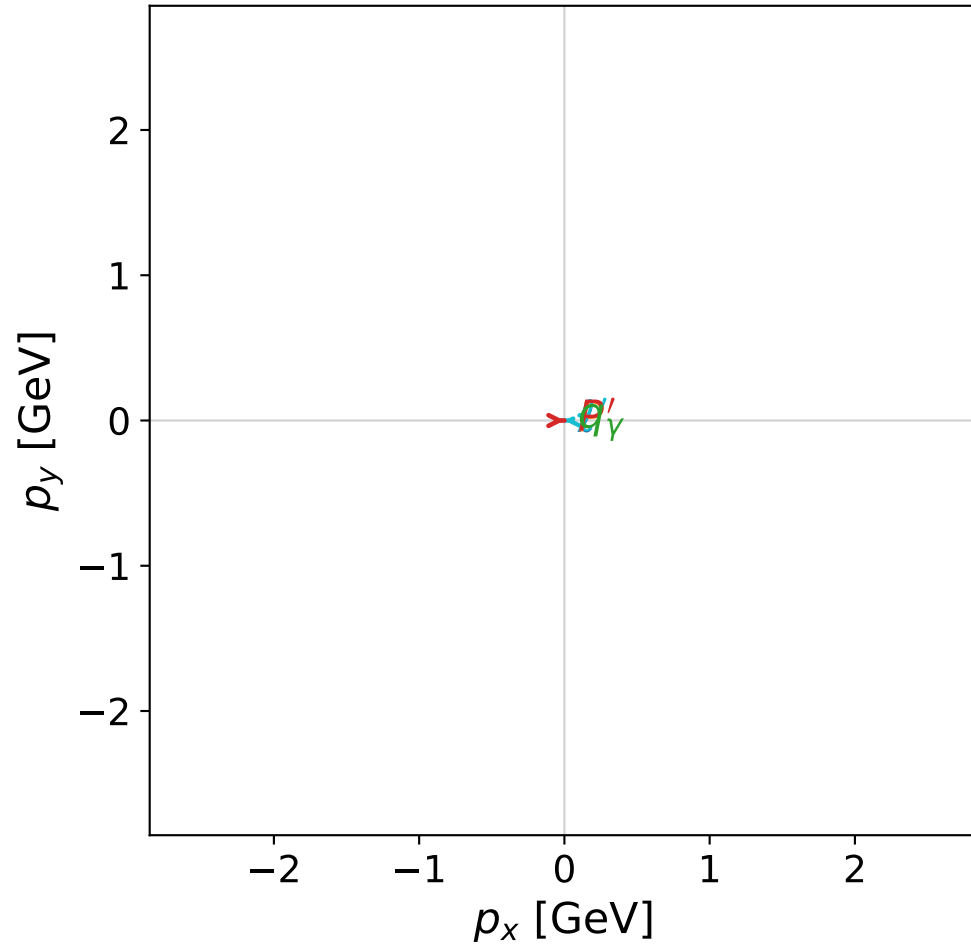


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

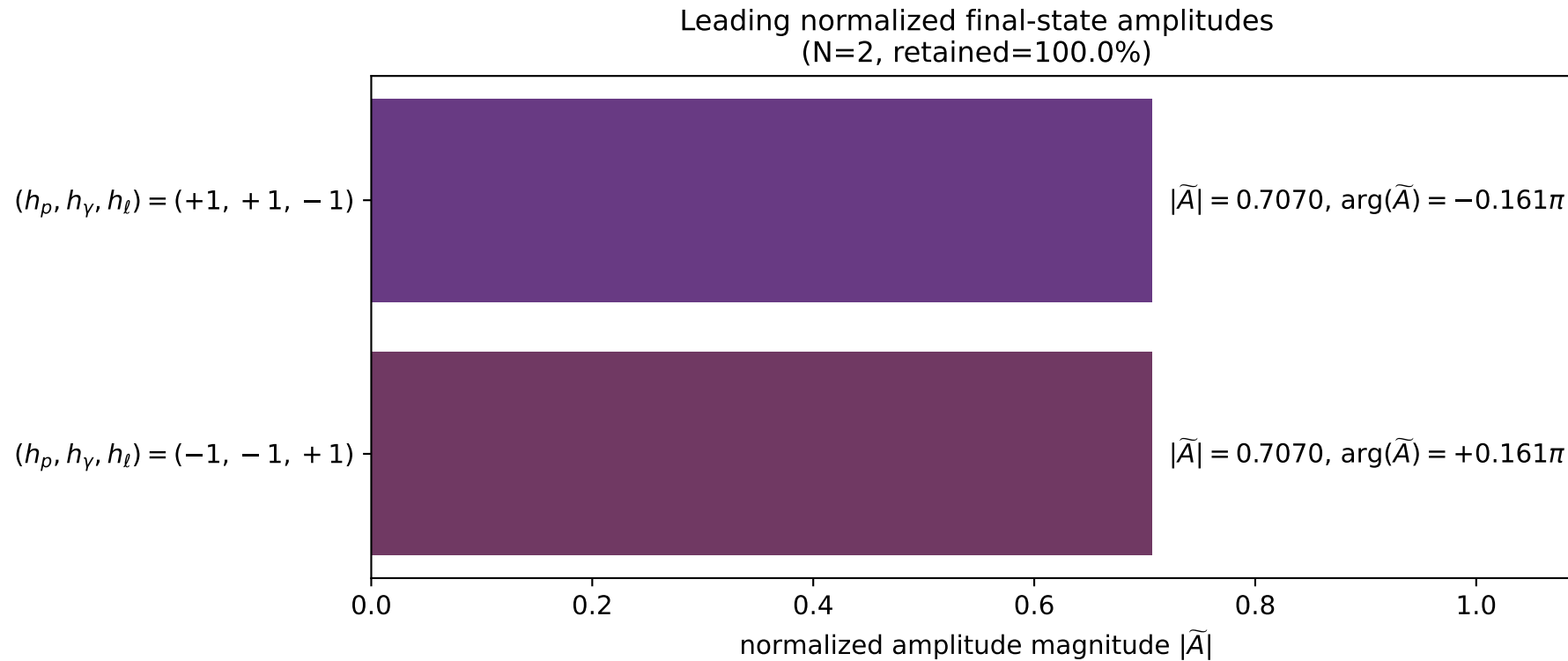


dghz_mixing_angles_polarization_01_local_minimum_135
 $d_{\text{GHZ}}=5.61544\text{e-}08$
 region: polarization_cluster_01_local_minimum_135
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0135
 lepton mixing angle: $\alpha_e=1.3218649$ rad
 incoming lepton: $0.246369|+\rangle + 0.969176|-\rangle$
 proton mixing angle: $\alpha_p=1.3218678$ rad
 incoming proton: $0.246366|+\rangle + 0.969177|-\rangle$

$s=24.3709$, $\sqrt{s}=4.93669$, $|\vec{P}|=2.37923$, $|\vec{P}'|=0.000777917$
 $\theta_{P'}=0.0397154$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.000309928$, $\phi_\gamma=3.64684$
 $E_\gamma=1.99973$, $\theta_{\ell'}=1.54516\text{e-}05$, $\phi_{\ell'}=3.1419$
 $Q^2=19.0239$, $x_B=0.835169$, $t=-3.0344$, $W^2=4.63445$, $y=0.969668$

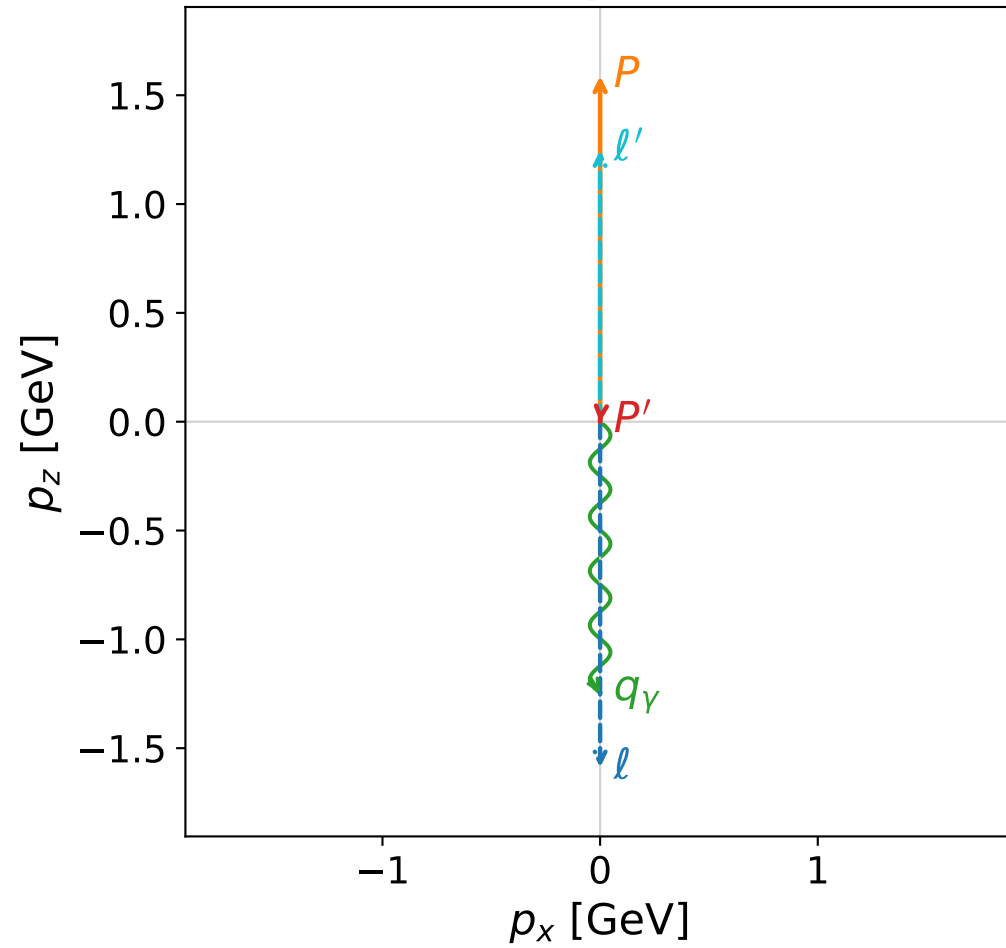
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3792$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3792$
 P $E= 2.5575$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3792$
 ℓ' $E= 1.9990$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9990$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0008$
 q_γ $E= 1.9997$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9997$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=6.201\text{e+}16$; $I_h=-3.633\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=4.105\text{e-}12$; $I_{TL}=+2.517\text{e-}06$

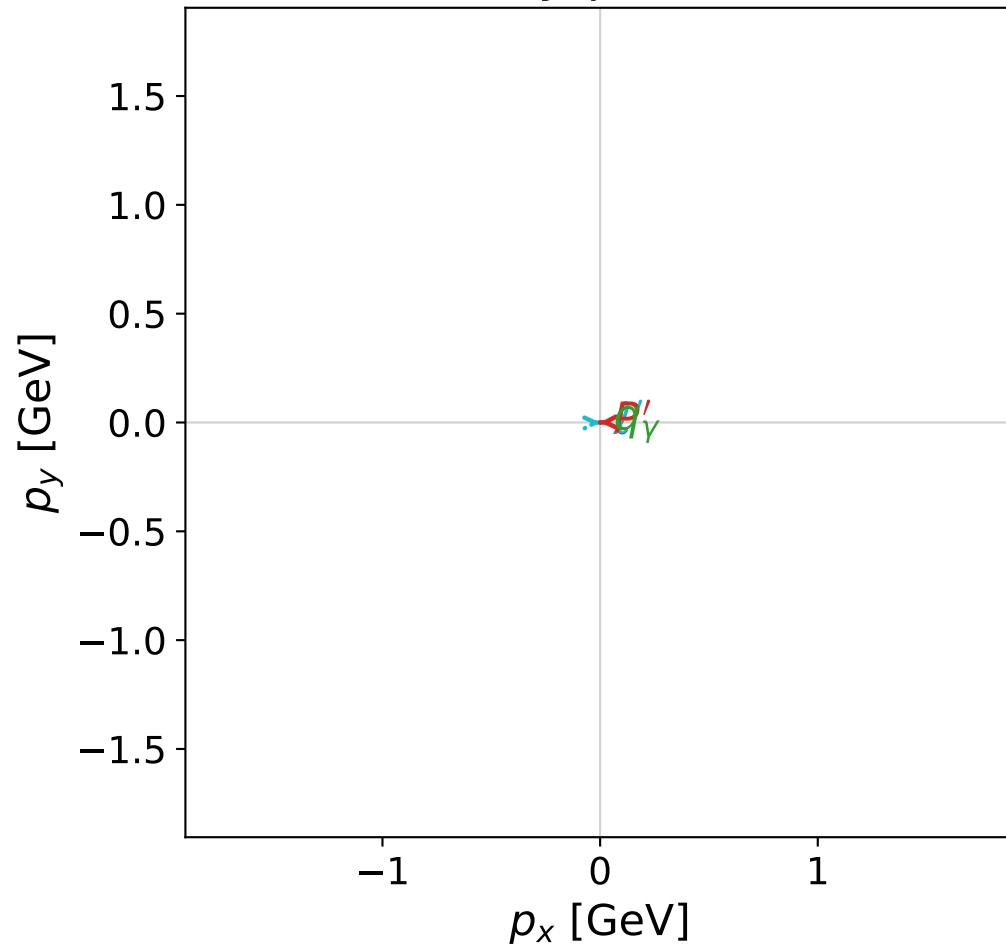


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



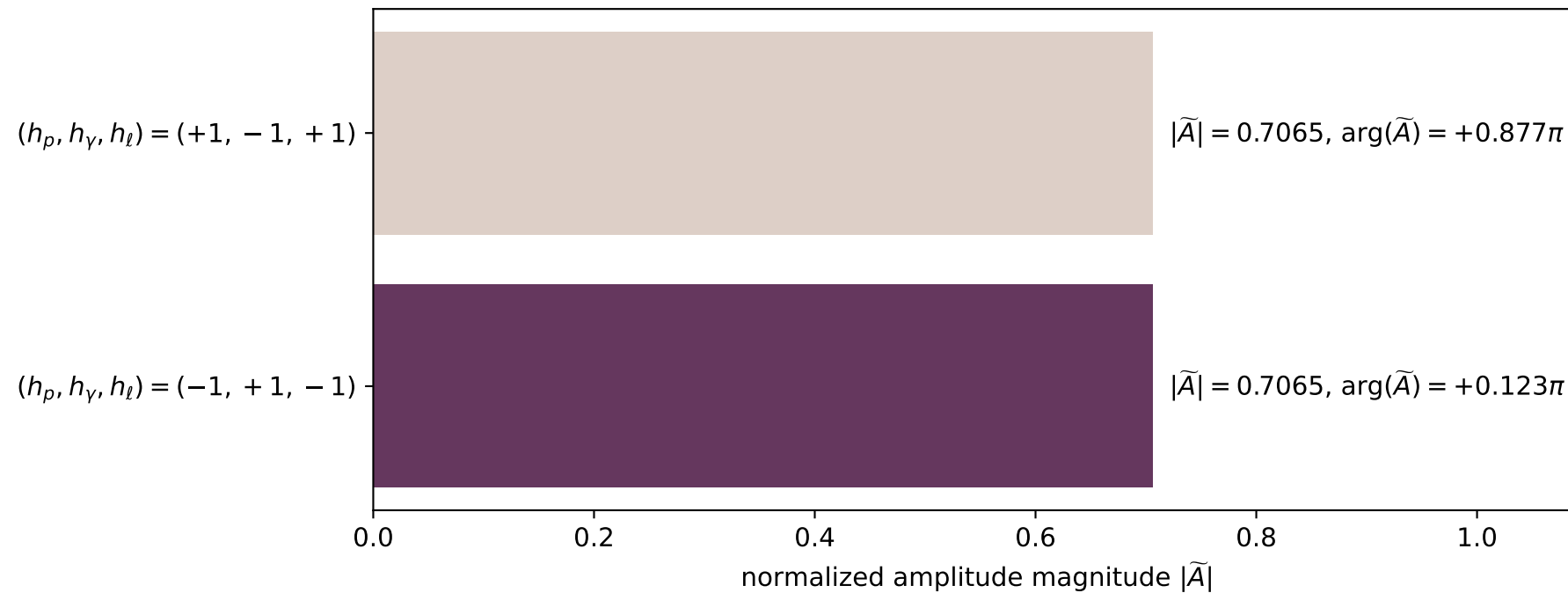
dghz_mixing_angles_polarization_01_local_minimum_138
 $d_{\text{GHZ}}=6.56334\text{e-}08$
 region: polarization_cluster_01_local_minimum_138
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0138
 lepton mixing angle: $\alpha_e=0.83474474$ rad
 incoming lepton: $0.671367|+\rangle + 0.741125|-\rangle$
 proton mixing angle: $\alpha_p=0.83473256$ rad
 incoming proton: $0.671376|+\rangle + 0.741117|-\rangle$

$s=11.7789$, $\sqrt{s}=3.43203$, $|\vec{P}|=1.58784$, $|\vec{P}'|=0.00100941$
 $\theta_{P'}=3.06084$, $\theta_Y=3.14159$, $\phi_{P'}=3.17093$, $\phi_Y=2.78545$
 $E_Y=1.24651$, $\theta_{\ell'}=6.52663\text{e-}05$, $\phi_{\ell'}=0.0293406$
 $Q^2=7.92343$, $x_B=0.772311$, $t=-1.70322$, $W^2=3.2158$, $y=0.941313$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5878$
 P $E= 1.8442$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5878$
 ℓ' $E= 1.2475$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 1.2475$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= -0.0010$
 q_Y $E= 1.2465$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.2465$

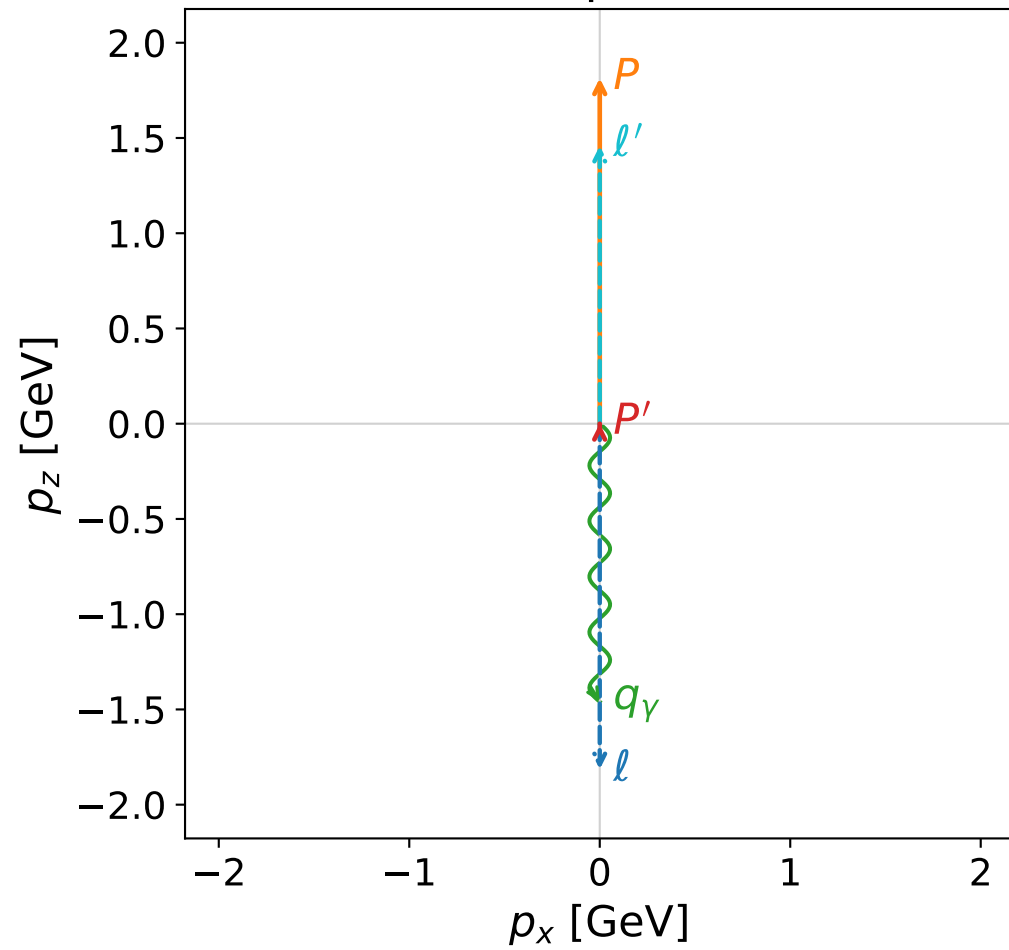
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.359\text{e+}15$; $I_h=+1.324\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=5.828\text{e-}11$; $I_{TL}=-1.061\text{e-}06$

Leading normalized final-state amplitudes (N=2, retained=99.8%)

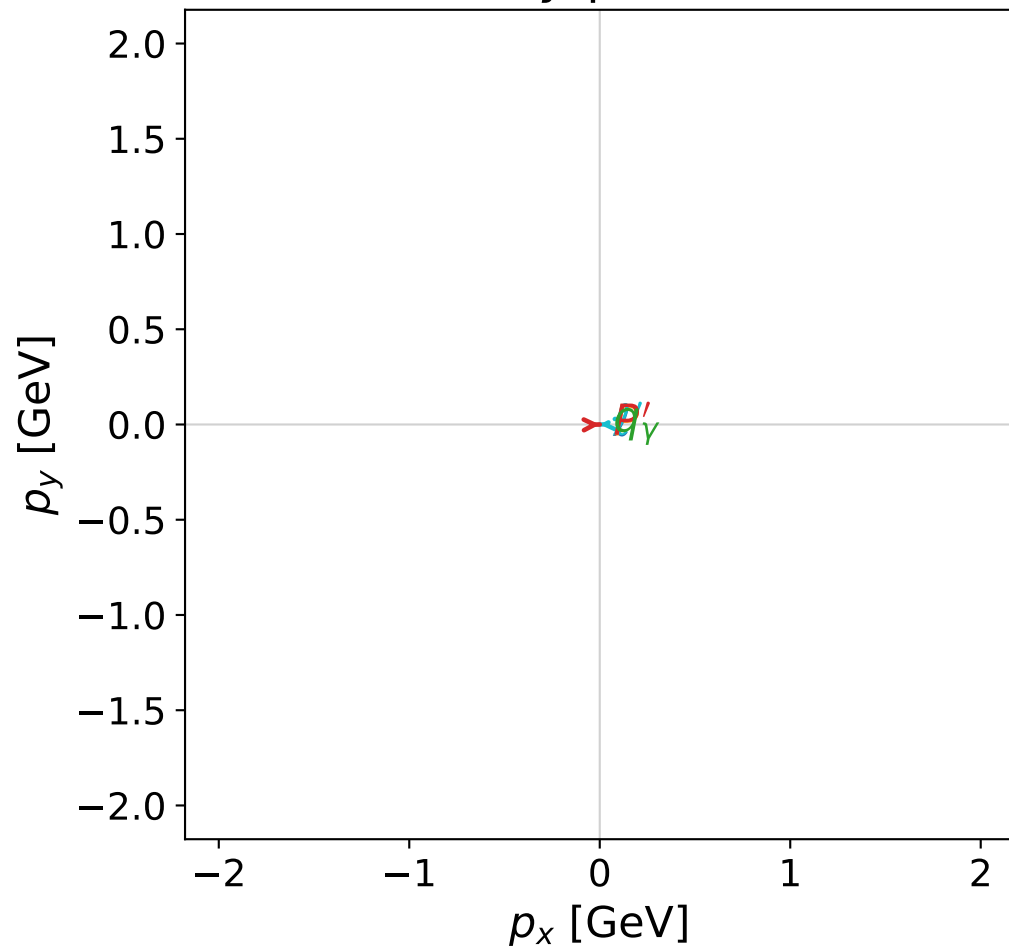


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

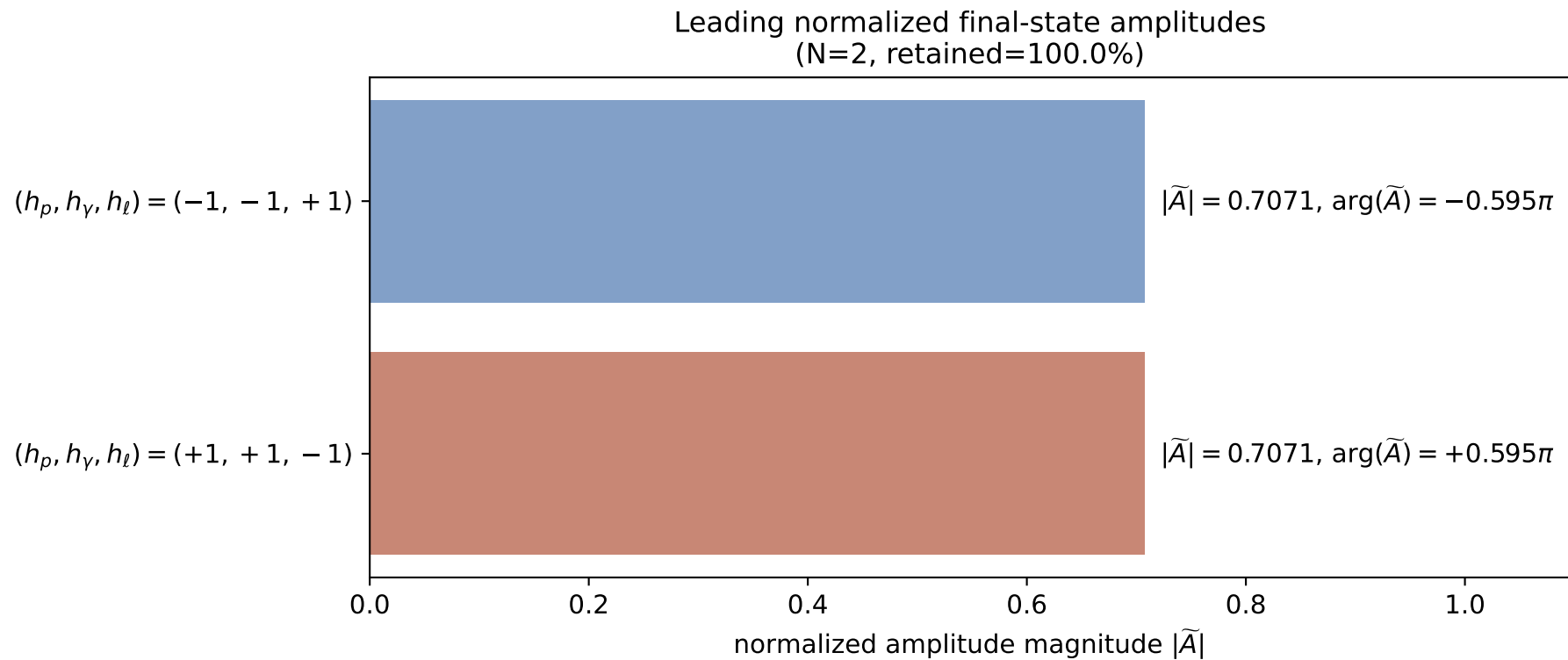


dghz_mixing_angles_polarization_01_local_minimum_139
 $d_{\text{GHZ}}=6.7347\text{e-}08$
 region: polarization_cluster_01_local_minimum_139
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0139
 lepton mixing angle: $\alpha_e=0.40504113$ rad
 incoming lepton: $0.919086|+\rangle + 0.394057|-\rangle$
 proton mixing angle: $\alpha_p=0.40503483$ rad
 incoming proton: $0.919089|+\rangle + 0.394051|-\rangle$

$s=14.8742$, $\sqrt{s}=3.85671$, $|\vec{P}|=1.81429$, $|\vec{P}'|=0.000584182$
 $\theta_{P'}=0.00490874$, $\theta_Y=3.14159$, $\phi_{P'}=0.0235349$, $\phi_Y=1.27295$
 $E_Y=1.45965$, $\theta_{\ell'}=1.96537\text{e-}06$, $\phi_{\ell'}=3.16513$
 $Q^2=10.5886$, $x_B=0.794427$, $t=-2.06977$, $W^2=3.61985$, $y=0.952429$

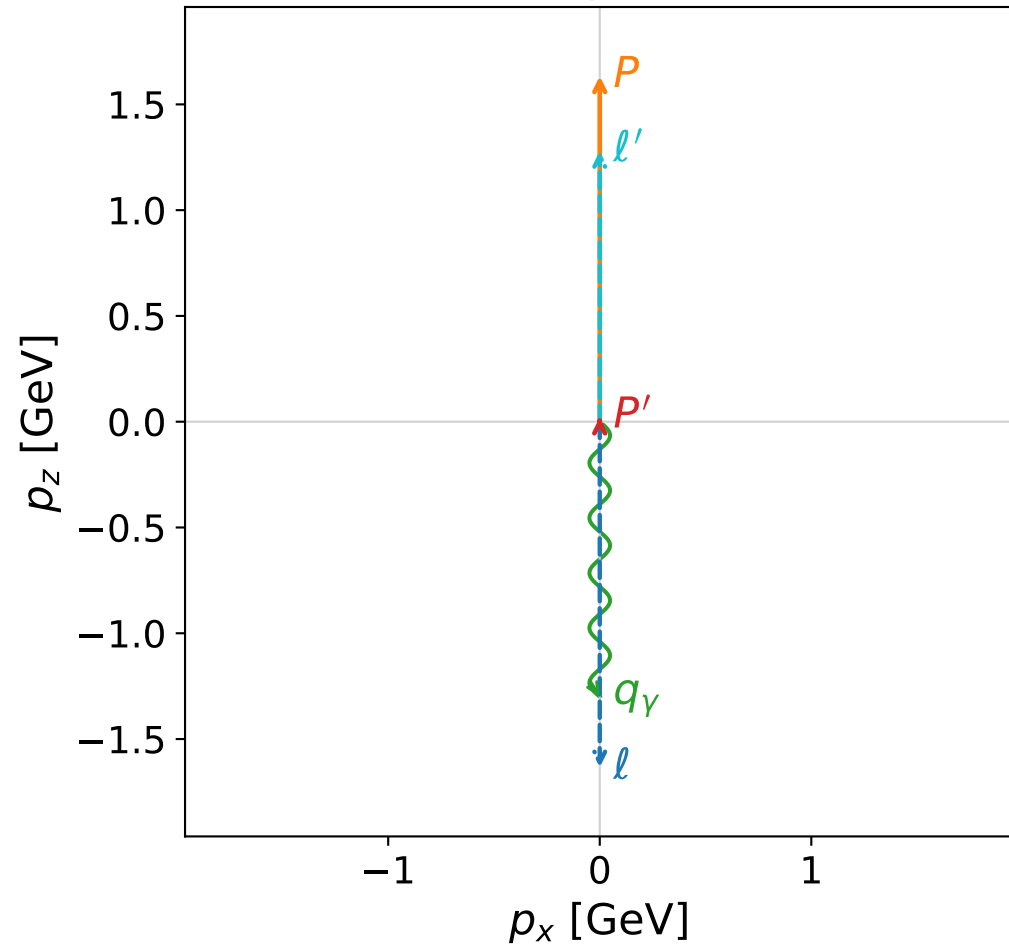
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8143$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8143$
 P $E= 2.0424$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8143$
 ℓ' $E= 1.4591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4591$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_Y $E= 1.4596$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4596$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=5.107\text{e+}18$; $I_h=+3.273\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=6.833\text{e-}14$; $I_{TL}=-2.548\text{e-}07$

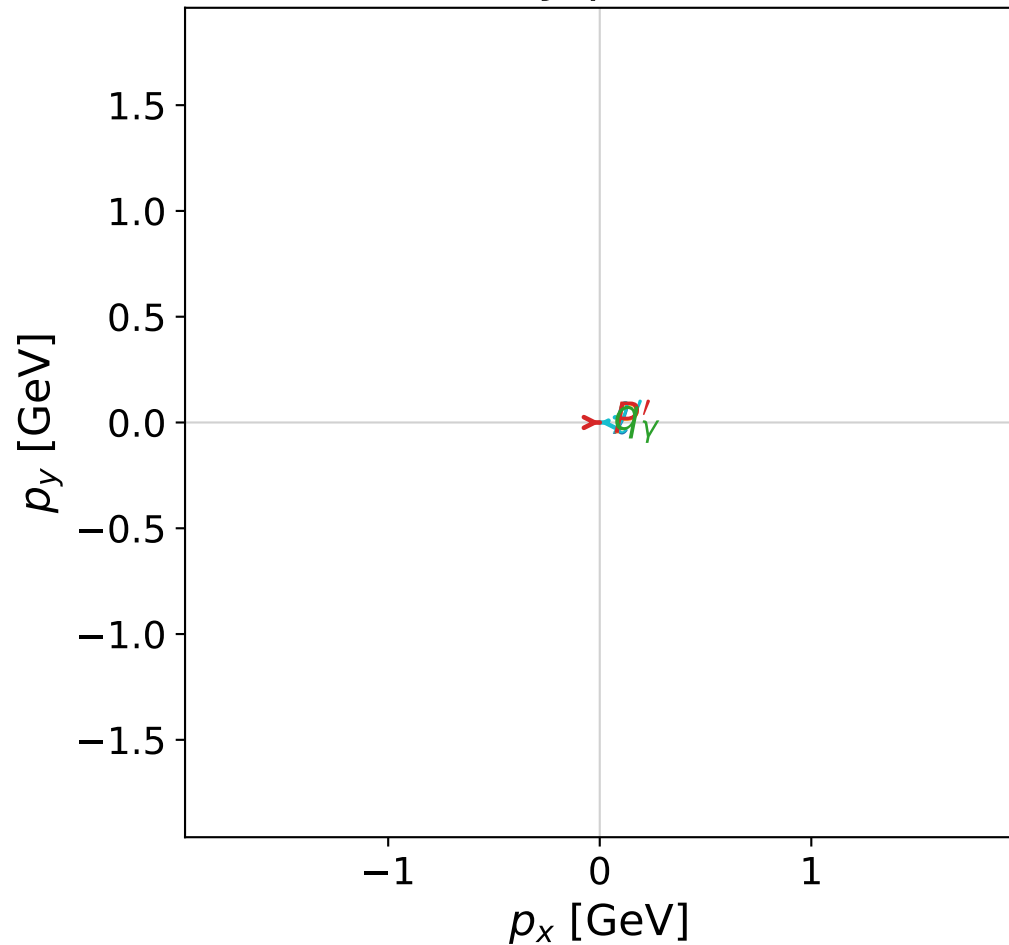


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



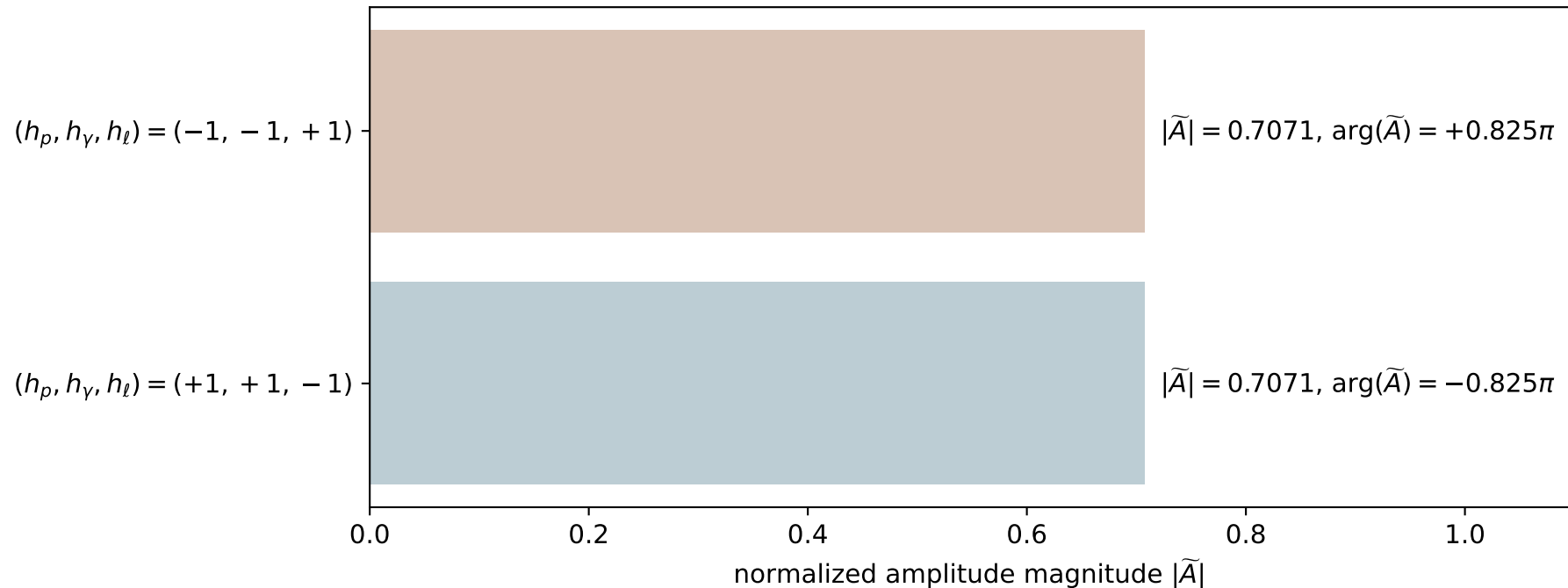
dghz_mixing_angles_polarization_01_local_minimum_140
 $d_{\text{GHZ}}=7.0516\text{e-}08$
 region: polarization_cluster_01_local_minimum_140
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0140
 lepton mixing angle: $\alpha_e=0.57080929$ rad
 incoming lepton: $0.841464|+\rangle +0.540313|-\rangle$
 proton mixing angle: $\alpha_p=0.57079935$ rad
 incoming proton: $0.841469|+\rangle +0.540305|-\rangle$

$s=12.3715$, $\sqrt{s}=3.51731$, $|\vec{P}|=1.63358$, $|\vec{P}'|=0.0205126$
 $\theta_{P'}=0.00635735$, $\theta_Y=3.14159$, $\phi_{P'}=6.28239$, $\phi_Y=5.7342$
 $E_Y=1.2998$, $\theta_{\ell'}=0.000101936$, $\phi_{\ell'}=3.1408$
 $Q^2=8.35928$, $x_B=0.770326$, $t=-1.70802$, $W^2=3.37217$, $y=0.944306$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.6336$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6336$
 P $E= 1.8837$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6336$
 ℓ' $E= 1.2793$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.2793$
 P' $E= 0.9382$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0205$
 q_Y $E= 1.2998$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.2998$

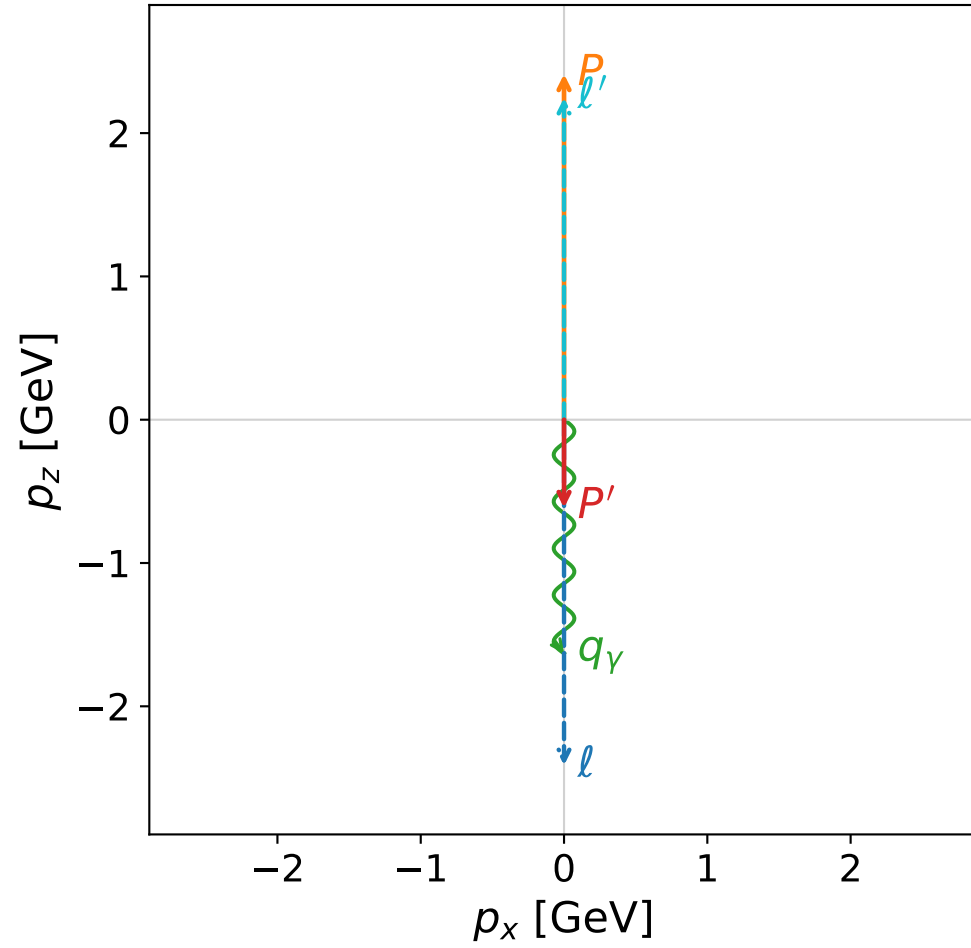
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.574\text{e+}15$; $I_h=+9.170\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.649\text{e-}10$; $I_{TL}=-7.552\text{e-}06$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

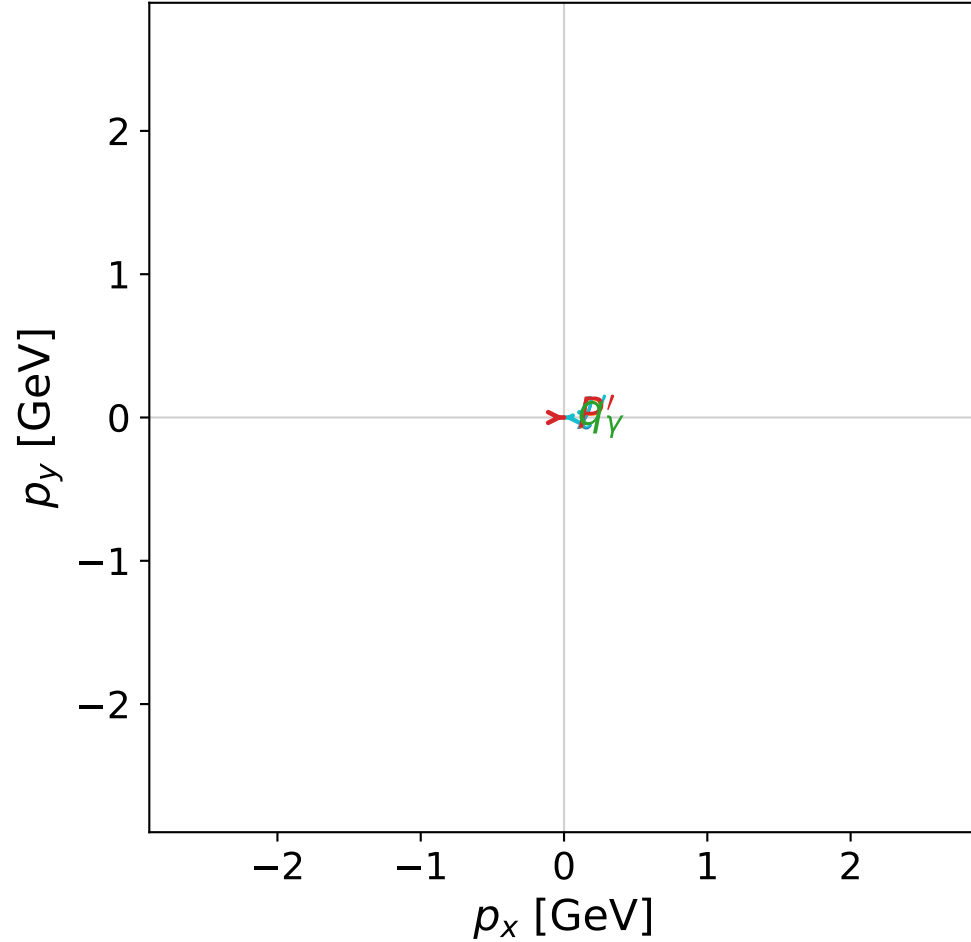


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

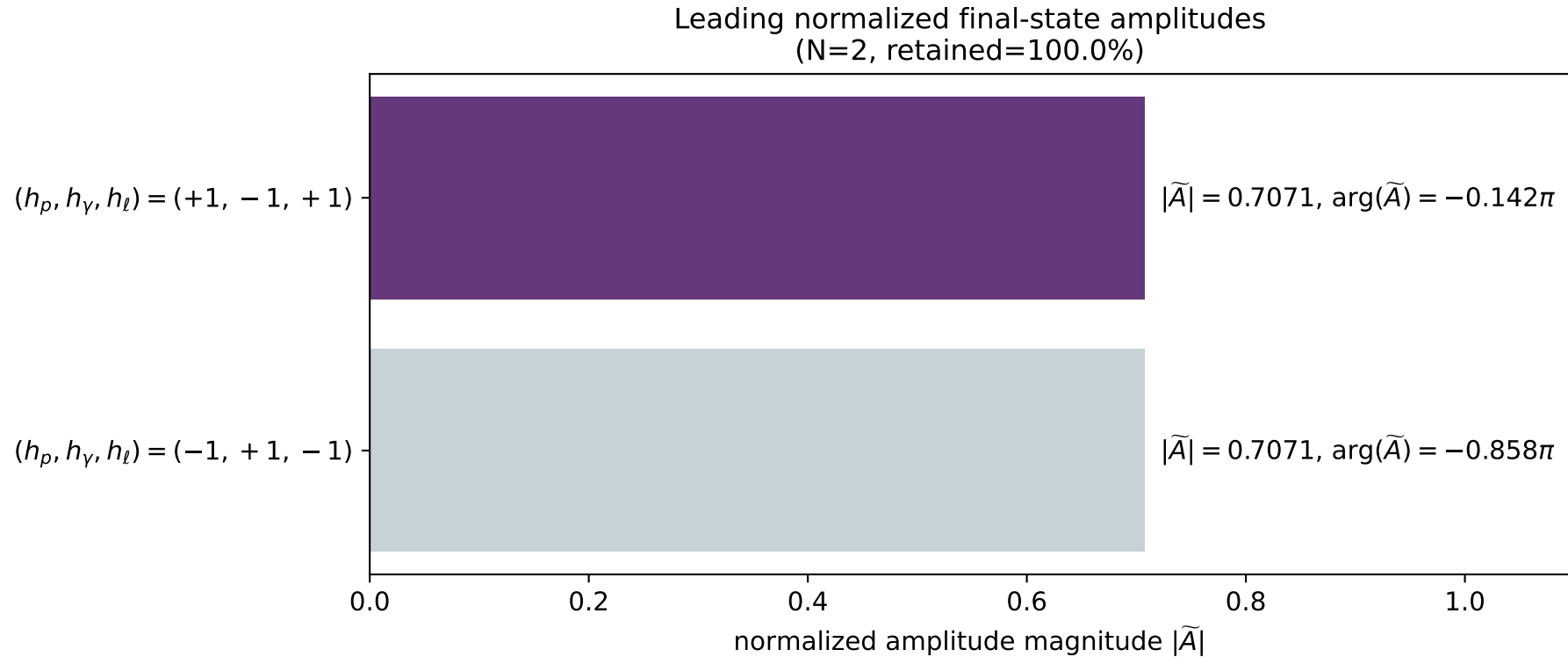


dghz_mixing_angles_polarization_01_local_minimum_146
 $d_{\text{GHZ}}=7.9425\text{e-}08$
 region: polarization_cluster_01_local_minimum_146
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0146
 lepton mixing angle: $\alpha_e=0.15440538$ rad
 incoming lepton: $0.988103|+\rangle +0.153793|-\rangle$
 proton mixing angle: $\alpha_p=0.15439576$ rad
 incoming proton: $0.988105|+\rangle +0.153783|-\rangle$

$s=24.9939$, $\sqrt{s}=4.99939$, $|\vec{P}|=2.4117$, $|\vec{P}'|=0.615551$
 $\theta_{p'}=3.14157$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.000416712$, $\phi_\gamma=2.69483$
 $E_\gamma=1.63095$, $\theta_{l'}=7.35473\text{e-}06$, $\phi_{l'}=3.14201$
 $Q^2=21.6715$, $x_B=0.929179$, $t=-7.01582$, $W^2=2.53163$, $y=0.967209$

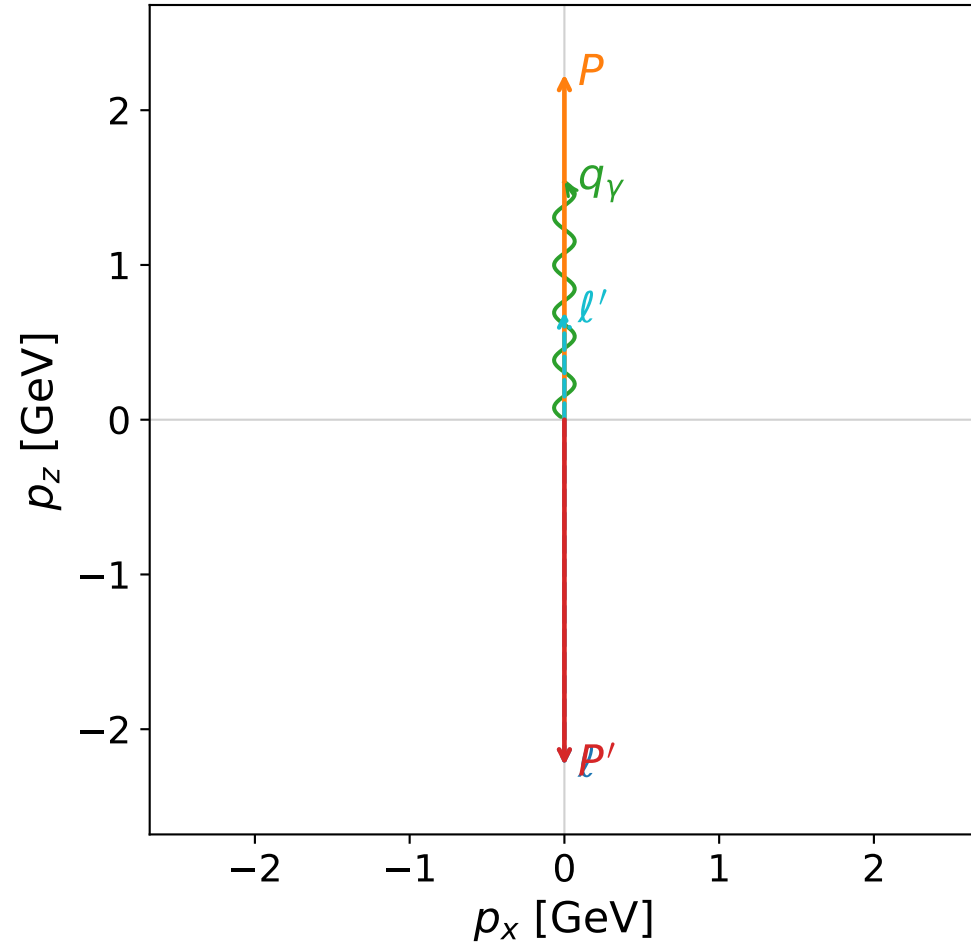
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4117$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4117$
 P $E= 2.5877$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4117$
 l' $E= 2.2465$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2465$
 P' $E= 1.1219$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6156$
 q_γ $E= 1.6309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.6309$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=7.190\text{e+}16$; $I_h=+3.613\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.567\text{e-}13$; $I_{TL}=-6.826\text{e-}07$

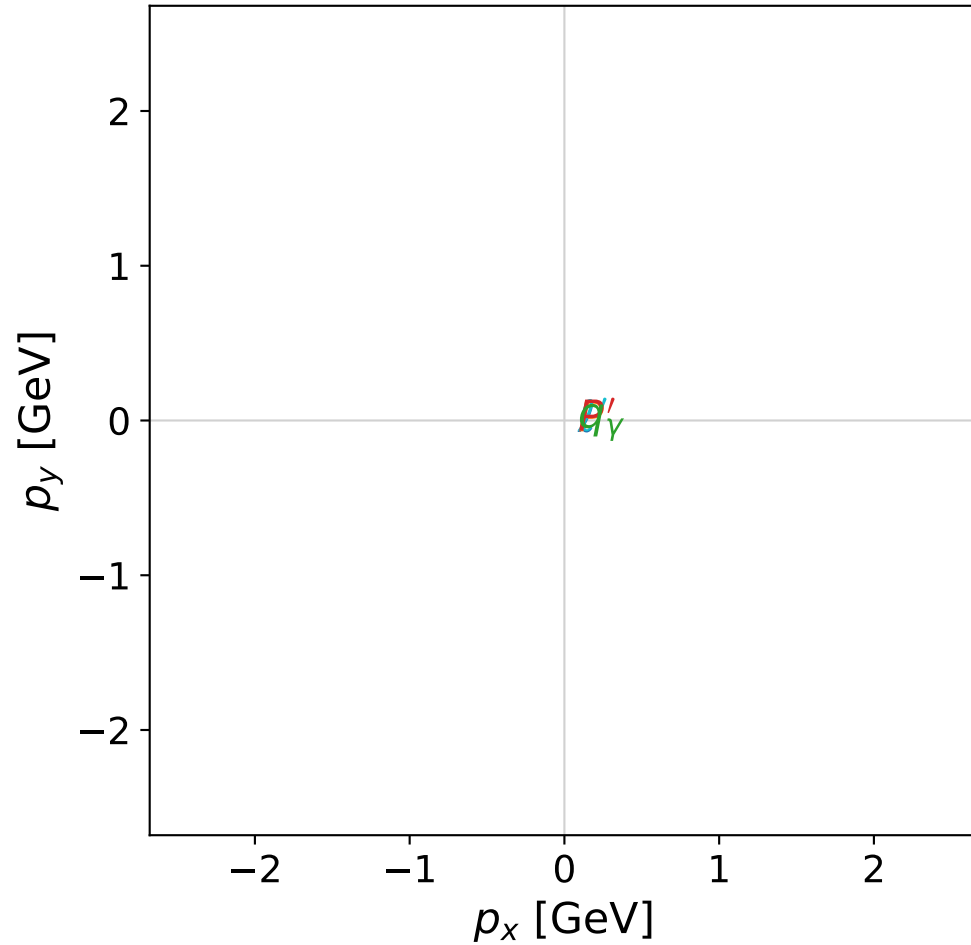


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

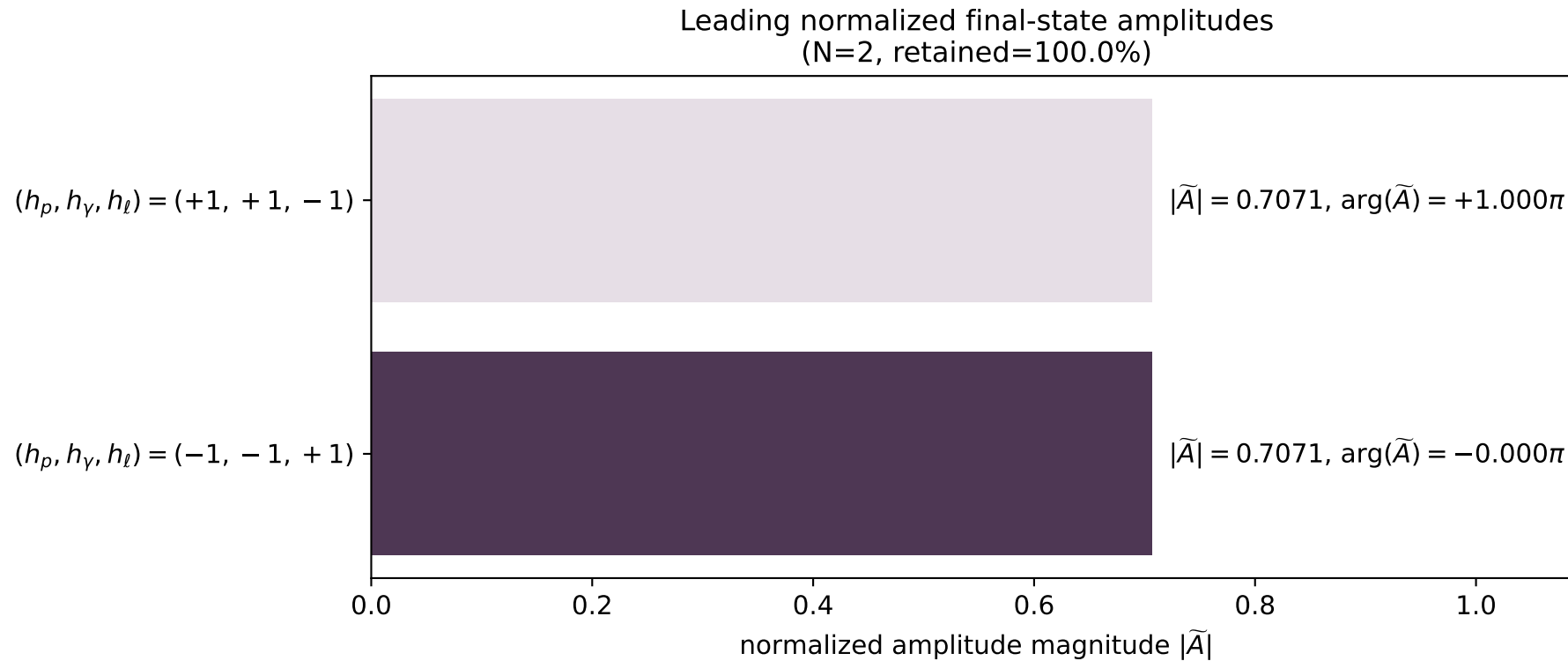


dghz_mixing_angles_polarization_01_local_minimum_149
 $d_{\text{GHZ}}=8.40053\text{e-}08$
 region: polarization_cluster_01_local_minimum_149
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0149
 lepton mixing angle: $\alpha_e=0.79062427$ rad
 incoming lepton: $0.703402|+\rangle + 0.710793|-\rangle$
 proton mixing angle: $\alpha_p=0.78016465$ rad
 incoming proton: $0.710798|+\rangle + 0.703396|-\rangle$

$s=21.6733$, $\sqrt{s}=4.65546$, $|\vec{P}|=2.23324$, $|\vec{P}'|=2.23324$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.62947$, $\phi_\gamma=0.0541682$
 $E_\gamma=1.53726$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.48787$
 $Q^2=6.2171$, $x_B=0.302824$, $t=-19.9494$, $W^2=15.1932$, $y=0.987349$

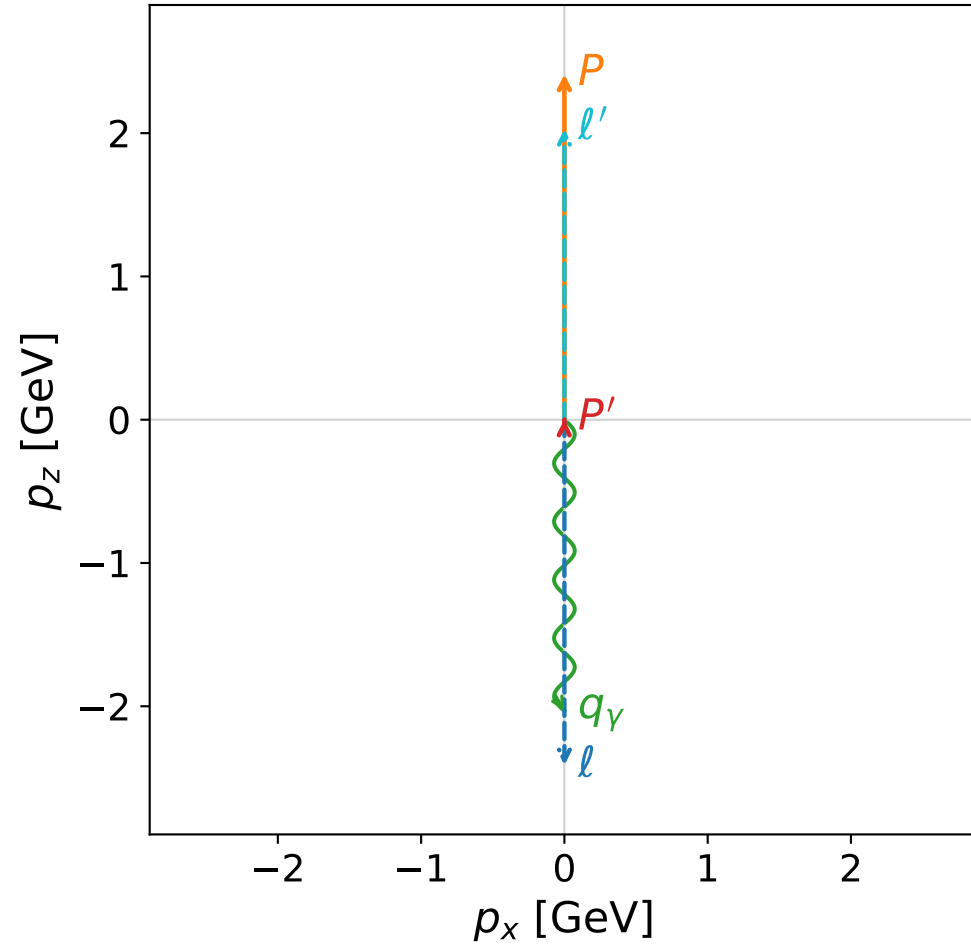
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2332$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2332$
 P $E= 2.4222$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2332$
 ℓ' $E= 0.6960$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6960$
 P' $E= 2.4222$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.2332$
 q_γ $E= 1.5373$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5373$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.507\text{e}+24$; $I_h=+1.772\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=1.570\text{e-}28$; $I_{TL}=+1.772\text{e-}14$

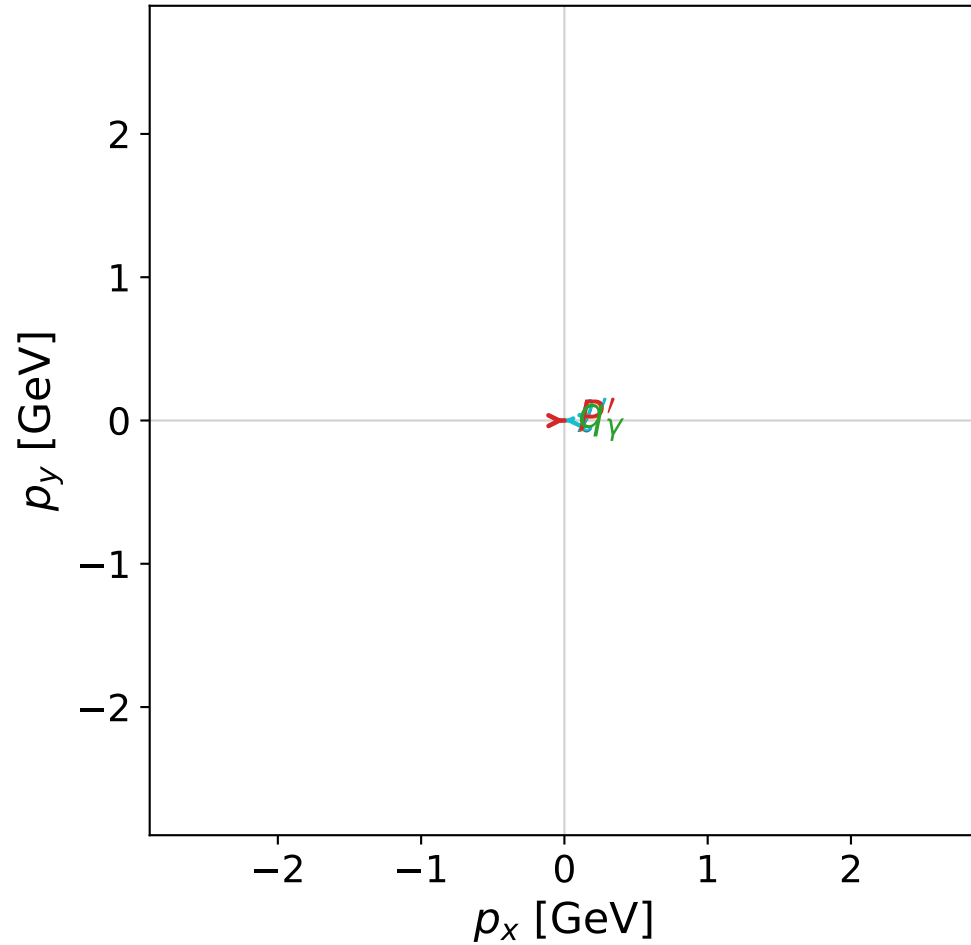


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



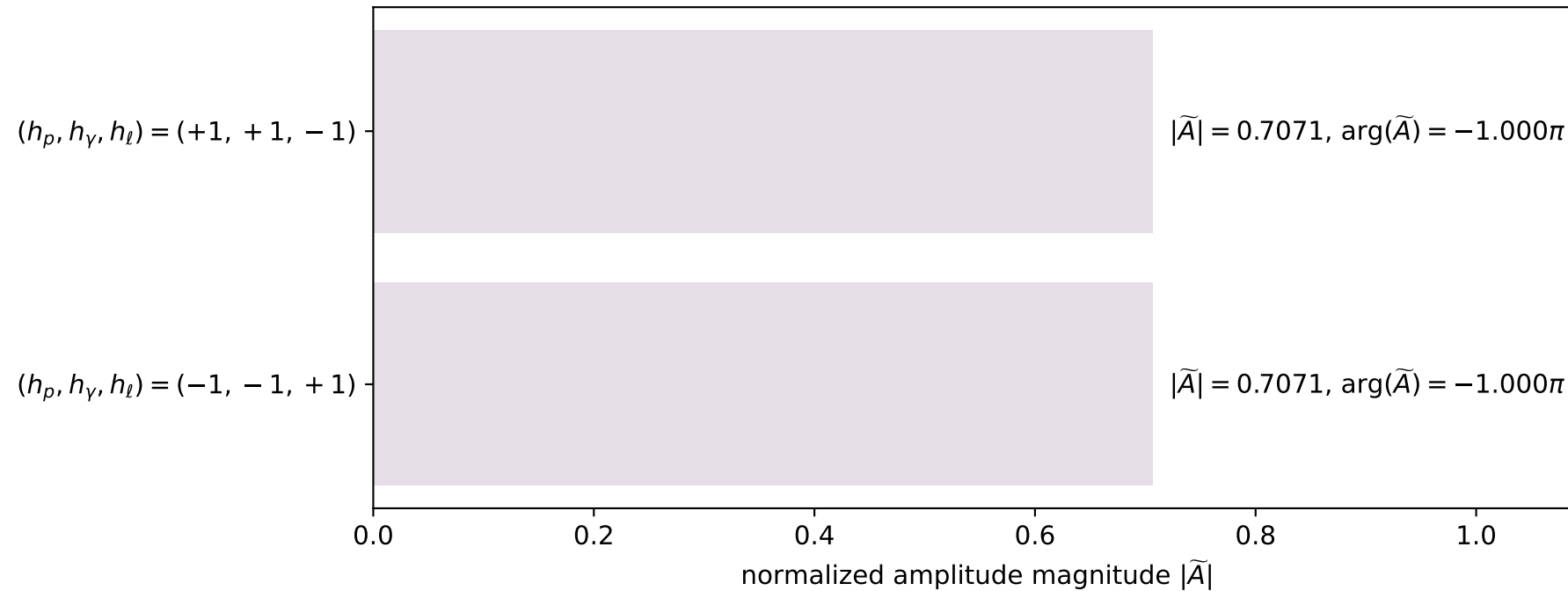
dghz_mixing_angles_polarization_01_local_minimum_151
 $d_{\text{GHZ}}=8.54757\text{e-}08$
 region: polarization_cluster_01_local_minimum_151
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0151
 lepton mixing angle: $\alpha_e=0.15707963$ rad
 incoming lepton: $0.987688|+\rangle +0.156434|-\rangle$
 proton mixing angle: $\alpha_p=0.15707963$ rad
 incoming proton: $0.987688|+\rangle +0.156434|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00283891$
 $\theta_{P'}=0.00981748$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00981748$, $\phi_\gamma=0$
 $E_\gamma=2.03242$, $\theta_{\ell'}=1.37321\text{e-}05$, $\phi_{\ell'}=3.15141$
 $Q^2=19.5815$, $x_B=0.836606$, $t=-3.0817$, $W^2=4.70422$, $y=0.970386$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0296$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0296$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0028$
 q_γ $E= 2.0324$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0324$

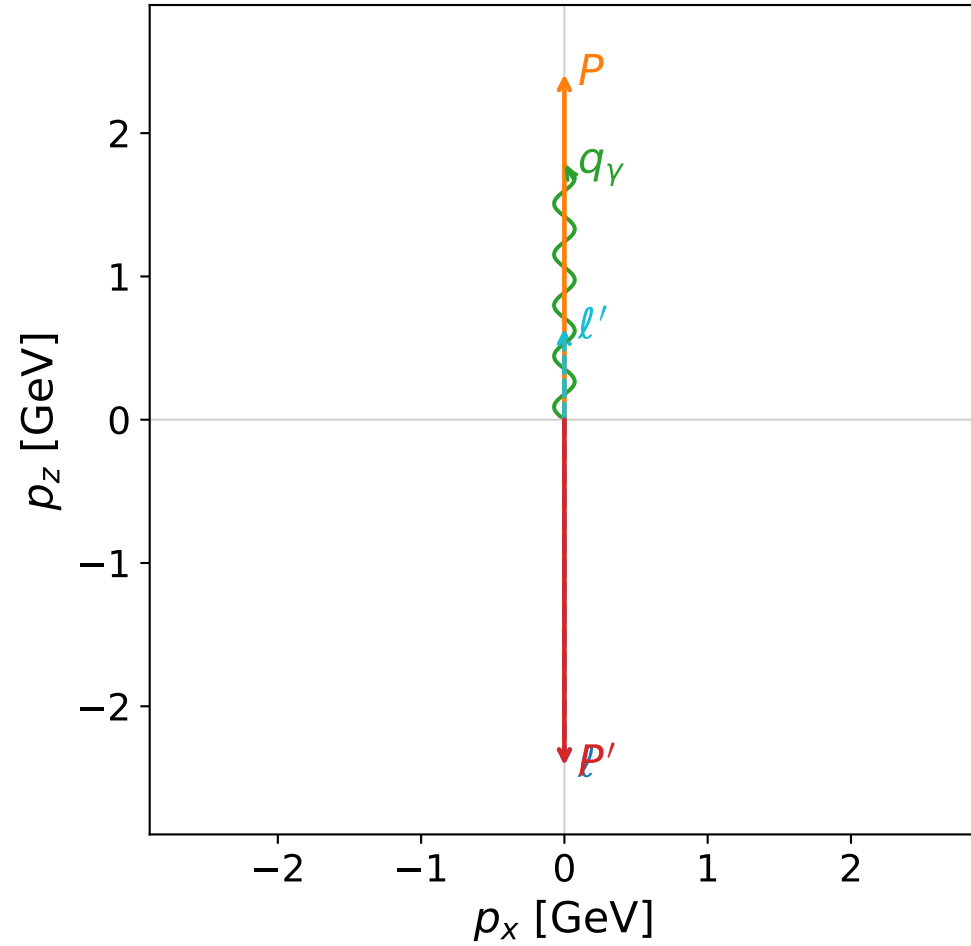
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.285\text{e+}16$; $I_h=+5.312\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=7.423\text{e-}12$; $I_{TL}=-3.664\text{e-}06$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

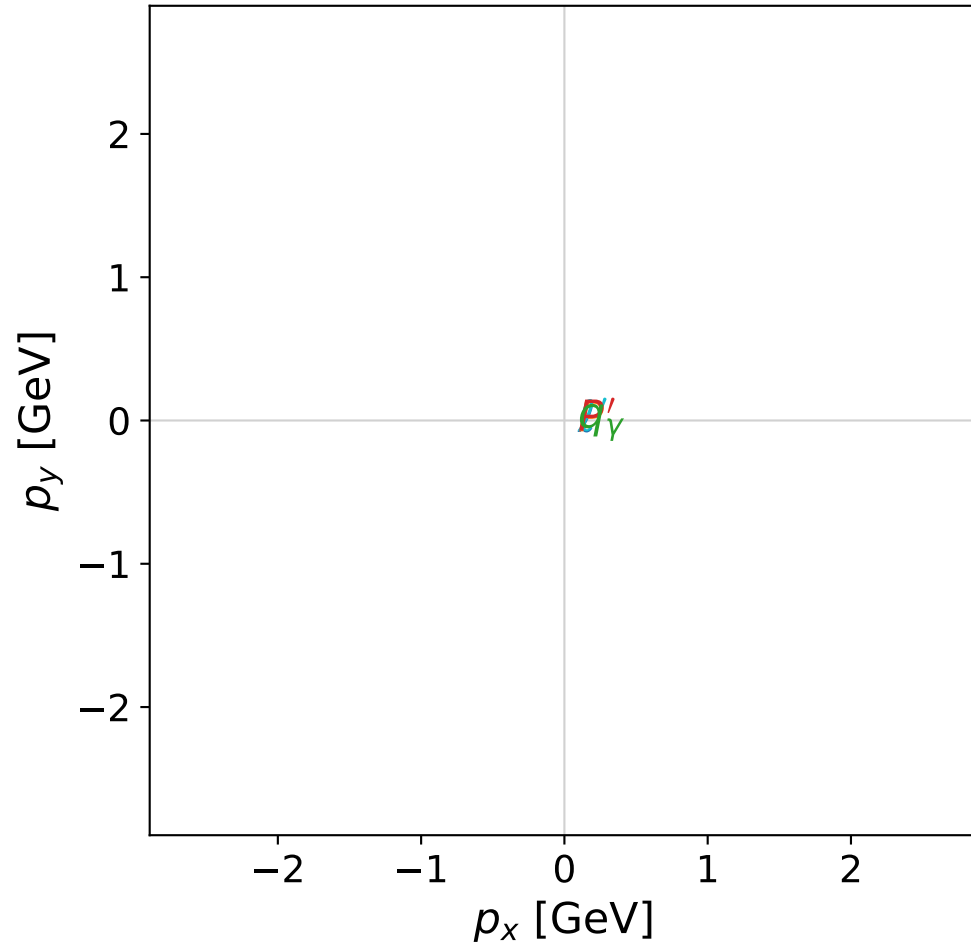


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

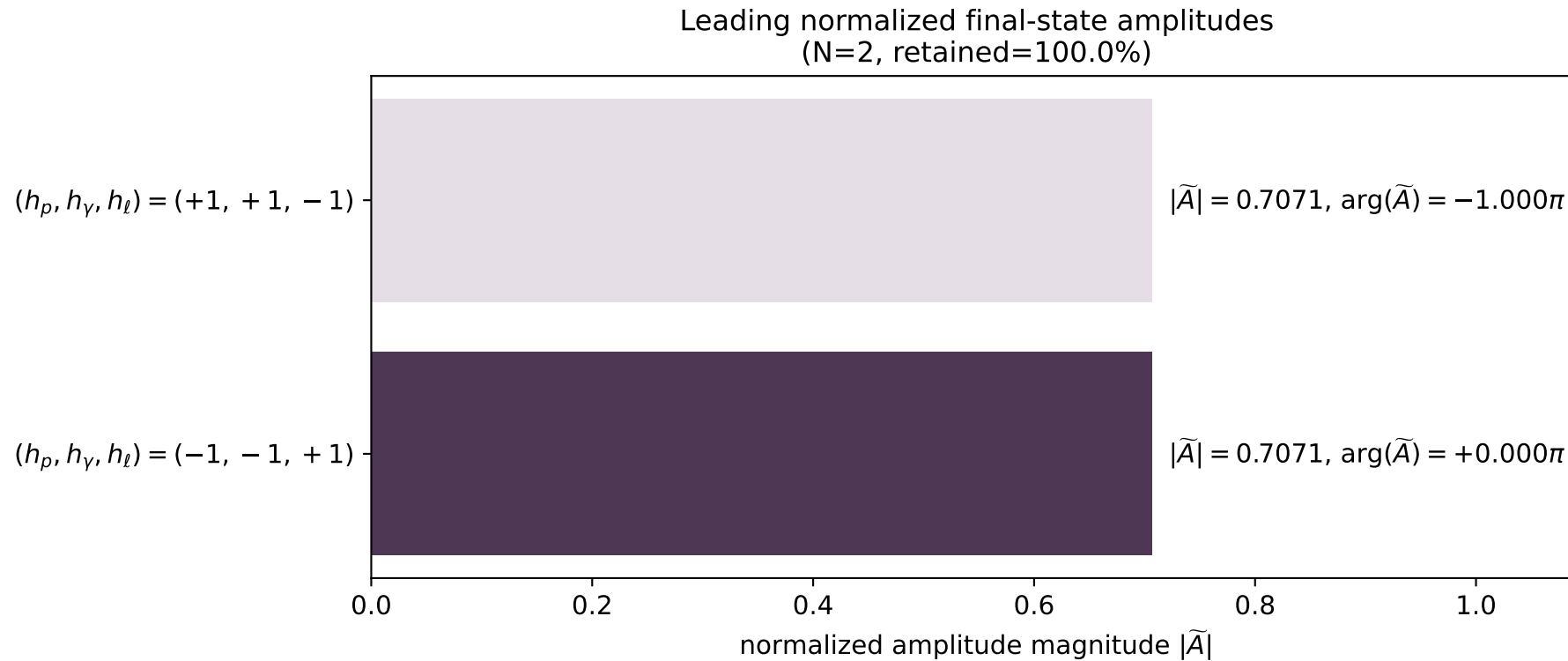


dghz_mixing_angles_polarization_01_local_minimum_152
 $d_{\text{GHZ}}=8.63787\text{e-}08$
 region: polarization_cluster_01_local_minimum_152
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0152
 lepton mixing angle: $\alpha_e=0.69075328$ rad
 incoming lepton: $0.770766|+\rangle +0.637118|-\rangle$
 proton mixing angle: $\alpha_p=0.88003373$ rad
 incoming proton: $0.637125|+\rangle +0.77076|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.42356$, $\phi_\gamma=5.97423$
 $E_\gamma=1.77397$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.56515$
 $Q^2=6.15594$, $x_B=0.257618$, $t=-23.2713$, $W^2=18.6195$, $y=0.99069$

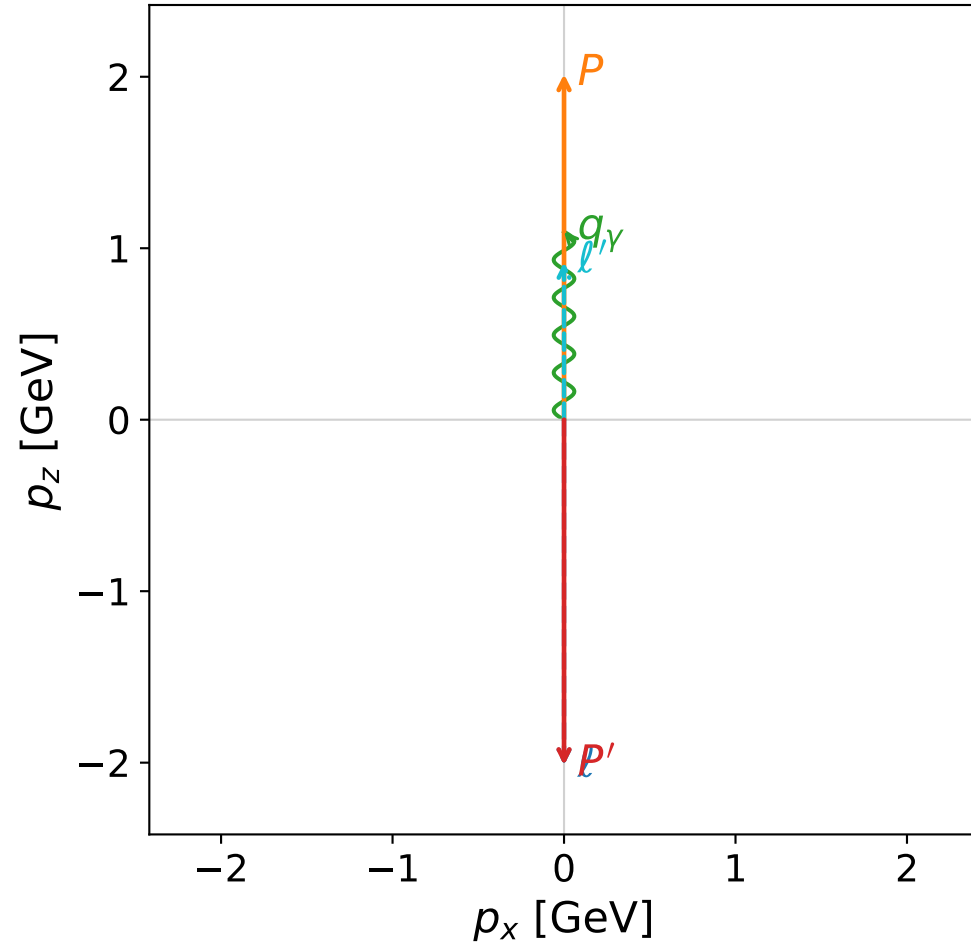
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 0.6380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.6380$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 1.7740$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7740$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.513\text{e}+24$; $I_h=-8.297\text{e-}15$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=3.568\text{e-}29$; $I_{TL}=-8.296\text{e-}15$

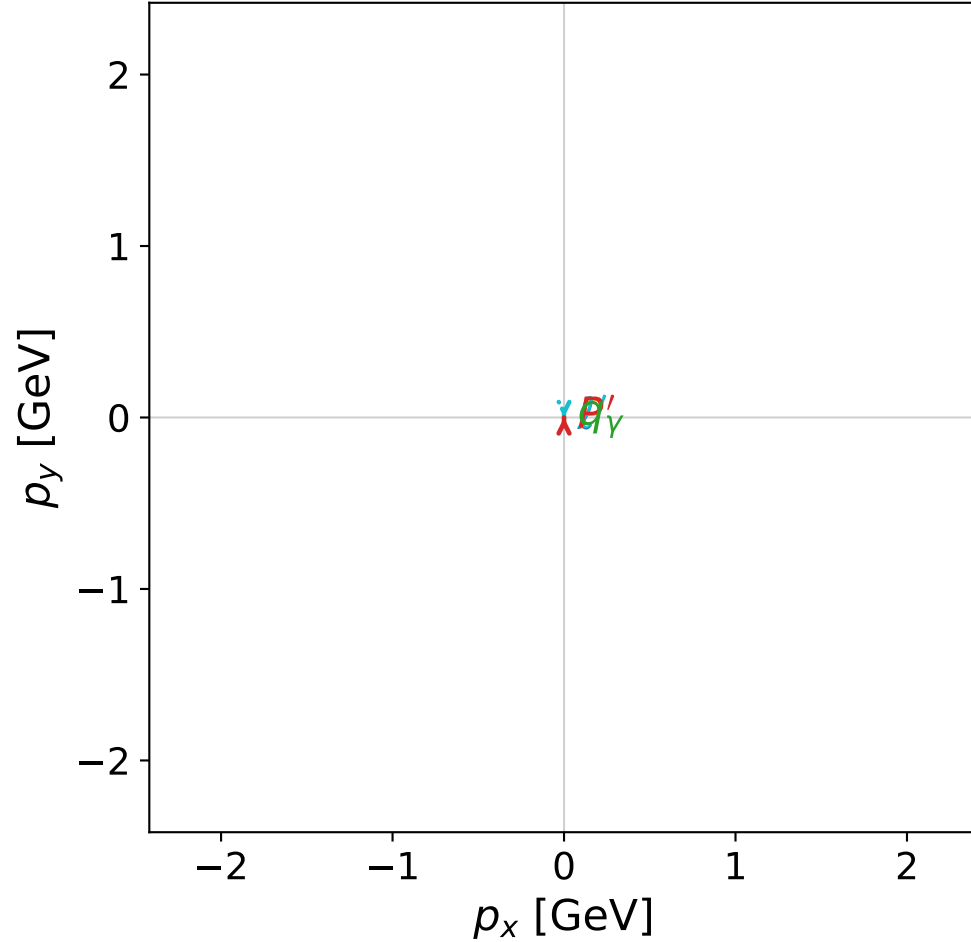


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

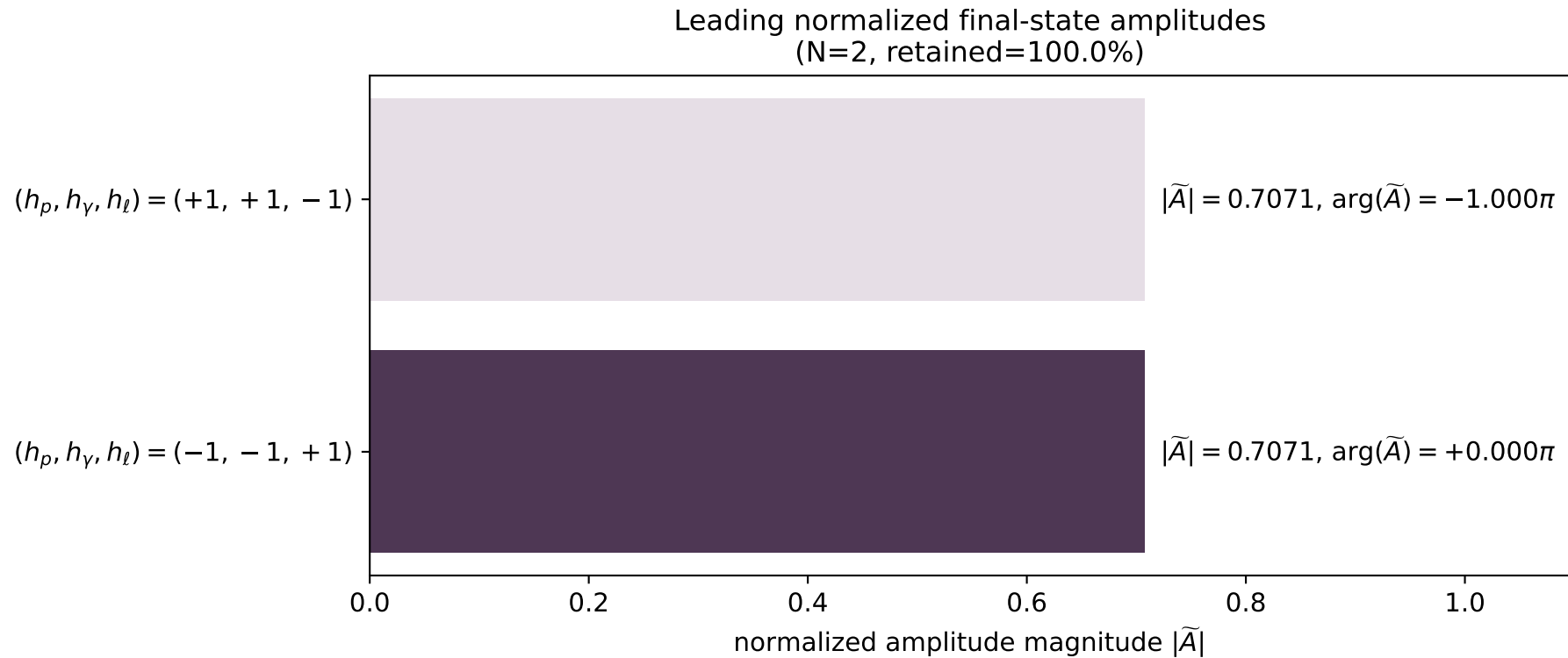


dghz_mixing_angles_polarization_01_local_minimum_163
 $d_{\text{GHZ}}=1.18968\text{e-}07$
 region: polarization_cluster_01_local_minimum_163
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0163
 lepton mixing angle: $\alpha_e=0.31723973$ rad
 incoming lepton: $0.9501|+\rangle + 0.311945|-\rangle$
 proton mixing angle: $\alpha_p=1.2535406$ rad
 incoming proton: $0.31196|+\rangle + 0.950095|-\rangle$

$s=17.9637$, $\sqrt{s}=4.23836$, $|\vec{P}|=2.01539$, $|\vec{P}'|=2.01539$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.26708$, $\phi_\gamma=1.45652$
 $E_\gamma=1.09647$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.40868$
 $Q^2=7.40786$, $x_B=0.443522$, $t=-16.2471$, $W^2=10.1743$, $y=0.977668$

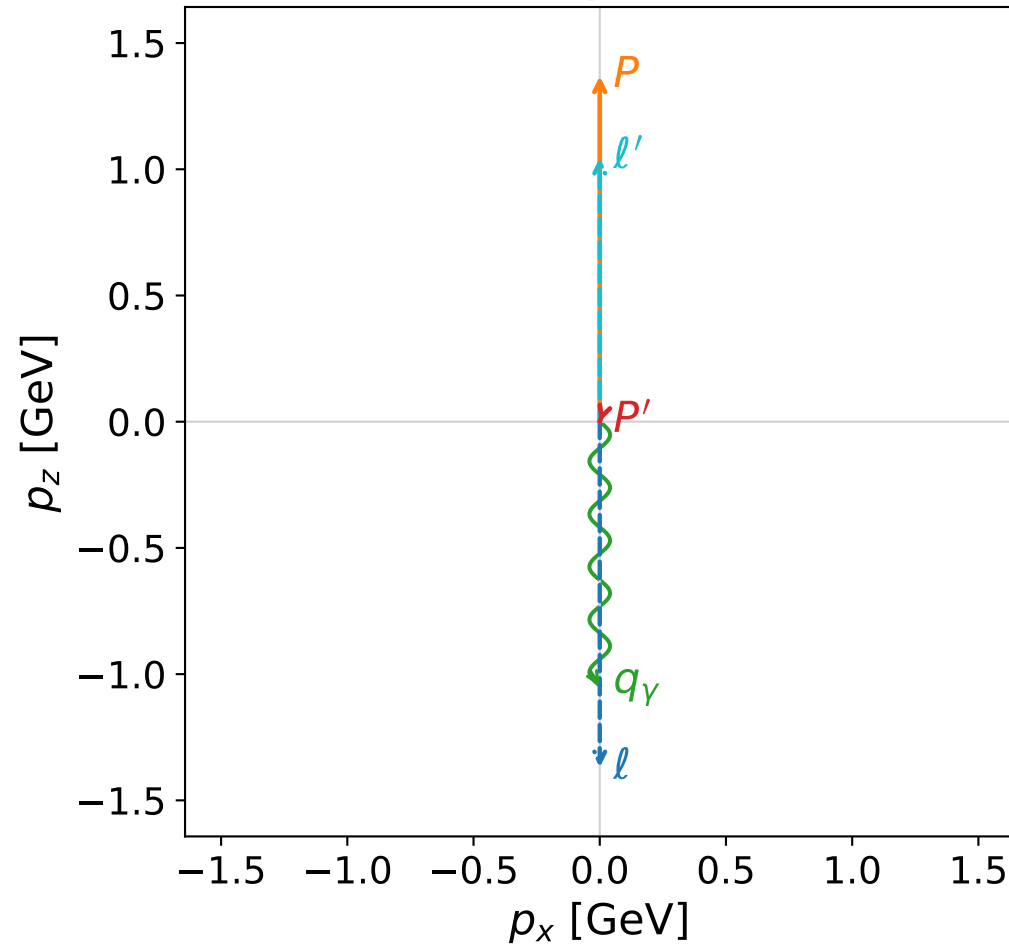
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0154$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0154$
 P $E= 2.2230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0154$
 ℓ' $E= 0.9189$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.9189$
 P' $E= 2.2230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0154$
 q_γ $E= 1.0965$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0965$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=1.054\text{e}+24$; $I_h=+2.596\text{e-}14$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.590\text{e-}28$; $I_{TL}=+2.596\text{e-}14$

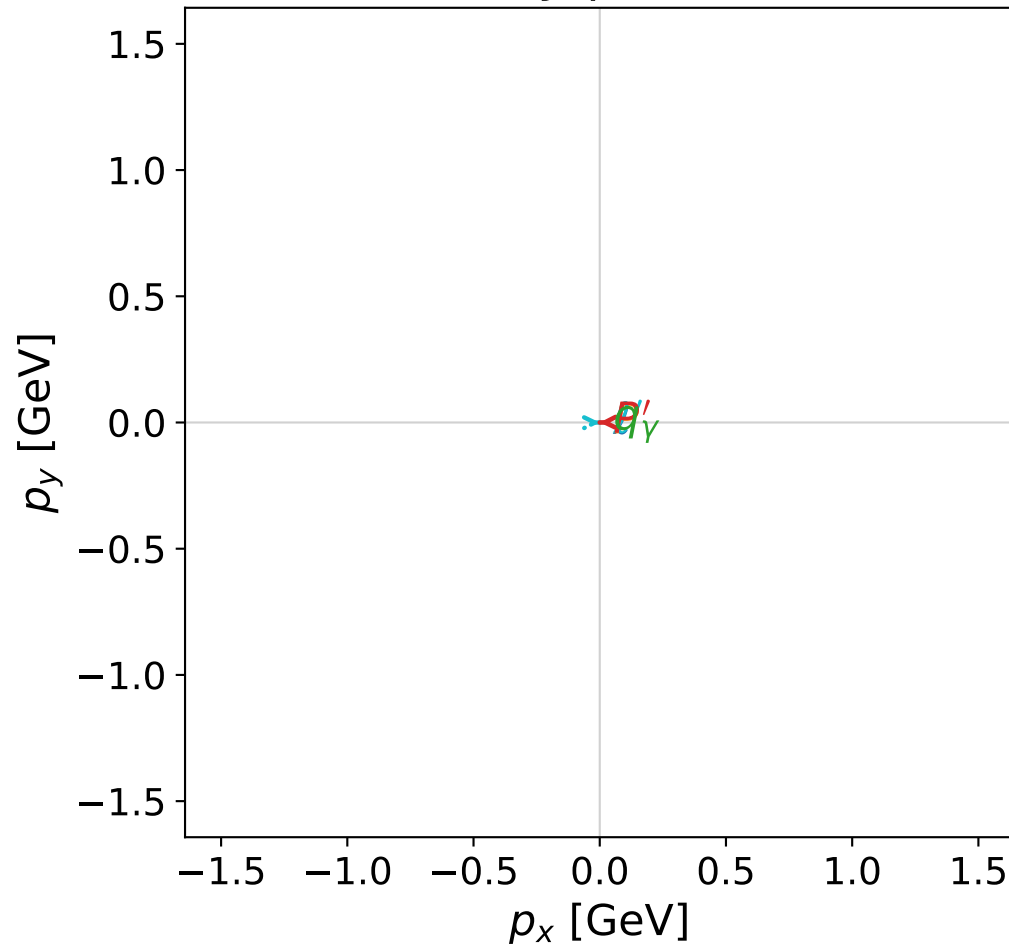


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



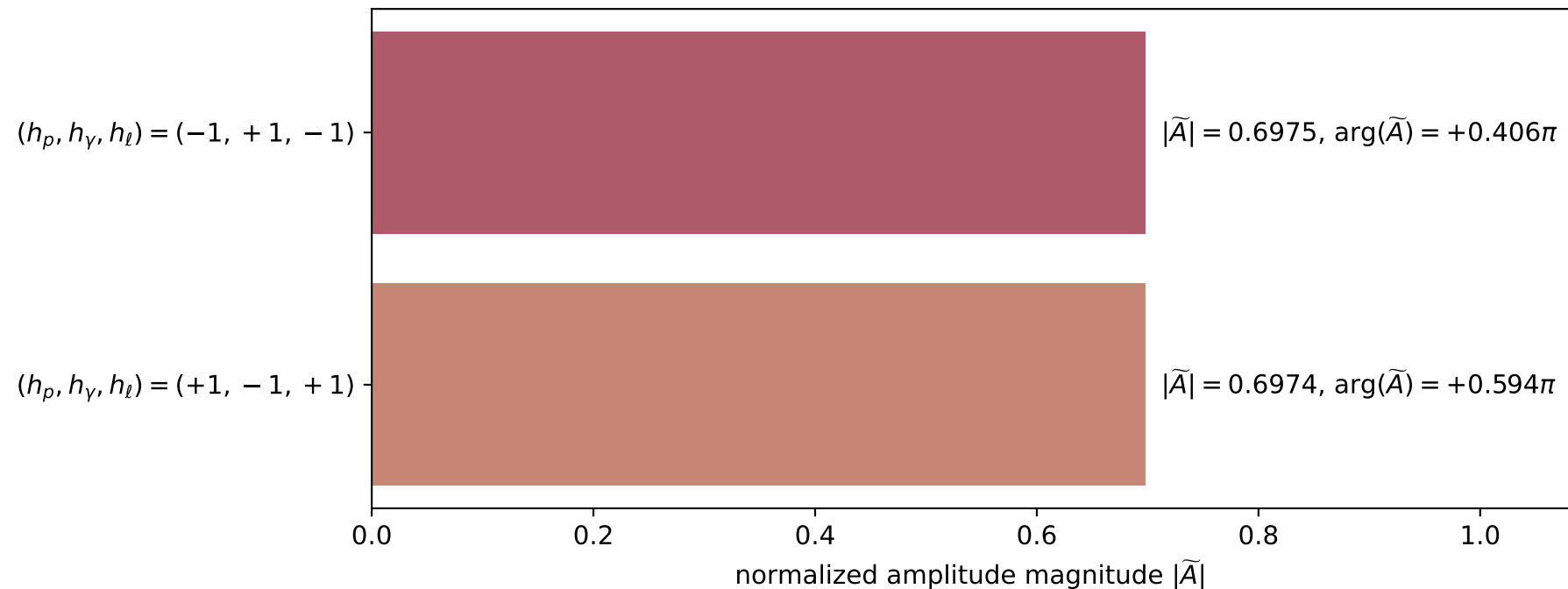
dghz_mixing_angles_polarization_01_local_minimum_165
 $d_{\text{GHZ}}=1.20835\text{e-}07$
 region: polarization_cluster_01_local_minimum_165
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0165
 lepton mixing angle: $\alpha_e=0.40930724$ rad
 incoming lepton: $0.917397|+\rangle + 0.397974|-\rangle$
 proton mixing angle: $\alpha_p=0.40933059$ rad
 incoming proton: $0.917387|+\rangle + 0.397995|-\rangle$

$s=9.17245$, $\sqrt{s}=3.0286$, $|\vec{P}|=1.36905$, $|\vec{P}'|=7.26171\text{e-}05$
 $\theta_{p'}=2.81094$, $\theta_Y=3.14159$, $\phi_{p'}=3.1542$, $\phi_Y=1.87819$
 $E_Y=1.04527$, $\theta_{\ell'}=2.25534\text{e-}05$, $\phi_{\ell'}=0.0126103$
 $Q^2=5.72446$, $x_B=0.744864$, $t=-1.35383$, $W^2=2.84062$, $y=0.926758$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3690$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3690$
 P $E= 1.6596$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3690$
 ℓ' $E= 1.0453$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0453$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0001$
 q_Y $E= 1.0453$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.0453$

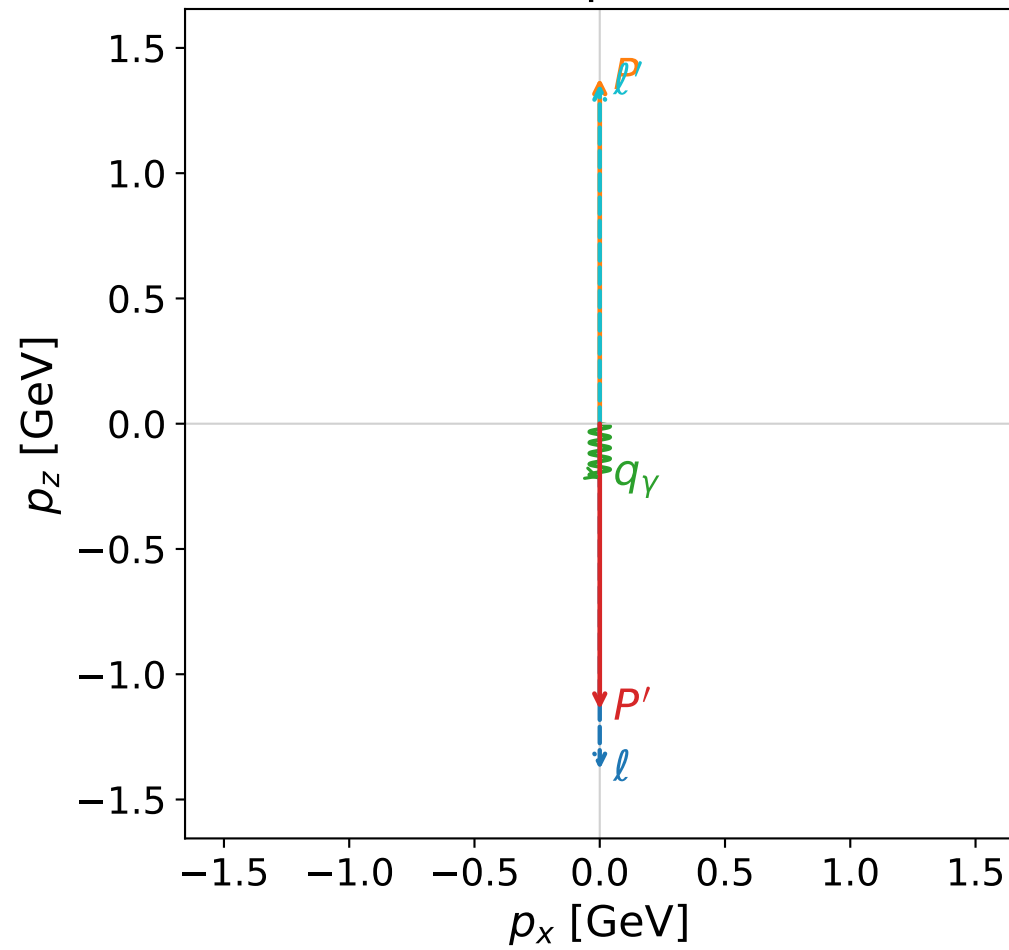
GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=2.024\text{e+}16$; $I_h=-5.157\text{e-}09$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{\text{L/T}}=1.933\text{e-}11$; $I_{\text{TL}}=+4.248\text{e-}06$

Leading normalized final-state amplitudes
 (N=2, retained=97.3%)

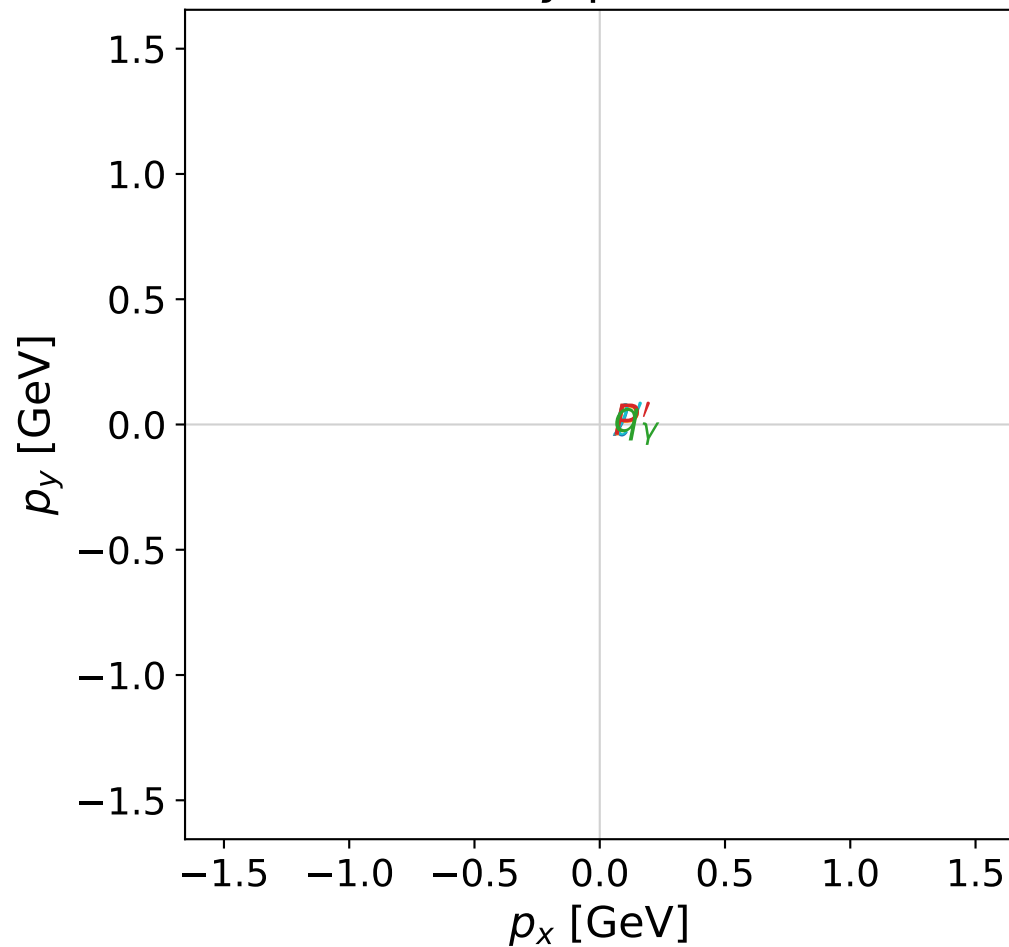


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

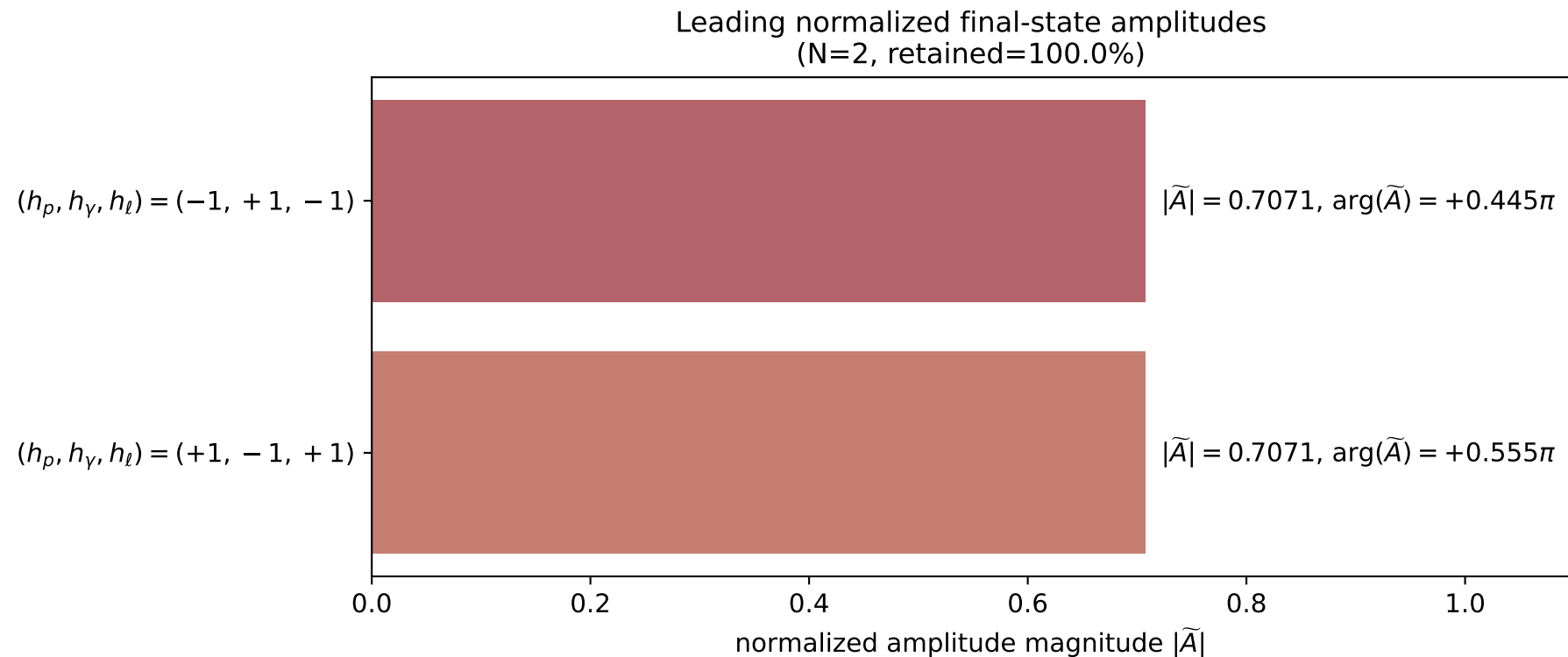


dghz_mixing_angles_polarization_01_local_minimum_170
 $d_{\{\rm GHZ\}}=1.94761\text{e-}07$
 region: polarization_cluster_01_local_minimum_170
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0170
 lepton mixing angle: $\alpha_e=0.18329382$ rad
 incoming lepton: $0.983249|+\rangle +0.182269|-\rangle$
 proton mixing angle: $\alpha_p=0.18329532$ rad
 incoming proton: $0.983248|+\rangle +0.182271|-\rangle$

$s=9.28794$, $\sqrt{s}=3.04761$, $|\vec{P}|=1.37946$, $|\vec{P}'|=1.14077$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.38865$, $\phi_\gamma=4.88405$
 $E_\gamma=0.214975$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.16901$
 $Q^2=7.48078$, $x_B=0.981048$, $t=-6.31497$, $W^2=1.02436$, $y=0.906898$

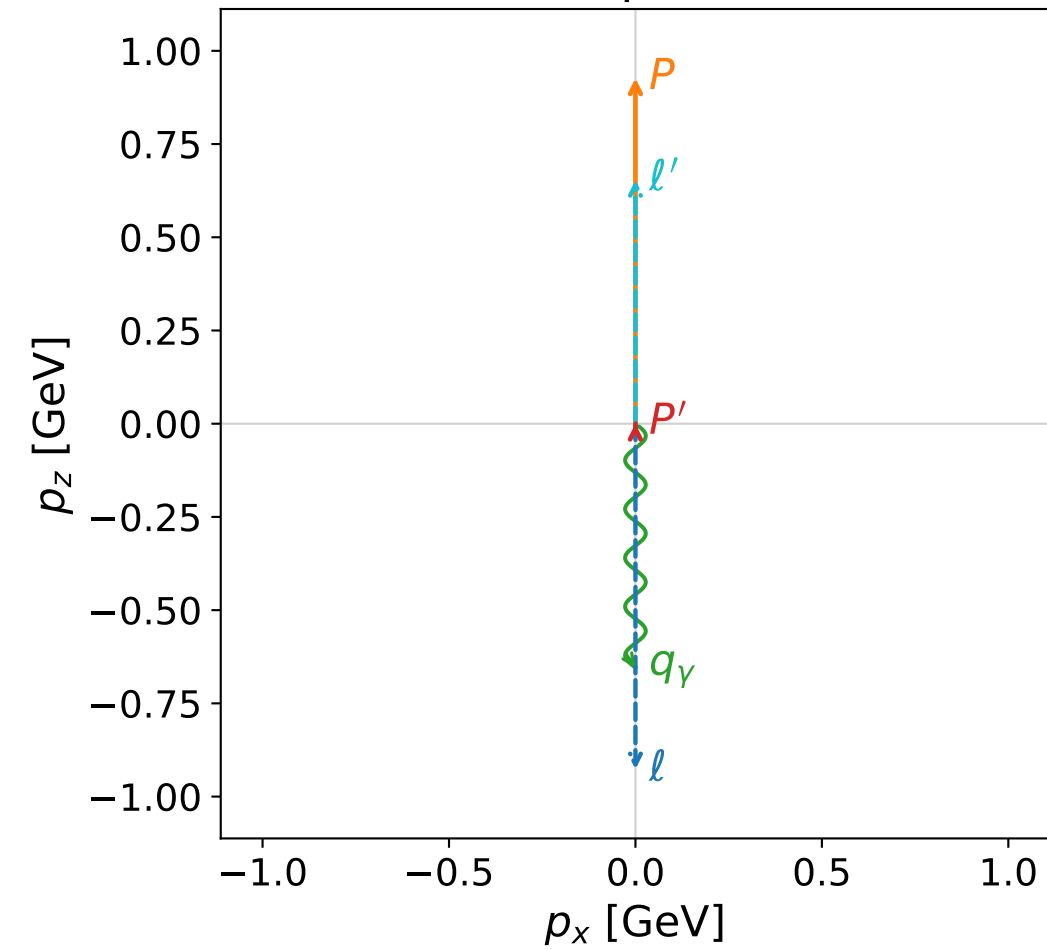
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3795$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3795$
 P $E= 1.6682$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3795$
 ℓ' $E= 1.3557$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.3557$
 P' $E= 1.4769$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.1408$
 q_γ $E= 0.2150$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.2150$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=8.902\text{e}+21$; $I_h=-1.593\text{e-}12$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=9.350\text{e-}27$; $I_{TL}=+2.488\text{e-}17$

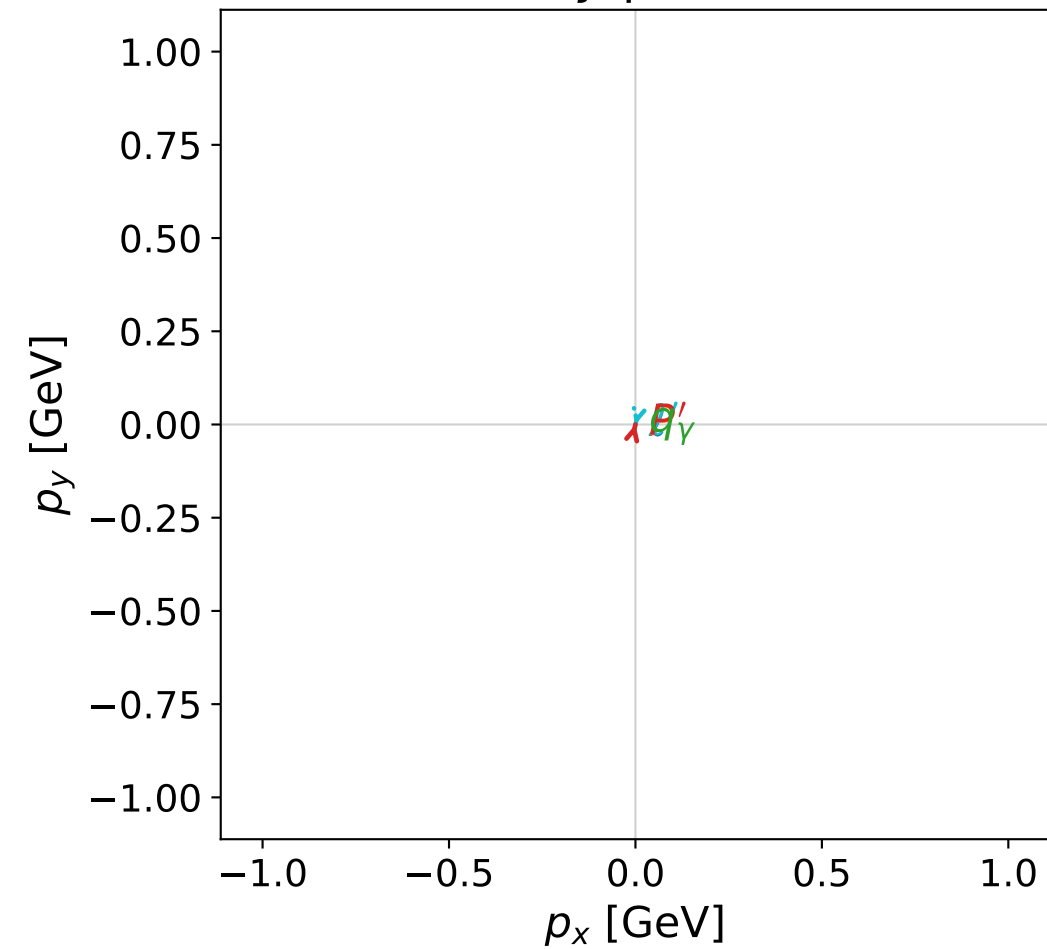


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_01_local_minimum_172
 $d_{\{\text{rm GHZ}\}}=2.15034\text{e-}07$
 region: polarization_cluster_01_local_minimum_172
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0172
 lepton mixing angle: $\alpha_e=0.74567333$ rad
 incoming lepton: $0.734631|+\rangle +0.678467|-\rangle$
 proton mixing angle: $\alpha_p=0.74572493$ rad
 incoming proton: $0.734596|+\rangle +0.678505|-\rangle$

$s=5.04216$, $\sqrt{s}=2.24548$, $|\vec{P}|=0.926823$, $|\vec{P}'|=0.000424397$
 $\theta_{p'}=0.0244936$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.33005$, $\phi_\gamma=2.86593$
 $E_\gamma=0.65395$, $\theta_{l'}=1.59044\text{e-}05$, $\phi_{l'}=4.47164$
 $Q^2=2.42281$, $x_B=0.663752$, $t=-0.713318$, $W^2=2.10721$, $y=0.876957$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.9268$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9268$
 P $E= 1.3187$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9268$
 l' $E= 0.6535$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.6535$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0004$
 q_γ $E= 0.6539$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.6539$

GHZ mechanism: helicity flipping (full H G C)
 helicity powers: flip 100.000%, conserve 0.000%
 $R_{\text{flip/conserve}}=3.423\text{e+}16$; $I_h=+1.178\text{e-}10$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=1.366\text{e-}11$; $I_{TL}=-9.879\text{e-}08$

